mielke-geometrodynamics: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE's — a formula has none of its own. A line's confidence is not a formula's.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
mielke-geometrodynamics_F00001	7	1.000	A	A	potential 1-form A via electron interference.
mielke-geometrodynamics_F00002	7	0.980	\operatorname{SL}(5, \mathbb{R})	$\operatorname{SL}(5, \mathbb{R})$	model with a primordial $\mathrm{SL}(5, \mathbb{R})$ gauge group is presented. After a tiny
mielke-geometrodynamics_F00003	15	0.898	\mathbb{R}	$\mathbb{R}$	**13 Topological SL(5, $\mathbb{R}$) Gauge-Invariant Action**
mielke-geometrodynamics_F00004	17	1.000	\beta	β	and Fermi's theory of weak interactions, being the cause for the β-decay of certain
mielke-geometrodynamics_F00005	17	0.965	\mathrm{W}^{ \pm}	$\mathrm{W}^{ \pm}$	empirical evidence by the detection of the intermediating $\mathrm{W}^{\pm}$ and Z^{o} bosons as the
mielke-geometrodynamics_F00006	17	0.965	\mathrm{Z}^{\mathrm{o}}	Z^{o}	empirical evidence by the detection of the intermediating $\mathrm{W}^{\pm}$ and Z^{o} bosons as the
mielke-geometrodynamics_F00007	19	1.000	1 \times 10^{-12}	1×10^{-12}	mass is measured nowadays with a precision better than 1×10^{-12}; this accuracy
mielke-geometrodynamics_F00008	20	1.000	\partial_{j}	∂_{j}	to the electromagnetic fields if the partial derivatives ∂_j in the Dirac equation are
mielke-geometrodynamics_F00009	21	1.000	D_{j}:=\partial_{j}+i A_{j}	$D_{j}:=\partial_{j}+i A_{j}$	replaced by covariant derivatives $D_j := \partial_j + i A_j$. In the latter concept, a "minimal"
mielke-geometrodynamics_F00010	21	1.000	A_{j}	A_{j}	substitution of the electromagnetic vector potential A_j is taken into consideration.
mielke-geometrodynamics_F00011	21	1.000	\psi \rightarrow \mathrm{e}^{\mathrm{i} \theta} \psi	$\psi \rightarrow \mathrm{e}^{\mathrm{i} \theta} \psi$	remains invariant under the *local* transformation $\psi \rightarrow \mathrm{e}^{\mathrm{i}\theta}\psi$ of the phase of the
mielke-geometrodynamics_F00012	21	1.000	\psi	ψ	wave function ψ and the simultaneous transformation $A_j \rightarrow A_j - \partial_j\theta$ of the four-
mielke-geometrodynamics_F00013	21	1.000	A_{j} \rightarrow A_{j}-\partial_{j} \theta	$A_{j} \rightarrow A_{j}-\partial_{j} \theta$	wave function ψ and the simultaneous transformation $A_j \rightarrow A_j - \partial_j\theta$ of the four-
mielke-geometrodynamics_F00014	22	0.526	\mathrm{n}>4	$\mathrm{n}>4$	($\mathrm{n} > 4$) spaces and was, to a certain extent, anticipated by the pioneering works of
mielke-geometrodynamics_F00015	23	0.343	{ }^{+}	${ }^{+}$	an SU(3) nonet that consists of the massive spin-2^{+} mesons f, f$'$, A_2, $\mathrm{K}^*(1420)$.
mielke-geometrodynamics_F00016	23	0.343	\mathrm{A}_{2}, \mathrm{~K}^{*}(1420)	$\mathrm{A}_{2}, \mathrm{~K}^{*}(1420)$	an SU(3) nonet that consists of the massive spin-2^{+} mesons f, f$'$, A_2, $\mathrm{K}^*(1420)$.
mielke-geometrodynamics_F00017	23	0.987	\operatorname{SL}(2, \mathbb{C})	$\operatorname{SL}(2, \mathbb{C})$	$\mathrm{SL}(2, \mathbb{C})$, respectively. If we apply these findings to *strong gravity*, then it is obvious
mielke-geometrodynamics_F00018	23	0.697	\operatorname{SU}(3)	$\operatorname{SU}(3)$	to tie together the internal symmetries, such as the SU(3) one, with the spacetime
mielke-geometrodynamics_F00019	23	0.997	\operatorname{SL}(2, \mathbb{C}) \subset \operatorname{SL}(2, \mathbb{C}) \otimes \operatorname{SU}(3) \subset \operatorname{SL}(6, \mathbb{C})	$\operatorname{SL}(2, \mathbb{C}) \subset \operatorname{SL}(2, \mathbb{C}) \otimes \operatorname{SU}(3) \subset \operatorname{SL}(6, \mathbb{C})$	chain $\mathrm{SL}(2, \mathbb{C}) \subset \mathrm{SL}(2, \mathbb{C}) \otimes \mathrm{SU}(3) \subset \mathrm{SL}(6, \mathbb{C})$. If these symmetries are regarded
mielke-geometrodynamics_F00020	24	0.834	\operatorname{SL}(6, \mathbb{C})	$\operatorname{SL}(6, \mathbb{C})$	as local ones, we get an $\mathrm{SL}(6, \mathbb{C})$ gauge theory of strong interaction (SALAM 1973)
mielke-geometrodynamics_F00021	24	1.000	\ell:=\ell^{*} / \sqrt{\kappa}	$\ell:=\ell^{*} / \sqrt{\kappa}$	a further length $\ell := \ell^*/\sqrt{\kappa}$ is introduced into particle physics that corresponds to
mielke-geometrodynamics_F00022	24	0.995	\mathrm{G}_{\mathrm{S}}=\mathrm{G}_{\mathrm{N}} / \kappa	$\mathrm{G}_{\mathrm{S}}=\mathrm{G}_{\mathrm{N}} / \kappa$	the scaled "Newtonian" coupling constant $\mathrm{G_S} = \mathrm{G_N}/\kappa$ of strong gravity. SALAM &

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mielke-geometrodynamics_F00023	25	1.000	`\sqrt{\mathrm{s}}=540 \mathrm{GeV}`	$\sqrt{\mathrm{s}}=540 \mathrm{GeV}$	associated missing energy at $\sqrt{s}=540$ GeV. Phys Lett B 122:103–116
mielke-geometrodynamics_F00024	30	1.000	`\mathrm{x}: \mathrm{M} \rightarrow \mathbb{R}^{\mathrm{n}}`	$\mathrm{x}: \mathrm{M} \rightarrow \mathbb{R}^{\mathrm{n}}$	achieve this, we consider injective (reversible) mappings $\mathrm{x}: \mathrm{M} \rightarrow \mathbb{R}^{\mathrm{n}}$, whose range
mielke-geometrodynamics_F00025	30	1.000	`\mathrm{E}^{\mathrm{n}}`	E^{n}	extends to an open subset of the usual Euclidean space E^{n}. Such a map together with
mielke-geometrodynamics_F00026	30	0.993	`\mathrm{U}_{\mathrm{i}}`	U_{i}	its domain U_{i} is called an n-dimensional chart. By the projection π_i onto the single
mielke-geometrodynamics_F00027	30	0.993	`\pi_{i}`	π_{i}	its domain U_{i} is called an n-dimensional chart. By the projection π_i onto the single
mielke-geometrodynamics_F00028	30	1.000	`\mathbb{R}^{\mathrm{n}}`	$\mathbb{R}^{\mathrm{n}}$	axes of the coordinate system of $\mathbb{R}^{\mathrm{n}}$, the *local coordinates* $\mathrm{x}_i=\pi_i \circ \mathrm{x}, i=1, \ldots, \mathrm{n}$
mielke-geometrodynamics_F00029	30	1.000	`\mathrm{x}_{i}=\pi_{i} \circ \mathrm{x}, i=1, \ldots, \mathrm{n}`	$\mathrm{x}_{i}=\pi_{i} \circ \mathrm{x}, i=1, \ldots, \mathrm{n}$	axes of the coordinate system of $\mathbb{R}^{\mathrm{n}}$, the *local coordinates* $\mathrm{x}_i=\pi_i \circ \mathrm{x}, i=1, \ldots, \mathrm{n}$
mielke-geometrodynamics_F00030	30	1.000	`\mathrm{S}^{\mathrm{n}}`	S^{n}	sphere S^{n}). However, it is possible to construct a collection of charts of identical
mielke-geometrodynamics_F00031	31	0.996	`\mathrm{U}_{\mathrm{i}} \subset \mathrm{M}`	$\mathrm{U}_{\mathrm{i}} \subset \mathrm{M}$	dimensions so that the set-theoretic union of their domains $\mathrm{U}_{\mathrm{i}} \subset \mathrm{M}$ cover M com-
mielke-geometrodynamics_F00032	31	0.999	`\mathscr{D}(\mathrm{M})`	$\mathscr{D}(\mathrm{M})$	form an infinite-dimensional group with respect to composition, i.e., the group $\mathscr{D}(\mathrm{M})$
mielke-geometrodynamics_F00033	31	1.000	`\left(\mathrm{U}_{\mathrm{i}}, \mathrm{x}_{\mathrm{i}} \right)`	$\left(\mathrm{U}_{\mathrm{i}}, \mathrm{x}_{\mathrm{i}}\right)$	A collection $(\mathrm{U}_{\mathrm{i}}, \mathrm{x}_{\mathrm{i}})$ of compatible charts is called a C^{∞}-atlas of M. (The common
mielke-geometrodynamics_F00034	31	1.000	`\mathrm{C}^{\infty}`	C^{∞}	A collection $(\mathrm{U}_{\mathrm{i}}, \mathrm{x}_{\mathrm{i}})$ of compatible charts is called a C^{∞}-atlas of M. (The common
mielke-geometrodynamics_F00035	31	0.997	`\mathrm{C}^{\mathrm{r}}`	C^{r}	of class C^{r}). This atlas again can be extended unequivocally to a *complete* one, which
mielke-geometrodynamics_F00036	32	1.000	`\mathrm{E}^{\mathrm{n}(\mathrm{n}+1) / 2}`	$\mathrm{E}^{\mathrm{n}(\mathrm{n}+1) / 2}$	locally isometric embeddings of analytic Riemannian manifolds into $\mathrm{E}^{\mathrm{n}(\mathrm{n}+1)/2}$ are
mielke-geometrodynamics_F00037	32	0.936	`N=n(3 n+11) / 2`	$N=n(3 n+11) / 2$	dimension $N=n(3n+11)/2$ was proven by NASH (1956); for a noncompact one,
mielke-geometrodynamics_F00038	32	1.000	`N=n(n+1)(3 n+11) / 2`	$N=n(n+1)(3 n+11) / 2$	the extravagant dimension $N=n(n+1)(3n+11)/2$ results; cf. CHEN (2000) for a
mielke-geometrodynamics_F00039	32	1.000	`\mathbb{E}^{4}`	$\mathbb{E}^{4}$	surface of a rotating Kerr–Newman black hole into $\mathbb{E}^4$.
mielke-geometrodynamics_F00040	32	0.997	`\mathrm{T}_{\mathrm{m}}(\mathrm{M})`	$\mathrm{T}_{\mathrm{m}}(\mathrm{M})$	*space* $\mathrm{T}_{\mathrm{m}}(\mathrm{M})$ attached to a point $\mathrm{m} \in \mathrm{M}$. Contrary to the given illustration (see
mielke-geometrodynamics_F00041	32	0.997	`\mathrm{m} \in \mathrm{M}`	$\mathrm{m} \in \mathrm{M}$	*space* $\mathrm{T}_{\mathrm{m}}(\mathrm{M})$ attached to a point $\mathrm{m} \in \mathrm{M}$. Contrary to the given illustration (see
mielke-geometrodynamics_F00042	32	0.573	`\mathrm{s}(\mathrm{t})`	$\mathrm{s}(\mathrm{t})$	to achieve this, a curve s(t) on the manifold is chosen that passes through a point
mielke-geometrodynamics_F00043	33	0.985	`\mathrm{e}(\mathrm{m})`	$\mathrm{e}(\mathrm{m})$	tiable functions f to the space $\mathbb{R}$ of real numbers, defines the *tangent vector* e(m) or
mielke-geometrodynamics_F00044	33	0.999	`\mathrm{e}_{\alpha}(\mathrm{m}), \alpha=1, \ldots, \mathrm{n}`	$\mathrm{e}_{\alpha}(\mathrm{m}), \alpha=1, \ldots, \mathrm{n}$	independent vectors $\mathrm{e}_{\alpha}(\mathrm{m}), \alpha=1, \ldots, \mathrm{n}$. It follows from (2.1.2) that the partial
mielke-geometrodynamics_F00045	33	1.000	`\partial_{i}:=\partial / \partial \mathrm{x}^{i}`	$\partial_{i}:=\partial / \partial \mathrm{x}^{i}$	derivates $\partial_i := \partial/\partial \mathrm{x}^i$ constitute a local basis for $\mathrm{T}_{\mathrm{m}}(\mathrm{M})$ with respect to a *holonomic*
mielke-geometrodynamics_F00046	33	1.000	`\left[\mathrm{e}_{i}(\mathrm{~m}), \mathrm{e}_{j}(\mathrm{~m})\right]=\left[\partial_{i}, \partial_{j} \right]=0`	$\left[\mathrm{e}_{i}(\mathrm{~m}), \mathrm{e}_{j}(\mathrm{~m})\right]=\left[\partial_{i}, \partial_{j}\right]=0$	coordinate system. By definition, the relation $[\mathrm{e}_i(\mathrm{m}), \mathrm{e}_j(\mathrm{m})]=[\partial_i, \partial_j]=0$ is satis-
mielke-geometrodynamics_F00047	33	1.000	`\mathrm{e}_{\alpha}^{i}(m)`	$\mathrm{e}_{\alpha}^{i}(m)$	The occurring nonsingular matrices $\mathrm{e}_{\alpha}^{i}(m)$ depend on the point in question and
mielke-geometrodynamics_F00048	33	0.628	`\mathrm{T}(\mathrm{M})`	$\mathrm{T}(\mathrm{M})$	In addition to the tangent bundle T(M), a so-called *dual* bundle $\mathrm{T}^*(\mathrm{M})=\bigcup_{\mathrm{m} \in \mathrm{M}} \mathrm{T}_{\mathrm{m}}^*(\mathrm{M})$
mielke-geometrodynamics_F00049	33	0.628	`\mathrm{T}^{*}(\mathrm{M})=\bigcup_{\mathrm{m} \in \mathrm{M}} \mathrm{T}_{\mathrm{m}}^{*}(\mathrm{M})`	$\mathrm{T}^{*}(\mathrm{M})=\bigcup_{\mathrm{m} \in \mathrm{M}} \mathrm{T}_{\mathrm{m}}^{*}(\mathrm{M})$	In addition to the tangent bundle T(M), a so-called *dual* bundle $\mathrm{T}^*(\mathrm{M})=\bigcup_{\mathrm{m} \in \mathrm{M}} \mathrm{T}_{\mathrm{m}}^*(\mathrm{M})$

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mielke-geometrodynamics_F00050	33	1.000	`\mathrm{T}_{\mathrm{m}}^{*}(\mathrm{M})`	$\mathrm{T}_{\mathrm{m}}^{*}(\mathrm{M})$	consisting of all possible cotangent spaces $\mathrm{T}_{\mathrm{m}}^{*}(\mathrm{M})$ on the manifold can be constructed.
mielke-geometrodynamics_F00051	33	1.000	`\vartheta^{\alpha}, \alpha=1, \ldots, n`	$\vartheta^{\alpha}, \alpha=1, \ldots, n$	Let ϑ^{α}, $\alpha=1,\ldots,n$, be a basis of $\mathrm{T}_{\mathrm{m}}^{*}(\mathrm{M})$. Then the "duality"[3] requires that ϑ^{α}
mielke-geometrodynamics_F00052	33	1.000	`\vartheta^{\alpha}`	ϑ^{α}	Let ϑ^{α}, $\alpha=1,\ldots,n$, be a basis of $\mathrm{T}_{\mathrm{m}}^{*}(\mathrm{M})$. Then the "duality"[3] requires that ϑ^{α}
mielke-geometrodynamics_F00053	33	0.999	`\mathrm{dx}^{i}`	dx^{i}	by dx^{i}. In terms of the rigid basis of E^{n}, the 1-forms
mielke-geometrodynamics_F00054	34	0.998	`\mathrm{d} \vartheta^{\alpha}`	$\mathrm{d} \vartheta^{\alpha}$	derivative $\mathrm{d}\vartheta^{\alpha}$, in contrast to that of dx^{j}, is generally different from zero. Following
mielke-geometrodynamics_F00055	34	0.998	`\mathrm{dx}^{j}`	dx^{j}	derivative $\mathrm{d}\vartheta^{\alpha}$, in contrast to that of dx^{j}, is generally different from zero. Following
mielke-geometrodynamics_F00056	34	0.599	`\mathrm{E}_{.}^{\alpha}{ }_{j}`	$\mathrm{E}_{.}^{\alpha}{ }_{j}$	(2.2.1), $\mathrm{E}^{\alpha}_{\cdot\ j}$ can be understood as the dual tetrad field "reciprocal" to e_{β}^{i} (cf. (2.1.3)).
mielke-geometrodynamics_F00057	34	0.599	`\mathrm{e}_{\beta}^{i}`	e_{β}^{i}	(2.2.1), $\mathrm{E}^{\alpha}_{\cdot\ j}$ can be understood as the dual tetrad field "reciprocal" to e_{β}^{i} (cf. (2.1.3)).
mielke-geometrodynamics_F00058	34	0.979	`\otimes_{\mathrm{s}, \mathrm{a}}`	$\otimes_{\mathrm{s}, \mathrm{a}}$	Completely symmetric or antisymmetric products are denoted by $\otimes_{\mathrm{s,a}}$ or, even more
mielke-geometrodynamics_F00059	34	1.000	`\vee`	$\vee$	commonly, by the symbols $\vee$ and $\wedge$, respectively. Note that a tensor is a geometric
mielke-geometrodynamics_F00060	34	1.000	`\wedge`	$\wedge$	commonly, by the symbols $\vee$ and $\wedge$, respectively. Note that a tensor is a geometric
mielke-geometrodynamics_F00061	34	0.962	`\mathrm{T}_{\alpha_{1} \cdots \alpha_{\mathrm{p}}}^{\beta_{1} \cdots \beta_{\mathrm{q}}}(\mathrm{m})`	$\mathrm{T}_{\alpha_{1} \cdots \alpha_{\mathrm{p}}}^{\beta_{1} \cdots \beta_{\mathrm{q}}}(\mathrm{m})$	The quantities $\mathrm{T}_{\alpha_{1} \cdots \alpha_{\mathrm{p}}}^{\beta_{1} \cdots \beta_{\mathrm{q}}}(\mathrm{m})$ are called its p covariant and q contravariant components,
mielke-geometrodynamics_F00062	34	0.919	`\mathrm{T}_{\mathrm{p}}^{\mathrm{q}}(\mathrm{M})`	$\mathrm{T}_{\mathrm{p}}^{\mathrm{q}}(\mathrm{M})$	and $\mathrm{T}_{\mathrm{p}}^{\mathrm{q}}(\mathrm{M})$ may be regarded as a bundle associated with $\mathrm{L}(\mathrm{M})$ having the manifold
mielke-geometrodynamics_F00063	34	0.919	`\mathrm{L}(\mathrm{M})`	$\mathrm{L}(\mathrm{M})$	and $\mathrm{T}_{\mathrm{p}}^{\mathrm{q}}(\mathrm{M})$ may be regarded as a bundle associated with $\mathrm{L}(\mathrm{M})$ having the manifold
mielke-geometrodynamics_F00064	34	0.708	`\mathrm{D}^{(\mathrm{p}, \mathrm{q})}`	$\mathrm{D}^{(\mathrm{p}, \mathrm{q})}$	M as a base, the tensor representation $\mathrm{D}^{(\mathrm{p,q})}$ of $\mathrm{GL}(\mathrm{n},\mathbb{R})$, i.e., $\rho^{(\mathrm{p,q})}(\mathrm{GL}(\mathrm{n},\mathbb{R}))$ as a
mielke-geometrodynamics_F00065	34	0.708	`\rho^{(\mathrm{p}, \mathrm{q})}(\mathrm{GL}(\mathrm{n}, \mathbb{R}))`	$\rho^{(\mathrm{p}, \mathrm{q})}(\mathrm{GL}(\mathrm{n}, \mathbb{R}))$	M as a base, the tensor representation $\mathrm{D}^{(\mathrm{p,q})}$ of $\mathrm{GL}(\mathrm{n},\mathbb{R})$, i.e., $\rho^{(\mathrm{p,q})}(\mathrm{GL}(\mathrm{n},\mathbb{R}))$ as a
mielke-geometrodynamics_F00066	34	0.907	`\otimes^{\mathrm{p} *} \mathbb{R}^{\mathrm{n}} \otimes^{\mathrm{q}} \mathbb{R}^{\mathrm{n}}`	$\otimes^{\mathrm{p} *} \mathbb{R}^{\mathrm{n}} \otimes^{\mathrm{q}} \mathbb{R}^{\mathrm{n}}$	structure group and the (Cartesian) product $\otimes^{\mathrm{p}*}\mathbb{R}^{\mathrm{n}} \otimes^{\mathrm{q}}\mathbb{R}^{\mathrm{n}}$ as a typical fiber.
mielke-geometrodynamics_F00067	34	1.000	`\mathrm{g}_{\mu \nu}`	$\mathrm{g}_{\mu \nu}$	the scale of the manifold. Due to the postulated symmetry of the components $\mathrm{g}_{\mu\nu}$,
mielke-geometrodynamics_F00068	35	0.651	`(\mathrm{p}, 0)`	$(\mathrm{p}, 0)$	(p, 0). Usually, $\wedge := \otimes_{\mathrm{a}}$ is an abbreviation for the antisymmetrized tensor product,
mielke-geometrodynamics_F00069	35	0.651	`\wedge:=\otimes_{\mathrm{a}}`	$\wedge:=\otimes_{\mathrm{a}}$	(p, 0). Usually, $\wedge := \otimes_{\mathrm{a}}$ is an abbreviation for the antisymmetrized tensor product,
mielke-geometrodynamics_F00070	35	1.000	`(1939,1957)`	$(1939,1957)$	(1939, 1957) of the representations of the transformation group in flat spacetime, i.e.,
mielke-geometrodynamics_F00071	36	1.000	`\mathscr{H}`	$\mathscr{H}$	rays in a Hilbert space $\mathscr{H}$, which is to be transformed according to an irreducible
mielke-geometrodynamics_F00072	36	0.997	`\varphi`	φ	assumed that φ at a point m is equal to the value $\varphi(\mathrm{m})$ in a complex vector space
mielke-geometrodynamics_F00073	36	0.997	`\varphi(\mathrm{m})`	$\varphi(\mathrm{m})$	assumed that φ at a point m is equal to the value $\varphi(\mathrm{m})$ in a complex vector space
mielke-geometrodynamics_F00074	36	1.000	`\mathrm{V}_{\mathrm{m}}`	V_{m}	V_{m}. Additionally, it can be postulated that a vector space of this kind is attached
mielke-geometrodynamics_F00075	36	0.941	`\pi`	π	**Definition** A *fiber bundle* $(\mathrm{F},\mathrm{M},\pi)$ consists of two C^{∞}-differentiable manifolds F
mielke-geometrodynamics_F00076	36	0.362	`\mathrm{F}_{\mathrm{m}}:=-\pi^{-1}(\mathrm{~m})`	$\mathrm{F}_{\mathrm{m}}:=-\pi^{-1}(\mathrm{~m})$	and π the *projection*. The closed submanifold $\mathrm{F}_{\mathrm{m}} := \overset{-1}{\pi}(\mathrm{m})$ will be called a *typical*

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mielke-geometrodynamics_F00077	37	0.980	`\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}_{\mathrm{m}}`	$\pi^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}_{\mathrm{m}}$	point m $\in$ M has a neighborhood U such that $\overset{-1}{\pi}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}_{\mathrm{m}}$ is isomorphic to the
mielke-geometrodynamics_F00078	37	0.686	`\left(\mathrm{U} \times \mathrm{F}_{\mathrm{m}}, \mathrm{U}, \pi_{\mathrm{u}}\right)`	$\left(\mathrm{U} \times \mathrm{F}_{\mathrm{m}}, \mathrm{U}, \pi_{\mathrm{u}}\right)$	product bundle $(\mathrm{U} \times \mathrm{F}_{\mathrm{m}}, \mathrm{U}, \pi_{\mathrm{u}})$. The mapping
mielke-geometrodynamics_F00079	37	0.637	`\mathrm{P}(\mathrm{M}, \mathrm{G}, \pi, \delta)`	$\mathrm{P}(\mathrm{M}, \mathrm{G}, \pi, \delta)$	group,then the collection P(M, G, π, δ) is called a *principal fiber bundle* if
mielke-geometrodynamics_F00080	37	0.790	`(\mathrm{P}, \mathrm{M}, \pi)`	$(\mathrm{P}, \mathrm{M}, \pi)$	(i) (P, M, π) is a fiber bundle with typical fiber G;
mielke-geometrodynamics_F00081	37	0.866	`\mathrm{P}, \mathrm{G}, \delta`	$\mathrm{P}, \mathrm{G}, \delta$	(ii) (P, G, δ) is a G-manifold (whereby G is acting on P from the right);
mielke-geometrodynamics_F00082	37	0.662	`\mathrm{U} \subset`	$\mathrm{U} \subset$	(iii) P is *locally* trivial, which means that every point of M has a neighborhood U $\subset$
mielke-geometrodynamics_F00083	37	0.941	`\iota: \mathrm{U} \times \mathrm{G} \rightarrow{ }^{-1}(\mathrm{U})`	$\iota: \mathrm{U} \times \mathrm{G} \rightarrow{ }^{-1}(\mathrm{U})$	M together with an isomorphism $\iota : \mathrm{U} \times \mathrm{G} \rightarrow \overset{-1}{\pi}(\mathrm{U})$ for which the following
mielke-geometrodynamics_F00084	37	0.857	`G`	G	A manifold is called a *G-manifold* (P, G, δ) if G acts on it as a free transformation
mielke-geometrodynamics_F00085	37	0.970	`\mathrm{p}_{2}=\mathrm{g} \mathrm{p}_{1}`	$\mathrm{p}_{2}=\mathrm{g} \mathrm{p}_{1}$	there always existed a g that would provide a relation $\mathrm{p}_2 = \mathrm{g}\ \mathrm{p}_1$ for given p_1
mielke-geometrodynamics_F00086	37	0.970	`\mathrm{p}_{1}`	p_{1}	there always existed a g that would provide a relation $\mathrm{p}_2 = \mathrm{g}\ \mathrm{p}_1$ for given p_1
mielke-geometrodynamics_F00087	37	0.666	`\mathrm{p}_{2}`	p_{2}	and p_2, then G considered as a manifold would be isomorphic to the base M.)
mielke-geometrodynamics_F00088	38	0.985	`\pi, \delta`	π, δ	this purpose, the fiber bundles V(M, F; G, P) associated with P(M, G, π, δ) are
mielke-geometrodynamics_F00089	38	0.999	`\mathrm{P} \times \mathrm{F}`	$\mathrm{P} \times \mathrm{F}$	i.e., on the product space P $\times$ F, an action from the right-hand side is defined by
mielke-geometrodynamics_F00090	38	0.678	`\mathrm{V}=\mathrm{P} \times_{\mathrm{G}} \mathrm{F}`	$\mathrm{V}=\mathrm{P} \times_{\mathrm{G}} \mathrm{F}$	by $\mathrm{V} = \mathrm{P} \times_{\mathrm{G}} \mathrm{F}$. The isomorphism $\overset{-1}{\pi}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{G}$ concerning the original domain
mielke-geometrodynamics_F00091	38	0.678	`{ }^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{G}`	${ }^{-1}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{G}$	by $\mathrm{V} = \mathrm{P} \times_{\mathrm{G}} \mathrm{F}$. The isomorphism $\overset{-1}{\pi}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{G}$ concerning the original domain
mielke-geometrodynamics_F00092	38	0.307	`{ }^{-1}{ }_{\mathrm{v}}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}`	${ }^{-1}{ }_{\mathrm{v}}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}$	induce the isomorphism $\overset{-1}{\pi}_{\mathrm{v}}(\mathrm{U}) \approx \mathrm{U} \times \mathrm{F}$ in the associated bundle. Provided that
mielke-geometrodynamics_F00093	38	0.907	`{ }^{-1} \pi_{\mathrm{v}}(\mathrm{U})`	${ }^{-1} \pi_{\mathrm{v}}(\mathrm{U})$	$\overset{-1}{\pi}_{\mathrm{v}}(\mathrm{U})$ is an open submanifold of V, then V can be equipped with a differentiable
mielke-geometrodynamics_F00094	38	0.998	`\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})`	$\rho: \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$	occurs, but its linear representation $\rho : \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$, i.e., the Lie homomorphism
mielke-geometrodynamics_F00095	38	0.852	`\mathrm{N} \times \mathrm{N}`	$\mathrm{N} \times \mathrm{N}$	of G into the general linear group of complex N $\times$ N matrices. If their representation
mielke-geometrodynamics_F00096	38	0.780	`\mathbb{C}^{\mathrm{N}}`	$\mathbb{C}^{\mathrm{N}}$	space, the N-dimensional vector space $\mathbb{C}^{\mathrm{N}}$ over the field of complex numbers, is
mielke-geometrodynamics_F00097	38	1.000	`\mathrm{T}_{\mathbb{C}}^{*}(\mathrm{M})`	$\mathrm{T}_{\mathbb{C}}^{*}(\mathrm{M})$	with the complexified[4] cotangent bundle $\mathrm{T}_{\mathbb{C}}^{*}(\mathrm{M})$, further vector bundles will come
mielke-geometrodynamics_F00098	38	1.000	`\rho`	ρ	ρ, i.e., the cross section
mielke-geometrodynamics_F00099	38	0.986	`\mathrm{N} \geq 1`	$\mathrm{N} \geq 1$	For N $\geq$ 1, these fields form an infinite-dimensional vector space over $\mathbb{C}^{\mathrm{N}}$. In
mielke-geometrodynamics_F00100	38	1.000	`\mathrm{b}=\left(\mathrm{b}_{\mathrm{A}}\right): \mathrm{U} \subset \mathrm{M} \rightarrow`	$\mathrm{b}=\left(\mathrm{b}_{\mathrm{A}}\right): \mathrm{U} \subset \mathrm{M} \rightarrow$	(2.4.3), the bundle coordinates relative to the local basis fields $\mathrm{b} = (\mathrm{b}_{\mathrm{A}}) : \mathrm{U} \subset \mathrm{M} \rightarrow$
mielke-geometrodynamics_F00101	38	1.000	`\mathrm{V}^{\rho}`	V^{ρ}	V^{ρ} are denoted by
mielke-geometrodynamics_F00102	39	1.000	`\rho: \mathrm{G} \rightarrow \operatorname{GL}(N, \mathbb{C})`	$\rho: \mathrm{G} \rightarrow \operatorname{GL}(N, \mathbb{C})$	Let the Lie homomorphism $\rho : \mathrm{G} \rightarrow \mathrm{GL}(N, \mathbb{C})$ be explicitly given by the nonsin-

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mielke-geometrodynamics_F00103	39	0.964	`\rho_{\mathrm{A}}{ }^{\mathrm{B}}(\mathrm{g}(\xi))`	$\rho_{\mathrm{A}}{ }^{\mathrm{B}}(\mathrm{g}(\xi))$	gular complex N $\times$ N matrices $\rho_{\mathrm{A}}{}^{\mathrm{B}}(\mathrm{g}(\xi))$, where $\xi^{\mathrm{j}}(\mathrm{j}=1,\ldots,\dim \mathrm{G})$ prescribes a
mielke-geometrodynamics_F00104	39	0.964	`\xi^{\mathrm{j}}(\mathrm{j}=1, \ldots, \operatorname{dim} \mathrm{G})`	$\xi^{\mathrm{j}}(\mathrm{j}=1, \ldots, \operatorname{dim} \mathrm{G})$	gular complex N $\times$ N matrices $\rho_{\mathrm{A}}{}^{\mathrm{B}}(\mathrm{g}(\xi))$, where $\xi^{\mathrm{j}}(\mathrm{j}=1,\ldots,\dim \mathrm{G})$ prescribes a
mielke-geometrodynamics_F00105	39	1.000	`\rho=\mathrm{id}`	$\rho=\mathrm{id}$	identical representation, i.e., for $\rho=\mathrm{id}$, the infinitesimal generators I_{j} form a basis
mielke-geometrodynamics_F00106	39	1.000	`\mathrm{I}_{\mathrm{j}}`	I_{j}	identical representation, i.e., for $\rho=\mathrm{id}$, the infinitesimal generators I_{j} form a basis
mielke-geometrodynamics_F00107	39	0.938	`\mathfrak{g}`	$\mathfrak{g}$	of the Lie algebra $\mathfrak{g}$ of G. As a representation of I_{j}, the infinitesimal operators inherit
mielke-geometrodynamics_F00108	39	1.000	`\mathrm{c}_{\mathrm{ij}}{ }^{\mathrm{k}}`	$\mathrm{c}_{\mathrm{ij}}{ }^{\mathrm{k}}$	These algebraic relations are determined by the structure constants $\mathrm{c}_{\mathrm{ij}}{}^{\mathrm{k}}$ of the Lie
mielke-geometrodynamics_F00109	39	0.960	`\mathrm{V}^{\mathrm{Ad}}`	V^{Ad}	vector bundle V^{Ad} respecting the adjoint representation of G as a structure group.
mielke-geometrodynamics_F00110	39	0.983	`\mathrm{s}(\mathrm{t}) \subset \mathrm{M}`	$\mathrm{s}(\mathrm{t}) \subset \mathrm{M}$	in a differentiable manner along a given curve $\mathrm{s}(\mathrm{t}) \subset \mathrm{M}$ in the base manifold. In order
mielke-geometrodynamics_F00111	39	1.000	`\mathrm{T}_{\mathrm{p}}{ }^{*}(\mathrm{P})`	$\mathrm{T}_{\mathrm{p}}{ }^{*}(\mathrm{P})$	cotangent bundle $\mathrm{T}_{\mathrm{p}}{}^{*}(\mathrm{P})$ at the point $\mathrm{p} \in \mathrm{P}$ (see KN I, Chap. 1). Then decompose it
mielke-geometrodynamics_F00112	39	1.000	`\mathrm{p} \in \mathrm{P}`	$\mathrm{p} \in \mathrm{P}$	cotangent bundle $\mathrm{T}_{\mathrm{p}}{}^{*}(\mathrm{P})$ at the point $\mathrm{p} \in \mathrm{P}$ (see KN I, Chap. 1). Then decompose it
mielke-geometrodynamics_F00113	40	1.000	`\mathrm{T}_{\mathrm{p}}{ }^{*}(\mathrm{P})=\mathrm{H}_{ \mathrm{p}} \oplus \mathrm{V}_{\mathrm{p}}`	$\mathrm{T}_{\mathrm{p}}{ }^{*}(\mathrm{P})=\mathrm{H}_{\mathrm{p}} \oplus \mathrm{V}_{\mathrm{p}}$	into the direct sum of horizontal and the vertical subspaces, i.e., $\mathrm{T}_{\mathrm{p}}{}^{*}(\mathrm{P})=\mathrm{H}_{\mathrm{p}} \oplus \mathrm{V}_{\mathrm{p}}$.
mielke-geometrodynamics_F00114	40	1.000	`\mathrm{F}_{\mathrm{m}}(\mathrm{M})`	$\mathrm{F}_{\mathrm{m}}(\mathrm{M})$	i.e., in the cotangent space of the bundle $\mathrm{F}_{\mathrm{m}}(\mathrm{M})$ of typical fibers on M.
mielke-geometrodynamics_F00115	40	0.999	`\omega(\mathrm{e}) \in \mathscr{C}`	$\omega(\mathrm{e}) \in \mathscr{C}$	This will be achieved by the assertion of a 1-form $\omega(\mathrm{e}) \in \mathscr{C}$ with values in the
mielke-geometrodynamics_F00116	40	1.000	`\mathrm{e}_{\mathrm{v}}=\mathrm{pdg} \in \mathrm{V}_{ \mathrm{p}}`	$\mathrm{e}_{\mathrm{v}}=\mathrm{pdg} \in \mathrm{V}_{\mathrm{p}}$	(ii) For vertical elements $\mathrm{e}_{\mathrm{v}}=\mathrm{pdg} \in \mathrm{V}_{\mathrm{p}}$, it will be mapped to the left-invariant
mielke-geometrodynamics_F00117	40	1.000	`\bigwedge^{\mathrm{p}} \mathrm{T}_{\mathbb{C}}^{*}( \mathrm{M}) \otimes V^{\rho}`	$\bigwedge^{\mathrm{p}} \mathrm{T}_{\mathbb{C}}^{*}(\mathrm{M}) \otimes V^{\rho}$	connection in an associated vector bundle $\bigwedge^{\mathrm{p}} \mathrm{T}_{\mathbb{C}}^{*}(\mathrm{M}) \otimes V^{\rho}$ is defined via the *covari-*
mielke-geometrodynamics_F00118	40	1.000	`\mathrm{T}_{\mathbb{C}}^{*}(\mathrm{M}) \otimes \mathrm{V}^{\rho}`	$\mathrm{T}_{\mathbb{C}}^{*}(\mathrm{M}) \otimes \mathrm{V}^{\rho}$	onto those of the product bundle $\mathrm{T}_{\mathbb{C}}^{*}(\mathrm{M}) \otimes \mathrm{V}^{\rho}$. This operator will be defined in such
mielke-geometrodynamics_F00119	41	1.000	`V^{\rho}`	V^{ρ}	expressed as a linear transformation of the selfsame basis of V^{ρ}, i.e.,
mielke-geometrodynamics_F00120	41	0.856	`b`	b	The image of Db will be referred to as the *(G-)covariant exterior derivative* of b.
mielke-geometrodynamics_F00121	41	0.641	`\mathrm{b}_{\mathrm{A}}, \mathrm{A}=1, \ldots, \mathrm{~N}`	$\mathrm{b}_{\mathrm{A}}, \mathrm{A}=1, \ldots, \mathrm{~N}$	restriction to a neighborhood U of M. Let $\mathrm{b}_{\mathrm{A}}, \mathrm{A}=1,\ldots,\mathrm{N}$ be a local basis for the
mielke-geometrodynamics_F00122	41	0.996	`\mathrm{V}_{\mathbb{U}}^{\rho}`	$\mathrm{V}_{\mathbb{U}}^{\rho}$	cross section $\mathrm{V}_{\mathrm{U}}^{\rho}$ restricted to U. With respect to this local system of reference, the
mielke-geometrodynamics_F00123	41	0.998	`\mathrm{b}_{\mathrm{A}}`	b_{A}	resulting effect of D on b_{A} can be expanded as follows:
mielke-geometrodynamics_F00124	41	0.998	`\left[\omega_{\mathrm{A}}{ }^{\mathrm{B}}\right]`	$\left[\omega_{\mathrm{A}}{ }^{\mathrm{B}}\right]$	matrix $[\omega_{\mathrm{A}}{}^{\mathrm{B}}]$ acquires values in $\mathrm{T}_{\mathbb{C}}^{*}(\mathrm{M})$ and will consequently be called a *connection*
mielke-geometrodynamics_F00125	41	1.000	`\Gamma_{\alpha}^{\mathrm{j}}(\mathrm{m})`	$\Gamma_{\alpha}^{\mathrm{j}}(\mathrm{m})$	The coefficients $\Gamma_{\alpha}^{\mathrm{j}}(\mathrm{m})$ will turn out to be generalized gauge potentials as they
mielke-geometrodynamics_F00126	42	0.997	`\mathrm{dd} \equiv 0`	$\mathrm{dd} \equiv 0$	For differential forms in general, the identity $\mathrm{dd} \equiv 0$ holds, which may be regarded as
mielke-geometrodynamics_F00127	42	1.000	`\partial \partial \equiv 0`	$\partial \partial \equiv 0$	a counterpart to the homological relation $\partial\partial \equiv 0$. Figuratively speaking, this means
mielke-geometrodynamics_F00128	42	0.905	`\Omega`	Ω	that the curvature 2-form Ω satisfies the *second structure equation* of É. CARTAN:
mielke-geometrodynamics_F00129	42	1.000	`\mathrm{G}=\mathrm{U}(1)`	$\mathrm{G}=\mathrm{U}(1)$	which can be formulated in a fiber bundle with abelian structure group $\mathrm{G}=\mathrm{U}(1)$,

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mielke-geometrodynamics_F00130	43	1.000	`\circlearrowleft=\partial \mathrm{U} \subset \mathrm{M}`	$\circlearrowleft=\partial \mathrm{U} \subset \mathrm{M}$	along an *infinitesimal* closed curve $\circlearrowleft= \partial \mathrm{U} \subset \mathrm{M}$. The very concept of a connection
mielke-geometrodynamics_F00131	43	1.000	`\partial \mathrm{U}`	$\partial \mathrm{U}$	closed curve $\partial \mathrm{U}$. Figure 2.2 shows this for a parallel displacement of a pair of tangent
mielke-geometrodynamics_F00132	43	1.000	`\mathrm{e}(\mathrm{m}) \in \mathrm{T}_{\mathrm{m}}(\mathrm{M})`	$\mathrm{e}(\mathrm{m}) \in \mathrm{T}_{\mathrm{m}}(\mathrm{M})$	$\mathrm{T_m(M)}$ onto itself. These are generated by displacements of $\mathrm{e(m) \in T_m(M)}$ along
mielke-geometrodynamics_F00133	44	0.996	`\mathrm{G}(\mathrm{p})`	$\mathrm{G}(\mathrm{p})$	(i) $\mathrm{G(p)}$ is equivariant, i.e.,
mielke-geometrodynamics_F00134	44	0.897	`\mathrm{F}_{\mathrm{m}}=\pi^{-1}(\mathrm{~m})`	$\mathrm{F}_{\mathrm{m}}=\pi^{-1}(\mathrm{~m})$	(ii) $\mathrm{G(p)}$ preserves each fiber $\mathrm{F_m} = \overset{-1}{\pi}(\mathrm{m})$, i.e., acts trivially on the base space,
mielke-geometrodynamics_F00135	44	1.000	`\mathscr{G}_{\mathrm{P}}`	$\mathscr{G}_{\mathrm{P}}$	position, they constitute an infinite-dimensional group, the group $\mathscr{G}_{\mathrm{P}}$ of *local*
mielke-geometrodynamics_F00136	45	0.513	`\exp : \mathfrak{g} \rightarrow \mathrm{G}`	$\exp : \mathfrak{g} \rightarrow \mathrm{G}$	The exponential mapping $\exp : \mathfrak{g} \rightarrow \mathrm{G}$ from the Lie algebra $\mathfrak{g}$ into the structure
mielke-geometrodynamics_F00137	45	0.614	`P \times{ }_{\mathrm{Ad}} \mathrm{G}`	$P \times{ }_{\mathrm{Ad}} \mathrm{G}$	group G of P induces a corresponding mapping within $P \times_{\mathrm{Ad}} \mathrm{G}$. As a result,
mielke-geometrodynamics_F00138	45	1.000	`\theta^{k}(m) \in C^{\infty}(M)`	$\theta^{k}(m) \in C^{\infty}(M)$	Here $\theta^{k}(m) \in C^{\infty}(M)$ denote real functions on the base space. Gauge trans-
mielke-geometrodynamics_F00139	45	0.984	`\mathrm{V}^{\rho}\left(\mathrm{M}, \mathbb{C}^{\mathrm{N}}, \operatorname{GL}(\mathrm{N}, \mathbb{C}), \mathrm{P}\right)`	$\mathrm{V}^{\rho}\left(\mathrm{M}, \mathbb{C}^{\mathrm{N}}, \operatorname{GL}(\mathrm{N}, \mathbb{C}), \mathrm{P}\right)$	formations within an associate vector bundle $\mathrm{V}^{\rho}(\mathrm{M}, \mathbb{C}^{\mathrm{N}}, \mathrm{GL}(\mathrm{N}, \mathbb{C}), \mathrm{P})$, i.e.,
mielke-geometrodynamics_F00140	45	1.000	`\mathscr{G}_{\mathrm{V}}`	$\mathscr{G}_{\mathrm{V}}$	elements of $\mathscr{G}_{\mathrm{V}}$, are represented by the same expression to the extent that L^{k}
mielke-geometrodynamics_F00141	45	1.000	`\mathrm{L}^{\mathrm{k}}`	L^{k}	elements of $\mathscr{G}_{\mathrm{V}}$, are represented by the same expression to the extent that L^{k}
mielke-geometrodynamics_F00142	45	0.482	`\rho: \mathrm{G} \rightarrow \operatorname{GL}(\mathrm{N}, \mathbb{C})`	$\rho: \mathrm{G} \rightarrow \operatorname{GL}(\mathrm{N}, \mathbb{C})$	$\rho : \mathrm{G} \rightarrow \mathrm{GL}(\mathrm{N}, \mathbb{C})$.
mielke-geometrodynamics_F00143	45	1.000	`\bar{\varphi}`	$\bar{\varphi}$	It has to be emphasized that the bundle coordinates $\overline{\varphi}$ of the vector bundle constructed
mielke-geometrodynamics_F00144	45	1.000	`\bar{\rho}`	$\bar{\rho}$	on the conjugate ("Dirac adjoint") representation $\overline{\rho}$ transforms according to
mielke-geometrodynamics_F00145	46	0.798	`\mathrm{V}\left(\mathrm{M}, \mathbb{C}^{\mathrm{N}}, \mathrm{GL}(\mathrm{N}, \mathbb{C}), \mathrm{P}\right)`	$\mathrm{V}\left(\mathrm{M}, \mathbb{C}^{\mathrm{N}}, \mathrm{GL}(\mathrm{N}, \mathbb{C}), \mathrm{P}\right)$	Let $\mathrm{V}(\mathrm{M}, \mathbb{C}^{\mathrm{N}}, \mathrm{GL}(\mathrm{N}, \mathbb{C}), \mathrm{P})$ be a complex vector bundle that is associated with
mielke-geometrodynamics_F00146	46	0.438	`\gamma_{\mathrm{k}}`	γ_{k}	2k-forms γ_{k} on P whose inverse images after the projection on the base space M read
mielke-geometrodynamics_F00147	46	1.000	`\mathrm{k} \leq n / 2`	$\mathrm{k} \leq n / 2$	The number of curvature 2-forms Ω in the exterior product is $\mathrm{k} \leq n/2$. From the
mielke-geometrodynamics_F00148	46	1.000	`\omega`	ω	of the latter is not determined by the particular choice of the connection ω, but depends
mielke-geometrodynamics_F00149	46	0.959	`\mathrm{k}^{\text {th }}`	$\mathrm{k}^{\text {th }}$	precisely (see: KN II, Theorem 3.1), it can be stated that the abstractly defined k^{th}
mielke-geometrodynamics_F00150	46	0.570	`\mathrm{c}_{\mathrm{k}}(\mathrm{V})`	$\mathrm{c}_{\mathrm{k}}(\mathrm{V})$	Chern class $\mathrm{c_k(V)}$ of a complex vector bundle V is represented by the closed 2k-form
mielke-geometrodynamics_F00151	46	0.546	`V^{\mathbb{R}}`	$V^{\mathbb{R}}$	In order to obtain the characteristic classes of a real vector bundle $V^{\mathbb{R}}$ over M with
mielke-geometrodynamics_F00152	46	0.983	`\mathbb{R}^{\mathrm{N}}`	$\mathbb{R}^{\mathrm{N}}$	the typical fiber $\mathbb{R}^{\mathrm{N}}$, it will be enlarged to a complex vector bundle V with the typical
mielke-geometrodynamics_F00153	46	0.753	`\mathbb{C}^{N}`	$\mathbb{C}^{N}$	fiber $\mathbb{C}^{\mathrm{N}}$. The latter arises from the complexification of each fiber of $\mathrm{V}^{\mathbb{R}}$. Then the
mielke-geometrodynamics_F00154	46	0.753	`\mathrm{V}^{\mathbb{R}}`	$\mathrm{V}^{\mathbb{R}}$	fiber $\mathbb{C}^{\mathrm{N}}$. The latter arises from the complexification of each fiber of $\mathrm{V}^{\mathbb{R}}$. Then the
mielke-geometrodynamics_F00155	47	1.000	`\mathrm{s}=0`	$\mathrm{s}=0$	the Euclidean signature $\mathrm{s} = 0$, we find locally, cf. DANIEL & VIALLET (1980),

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mielke-geometrodynamics_F00156	47	1.000	`\mathrm{P}\left(\mathrm{S}^{4}, \mathrm{SU}(\mathrm{f}), \pi, \delta\right)`	$\mathrm{P}\left(\mathrm{S}^{4}, \mathrm{SU}(\mathrm{f}), \pi, \delta\right)$	It is typical for the principal fiber bundles $\mathrm{P}(\mathrm{S}^4, \mathrm{SU}(\mathrm{f}), \pi, \delta)$, which are to be found in
mielke-geometrodynamics_F00157	47	1.000	`\operatorname{det} \mathrm{G}=1`	$\operatorname{det} \mathrm{G}=1$	for which $\det \mathrm{G} = 1$ holds. Inasmuch as the trace of the Lie-algebra-valued curvature
mielke-geometrodynamics_F00158	47	0.565	`\operatorname{Tr}(\Omega \wedge \Omega)`	$\operatorname{Tr}(\Omega \wedge \Omega)$	2-form Ω vanishes in such instances, the gauge-invariant 4-form $\mathrm{Tr}(\Omega \wedge \Omega)$ suffices
mielke-geometrodynamics_F00159	47	0.892	`\stackrel{*}{\pi}\left(\gamma_{2}\right)`	$\stackrel{*}{\pi}\left(\gamma_{2}\right)$	The integration of the second Chern class $\stackrel{*}{\pi}(\gamma_2)$ over the base space yields a
mielke-geometrodynamics_F00160	47	0.992	`\mathrm{G}=\mathrm{SU}(\mathrm{f})`	$\mathrm{G}=\mathrm{SU}(\mathrm{f})$	index"; see JACKIW (1977). Concerning bundles with structure group $\mathrm{G} = \mathrm{SU}(\mathrm{f})$,
mielke-geometrodynamics_F00161	47	0.827	`\mathrm{SU}(2)`	$\mathrm{SU}(2)$	et al. (1975) is correct, since in their paper, the isospin group $\mathrm{SU}(2)$ has been enlarged
mielke-geometrodynamics_F00162	47	0.749	`\widetilde{\mathrm{SO}}(4) \approx \mathrm{SU}(2) \otimes \mathrm{SU}(2)`	$\widetilde{\mathrm{SO}}(4) \approx \mathrm{SU}(2) \otimes \mathrm{SU}(2)$	to the real structure group $\widetilde{\mathrm{SO}}(4) \approx \mathrm{SU}(2) \otimes \mathrm{SU}(2)$, due to a special isomorphism
mielke-geometrodynamics_F00163	47	0.995	`\gamma_{2}`	γ_{2}	of the closed form γ_2 onto M is even an exact one. This is a property that it shares
mielke-geometrodynamics_F00164	48	1.000	`\mathrm{F}_{\alpha \beta}{ }^{\mathrm{j}}`	$\mathrm{F}_{\alpha \beta}{ }^{\mathrm{j}}$	that the field strengths $\mathrm{F}_{\alpha\beta}{}^{\mathrm{j}}$ (components of the curvature form) have to vanish faster
mielke-geometrodynamics_F00165	48	1.000	`|\mathrm{x}|^{-2}`	$\lvert\mathrm{x}\rvert^{-2}$	than $\lvert\mathrm{x}\rvert^{-2}$ at infinity, since otherwise, the integral (2.8.13) would not exist. In order to
mielke-geometrodynamics_F00166	49	1.000	`|x| \rightarrow \infty`	$\lvert x\rvert \rightarrow \infty$	should hold for $\lvert x\rvert \rightarrow \infty$. Here $\stackrel{\infty}{\omega}$ denotes a Maurer–Cartan connection, i.e., a left-
mielke-geometrodynamics_F00167	49	1.000	`\stackrel{\infty}{\omega}`	$\stackrel{\infty}{\omega}$	should hold for $\lvert x\rvert \rightarrow \infty$. Here $\stackrel{\infty}{\omega}$ denotes a Maurer–Cartan connection, i.e., a left-
mielke-geometrodynamics_F00168	49	0.978	`\partial \mathrm{M}`	$\partial \mathrm{M}$	If the boundary $\partial\mathrm{M}$ is chosen in such a way that only a pure gauge connection $\stackrel{\infty}{\omega}$
mielke-geometrodynamics_F00169	49	0.999	`\mathrm{G}=\mathrm{SU}(2)`	$\mathrm{G}=\mathrm{SU}(2)$	in full compliance with (2.8.15); cf. JACKIW (1980). If $\mathrm{G} = \mathrm{SU}(2)$ serves as a struc-
mielke-geometrodynamics_F00170	49	1.000	`\mathrm{S}^{3}`	S^{3}	S^3 topologically, then it can be shown that $\mathrm{c}_2(\mathrm{M})$ determines the "mapping degree"
mielke-geometrodynamics_F00171	49	1.000	`\mathrm{c}_{2}(\mathrm{M})`	$\mathrm{c}_{2}(\mathrm{M})$	S^3 topologically, then it can be shown that $\mathrm{c}_2(\mathrm{M})$ determines the "mapping degree"
mielke-geometrodynamics_F00172	49	0.954	`\mathrm{S}^{3} \approx \mathrm{SU}(2)`	$\mathrm{S}^{3} \approx \mathrm{SU}(2)$	or the *winding number*[6] of the mapping of S^3 in $\mathrm{S}^3 \approx \mathrm{SU}(2)$. These considerations
mielke-geometrodynamics_F00173	49	0.995	`\mathrm{n}^{\text {th }}`	$\mathrm{n}^{\text {th }}$	[6]Speaking more generally, the n^{th}-homotopy group of the n-dimensional sphere S^n is given by the
mielke-geometrodynamics_F00174	49	1.000	`\mathbb{Z}`	$\mathbb{Z}$	group $\mathbb{Z}$ of integers, i.e., by $\pi_n(S^n) = \mathbb{Z}$. As such, it determines the winding number of the mapping
mielke-geometrodynamics_F00175	49	1.000	`\pi_{n}\left(S^{n}\right)=\mathbb{Z}`	$\pi_{n}\left(S^{n}\right)=\mathbb{Z}$	group $\mathbb{Z}$ of integers, i.e., by $\pi_n(S^n) = \mathbb{Z}$. As such, it determines the winding number of the mapping
mielke-geometrodynamics_F00176	49	1.000	`\mathrm{S}^{\mathrm{n}} \rightarrow \mathrm{S}^{\mathrm{n}}`	$\mathrm{S}^{\mathrm{n}} \rightarrow \mathrm{S}^{\mathrm{n}}$	$\mathrm{S}^{\mathrm{n}} \rightarrow \mathrm{S}^{\mathrm{n}}$.
mielke-geometrodynamics_F00177	52	1.000	`\delta \varphi`	$\delta \varphi$	The operator $\delta\varphi$ of *small variations* $\varphi + \delta\varphi$ of the fields φ is satisfying the Leibniz
mielke-geometrodynamics_F00178	52	1.000	`\varphi+\delta \varphi`	$\varphi+\delta \varphi$	The operator $\delta\varphi$ of *small variations* $\varphi + \delta\varphi$ of the fields φ is satisfying the Leibniz
mielke-geometrodynamics_F00179	53	0.937	`(n-1)`	$(n-1)$	be the $(n-1)$-form of the *field momenta* which are *canonically conjugated* to $d\varphi$. It
mielke-geometrodynamics_F00180	53	0.937	`d \varphi`	$d \varphi$	be the $(n-1)$-form of the *field momenta* which are *canonically conjugated* to $d\varphi$. It
mielke-geometrodynamics_F00181	53	1.000	`\delta`	δ	after δ has been interchanged with the integration. The last term is an exact form.
mielke-geometrodynamics_F00182	53	1.000	`s`	s	where s is the signature of the pseudo-Riemannian manifold.

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mielke-geometrodynamics_F00183	53	1.000	`d d \equiv 0`	$d d \equiv 0$	where the last term vanishes identically, due to $dd \equiv 0$. Following the physically
mielke-geometrodynamics_F00184	53	0.913	`\mathrm{x}^{i}`	x^{i}	quently does not depend upon the coordinates x^i explicitly, we get the well-known
mielke-geometrodynamics_F00185	54	1.000	`\varphi \rightarrow \psi:=\left(\varphi, \Theta^{(1)}:=d \varphi\right)`	$\varphi \rightarrow \psi:=\left(\varphi, \Theta^{(1)}:=d \varphi\right)$	via the substitution $\varphi \rightarrow \psi := (\varphi, \Theta^{(1)} := d\varphi)$ into a first order system of the
mielke-geometrodynamics_F00186	54	1.000	`\mathscr{D}(M)`	$\mathscr{D}(M)$	respect to the group $\mathscr{D}(M)$ of general coordinate-transformations of the spacetime
mielke-geometrodynamics_F00187	54	1.000	`\mathscr{G}`	$\mathscr{G}$	labels, but as well with respect to the "internal covariance group" $\mathscr{G}$ of gauge trans-
mielke-geometrodynamics_F00188	54	0.989	`L_{\text {mat }}`	$L_{\text {mat }}$	The invariance of L_{mat} with respect to $\mathscr{D}(\mathrm{M})$ is already guaranteed by the use
mielke-geometrodynamics_F00189	55	1.000	`\bar{\varphi} \mathscr{O} \varphi`	$\bar{\varphi} \mathscr{O} \varphi$	depend upon $\overline{\varphi}\mathscr{O}\varphi$, where $\mathscr{O}$ denotes a gauge-covariant operator
mielke-geometrodynamics_F00190	55	1.000	`\mathscr{O}`	$\mathscr{O}$	depend upon $\overline{\varphi}\mathscr{O}\varphi$, where $\mathscr{O}$ denotes a gauge-covariant operator
mielke-geometrodynamics_F00191	55	0.996	`\omega^{\prime}`	$\omega^{\prime}$	by means of a 1-form ω' in such a way that the term GdG^{-1} resulting from gauge
mielke-geometrodynamics_F00192	55	0.996	`\mathrm{GdG}^{-1}`	GdG^{-1}	by means of a 1-form ω' in such a way that the term GdG^{-1} resulting from gauge
mielke-geometrodynamics_F00193	55	1.000	`\mathrm{G} \in \mathscr{G}_{\mathrm{V}}`	$\mathrm{G} \in \mathscr{G}_{\mathrm{V}}$	concerning the "actively" applied gauge transformation $\mathrm{G} \in \mathscr{G}_{\mathrm{V}}$. For the derivation
mielke-geometrodynamics_F00194	55	1.000	`{ }^{\mathrm{G}} \omega`	${ }^{\mathrm{G}} \omega$	matrix ${}^{\mathrm{G}}\omega$ of the connection relative to a basis ${}^{\mathrm{G}}\mathrm{b}$, which is locally "rotated" by
mielke-geometrodynamics_F00195	55	1.000	`{ }^{\mathrm{G}} \mathrm{b}`	${ }^{\mathrm{G}} \mathrm{b}$	matrix ${}^{\mathrm{G}}\omega$ of the connection relative to a basis ${}^{\mathrm{G}}\mathrm{b}$, which is locally "rotated" by
mielke-geometrodynamics_F00196	55	1.000	`\mathrm{G}^{-1}`	G^{-1}	If G ist substituted by G^{-1} exactly the relation (3.2.3) comes into existence, which
mielke-geometrodynamics_F00197	56	0.749	`{ }_{\sigma_{\alpha}}^{*} \omega`	${ }_{\sigma_{\alpha}}^{*} \omega$	image or pull-back $\overset{*}{\sigma}_\alpha\omega$ due to the defining properties of a connection (see: KN I,
mielke-geometrodynamics_F00198	56	0.818	`G^{-1} \omega`	$G^{-1} \omega$	The transformed connection ${}^{G^{-1}}\omega$ deviates from the original one solely by a gauge-
mielke-geometrodynamics_F00199	56	1.000	`\bar{\omega}`	$\bar{\omega}$	For a so-called "copy" $\overline{\omega}$ of a connection ω, this expression is to be taken as different
mielke-geometrodynamics_F00200	56	1.000	`\bar{\Omega}=\mathrm{G}^{-1} \Omega \mathrm{G}`	$\bar{\Omega}=\mathrm{G}^{-1} \Omega \mathrm{G}$	from zero, although - according to the definition - its curvature $\overline{\Omega} = \mathrm{G}^{-1}\Omega\mathrm{G}$ is
mielke-geometrodynamics_F00201	56	1.000	`d^{*} \omega \simeq 0`	$d^{*} \omega \simeq 0$	transversal or Coulomb condition $d^*\omega \simeq 0$ does not fix uniquely (large) gauge trans-
mielke-geometrodynamics_F00202	56	1.000	`A=A_{i} d x^{i}`	$A=A_{i} d x^{i}$	tions of Maxwell, the electromagnetic four-potential $A = A_i dx^i$, a one-form, is the
mielke-geometrodynamics_F00203	56	1.000	`F`	F	fundamental variable, whereas Faraday's field strength F is a derived concept and
mielke-geometrodynamics_F00204	57	0.992	`d d \Phi \equiv 0`	$d d \Phi \equiv 0$	On the other hand, $dd\Phi \equiv 0$ for any exterior p-form Φ, due to the Poincaré lemma.
mielke-geometrodynamics_F00205	57	0.992	`\Phi`	Φ	On the other hand, $dd\Phi \equiv 0$ for any exterior p-form Φ, due to the Poincaré lemma.
mielke-geometrodynamics_F00206	57	1.000	`d F \equiv 0`	$d F \equiv 0$	Thus if $dF \equiv 0$, there exists locally a one-form A such that $F = dA$. However, the
mielke-geometrodynamics_F00207	57	1.000	`F=d A`	$F=d A$	Thus if $dF \equiv 0$, there exists locally a one-form A such that $F = dA$. However, the
mielke-geometrodynamics_F00208	57	0.981	`U(1)`	$U(1)$	local $U(1)$ *gauge transformation*
mielke-geometrodynamics_F00209	57	1.000	`D \rightarrow \mathscr{D}:=D+i e A`	$D \rightarrow \mathscr{D}:=D+i e A$	Although, A is needed for the standard minimal coupling $D \rightarrow \mathscr{D} := D + ieA$ to

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mielke-geometrodynamics_F00210	57	1.000	`L`	L	For a Lagrangian formulation, let us depart from the four-form L which consist of
mielke-geometrodynamics_F00211	57	0.980	`\Psi`	Ψ	the gauge field part and L_{mat} governing the matter field Ψ and its minimal coupling
mielke-geometrodynamics_F00212	57	1.000	`S=\int L`	$S=\int L$	usually be assumed to be of first order in the fields. Stationarity of the action $S=\int L$
mielke-geometrodynamics_F00213	57	1.000	`H`	H	The excitation H is the field momentum conjugated to A, i.e.
mielke-geometrodynamics_F00214	58	1.000	`d d H \equiv 0`	$d d H \equiv 0$	$ddH \equiv 0$, the field equations imposes 'on shell' that the electric current j is con-
mielke-geometrodynamics_F00215	58	1.000	`j`	j	$ddH \equiv 0$, the field equations imposes 'on shell' that the electric current j is con-
mielke-geometrodynamics_F00216	58	1.000	`C:=A \wedge F=A \wedge d A`	$C:=A \wedge F=A \wedge d A$	where $C := A \wedge F = A \wedge dA$ is the abelian Chern–Simons term, known to violate
mielke-geometrodynamics_F00217	58	1.000	`P`	P	parity P.
mielke-geometrodynamics_F00218	58	0.885	`\stackrel{\theta}{L}=-\left(\cos \theta F \wedge{ }^{*} F+\sin \theta F \wedge F\right) / 2`	$\stackrel{\theta}{L}=-\left(\cos \theta F \wedge{ }^{*} F+\sin \theta F \wedge F\right) / 2$	$\stackrel{\theta}{L} = -(\cos\theta\, F \wedge {}^*F + \sin\theta\, F \wedge F)/2$, where θ is the'vacuum' angle of duality rotations or
mielke-geometrodynamics_F00219	58	0.885	`\theta`	θ	$\stackrel{\theta}{L} = -(\cos\theta\, F \wedge {}^*F + \sin\theta\, F \wedge F)/2$, where θ is the'vacuum' angle of duality rotations or
mielke-geometrodynamics_F00220	59	0.950	`*`	$*$	The Hodge dual $*$ depends on the metric g and on the orientation of the manifold, as
mielke-geometrodynamics_F00221	59	0.950	`g`	g	The Hodge dual $*$ depends on the metric g and on the orientation of the manifold, as
mielke-geometrodynamics_F00222	59	1.000	`\eta`	η	where η is the standard volume four-form. The Pontryagin four-form can be written
mielke-geometrodynamics_F00223	59	1.000	`\partial \mathfrak{D} / \partial t`	$\partial \mathfrak{D} / \partial t$	[3]The displacement current $\partial\mathfrak{D}/\partial t$, where $\mathfrak{D} = \varepsilon\mathbf{E}$, was anticipated 1839 by James Mac Cullagh, cf.
mielke-geometrodynamics_F00224	59	1.000	`\mathfrak{D}=\varepsilon \mathbf{E}`	$\mathfrak{D}=\varepsilon \mathbf{E}$	[3]The displacement current $\partial\mathfrak{D}/\partial t$, where $\mathfrak{D} = \varepsilon\mathbf{E}$, was anticipated 1839 by James Mac Cullagh, cf.
mielke-geometrodynamics_F00225	60	1.000	`\vartheta_{[\alpha} \wedge \Sigma_{\beta]}=0`	$\vartheta_{[\alpha} \wedge \Sigma_{\beta]}=0$	and turns out to be symmetric, i.e., $\vartheta_{[\alpha} \wedge \Sigma_{\beta]} = 0$. For non-vanishing charge current
mielke-geometrodynamics_F00226	60	1.000	`B=\frac{1}{2} B_{i j} d x^{i} \wedge d x^{j}`	$B=\frac{1}{2} B_{i j} d x^{i} \wedge d x^{j}$	two-form $B = \frac{1}{2}B_{ij}dx^i \wedge dx^j$ are varied *independently*, slightly reminiscent of the
mielke-geometrodynamics_F00227	60	1.000	`B`	B	Independent variations with respect to A and B lead to $dB \cong 0$ and to the *constraint* of
mielke-geometrodynamics_F00228	60	1.000	`d B \cong 0`	$d B \cong 0$	Independent variations with respect to A and B lead to $dB \cong 0$ and to the *constraint* of
mielke-geometrodynamics_F00229	60	1.000	`F:=d A \cong 0`	$F:=d A \cong 0$	vanishing field strength $F := dA \cong 0$. This implies, however, that such a primordial
mielke-geometrodynamics_F00230	60	1.000	`d C`	$d C$	(3.3.22) only by a boundary term dC derived from a *Chern–Simons* (CS) three-form
mielke-geometrodynamics_F00231	60	1.000	`H=d B`	$H=d B$	excitation $H = dB$, the Kalb–Ramond *axion* three-form.
mielke-geometrodynamics_F00232	61	1.000	`\delta d C / \delta A=d F \equiv 0`	$\delta d C / \delta A=d F \equiv 0$	can be recovered via the variation of the Pontryagin term, e.g., $\delta dC/\delta A = dF \equiv 0$
mielke-geometrodynamics_F00233	61	0.998	`\psi=\psi_{i} d x^{i}`	$\psi=\psi_{i} d x^{i}$	where $\psi = \psi_i dx^i$ is a one-form or a covector; cf. LUCCHESI et al. (1993). In a
mielke-geometrodynamics_F00234	61	1.000	`s \psi=D \phi`	$s \psi=D \phi$	BRST quantization, due to the zero mode $s\psi = D\phi$ with $s\phi = 0$, the "topological"
mielke-geometrodynamics_F00235	61	1.000	`s \phi=0`	$s \phi=0$	BRST quantization, due to the zero mode $s\psi = D\phi$ with $s\phi = 0$, the "topological"
mielke-geometrodynamics_F00236	61	1.000	`B F`	$B F$	to topological field theory is the BF formalism which, in the case of gravity, was

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mielke-geometrodynamics_F00237	61	1.000	`F={ }^{*} B`	$F={ }^{*} B$	the definition of the field strength (3.3.24), but as a bonus, from the relation $F = {}^*B$
mielke-geometrodynamics_F00238	61	0.998	`j:=\delta L_{\text {matter }} / \delta A`	$j:=\delta L_{\text {matter }} / \delta A$	where $j := \delta L_{\text{matter}}/\delta A$ is the matter current.
mielke-geometrodynamics_F00239	61	1.000	`d j \cong 0`	$d j \cong 0$	which "on shell," is conserved classically, i.e., $dj \cong 0$. In general, this includes
mielke-geometrodynamics_F00240	61	0.999	`d A`	$d A$	*Pauli-type terms* generated by the variation of the Lagrangian with respect to dA; cf.
mielke-geometrodynamics_F00241	62	1.000	`V(B)`	$V(B)$	The quadratic term in (3.3.28) can be generalized to a four-form potential $V(B)$.
mielke-geometrodynamics_F00242	62	1.000	`d B+j \cong 0`	$d B+j \cong 0$	Then the field equation $dB + j \cong 0$ together with the relation $F \cong \partial V/\partial B$ for
mielke-geometrodynamics_F00243	62	1.000	`F \cong \partial V / \partial B`	$F \cong \partial V / \partial B$	Then the field equation $dB + j \cong 0$ together with the relation $F \cong \partial V/\partial B$ for
mielke-geometrodynamics_F00244	62	1.000	`-B \wedge F+B \wedge B \wedge B / 3`	$-B \wedge F+B \wedge B \wedge B / 3$	systems with a cubic term $-B\wedge F+B\wedge B\wedge B/3$ are directly related to Chern–Simons
mielke-geometrodynamics_F00245	62	0.999	`C:=A \wedge F-A \wedge A \wedge A / 3`	$C:=A \wedge F-A \wedge A \wedge A / 3$	theories departing from the noninvariant three-form $C := A \wedge F - A \wedge A \wedge A/3$; cf.
mielke-geometrodynamics_F00246	62	1.000	`F=-B \wedge B`	$F=-B \wedge B$	$F = -B \wedge B$ together with the Bianchi identity $DB = DF \equiv 0$ correspond, in 3D
mielke-geometrodynamics_F00247	62	1.000	`D B=D F \equiv 0`	$D B=D F \equiv 0$	$F = -B \wedge B$ together with the Bianchi identity $DB = DF \equiv 0$ correspond, in 3D
mielke-geometrodynamics_F00248	62	0.999	`\Gamma_{\alpha}{ }^{\mathrm{j}}(\mathrm{m})`	$\Gamma_{\alpha}{ }^{\mathrm{j}}(\mathrm{m})$	Speaking a priori, the connection ω and the corresponding gauge potentials $\Gamma_\alpha{}^{\mathrm{j}}(\mathrm{m})$
mielke-geometrodynamics_F00249	62	1.000	`d \omega`	$d \omega$	only of its first derivative $d\omega$. The only *gauge-covariant* object of this kind that meets
mielke-geometrodynamics_F00250	62	0.668	`\mathscr{F}`	$\mathscr{F}$	4-forms into 0-forms, i.e., scalar functions. In quantized gauge theories, $\mathscr{F}$ and ${}^*\mathscr{F}$
mielke-geometrodynamics_F00251	62	0.668	`{ }^{*} \mathscr{F}`	${ }^{*} \mathscr{F}$	4-forms into 0-forms, i.e., scalar functions. In quantized gauge theories, $\mathscr{F}$ and ${}^*\mathscr{F}$
mielke-geometrodynamics_F00252	63	0.731	`\Pi^{\Omega}`	Π^{Ω}	The 2-form Π^Ω of the *field momenta* being *canonically conjugated* to Ω are formally
mielke-geometrodynamics_F00253	63	0.894	`n-1`	$n-1$	$(n-1)$-form τ of the matter current as a source. The latter is canonically conjugate
mielke-geometrodynamics_F00254	63	0.894	`\tau`	τ	$(n-1)$-form τ of the matter current as a source. The latter is canonically conjugate
mielke-geometrodynamics_F00255	64	1.000	`\delta L / \delta \omega`	$\delta L / \delta \omega$	the variation $\delta L/\delta\omega$ of the total system. They are complemented by the Bianchi
mielke-geometrodynamics_F00256	64	1.000	`(1978,1980)`	$(1978,1980)$	it has been shown by HORNDESKI (1978, 1980) that a field theory of Yang–Mills
mielke-geometrodynamics_F00257	65	1.000	`\mathrm{H}^{3}`	H^{3}	hypersurface H^3 or its boundary $\partial\mathrm{H}^3$, respectively. Note that ι involves not only
mielke-geometrodynamics_F00258	65	1.000	`\partial \mathrm{H}^{3}`	$\partial \mathrm{H}^{3}$	hypersurface H^3 or its boundary $\partial\mathrm{H}^3$, respectively. Note that ι involves not only
mielke-geometrodynamics_F00259	65	1.000	`\iota`	ι	hypersurface H^3 or its boundary $\partial\mathrm{H}^3$, respectively. Note that ι involves not only
mielke-geometrodynamics_F00260	65	0.879	`\mathrm{SU}(\mathrm{N})`	$\mathrm{SU}(\mathrm{N})$	In more generalized versions, these theories make use of SU(N) as a structure group,
mielke-geometrodynamics_F00261	66	0.971	`L\left(\mathscr{F},{ }^{*} \mathscr{F}\right)`	$L\left(\mathscr{F},{ }^{*} \mathscr{F}\right)$	$L(\mathscr{F}, {}^*\mathscr{F})$ is of course a mere empirical one and can be answered only by means of
mielke-geometrodynamics_F00262	66	0.435	`\mathrm{I}^{\mathrm{j}}=\mathrm{i} \sigma^{\mathrm{j}}`	$\mathrm{I}^{\mathrm{j}}=\mathrm{i} \sigma^{\mathrm{j}}$	with the infinitesimal generators $\mathrm{I}^{\mathrm{j}} = \mathrm{i}\sigma^{\mathrm{j}}$, thus obtaining the group $\mathrm{SU}(2)\times\mathrm{SU}(2) \approx$
mielke-geometrodynamics_F00263	66	0.435	`\mathrm{SU}(2) \times \mathrm{SU}(2) \approx`	$\mathrm{SU}(2) \times \mathrm{SU}(2) \approx$	with the infinitesimal generators $\mathrm{I}^{\mathrm{j}} = \mathrm{i}\sigma^{\mathrm{j}}$, thus obtaining the group $\mathrm{SU}(2)\times\mathrm{SU}(2) \approx$

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mielke-geometrodynamics_F00264	66	0.765	`\mathrm{SO}(4)`	$\mathrm{SO}(4)$	$\mathrm{SO}(4)$ by proceeding from the array of matrices
mielke-geometrodynamics_F00265	67	0.998	`\mathrm{h}=\mathrm{h}(|\mathrm{x}|)`	$\mathrm{h}=\mathrm{h}(\vert\mathrm{x}\vert)$	i.e., $\mathrm{h}=\mathrm{h}(\vert\mathrm{x}\vert)$ in the spherically symmetric case. A similar nonlinear equation occurs
mielke-geometrodynamics_F00266	67	0.499	`\mathrm{M}^{4}`	M^{4}	cf. JACKIW et al. (1977). In a flat space M^4 with Euclidean signature $\mathrm{s}=0$, this is a
mielke-geometrodynamics_F00267	67	0.999	`\mathrm{k}=1`	$\mathrm{k}=1$	in the case of $\mathrm{k}=1$ ('T HOOFT 1977). The configuration as given above depends
mielke-geometrodynamics_F00268	67	0.512	`5 \mathrm{k}+4`	$5 \mathrm{k}+4$	on $5\mathrm{k}+4$ parameters. From studies of related objects in algebraic topology it is
mielke-geometrodynamics_F00269	67	1.000	`\mathbb{R}^{4} \cup\{\infty\} \approx \mathrm{S}^{4}`	$\mathbb{R}^{4} \cup\{\infty\} \approx \mathrm{S}^{4}$	of the Yang–Mills equation in a Euclidean space $\mathbb{R}^4 \cup \{\infty\} \approx \mathrm{S}^4$, which is confor-
mielke-geometrodynamics_F00270	68	1.000	`|-\infty\rangle`	$\vert-\infty\rangle$	transition amplitude between the asymptotic states $\vert-\infty\rangle$ and $\langle\infty\vert$. This ampli-
mielke-geometrodynamics_F00271	68	1.000	`\langle\infty|`	$\langle\infty\vert$	transition amplitude between the asymptotic states $\vert-\infty\rangle$ and $\langle\infty\vert$. This ampli-
mielke-geometrodynamics_F00272	68	1.000	`\mathscr{M}_{\omega}`	$\mathscr{M}_{\omega}$	configuration space $\mathscr{M}_\omega$ of all inequivalent gauge fields, i.e., of all possible connec-
mielke-geometrodynamics_F00273	68	0.991	`\mathscr{M}_{\omega}=\mathscr{C} / \mathscr{G}_{p}`	$\mathscr{M}_{\omega}=\mathscr{C} / \mathscr{G}_{p}$	$\mathscr{M}_\omega=\mathscr{C}/\mathscr{G}_p$ has to be taken as the configuration space proper. Under not too stringent
mielke-geometrodynamics_F00274	68	0.999	`\mu[\omega]`	$\mu[\omega]$	can be shown that the “measure” $\mu[\omega]$ for the functional integration, as given by the
mielke-geometrodynamics_F00275	69	0.976	`S`	S	regime of gauge fields, the so-called *S-duality*.
mielke-geometrodynamics_F00276	69	1.000	`\operatorname{Spin}^{c}(4)`	$\operatorname{Spin}^{c}(4)$	2006) coupled to spinors ψ regarded as a lift from the frame bundle to $\mathrm{Spin}^c(4)$.
mielke-geometrodynamics_F00277	69	0.988	`\psi=\psi_{L}+\psi_{R}=P_{-} \psi+P_{+} \psi`	$\psi=\psi_{L}+\psi_{R}=P_{-} \psi+P_{+} \psi$	cf. JOST et al. (1996), where here the decomposition $\psi=\psi_L+\psi_R=P_-\psi+P_+\psi$
mielke-geometrodynamics_F00278	70	1.000	`\bar{\psi}`	$\bar{\psi}$	The variation with respect to $\overline{\psi}$ and A leads to the *convective*-type spinor equation
mielke-geometrodynamics_F00279	70	1.000	`R`	R	(3.6.1). For a nonnegative scalar curvature R in the Weizenböck formula, all solutions
mielke-geometrodynamics_F00280	70	1.000	`\psi=0`	$\psi=0$	satisfy $\psi=0$ and then are $U(1)$ instantons.
mielke-geometrodynamics_F00281	70	0.996	`F^{ \pm}`	$F^{ \pm}$	$F^\pm$ resembles (MIELKE 1984) the modified (double) duality ansatz
mielke-geometrodynamics_F00282	71	1.000	`\mathrm{A}_{\alpha}{ }^{\mathrm{j}}`	$\mathrm{A}_{\alpha}{ }^{\mathrm{j}}$	gauge potentials $\mathrm{A}_\alpha{}^j$ in the Lagrangian n-form. However, there is only one massless
mielke-geometrodynamics_F00283	71	1.000	`\varphi^{j}`	φ^{j}	fields φ^j with f components but also the vector fields $\Theta^{(1)j}$ will be coupled to the
mielke-geometrodynamics_F00284	71	1.000	`\Theta^{(1) j}`	$\Theta^{(1) j}$	fields φ^j with f components but also the vector fields $\Theta^{(1)j}$ will be coupled to the
mielke-geometrodynamics_F00285	71	0.902	`\phi^{(0)}`	$\phi^{(0)}$	the 0-form $\phi^{(0)}$ and the 1-form $\Theta^{(1)}$ with values in the adjoint representation of G.
mielke-geometrodynamics_F00286	71	0.902	`\Theta^{(1)}`	$\Theta^{(1)}$	the 0-form $\phi^{(0)}$ and the 1-form $\Theta^{(1)}$ with values in the adjoint representation of G.
mielke-geometrodynamics_F00287	71	1.000	`\mathrm{A}_{\alpha}{ }^{\mathrm{j}}(\mathrm{m})`	$\mathrm{A}_{\alpha}{ }^{\mathrm{j}}(\mathrm{m})$	gies to the Higgs model. The coupling of the gauge potentials $\mathrm{A}_\alpha{}^j(\mathrm{m})$ to the Higgs
mielke-geometrodynamics_F00288	71	1.000	`\delta L_{\mathrm{H}} / \delta \bar{\phi}^{(0)}`	$\delta L_{\mathrm{H}} / \delta \bar{\phi}^{(0)}$	the G-equivalence principle. From the variations and $\delta L_{\mathrm{H}}/\delta\overline{\phi}^{(0)}$ and $\delta L_{\mathrm{H}}/\delta\overline{\Theta}^{(1)}$, the
mielke-geometrodynamics_F00289	71	1.000	`\delta L_{\mathrm{H}} / \delta \bar{\Theta}^{(1)}`	$\delta L_{\mathrm{H}} / \delta \bar{\Theta}^{(1)}$	the G-equivalence principle. From the variations and $\delta L_{\mathrm{H}}/\delta\overline{\phi}^{(0)}$ and $\delta L_{\mathrm{H}}/\delta\overline{\Theta}^{(1)}$, the
mielke-geometrodynamics_F00290	72	1.000	`\mu`	μ	It should be noted that the parameter μ in (3.7.5) complies formally with an imaginary

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mielke-geometrodynamics_F00291	72	1.000	`\varphi=0`	$\varphi=0$	mass of the Klein–Gordon-type field. However, the trivial field configuration $\varphi=0$
mielke-geometrodynamics_F00292	72	1.000	`|\varphi|`	$\lvert\varphi\rvert$	This is a consequence of the *nonlinear term* in (3.7.1), which is quartic in $\lvert\varphi\rvert$.
mielke-geometrodynamics_F00293	72	1.000	`{ }^{\lambda} \varphi`	${ }^{\lambda} \varphi$	quantum-mechanical state ${ }^{\lambda}\varphi$, for which the vacuum expectation value
mielke-geometrodynamics_F00294	72	1.000	`\varphi_{\mathrm{H}}:=\varphi-\mu / \sqrt{\lambda}`	$\varphi_{\mathrm{H}}:=\varphi-\mu / \sqrt{\lambda}$	$\varphi_{\mathrm{H}}:=\varphi-\mu/\sqrt{\lambda}$ along with the gauge fields $\mathrm{A}_{\alpha}{}^{j}$ acquire a real, i.e., physical, mass
mielke-geometrodynamics_F00295	72	1.000	`\mathrm{m}_{\mathrm{H}}=\sqrt{2} \mu`	$\mathrm{m}_{\mathrm{H}}=\sqrt{2} \mu$	that is determined by the curvature of the potential at the bottom, i.e., by $\mathrm{m}_{\mathrm{H}}=\sqrt{2}\mu$,
mielke-geometrodynamics_F00296	72	0.882	`\mathrm{G} / \mathrm{H}`	G / H	field φ such that its range is an orbit G/H of G in V^{ρ} with respect to a subgroup H,
mielke-geometrodynamics_F00297	72	1.000	`\mathrm{v}_{0} \in V^{\rho}`	$\mathrm{v}_{0} \in V^{\rho}$	Let $\mathrm{v}_0 \in V^{\rho}$ be a fixed vector for which the cross section $\stackrel{\circ}{\varphi}_{\text{min}} := \sigma(\mathrm{v}_0) \in \mathrm{C}^{\infty}(\mathrm{V}^{\rho})$
mielke-geometrodynamics_F00298	72	1.000	`\stackrel{\circ}{\varphi}_{\text {min }}:=\sigma\left( \mathrm{v}_{0}\right) \in \mathrm{C}^{\infty}\left( \mathrm{V}^{\rho}\right)`	$\stackrel{\circ}{\varphi}_{\text {min }}:=\sigma\left(\mathrm{v}_{0}\right) \in \mathrm{C}^{\infty}\left(\mathrm{V}^{\rho}\right)$	Let $\mathrm{v}_0 \in V^{\rho}$ be a fixed vector for which the cross section $\stackrel{\circ}{\varphi}_{\text{min}} := \sigma(\mathrm{v}_0) \in \mathrm{C}^{\infty}(\mathrm{V}^{\rho})$
mielke-geometrodynamics_F00299	72	0.683	`\mathrm{a} \mathrm{g} \in \mathrm{G}`	$\mathrm{a} \mathrm{g} \in \mathrm{G}$	to find a $\mathrm{g} \in \mathrm{G}$ that connects any two vectors v and v_0 from V^{ρ} via the relation
mielke-geometrodynamics_F00300	72	0.683	`\mathrm{v}_{0}`	v_{0}	to find a $\mathrm{g} \in \mathrm{G}$ that connects any two vectors v and v_0 from V^{ρ} via the relation
mielke-geometrodynamics_F00301	72	1.000	`\mathrm{v}=\rho(\mathrm{g}) \cdot \mathrm{v}_{0}`	$\mathrm{v}=\rho(\mathrm{g}) \cdot \mathrm{v}_{0}$	$\mathrm{v}=\rho(\mathrm{g})\cdot\mathrm{v}_0$. An invariant subgroup of G, the so-called isotropy group
mielke-geometrodynamics_F00302	73	0.996	`\stackrel{\circ}{\varphi}_{\text {min }}=\sigma\left( \mathrm{v}_{0}\right)`	$\stackrel{\circ}{\varphi}_{\text {min }}=\sigma\left(\mathrm{v}_{0}\right)$	can be *uniquely* adjoined to the fixed $\stackrel{\circ}{\varphi}_{\text{min}}=\sigma(\mathrm{v}_0)$. Then
mielke-geometrodynamics_F00303	73	1.000	`\mathrm{P}(\mathrm{G}, \mathrm{M}, \pi, \delta)`	$\mathrm{P}(\mathrm{G}, \mathrm{M}, \pi, \delta)$	is a subbundle of $\mathrm{P}(\mathrm{G},\mathrm{M},\pi,\delta)$ over the same base space; its structure group is H
mielke-geometrodynamics_F00304	73	0.702	`\mathfrak{h}`	$\mathfrak{h}$	**Proposition** *A connection ω in P is reducible to an $\mathfrak{h}$-valued connection ω_{H} in Q if*
mielke-geometrodynamics_F00305	73	0.702	`\omega_{\mathrm{H}}`	ω_{H}	**Proposition** *A connection ω in P is reducible to an $\mathfrak{h}$-valued connection ω_{H} in Q if*
mielke-geometrodynamics_F00306	73	0.702	`Q`	Q	**Proposition** *A connection ω in P is reducible to an $\mathfrak{h}$-valued connection ω_{H} in Q if*
mielke-geometrodynamics_F00307	73	1.000	`\omega_{0}`	ω_{0}	theories of gravity that will be developed later on. Let ω_0 be a timelike component
mielke-geometrodynamics_F00308	74	1.000	`\phi`	ϕ	of the connection can be identified with the Higgs field ϕ without self-interaction:
mielke-geometrodynamics_F00309	74	1.000	`\lambda \rightarrow 0`	$\lambda \rightarrow 0$	limit $\lambda \rightarrow 0$ may be obtained by identifying ω_0 in the interaction-free Yang–Mills
mielke-geometrodynamics_F00310	74	1.000	`\stackrel{*}{\sigma} \vec{\omega}`	$\stackrel{*}{\sigma} \vec{\omega}$	provides the static Yang–Mills potential $\stackrel{*}{\sigma}\vec{\omega}$ of the monopole. After this identifica-
mielke-geometrodynamics_F00311	74	0.503	`(1981,1983)`	$(1981,1983)$	(1977), TCHRAKIAN (1981, 1983), PRASAD & ROSSI (1981a, b). The topological
mielke-geometrodynamics_F00312	75	1.000	`\sigma`	σ	In this context, it is worthwhile to mention that in the nonlinear σ-model, a variant

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mielke-geometrodynamics_F00313	75	0.657	`\mathbb{R}^{2} \times \mathrm{S}^{2}`	$\mathbb{R}^{2} \times \mathrm{S}^{2}$	**Gauge Field Terminology** | **Fiber Bundle Terminology** spacetime | base space space of phase factors | bundle space symmetry or gauge group | structure group gauge transformations | inner automorphisms of the bundle gauge principle | G-equivalence principle classical fields | cross section of a vector bundle gauge potential | connection 1-form gauge field strength | curvature 2-form electromagnetism | dynamical theory with a connection in a U(1)-bundle electromagnetism with monopoles | connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times S^2$ Dirac's monopole quantization | classification of U(1)-bundles according to the first Chern class Yang–Mills theory | dynamical theory with a connection in a SU(2)-bundle instanton number | classification of SU(2)-bundles over S^4 according to the second Chern class
mielke-geometrodynamics_F00314	75	0.657	`\mathrm{S}^{4}`	S^{4}	**Gauge Field Terminology** | **Fiber Bundle Terminology** spacetime | base space space of phase factors | bundle space symmetry or gauge group | structure group gauge transformations | inner automorphisms of the bundle gauge principle | G-equivalence principle classical fields | cross section of a vector bundle gauge potential | connection 1-form gauge field strength | curvature 2-form electromagnetism | dynamical theory with a connection in a U(1)-bundle electromagnetism with monopoles | connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times S^2$ Dirac's monopole quantization | classification of U(1)-bundles according to the first Chern class Yang–Mills theory | dynamical theory with a connection in a SU(2)-bundle instanton number | classification of SU(2)-bundles over S^4 according to the second Chern class
mielke-geometrodynamics_F00315	78	1.000	`\mathrm{N}=2`	$\mathrm{N}=2$	N= 2 supersymmetric Yang-Mills theory. Nuclear Phys B 426(1):19–52
mielke-geometrodynamics_F00316	79	1.000	`\ell^{*}`	ℓ^{*}	diminutive coupling constant given by the Planck length ℓ^*. It may even provide the
mielke-geometrodynamics_F00317	80	0.998	`(1980,1981)`	$(1980,1981)$	those theories of gravity, was worked out in more detail by HEHL (1980, 1981) using
mielke-geometrodynamics_F00318	81	0.951	`\mathrm{A}(\mathrm{n}, \mathbb{R})`	$\mathrm{A}(\mathrm{n}, \mathbb{R})$	and to start off deductively from the affine group $A(n, \mathbb{R})$ as structure group. As a
mielke-geometrodynamics_F00319	81	1.000	`\mathrm{A}^{\mathrm{n}}`	A^{n}	Euclidean space E^n, regarded as an affine space A^n.
mielke-geometrodynamics_F00320	81	0.999	`\mathrm{a}=\left(\mathrm{a}^{\alpha}\right) \in \mathbb{R}^{\mathrm{n}}`	$\mathrm{a}=\left(\mathrm{a}^{\alpha}\right) \in \mathbb{R}^{\mathrm{n}}$	Let $a = (a^\alpha) \in \mathbb{R}^n$ be a row vector. According to its definition, the action of an
mielke-geometrodynamics_F00321	81	0.992	`\mathrm{x} \in \mathrm{A}^{\mathrm{n}}`	$\mathrm{x} \in \mathrm{A}^{\mathrm{n}}$	element of $A(n, \mathbb{R})$ on a vector $x \in A^n$ can expressed as the composition of a general
mielke-geometrodynamics_F00322	81	1.000	`\tilde{x}:=\binom{x}{1} \in \mathbb{R}^{\mathrm{n}+1}`	$\tilde{x}:=\binom{x}{1} \in \mathbb{R}^{\mathrm{n}+1}$	In the following, it will be convenient to introduce a vector $\tilde{x} := \binom{x}{1} \in \mathbb{R}^{n+1}$ with
mielke-geometrodynamics_F00323	81	1.000	`\mathrm{n}+1`	$\mathrm{n}+1$	$n + 1$ components and to replace the group action (4.1.1) by the abstract transfor-
mielke-geometrodynamics_F00324	81	0.998	`\tilde{\mathrm{x}}`	$\tilde{\mathrm{x}}$	mations of $\tilde{x}$ via the matrices
mielke-geometrodynamics_F00325	81	0.890	`\mathrm{GL}(\mathrm{n}+1, \mathbb{R})`	$\mathrm{GL}(\mathrm{n}+1, \mathbb{R})$	This faithful representation of $A(n,\mathbb{R})$ by a subgroup of $GL(n + 1, \mathbb{R})$ shows clearly
mielke-geometrodynamics_F00326	81	0.623	`\mathrm{a} \in \mathbb{R}^{\mathrm{n}}`	$\mathrm{a} \in \mathbb{R}^{\mathrm{n}}$	of the translations $a \in \mathbb{R}^n$ and the linear transformations $g \in GL(n, \mathbb{R})$. The rule for
mielke-geometrodynamics_F00327	81	0.623	`\mathrm{g} \in \mathrm{GL}(\mathrm{n}, \mathbb{R})`	$\mathrm{g} \in \mathrm{GL}(\mathrm{n}, \mathbb{R})$	of the translations $a \in \mathbb{R}^n$ and the linear transformations $g \in GL(n, \mathbb{R})$. The rule for
mielke-geometrodynamics_F00328	81	1.000	`0(1, \mathrm{n}-1) \subset \mathrm{GL}(\mathrm{n}, \mathbb{R})`	$0(1, \mathrm{n}-1) \subset \mathrm{GL}(\mathrm{n}, \mathbb{R})$	$0(1, n - 1) \subset GL(n, \mathbb{R})$. The invariance of the (flat) metric ground form of the

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mielke-geometrodynamics_F00329	81	1.000	`\mathfrak{a}(\mathrm{n}, \mathbb{R})`	$\mathfrak{a}(\mathrm{n}, \mathbb{R})$	$\mathfrak{a}(\mathrm{n}, \mathbb{R})$ of the affine group, the analogous decomposition
mielke-geometrodynamics_F00330	82	0.999	`L_{\alpha \beta} \in \mathfrak{g l}(\mathrm{n}, \mathbb{R})`	$L_{\alpha \beta} \in \mathfrak{g l}(\mathrm{n}, \mathbb{R})$	In these formulas, $L_{\alpha\beta} \in \mathfrak{gl}(\mathrm{n}, \mathbb{R})$ and $\mathrm{P}_{\alpha} \in \mathbb{R}^{\mathrm{n}}$ denote the infinitesimal generators of
mielke-geometrodynamics_F00331	82	0.999	`\mathrm{P}_{\alpha} \in \mathbb{R}^{\mathrm{n}}`	$\mathrm{P}_{\alpha} \in \mathbb{R}^{\mathrm{n}}$	In these formulas, $L_{\alpha\beta} \in \mathfrak{gl}(\mathrm{n}, \mathbb{R})$ and $\mathrm{P}_{\alpha} \in \mathbb{R}^{\mathrm{n}}$ denote the infinitesimal generators of
mielke-geometrodynamics_F00332	82	1.000	`\mathrm{GL}(\mathrm{n}, \mathbb{R})`	$\mathrm{GL}(\mathrm{n}, \mathbb{R})$	for the tangent space. The general linear group $\mathrm{GL}(\mathrm{n},\mathbb{R})$ acts on this basis as follows:
mielke-geometrodynamics_F00333	82	0.833	`\mathrm{L}_{\mathrm{m}}(\mathrm{M})`	$\mathrm{L}_{\mathrm{m}}(\mathrm{M})$	It thus generates a space $\mathrm{L}_{\mathrm{m}}(\mathrm{M})$ of *linear frames of reference* at the point m. Let
mielke-geometrodynamics_F00334	82	0.993	`\mathrm{U} \subset \mathrm{M}`	$\mathrm{U} \subset \mathrm{M}$	remember that in a neighborhood $\mathrm{U} \subset \mathrm{M}$ of a point $\mathrm{m} \in \mathrm{M}$ with coordinates x^{i},
mielke-geometrodynamics_F00335	82	0.994	`\mathrm{e}_{\mathrm{a}}(\mathrm{m})`	$\mathrm{e}_{\mathrm{a}}(\mathrm{m})$	a basis vector $\mathrm{e}_{\mathrm{a}}(\mathrm{m})$ of $\mathrm{T}_{\mathrm{m}}(\mathrm{M})$ may always be written in the way given above.
mielke-geometrodynamics_F00336	82	0.852	`\mathbb{R}^{\mathrm{n}^{2}}`	$\mathbb{R}^{\mathrm{n}^{2}}$	an open neighborhood of $\mathbb{R}^{\mathrm{n}^2}$. Since $\mathrm{e}_{\alpha}^{j}(\mathrm{m})$ represents a nonsingular $n \times n$ matrix,
mielke-geometrodynamics_F00337	82	0.852	`\mathrm{e}_{\alpha}^{j}(\mathrm{~m})`	$\mathrm{e}_{\alpha}^{j}(\mathrm{~m})$	an open neighborhood of $\mathbb{R}^{\mathrm{n}^2}$. Since $\mathrm{e}_{\alpha}^{j}(\mathrm{m})$ represents a nonsingular $n \times n$ matrix,
mielke-geometrodynamics_F00338	82	0.852	`n \times n`	$n \times n$	an open neighborhood of $\mathbb{R}^{\mathrm{n}^2}$. Since $\mathrm{e}_{\alpha}^{j}(\mathrm{m})$ represents a nonsingular $n \times n$ matrix,
mielke-geometrodynamics_F00339	82	0.987	`\left(\mathrm{x}^{i}, \mathrm{e}_{\alpha}^{-j}\right)`	$\left(\mathrm{x}^{i}, \mathrm{e}_{\alpha}^{-j}\right)$	the collection $(\mathrm{x}^{i}, \mathrm{e}_{\alpha}^{j})$ may be regarded as a choice of coordinates of the Cartesian
mielke-geometrodynamics_F00340	82	0.840	`\mathrm{U} \times \mathrm{GL}(\mathrm{n}, \mathbb{R})`	$\mathrm{U} \times \mathrm{GL}(\mathrm{n}, \mathbb{R})$	product $\mathrm{U} \times \mathrm{GL}(\mathrm{n}, \mathbb{R})$. These coordinates arise from the original domain $\overset{-1}{\pi}(\mathrm{U})$, for
mielke-geometrodynamics_F00341	82	0.840	`\pi^{-1}(\mathrm{U})`	$\pi^{-1}(\mathrm{U})$	product $\mathrm{U} \times \mathrm{GL}(\mathrm{n}, \mathbb{R})$. These coordinates arise from the original domain $\overset{-1}{\pi}(\mathrm{U})$, for
mielke-geometrodynamics_F00342	83	0.858	`\operatorname{GL}(\mathrm{n}`	$\operatorname{GL}(\mathrm{n}$	which the projection π of the principal fiber bundle with the structure group GL(n,
mielke-geometrodynamics_F00343	83	0.899	`\vartheta`	ϑ	of linear frames. This is the case if an $\mathbb{R}^{\mathrm{n}}$-valued 1-form ϑ exists that not only is
mielke-geometrodynamics_F00344	83	0.825	`\mathrm{n}=\operatorname{dim} \mathrm{M}`	$\mathrm{n}=\operatorname{dim} \mathrm{M}$	this close contact and due to the fact that ϑ has rank $\mathrm{n} = \dim \mathrm{M}$, it is natural to call
mielke-geometrodynamics_F00345	83	0.394	`\mathbb{R}^{n}`	$\mathbb{R}^{n}$	associated with L(M). Its standard fiber is $\mathbb{R}^{\mathrm{n}}$, and thus T(M) has the same dimension
mielke-geometrodynamics_F00346	83	0.569	`\mathrm{A}(\mathrm{M}) / \mathbb{R}^{\mathrm{n}}`	$\mathrm{A}(\mathrm{M}) / \mathbb{R}^{\mathrm{n}}$	The possibility of identifying L(M) with the quotient bundle $\mathrm{A}(\mathrm{M})/\mathbb{R}^{\mathrm{n}}$ results from
mielke-geometrodynamics_F00347	84	0.999	`\mathrm{e}_{\alpha}(\mathrm{m})`	$\mathrm{e}_{\alpha}(\mathrm{m})$	the basis vectors $\mathrm{e}_{\alpha}(\mathrm{m})$ in the tangent space $\mathrm{T}_{\mathrm{m}}(\mathrm{M})$ at the point $\mathrm{m} \in \mathrm{M}$ appears to
mielke-geometrodynamics_F00348	84	0.593	`\mathrm{T}=\mathbb{R}^{\mathrm{n}}`	$\mathrm{T}=\mathbb{R}^{\mathrm{n}}$	group $\mathrm{T} = \mathbb{R}^{\mathrm{n}}$ of translations can be regarded as the quotient space $\mathrm{A}(\mathrm{n},\mathbb{R})/\mathrm{GL}(\mathrm{n},\mathbb{R})$,
mielke-geometrodynamics_F00349	84	0.593	`\mathrm{A}(\mathrm{n}, \mathbb{R}) / \mathrm{GL}(\mathrm{n}, \mathbb{R})`	$\mathrm{A}(\mathrm{n}, \mathbb{R}) / \mathrm{GL}(\mathrm{n}, \mathbb{R})$	group $\mathrm{T} = \mathbb{R}^{\mathrm{n}}$ of translations can be regarded as the quotient space $\mathrm{A}(\mathrm{n},\mathbb{R})/\mathrm{GL}(\mathrm{n},\mathbb{R})$,
mielke-geometrodynamics_F00350	84	1.000	`\mathrm{T}=\mathrm{P} / \mathrm{L}`	$\mathrm{T}=\mathrm{P} / \mathrm{L}$	realized by translations acting on the coset space $\mathrm{T} = \mathrm{P}/\mathrm{L}$, which is isomorphic to the
mielke-geometrodynamics_F00351	84	0.550	`\widetilde{\widetilde{\omega}}`	$\widetilde{\widetilde{\omega}}$	connection. Let $\widetilde{\widetilde{\omega}}$ be the 1-form of such a *generalized affine connection*, and let
mielke-geometrodynamics_F00352	84	0.999	`\chi: \mathrm{L}(\mathrm{M}) \rightarrow 0_{\mathrm{m}} \times \mathrm{L}(\mathrm{M}) \subset \mathrm{A}(\mathrm{M})`	$\chi: \mathrm{L}(\mathrm{M}) \rightarrow 0_{\mathrm{m}} \times \mathrm{L}(\mathrm{M}) \subset \mathrm{A}(\mathrm{M})$	$\chi : \mathrm{L}(\mathrm{M}) \to 0_{\mathrm{m}} \times \mathrm{L}(\mathrm{M}) \subset \mathrm{A}(\mathrm{M})$ be the natural injection of linear frames into the
mielke-geometrodynamics_F00353	85	0.448	`\stackrel{*}{\chi} \widetilde{\omega}`	$\stackrel{*}{\chi} \widetilde{\omega}$	image $\stackrel{*}{\chi}\widetilde{\widetilde{\omega}}$ of the connection $\widetilde{\widetilde{\omega}}$ is an $\mathfrak{a}(\mathrm{n}, \mathbb{R})$-valued 1-form in L(M) that decomposes
mielke-geometrodynamics_F00354	85	0.507	`\omega^{\mathrm{L}}`	ω^{L}	tation (4.1.2) of $\mathrm{A}(\mathrm{n},\mathbb{R})$. In the above formula, ω^{L} denotes a $\mathfrak{gl}(\mathrm{n},\mathbb{R})$-valued 1-form

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mielke-geometrodynamics_F00355	85	0.507	`\mathfrak{g l}(\mathrm{n}, \mathbb{R})`	$\mathfrak{g l}(\mathrm{n}, \mathbb{R})$	tation (4.1.2) of A(n,$\mathbb{R}$). In the above formula, ω^{L} denotes a $\mathfrak{gl}$(n,$\mathbb{R}$)-valued 1-form
mielke-geometrodynamics_F00356	85	0.912	`\omega^{\mathrm{T}}`	ω^{T}	in M, while ω^{T} is an $\mathbb{R}^{n}$-valued 1-form, i.e., a tensorial form of the representation
mielke-geometrodynamics_F00357	85	0.990	`\mathrm{H} \subset \mathrm{G}`	$\mathrm{H} \subset \mathrm{G}$	structure with respect to a subgroup $\mathrm{H} \subset \mathrm{G}$.
mielke-geometrodynamics_F00358	85	0.944	`\widetilde{\widetilde{\Omega}}`	$\widetilde{\widetilde{\Omega}}$	The curvature $\widetilde{\widetilde{\Omega}}$ in A(M) can be derived from $\widetilde{\widetilde{\omega}}$ by an analogous application of
mielke-geometrodynamics_F00359	85	1.000	`\left[\omega^{\mathrm{T}}, \omega^{\mathrm{T}}\right]`	$\left[\omega^{\mathrm{T}}, \omega^{\mathrm{T}}\right]$	The commutator $[\omega^{\mathrm{T}}, \omega^{\mathrm{T}}]$ occurring in the calculation
mielke-geometrodynamics_F00360	85	0.989	`\omega^{\mathrm{T}}, \omega^{\mathrm{L}}`	$\omega^{\mathrm{T}}, \omega^{\mathrm{L}}$	which is induced by ω^{L}. It is of importance to realize that the commutator $[\omega^{\mathrm{T}}, \omega^{\mathrm{L}}]$
mielke-geometrodynamics_F00361	86	1.000	`\mathscr{A}(n, \mathbb{R})`	$\mathscr{A}(n, \mathbb{R})$	of local translations are subgroups of $\mathscr{A}(n, \mathbb{R})$. Taking the cross section in the asso-
mielke-geometrodynamics_F00362	86	0.996	`C^{\infty}`	C^{∞}	ciated bundle is abbreviated by C^{∞}, and Ad denotes the adjoint representation with
mielke-geometrodynamics_F00363	86	0.998	`G L(n, \mathbb{R})`	$G L(n, \mathbb{R})$	respect to $GL(n, \mathbb{R})$. Due to its construction, the group of local translations $\mathscr{T}(n, \mathbb{R})$
mielke-geometrodynamics_F00364	86	0.998	`\mathscr{T}(n, \mathbb{R})`	$\mathscr{T}(n, \mathbb{R})$	respect to $GL(n, \mathbb{R})$. Due to its construction, the group of local translations $\mathscr{T}(n, \mathbb{R})$
mielke-geometrodynamics_F00365	86	1.000	`\operatorname{Diff}(n, \mathbb{R})`	$\operatorname{Diff}(n, \mathbb{R})$	is *locally* isomorphic to the group of active diffeomorphisms *Diff* $(n, \mathbb{R})$ of the man-
mielke-geometrodynamics_F00366	86	0.905	`(n, \mathbb{R})`	$(n, \mathbb{R})$	*Diff* $(n, \mathbb{R})$ contains the $(n+n^{2})$-dimensional group $A(n, \mathbb{R})_{\mathrm{H}}$ of *holonomic* affine
mielke-geometrodynamics_F00367	86	0.905	`\left(n+n^{2}\right)`	$\left(n+n^{2}\right)$	*Diff* $(n, \mathbb{R})$ contains the $(n+n^{2})$-dimensional group $A(n, \mathbb{R})_{\mathrm{H}}$ of *holonomic* affine
mielke-geometrodynamics_F00368	86	0.905	`A(n, \mathbb{R})_{\mathrm{H}}`	$A(n, \mathbb{R})_{\mathrm{H}}$	*Diff* $(n, \mathbb{R})$ contains the $(n+n^{2})$-dimensional group $A(n, \mathbb{R})_{\mathrm{H}}$ of *holonomic* affine
mielke-geometrodynamics_F00369	86	0.999	`P_{i}=\partial_{i}:=\partial / \partial x^{i}`	$P_{i}=\partial_{i}:=\partial / \partial x^{i}$	transformations as a subgroup, generated by the vector fields $P_{i}=\partial_{i}:=\partial / \partial x^{i}$ and
mielke-geometrodynamics_F00370	86	1.000	`L^{i}{ }_{j}=x^{i} \partial_{j}`	$L^{i}{ }_{j}=x^{i} \partial_{j}$	$L^{i}{}_{j}=x^{i} \partial_{j}$. Note that differentiable coordinate transformations, which leave exterior
mielke-geometrodynamics_F00371	86	1.000	`\chi`	χ	the natural injection χ, this formula decomposes into
mielke-geometrodynamics_F00372	87	1.000	`\mathrm{V}^{\text {id }}`	$\mathrm{V}^{\text {id }}$	in V^{id} consists of an orbit $\mathbb{R}^{n}$ of A(n,$\mathbb{R}$). The isotropy group of the zero section
mielke-geometrodynamics_F00373	87	0.928	`\stackrel{\circ}{\tilde{\varphi}}=\sigma\left(0_{\mathrm{m}}\right)`	$\stackrel{\circ}{\tilde{\varphi}}=\sigma\left(0_{\mathrm{m}}\right)$	$\stackrel{\circ}{\tilde{\varphi}}=\sigma(0_{\mathrm{m}})$ is $\mathrm{H}=\mathrm{GL}(\mathrm{n}, \mathbb{R})$, i.e., the general linear group. Then the subbundle
mielke-geometrodynamics_F00374	87	0.928	`\mathrm{H}=\mathrm{GL}(\mathrm{n}, \mathbb{R})`	$\mathrm{H}=\mathrm{GL}(\mathrm{n}, \mathbb{R})$	$\stackrel{\circ}{\tilde{\varphi}}=\sigma(0_{\mathrm{m}})$ is $\mathrm{H}=\mathrm{GL}(\mathrm{n}, \mathbb{R})$, i.e., the general linear group. Then the subbundle
mielke-geometrodynamics_F00375	87	0.863	`\widetilde{\vartheta}`	$\widetilde{\vartheta}$	bution $\widetilde{\widetilde{\vartheta}}$ transforms under $\mathscr{A}_{\mathrm{p}}$ like
mielke-geometrodynamics_F00376	87	0.863	`\mathscr{A}_{\mathrm{p}}`	$\mathscr{A}_{\mathrm{p}}$	bution $\widetilde{\widetilde{\vartheta}}$ transforms under $\mathscr{A}_{\mathrm{p}}$ like
mielke-geometrodynamics_F00377	88	0.968	`\mathrm{D}=(\stackrel{*}{\chi} \widetilde{\mathrm{D}})`	$\mathrm{D}=(\stackrel{*}{\chi} \widetilde{\mathrm{D}})$	to $\mathrm{D}=(\stackrel{*}{\chi}\widetilde{\widetilde{\mathrm{D}}})$, it follows that the torsion 2-form transforms like
mielke-geometrodynamics_F00378	88	0.999	`\mathscr{G}_{\mathrm{p}}`	$\mathscr{G}_{\mathrm{p}}$	i.e., as a vector with respect to the subgroup $\mathscr{G}_{\mathrm{p}}$ of linear gauge transformations. As
mielke-geometrodynamics_F00379	89	0.993	`\widetilde{\widetilde{\mathrm{D}}}=\mathrm{D}`	$\widetilde{\widetilde{\mathrm{D}}}=\mathrm{D}$	$\widetilde{\widetilde{\mathrm{D}}}=\mathrm{D}$ it remains valid) the affine connection to the linear $\mathfrak{gl}$(n,$\mathbb{R}$)-valued connection
mielke-geometrodynamics_F00380	90	1.000	`\mathscr{C} \subset \mathrm{M}`	$\mathscr{C} \subset \mathrm{M}$	to a smooth curve $\mathscr{C} \subset \mathrm{M}$ that is parametrized by s(t). In order to satisfy

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mielke-geometrodynamics_F00381	90	1.000	`\mathrm{F}(\mathrm{m}, \vartheta): \mathrm{M} \times \mathrm{T}^{*}(\mathrm{M}) \rightarrow \mathbb{R}^{+}`	$\mathrm{F}(\mathrm{m}, \vartheta): \mathrm{M} \times \mathrm{T}^{*}(\mathrm{M}) \rightarrow \mathbb{R}^{+}$	where $\mathrm{F}(\mathrm{m}, \vartheta): \mathrm{M} \times \mathrm{T}^{*}(\mathrm{M}) \rightarrow \mathbb{R}^{+}$ is a smooth, *positive*, and *homogeneous* funda-
mielke-geometrodynamics_F00382	91	1.000	`\mathscr{C}`	$\mathscr{C}$	ance of (4.3.4) versus admissible reparametrizations of the curve $\mathscr{C}$. Generally, these
mielke-geometrodynamics_F00383	91	1.000	`(2,0)`	$(2,0)$	Subsequently, a covariant symmetric tensor field of degree $(2, 0)$ is imprinted onto
mielke-geometrodynamics_F00384	91	1.000	`\mathrm{g}_{i j} \in \mathrm{C}^{\infty}(\mathrm{M})`	$\mathrm{g}_{i j} \in \mathrm{C}^{\infty}(\mathrm{M})$	Indeed, the functions $\mathrm{g}_{ij} \in \mathrm{C}^{\infty}(\mathrm{M})$ describe "concerning the chosen, arbitrary coor-
mielke-geometrodynamics_F00385	92	0.689	`\mathrm{e}_{\alpha}^{\cdot i}`	$\mathrm{e}_{\alpha}^{\cdot i}$	1-forms ϑ admitted. Since the $\mathrm{e}_{\alpha}^{\cdot i}$ have been considered as local bundle coordinates
mielke-geometrodynamics_F00386	92	1.000	`\mathfrak{O}(\mathrm{s}, \mathrm{n}-\mathrm{s})`	$\mathfrak{O}(\mathrm{s}, \mathrm{n}-\mathrm{s})$	$\mathfrak{O}(\mathrm{s}, \mathrm{n}-\mathrm{s})$-valued connection ω^{g}, the constraint
mielke-geometrodynamics_F00387	92	1.000	`\omega^{\mathrm{g}}`	ω^{g}	$\mathfrak{O}(\mathrm{s}, \mathrm{n}-\mathrm{s})$-valued connection ω^{g}, the constraint
mielke-geometrodynamics_F00388	92	0.532	`\phi^{(\circ)}`	$\phi^{(\circ)}$	is a necessary and sufficient condition. For tensor-valued 0-forms $\phi^{(\circ)}$, the exterior
mielke-geometrodynamics_F00389	92	0.651	`\Gamma_{\mathrm{i}}{ }^{\alpha \beta}`	$\Gamma_{\mathrm{i}}{ }^{\alpha \beta}$	where the $\Gamma_{\mathrm{i}}^{\cdot\alpha\beta}$ denote the (local) *Ricci rotation coefficients*. Due to the fact that
mielke-geometrodynamics_F00390	92	1.000	`L_{\alpha \beta}`	$L_{\alpha \beta}$	the infinitesimal generators $L_{\alpha\beta}$ become skew after the restriction to the orthogonal
mielke-geometrodynamics_F00391	92	1.000	`\omega^{\alpha \beta}=-\omega^{\beta \alpha}`	$\omega^{\alpha \beta}=-\omega^{\beta \alpha}$	subgroup, the constraint (4.3.14) analogously yields $\omega^{\alpha\beta} = -\omega^{\beta\alpha}$, i.e., the skew-
mielke-geometrodynamics_F00392	92	0.988	`\mathrm{m}_{\mathrm{o}} \in \mathrm{M}`	$\mathrm{m}_{\mathrm{o}} \in \mathrm{M}$	neighborhood of a point $\mathrm{m}_{\mathrm{o}} \in \mathrm{M}$ depends entirely on Minkowski's metric tensor o_{ij}
mielke-geometrodynamics_F00393	92	0.988	`o_{i j}`	$o_{i j}$	neighborhood of a point $\mathrm{m}_{\mathrm{o}} \in \mathrm{M}$ depends entirely on Minkowski's metric tensor o_{ij}
mielke-geometrodynamics_F00394	93	1.000	`\mathrm{H}=\mathrm{O}(\mathrm{s}, \mathrm{n}-\mathrm{s})`	$\mathrm{H}=\mathrm{O}(\mathrm{s}, \mathrm{n}-\mathrm{s})$	gauge theory has only the orthogonal "isotropy group" $\mathrm{H}=\mathrm{O}(\mathrm{s}, \mathrm{n}-\mathrm{s})$ as resid-
mielke-geometrodynamics_F00395	93	1.000	`\mathfrak{o}(\mathrm{s}, \mathrm{n}-\mathrm{s})`	$\mathfrak{o}(\mathrm{s}, \mathrm{n}-\mathrm{s})$	the reduction of the linear connection ω^{L} to an $\mathfrak{o}(\mathrm{s}, \mathrm{n}-\mathrm{s})$-valued connection ω^{g}
mielke-geometrodynamics_F00396	93	0.999	`\mathrm{L}^{\mathrm{g}}(\mathrm{M})`	$\mathrm{L}^{\mathrm{g}}(\mathrm{M})$	in $\mathrm{L}^{\mathrm{g}}(\mathrm{M})$. This consistency does not, as is implied by HANSON & REGGE (1979),
mielke-geometrodynamics_F00397	93	1.000	`\mathrm{D} \vartheta=0`	$\mathrm{D} \vartheta=0$	demand the vanishing of the torsion, i.e., $\mathrm{D}\vartheta = 0$. In a Lagrangian formulation of
mielke-geometrodynamics_F00398	93	0.995	`\Theta`	Θ	Higgs field ϑ, i.e., the canonical *orthogonal* 1-form,[4] and the resulting torsion Θ
mielke-geometrodynamics_F00399	93	0.913	`\vartheta^{\mathrm{g}}`	ϑ^{g}	[4]As for the nomenclature, the 1-form ϑ, which is restricted by (4.3.10), will not be denoted by ϑ^{g}
mielke-geometrodynamics_F00400	93	1.000	`\Omega^{\mathrm{g}}`	Ω^{g}	and thus differs from ω^{g} and Ω^{g}, since we prefer to stick to (4.3.10) or (4.3.9), respectively, as
mielke-geometrodynamics_F00401	94	0.997	`\omega^{\{ \}}`	$\omega^{\{ \}}$	Here $\omega^{\{\}}$ denotes the 1-form of the torsion-free Christoffel-type connection for which
mielke-geometrodynamics_F00402	94	0.890	`K`	K	between the 2-form Θ and the 1-form K of the so-called *contortion*, having values
mielke-geometrodynamics_F00403	94	1.000	`\mathrm{U}_{\mathrm{n}}`	U_{n}	denoted by U_{n}, in contrast to a purely Riemannian manifold, which is symbolized by V_{n}.
mielke-geometrodynamics_F00404	94	1.000	`\mathrm{V}_{\mathrm{n}}`	V_{n}	denoted by U_{n}, in contrast to a purely Riemannian manifold, which is symbolized by V_{n}.
mielke-geometrodynamics_F00405	95	0.456	`R_{i j c}^{\{. \ldots d}`	$R_{i j c}^{\{. \ldots d}$	In this formula, $R_{ijc}^{\{\}\ldots d}$ are the mixed components of the Riemannian curvature tensor,
mielke-geometrodynamics_F00406	95	0.999	`R_{i j k l}^{\{ \}}`	$R_{i j k l}^{\{ \}}$	holonomic components $R_{ijkl}^{\{\}}$ have the following symmetries with respect to an inter-
mielke-geometrodynamics_F00407	95	0.666	`S S B`	$S S B$	cation of a Higgs–Kibble-type mechanism of *SSB* that resulted in the construction of

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mielke-geometrodynamics_F00408	95	1.000	`\mathrm{O}(1, \mathrm{n}-1)`	$\mathrm{O}(1, \mathrm{n}-1)$	O(1, n-1) as the "residual" symmetry of the theory via the general linear group
mielke-geometrodynamics_F00409	95	0.778	`\operatorname{GL}(\mathrm{n}, \mathbb{R})`	$\operatorname{GL}(\mathrm{n}, \mathbb{R})$	GL(n, $\mathbb{R}$).
mielke-geometrodynamics_F00410	96	1.000	`\mathrm{R}_{\alpha \beta}^{. . \mathrm{cd}}(\mathrm{m})`	$\mathrm{R}_{\alpha \beta}^{. . \mathrm{cd}}(\mathrm{m})$	spacetime-dependent "structure constants" $\mathrm{R}_{\alpha\beta}^{\cdot\cdot\mathrm{cd}}(\mathrm{m})$ occur in the generic case. (The
mielke-geometrodynamics_F00411	96	1.000	`\vartheta, \Theta, \Omega^{\mathrm{g}}`	$\vartheta, \Theta, \Omega^{\mathrm{g}}$	However, contrasting with $\vartheta, \Theta, \Omega^{\mathrm{g}}$ (see (4.2.20), (4.2.14), and (4.2.13)), the
mielke-geometrodynamics_F00412	97	1.000	`L_{\mathrm{g}}`	L_{g}	only via of the forms Θ and Ω^{g}. The variation of L_{g}, regarded as a total differential,
mielke-geometrodynamics_F00413	97	1.000	`t`	t	defines the *t*ranslational and (*L*orentz-)rotational canonical gauge field momenta Π^{T}
mielke-geometrodynamics_F00414	97	1.000	`\Pi^{\mathrm{T}}`	Π^{T}	defines the *t*ranslational and (*L*orentz-)rotational canonical gauge field momenta Π^{T}
mielke-geometrodynamics_F00415	97	0.999	`\Pi^{\mathrm{L}}`	Π^{L}	and Π^{L}. Due to the special role of ϑ, there occurs additionally the (n-1)-form E,
mielke-geometrodynamics_F00416	97	0.999	`E`	E	and Π^{L}. Due to the special role of ϑ, there occurs additionally the (n-1)-form E,
mielke-geometrodynamics_F00417	97	0.998	`\Pi_{[]}^{\mathrm{T}}`	$\Pi_{[]}^{\mathrm{T}}$	where the "contortional" 1-form $\Pi_{[]}^{\mathrm{T}}$, having values in the Lie algebra of the
mielke-geometrodynamics_F00418	97	0.931	`\Pi^{\mathrm{T}}:=\left[\Pi_{[ }^{\mathrm{T}}, \vartheta \right]`	$\Pi^{\mathrm{T}}:=\left[\Pi_{[}^{\mathrm{T}}, \vartheta \right]$	$\Pi^{\mathrm{T}} := [\Pi_{[]}^{\mathrm{T}}, \vartheta]$. This is in complete analogy to the relation (4.3.24) for the contor-
mielke-geometrodynamics_F00419	97	1.000	`\delta \vartheta`	$\delta \vartheta$	The application of Hamilton's principle to the independent variations of S for $\delta\vartheta$
mielke-geometrodynamics_F00420	97	1.000	`\delta \omega^{\mathrm{g}}`	$\delta \omega^{\mathrm{g}}$	and $\delta\omega^{\mathrm{g}}$ leads to the *Euler–Lagrange* equations
mielke-geometrodynamics_F00421	98	0.844	`\Sigma`	Σ	fied with the 3-form Σ, the energy–momentum current, and the 3-form τ of the spin
mielke-geometrodynamics_F00422	98	0.999	`\lambda^{\varphi}=\mathrm{o}`	$\lambda^{\varphi}=\mathrm{o}$	If the Euler–Lagrange equations $\lambda^{\varphi} = \mathrm{o}$ of the matter fields are satisfied, it will
mielke-geometrodynamics_F00423	98	1.000	`\tau_{\mathrm{s}}`	τ_{s}	turn out that Σ and τ_{s} are the field momenta of matter that are *canonically conjugate*
mielke-geometrodynamics_F00424	98	1.000	`\Upsilon`	Υ	hypermomentum current Υ of the matter fields in such a generalization (HEHL &
mielke-geometrodynamics_F00425	100	1.000	`\Gamma^{\alpha \beta}=-\Gamma^{\beta \alpha}`	$\Gamma^{\alpha \beta}=-\Gamma^{\beta \alpha}$	symmetric connection 1-form $\Gamma^{\alpha\beta} = -\Gamma^{\beta\alpha}$ is defined by $\delta L = \delta\Gamma^{\alpha\beta} \wedge (\partial L/\partial\Gamma^{\alpha\beta})$.
mielke-geometrodynamics_F00426	100	1.000	`\delta L=\delta \Gamma^{\alpha \beta} \wedge\left( \partial L / \partial \Gamma^{\alpha \beta}\right)`	$\delta L=\delta \Gamma^{\alpha \beta} \wedge\left(\partial L / \partial \Gamma^{\alpha \beta}\right)$	symmetric connection 1-form $\Gamma^{\alpha\beta} = -\Gamma^{\beta\alpha}$ is defined by $\delta L = \delta\Gamma^{\alpha\beta} \wedge (\partial L/\partial\Gamma^{\alpha\beta})$.
mielke-geometrodynamics_F00427	100	0.629	`S=\int_{M} L_{\text {mat }}`	$S=\int_{M} L_{\text {mat }}$	The action $S = \int_M L_{\mathrm{mat}}$ for the matter Lagrangian is, by construction, invari-
mielke-geometrodynamics_F00428	100	1.000	`\operatorname{Diff}(M)`	$\operatorname{Diff}(M)$	ant under the group *Diff*(M) of coordinate transformations and local frame rota-
mielke-geometrodynamics_F00429	100	0.924	`\mathscr{T} \subset \operatorname{Diff}(M)`	$\mathscr{T} \subset \operatorname{Diff}(M)$	one-parameter group of *local* translations $\mathscr{T} \subset$ *Diff*(M), we employ the $SO(1,3)$-
mielke-geometrodynamics_F00430	100	0.924	`S O(1,3)`	$S O(1,3)$	one-parameter group of *local* translations $\mathscr{T} \subset$ *Diff*(M), we employ the $SO(1,3)$-
mielke-geometrodynamics_F00431	100	0.590	`\mathrm{Ł}_{\xi}:=\xi \jmath D+D \xi ⌟`	$Ł_{\xi}:=\xi \jmath D+D \xi \lrcorner$	covariant Lie derivative $Ł_{\xi} := \xi \rfloor D + D\xi \rfloor$ on M with respect to an arbitrary vector
mielke-geometrodynamics_F00432	100	0.590	`M`	M	covariant Lie derivative $Ł_{\xi} := \xi \rfloor D + D\xi \rfloor$ on M with respect to an arbitrary vector
mielke-geometrodynamics_F00433	100	1.000	`\xi`	ξ	field ξ. Since $Dg_{\alpha\beta} = 0$, we obtain
mielke-geometrodynamics_F00434	100	1.000	`D g_{\alpha \beta}=0`	$D g_{\alpha \beta}=0$	field ξ. Since $Dg_{\alpha\beta} = 0$, we obtain

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mielke-geometrodynamics_F00435	101	0.779	`\mathrm{L}_{\xi} L=D \xi \jmath L`	$\mathrm{L}_{\xi} L=D \xi \jmath L$	$Ł_\xi L = D\xi\rfloor L$. Comparing the boundary term, we can read off the identity
mielke-geometrodynamics_F00436	101	1.000	`\xi \rightarrow e_{\alpha}`	$\xi \rightarrow e_{\alpha}$	If we replace $\xi \rightarrow e_\alpha$ by the basis frame, then (4.5.4) yields directly the explicit
mielke-geometrodynamics_F00437	101	0.995	`\left.F_{\alpha}^{\text {Max }}=\left(e_{\alpha}\right \rfloor F\right) \wedge j`	$\left.F_{\alpha}^{\text {Max }}=\left(e_{\alpha}\right \rfloor F\right) \wedge j$	calculus $F_\alpha^{\mathrm{Max}} = (e_\alpha\rfloor F)\wedge j$.
mielke-geometrodynamics_F00438	101	1.000	`\simeq`	$\simeq$	no field equations are invoked. *Weak* identities, which are denoted by $\simeq$, hold only
mielke-geometrodynamics_F00439	101	0.999	`\delta L / \delta \Psi=0`	$\delta L / \delta \Psi=0$	if the matter field equation $\delta L/\delta\Psi = 0$ is satisfied.
mielke-geometrodynamics_F00440	101	0.972	`\delta D \Psi=D \delta \Psi+\delta \Gamma_{\alpha}{ }^{\beta} \wedge I^{\alpha}{ }_{\beta} \Psi`	$\delta D \Psi=D \delta \Psi+\delta \Gamma_{\alpha}{ }^{\beta} \wedge I^{\alpha}{ }_{\beta} \Psi$	Since $\delta D\Psi = D\delta\Psi + \delta\Gamma_\alpha{}^\beta \wedge I^\alpha{}_\beta\Psi$, this is equivalent to
mielke-geometrodynamics_F00441	102	1.000	`\varepsilon_{\alpha}{ }^{\beta}(x):=\Lambda_{\alpha}{ }^{\beta}(x)-\delta_{\alpha}^{\beta}`	$\varepsilon_{\alpha}{ }^{\beta}(x):=\Lambda_{\alpha}{ }^{\beta}(x)-\delta_{\alpha}^{\beta}$	Under an infinitesimal Lorentz rotation $\varepsilon_\alpha{}^\beta(x) := \Lambda_\alpha{}^\beta(x) - \delta_\alpha^\beta$ of the frames, we
mielke-geometrodynamics_F00442	102	0.950	`T^{\alpha}=D \vartheta^{\alpha}`	$T^{\alpha}=D \vartheta^{\alpha}$	potential; field strength; Bianchi identity; Noether identity ϑ^α; $T^\alpha = D\vartheta^\alpha$; $DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma$; $\overparen{D}\,\Sigma_\alpha \cong (e_\alpha\rfloor R_\beta{}^\gamma)\wedge\Delta^\beta{}_\gamma - \frac{1}{2}(e_\alpha\rfloor Q_{\beta\gamma})\wedge\sigma^{\beta\gamma}$ $\Gamma_\alpha{}^\beta$; $R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma\wedge\Gamma_\gamma{}^\beta$; $DR_\alpha{}^\beta \equiv 0$; $D\Delta^\alpha{}_\beta + \vartheta^\alpha\wedge\Sigma_\beta \cong \sigma^\alpha{}_\beta$
mielke-geometrodynamics_F00443	102	0.950	`D T^{\alpha} \equiv R_{\gamma}{ }^{\alpha} \wedge \vartheta^{\gamma}`	$D T^{\alpha} \equiv R_{\gamma}{ }^{\alpha} \wedge \vartheta^{\gamma}$	potential; field strength; Bianchi identity; Noether identity ϑ^α; $T^\alpha = D\vartheta^\alpha$; $DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma$; $\overparen{D}\,\Sigma_\alpha \cong (e_\alpha\rfloor R_\beta{}^\gamma)\wedge\Delta^\beta{}_\gamma - \frac{1}{2}(e_\alpha\rfloor Q_{\beta\gamma})\wedge\sigma^{\beta\gamma}$ $\Gamma_\alpha{}^\beta$; $R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma\wedge\Gamma_\gamma{}^\beta$; $DR_\alpha{}^\beta \equiv 0$; $D\Delta^\alpha{}_\beta + \vartheta^\alpha\wedge\Sigma_\beta \cong \sigma^\alpha{}_\beta$
mielke-geometrodynamics_F00444	102	—	`\begin{aligned} \widehat{D} \Sigma_{\alpha} & \left.\cong \left(e_{\alpha}\right\rfloor R_{\beta}^{\gamma}\right) \wedge \Delta^{\beta} \gamma \\ & \left.-\frac{1}{2} \left(e_{\alpha}\right\rfloor Q_{\beta \gamma}\right) \wedge \sigma^{\beta \gamma} \end{aligned}`	$\begin{aligned} \widehat{D} \Sigma_{\alpha} & \left.\cong \left(e_{\alpha}\right\rfloor R_{\beta}^{\gamma}\right) \wedge \Delta^{\beta} \gamma \\ & \left.-\frac{1}{2} \left(e_{\alpha}\right\rfloor Q_{\beta \gamma}\right) \wedge \sigma^{\beta \gamma} \end{aligned}$	—
mielke-geometrodynamics_F00445	102	0.950	`\Gamma_{\alpha}{ }^{\beta}`	$\Gamma_{\alpha}{ }^{\beta}$	potential; field strength; Bianchi identity; Noether identity ϑ^α; $T^\alpha = D\vartheta^\alpha$; $DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma$; $\overparen{D}\,\Sigma_\alpha \cong (e_\alpha\rfloor R_\beta{}^\gamma)\wedge\Delta^\beta{}_\gamma - \frac{1}{2}(e_\alpha\rfloor Q_{\beta\gamma})\wedge\sigma^{\beta\gamma}$ $\Gamma_\alpha{}^\beta$; $R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma\wedge\Gamma_\gamma{}^\beta$; $DR_\alpha{}^\beta \equiv 0$; $D\Delta^\alpha{}_\beta + \vartheta^\alpha\wedge\Sigma_\beta \cong \sigma^\alpha{}_\beta$
mielke-geometrodynamics_F00446	102	—	`\begin{aligned} & R_{\alpha}{ }^{\beta}=d \Gamma_{\alpha}{ }^{\beta} \\ & -\Gamma_{\alpha}{ }^{\gamma} \wedge \Gamma_{\gamma}{ }^{\beta} \end{aligned}`	$\begin{aligned} & R_{\alpha}{ }^{\beta}=d \Gamma_{\alpha}{ }^{\beta} \\ & -\Gamma_{\alpha}{ }^{\gamma} \wedge \Gamma_{\gamma}{ }^{\beta} \end{aligned}$	—
mielke-geometrodynamics_F00447	102	0.950	`D R_{\alpha}{ }^{\beta} \equiv 0`	$D R_{\alpha}{ }^{\beta} \equiv 0$	potential; field strength; Bianchi identity; Noether identity ϑ^α; $T^\alpha = D\vartheta^\alpha$; $DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma$; $\overparen{D}\,\Sigma_\alpha \cong (e_\alpha\rfloor R_\beta{}^\gamma)\wedge\Delta^\beta{}_\gamma - \frac{1}{2}(e_\alpha\rfloor Q_{\beta\gamma})\wedge\sigma^{\beta\gamma}$ $\Gamma_\alpha{}^\beta$; $R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma\wedge\Gamma_\gamma{}^\beta$; $DR_\alpha{}^\beta \equiv 0$; $D\Delta^\alpha{}_\beta + \vartheta^\alpha\wedge\Sigma_\beta \cong \sigma^\alpha{}_\beta$
mielke-geometrodynamics_F00448	102	0.950	`D \Delta^{\alpha}{ }_{\beta}+\vartheta^{\alpha} \wedge \Sigma_{\beta} \cong \sigma^{\alpha}{ }_{\beta}`	$D \Delta^{\alpha}{ }_{\beta}+\vartheta^{\alpha} \wedge \Sigma_{\beta} \cong \sigma^{\alpha}{ }_{\beta}$	potential; field strength; Bianchi identity; Noether identity ϑ^α; $T^\alpha = D\vartheta^\alpha$; $DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma$; $\overparen{D}\,\Sigma_\alpha \cong (e_\alpha\rfloor R_\beta{}^\gamma)\wedge\Delta^\beta{}_\gamma - \frac{1}{2}(e_\alpha\rfloor Q_{\beta\gamma})\wedge\sigma^{\beta\gamma}$ $\Gamma_\alpha{}^\beta$; $R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma\wedge\Gamma_\gamma{}^\beta$; $DR_\alpha{}^\beta \equiv 0$; $D\Delta^\alpha{}_\beta + \vartheta^\alpha\wedge\Sigma_\beta \cong \sigma^\alpha{}_\beta$
mielke-geometrodynamics_F00449	102	0.986	`\overparen{D}`	$\overparen{D}$	$\overparen{D}$ with respect to the *transposed* connection $\overparen{\Gamma}{}_\alpha{}^\beta := \Gamma_\alpha{}^\beta + e_\alpha\rfloor T^\beta$.

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mielke-geometrodynamics_F00450	102	0.986	`\left.\overparen{\Gamma}{ }_{\alpha}{ }^{\beta}:=\Gamma_{\alpha}{ }^{\beta}+e_{\alpha}\right\rfloor T^{\beta}`	$\left.\overparen{\Gamma}{ }_{\alpha}{ }^{\beta}:=\Gamma_{\alpha}{ }^{\beta}+e_{\alpha}\right\rfloor T^{\beta}$	$\widehat{D}$ with respect to the *transposed* connection $\widehat{\Gamma}_{\alpha}{}^{\beta} := \Gamma_{\alpha}{}^{\beta} + e_{\alpha} \rfloor T^{\beta}$.
mielke-geometrodynamics_F00451	102	0.999	`\tau_{\alpha \beta}=-\tau_{\beta \alpha}`	$\tau_{\alpha \beta}=-\tau_{\beta \alpha}$	identity, consists of an intrinsic or *spin* part $\tau_{\alpha\beta} = -\tau_{\beta\alpha}$ and an orbital part $x_{[\alpha} \wedge \Sigma_{\beta]}$,
mielke-geometrodynamics_F00452	102	0.999	`x_{[\alpha} \wedge \Sigma_{\beta]}`	$x_{[\alpha} \wedge \Sigma_{\beta]}$	identity, consists of an intrinsic or *spin* part $\tau_{\alpha\beta} = -\tau_{\beta\alpha}$ and an orbital part $x_{[\alpha} \wedge \Sigma_{\beta]}$,
mielke-geometrodynamics_F00453	102	1.000	`x^{i}`	x^{i}	coordinates x^i with $x^\alpha = \delta_i^\alpha x^i$.
mielke-geometrodynamics_F00454	102	1.000	`x^{\alpha}=\delta_{i}^{\alpha} x^{i}`	$x^{\alpha}=\delta_{i}^{\alpha} x^{i}$	coordinates x^i with $x^\alpha = \delta_i^\alpha x^i$.
mielke-geometrodynamics_F00455	103	1.000	`\xi=\partial / \partial t`	$\xi=\partial / \partial t$	solution, if $\xi = \partial/\partial t$ or $\xi = \partial/\partial\varphi$, respectively, is chosen. Here no “factor of two”
mielke-geometrodynamics_F00456	103	1.000	`\xi=\partial / \partial \varphi`	$\xi=\partial / \partial \varphi$	solution, if $\xi = \partial/\partial t$ or $\xi = \partial/\partial\varphi$, respectively, is chosen. Here no “factor of two”
mielke-geometrodynamics_F00457	104	1.000	`\pm(\Lambda / 3) r^{3}`	$\pm(\Lambda / 3) r^{3}$	complex each contribute *one-half* of the mass, so that the divergent terms $\pm(\Lambda/3)r^3$
mielke-geometrodynamics_F00458	104	1.000	`\Lambda`	Λ	proportional to the cosmological constant Λ exactly cancel each other.
mielke-geometrodynamics_F00459	104	0.539	`\Gamma^{(T) \alpha}`	$\Gamma^{(T) \alpha}$	includes the true translational potential $\Gamma^{(T)\alpha}$ and the $GL(n,\mathbb{R})$-gauge connection
mielke-geometrodynamics_F00460	104	1.000	`\Gamma_{\alpha}^{(L) \beta}`	$\Gamma_{\alpha}^{(L) \beta}$	$\Gamma_\alpha^{(L)\beta}$, as can be deduced from the Möbius-type five-dimensional representation of
mielke-geometrodynamics_F00461	104	1.000	`A(n, \mathbb{R}):=\mathbb{R}^{n} \otimes G L(n, \mathbb{R})`	$A(n, \mathbb{R}):=\mathbb{R}^{n} \otimes G L(n, \mathbb{R})$	the affine gauge group $A(n,\mathbb{R}) := \mathbb{R}^n \otimes GL(n,\mathbb{R})$.
mielke-geometrodynamics_F00462	104	0.559	`\widetilde{\widetilde{\xi}}`	$\widetilde{\widetilde{\xi}}$	view: the condition for a parallel transport of an affine vector $\widetilde{\widetilde{\xi}}$ around a small closed
mielke-geometrodynamics_F00463	104	0.993	`\widetilde{\widetilde{\Gamma}}`	$\widetilde{\widetilde{\Gamma}}$	loop by means of $\widetilde{\widetilde{\Gamma}}$ reads
mielke-geometrodynamics_F00464	104	0.272	`\left.Ł_{y}:=y\right\rfloor D+D y`	$Ł_{y}:=y\rfloor D+D y$	where $Ł_y := y\rfloor D + Dy\rfloor$ denotes the gauge-covariant Lie derivative and $R^{(T)\alpha}$ the
mielke-geometrodynamics_F00465	104	0.272	`R^{(T) \alpha}`	$R^{(T) \alpha}$	where $Ł_y := y\rfloor D + Dy\rfloor$ denotes the gauge-covariant Lie derivative and $R^{(T)\alpha}$ the
mielke-geometrodynamics_F00466	104	1.000	`y`	y	loop parametrized by y yields
mielke-geometrodynamics_F00467	105	1.000	`A=A_{i}^{J} \lambda_{J} d x^{i}`	$A=A_{i}^{J} \lambda_{J} d x^{i}$	where $A = A_i^J \lambda_J dx^i$ is a Yang–Mills-type connection and P is the principal value
mielke-geometrodynamics_F00468	105	0.996	`\gamma`	γ	of the integral. If the loop γ lies in a field-free region, i.e., one with $U(1)$ field
mielke-geometrodynamics_F00469	105	1.000	`F:=d A=0`	$F:=d A=0$	strength $F := dA = 0$ or Yang–Mills curvature $F = dA + A \wedge A = 0$, but encloses
mielke-geometrodynamics_F00470	105	1.000	`F=d A+A \wedge A=0`	$F=d A+A \wedge A=0$	strength $F := dA = 0$ or Yang–Mills curvature $F = dA + A \wedge A = 0$, but encloses
mielke-geometrodynamics_F00471	105	1.000	`F \neq 0`	$F \neq 0$	a “confined” region with nontrivial “magnetic” flux $F \neq 0$, the potential A can still
mielke-geometrodynamics_F00472	105	1.000	`A(4, \mathbb{R})`	$A(4, \mathbb{R})$	arising from a gauge theory of the affine structure group $A(4,\mathbb{R})$.
mielke-geometrodynamics_F00473	105	0.924	`G L(4, \mathbb{R})`	$G L(4, \mathbb{R})$	For a so-called manifield Ψ carrying no $GL(4,\mathbb{R})$-excitations, i.e., no spin, no
mielke-geometrodynamics_F00474	105	1.000	`\Gamma^{(T) \alpha} \stackrel{*}{=} 0`	$\Gamma^{(T) \alpha} \stackrel{*}{=} 0$	of the cone there is $\Gamma^{(T)\alpha} \stackrel{*}{=} 0$ locally.
mielke-geometrodynamics_F00475	105	0.651	`(1993,1994)`	(1993, 1994)	time manifold; cf. ANANDAN (1993, 1994). Because the coframe is nondegenerate

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mielke-geometrodynamics_F00476	105	1.000	`P_{\alpha}`	P_{α}	Since the physical dimension of P_α is $2\pi\hbar/\ell$, the gravitational analogue of Dirac's
mielke-geometrodynamics_F00477	105	1.000	`2 \pi \hbar / \ell`	$2\pi\hbar/\ell$	Since the physical dimension of P_α is $2\pi\hbar/\ell$, the gravitational analogue of Dirac's
mielke-geometrodynamics_F00478	105	1.000	`\mathscr{M}_{n}=n M_{\text {Planck }}`	$\mathscr{M}_{n}=nM_{\text{Planck}}$	i.e., the mass would turn out to be a multiple $\mathscr{M}_n = nM_{\text{Planck}}$ of the (huge) Planck
mielke-geometrodynamics_F00479	109	0.485	`\Gamma^{\alpha \beta}=\Gamma^{\{j \alpha \beta}-K^{\alpha \beta}`	$\Gamma^{\alpha\beta}=\Gamma^{\{j\alpha\beta}-K^{\alpha\beta}$	The Riemann–Cartan (RC) connection $\Gamma^{\alpha\beta} = \Gamma^{\{\}\alpha\beta} - K^{\alpha\beta}$, a one-form, can be
mielke-geometrodynamics_F00480	109	0.469	`\Gamma^{\{\alpha \beta\}}`	$\Gamma^{\{\alpha\beta\}}$	split into the unique Levi-Civita connection $\Gamma^{\{\}\alpha\beta}$ of Riemannian geometry and
mielke-geometrodynamics_F00481	109	1.000	`K_{\alpha \beta}=-K_{\beta \alpha}`	$K_{\alpha\beta}=-K_{\beta\alpha}$	a *contortion* one-form $K_{\alpha\beta} = -K_{\beta\alpha}$, a tensor-valued one-form $K = iK^{\alpha\beta}\sigma_{\alpha\beta}/4$,
mielke-geometrodynamics_F00482	109	1.000	`K=i K^{\alpha \beta} \sigma_{\alpha \beta} / 4`	$K=iK^{\alpha\beta}\sigma_{\alpha\beta}/4$	a *contortion* one-form $K_{\alpha\beta} = -K_{\beta\alpha}$, a tensor-valued one-form $K = iK^{\alpha\beta}\sigma_{\alpha\beta}/4$,
mielke-geometrodynamics_F00483	109	1.000	`T^{\alpha}=K^{\alpha}{ }_{\beta} \wedge \vartheta^{\beta}`	$T^{\alpha}=K^{\alpha}{}_{\beta}\wedge\vartheta^{\beta}$	implicitly related to torsion via $T^\alpha = K^\alpha{}_\beta \wedge \vartheta^\beta$. Recently, it has been stressed by
mielke-geometrodynamics_F00484	109	1.000	`\Gamma=\Gamma^{\{ \}}-K`	$\Gamma=\Gamma^{\{\}}-K$	WEINBERG (2005) that this RC connection $\Gamma = \Gamma^{\{\}} - K$ is just a *deformation* (or
mielke-geometrodynamics_F00485	109	1.000	`\Gamma^{\{ \}}`	$\Gamma^{\{\}}$	field redefinition) of the Christoffel connection $\Gamma^{\{\}}$ by the contortion, at least from
mielke-geometrodynamics_F00486	110	1.000	`D>2`	$D>2$	renormalizable in $D>2$ without Chern–Simons (CS) terms (ALVES et al. 1999),
mielke-geometrodynamics_F00487	110	1.000	`j_{5}`	j_{5}	equation relates torsion to spin, i.e., to the axial current j_5 in the case of Dirac fields,
mielke-geometrodynamics_F00488	110	0.972	`\bar{\psi}:=\psi^{\dagger} \gamma_{0}`	$\bar{\psi}:=\psi^{\dagger}\gamma_{0}$	$\overline{\psi} := \psi^\dagger\gamma_0$ denotes the Dirac adjoint and $D\psi := d\psi + \Gamma\wedge\psi$ is the exterior covari-
mielke-geometrodynamics_F00489	110	0.972	`D \psi:=d \psi+\Gamma \wedge \psi`	$D\psi:=d\psi+\Gamma\wedge\psi$	$\overline{\psi} := \psi^\dagger\gamma_0$ denotes the Dirac adjoint and $D\psi := d\psi + \Gamma\wedge\psi$ is the exterior covari-
mielke-geometrodynamics_F00490	110	1.000	`\gamma:=\gamma_{\alpha} \vartheta^{\alpha}`	$\gamma:=\gamma_{\alpha}\vartheta^{\alpha}$	where $\gamma := \gamma_\alpha\vartheta^\alpha$ is the Clifford-algebra-valued coframe satisfying $D\gamma = [\gamma, K]$
mielke-geometrodynamics_F00491	110	1.000	`D \gamma=[\gamma, K]`	$D\gamma=[\gamma,K]$	where $\gamma := \gamma_\alpha\vartheta^\alpha$ is the Clifford-algebra-valued coframe satisfying $D\gamma = [\gamma, K]$
mielke-geometrodynamics_F00492	110	1.000	`=\gamma_{\alpha} T^{\alpha}`	$=\gamma_{\alpha}T^{\alpha}$	$= \gamma_\alpha T^\alpha$, and $T^\alpha := D\vartheta^\alpha$ is the torsion two-form.
mielke-geometrodynamics_F00493	110	1.000	`T^{\alpha}:=D \vartheta^{\alpha}`	$T^{\alpha}:=D\vartheta^{\alpha}$	$= \gamma_\alpha T^\alpha$, and $T^\alpha := D\vartheta^\alpha$ is the torsion two-form.
mielke-geometrodynamics_F00494	110	1.000	`L_{\mathrm{D}}=\bar{L}_{\mathrm{D}}=L_{\mathrm{D}}^{\dagger}`	$L_{\mathrm{D}}=\bar{L}_{\mathrm{D}}=L_{\mathrm{D}}^{\dagger}$	Since $L_{\mathrm{D}} = \overline{L}_{\mathrm{D}} = L_{\mathrm{D}}^\dagger$ even in an unholonomic frame, the minimal coupling pro-
mielke-geometrodynamics_F00495	111	1.000	`\tau_{\alpha \beta \gamma}=\tau_{[\alpha \beta \gamma]}`	$\tau_{\alpha\beta\gamma}=\tau_{[\alpha\beta\gamma]}$	with *totally antisymmetric* components $\tau_{\alpha\beta\gamma} = \tau_{[\alpha\beta\gamma]}$. Eventually, torsion merely
mielke-geometrodynamics_F00496	111	0.975	`\mu_{\alpha}=\vartheta_{\alpha} \wedge{ }^{*} j_{5} / 4`	$\mu_{\alpha}=\vartheta_{\alpha}\wedge{}^{*}j_{5}/4$	couples to the *spin–energy potential* $\mu_\alpha = \vartheta_\alpha \wedge {}^*j_5/4$, i.e., to a two-form propor-
mielke-geometrodynamics_F00497	111	0.792	`\mathrm{GR}_{|}`	$\mathrm{GR}_{\|}$	Transitions from EC theory or GR to $\mathrm{GR}_{\|}$ can be *generated* via the Chern–Simons-
mielke-geometrodynamics_F00498	111	0.701	`C_{\mathrm{TT} *}:=\vartheta^{\alpha} \wedge{ }^{*} T_{\alpha}`	$C_{\mathrm{TT}*}:=\vartheta^{\alpha}\wedge{}^{*}T_{\alpha}$	type term $C_{\mathrm{TT}*} := \vartheta^\alpha \wedge {}^*T_\alpha$, involving the Hodge dual of the torsion. On the material
mielke-geometrodynamics_F00499	111	1.000	`L_{\mathrm{D}} \rightarrow L_{\mathrm{D}}+d U`	$L_{\mathrm{D}}\rightarrow L_{\mathrm{D}}+dU$	side, a related change of the Dirac Lagrangian can be induced via $L_{\mathrm{D}} \rightarrow L_{\mathrm{D}} + dU$,
mielke-geometrodynamics_F00500	111	1.000	`U=\vartheta^{\alpha} \wedge \mu_{\alpha}`	$U=\vartheta^{\alpha}\wedge\mu_{\alpha}$	where $U = \vartheta^\alpha\wedge\mu_\alpha$ features as a superpotential. Then the corresponding boundary
mielke-geometrodynamics_F00501	111	1.000	`\hat{\tau}_{\alpha \beta}`	$\hat{\tau}_{\alpha\beta}$	sions (R1) and (R2) of MIELKE et al. (1989), such that the relocalized spin $\hat{\tau}_{\alpha\beta}$
mielke-geometrodynamics_F00502	111	1.000	`U=\vartheta^{\alpha} \wedge \vartheta_{\alpha} \wedge{ }^{*} j_{5} / 4`	$U=\vartheta^{\alpha}\wedge\vartheta_{\alpha}\wedge{}^{*}j_{5}/4$	vanishes. Observe that for Dirac spinors, $U = \vartheta^\alpha\wedge\vartheta_\alpha\wedge{}^*j_5/4$ is trivial "on shell."

Identifier	Page	Conf.	LaTeX source	Rendered	Image
mielke-geometrodynamics_F00503	111	■1.000	`\kappa=8 \pi G_{\mathrm{N}}`	$\kappa=8 \pi G_{\mathrm{N}}$	and the orthogonal tetrads, whereas $\kappa = 8\pi G_{\mathrm{N}}$ is the gravitational constant in natural
mielke-geometrodynamics_F00504	112	■1.000	`\Sigma_{\alpha}`	Σ_{α}	coupled to the canonical energy–momentum current Σ_α of matter, is obtained by
mielke-geometrodynamics_F00505	112	■1.000	`L_{\mathrm{EC}}`	L_{EC}	varying the matter coupled Lagrangian L_{EC} with respect to the coframe ϑ^α.
mielke-geometrodynamics_F00506	112	■0.973	`\left.\widehat{\Gamma}_{\alpha}{ }^{\beta}:=\Gamma_{ \alpha}{ }^{\beta}+e_{\alpha}\right\rfloor T^{\beta}`	$\left.\widehat{\Gamma}_{\alpha}{ }^{\beta}:=\Gamma_{\alpha}{ }^{\beta}+e_{\alpha}\right\rfloor T^{\beta}$	with respect to the transposed connection $\widehat{\Gamma}_\alpha{}^\beta := \Gamma_\alpha{}^\beta + e_\alpha \rfloor T^\beta$. Observe that the
mielke-geometrodynamics_F00507	113	■1.000	`\sigma:=\frac{1}{2} \sigma_{\alpha \beta} \vartheta^{ \alpha} \wedge \vartheta^{\beta}=\frac{i}{2} \gamma \wedge \gamma`	$\sigma:=\frac{1}{2} \sigma_{\alpha \beta} \vartheta^{\alpha} \wedge \vartheta^{\beta}=\frac{i}{2} \gamma \wedge \gamma$	involving the unit-curvature two-form $\sigma := \frac{1}{2}\sigma_{\alpha\beta}\,\vartheta^\alpha\wedge\vartheta^\beta = \frac{i}{2}\,\gamma\wedge\gamma$, the equiva-
mielke-geometrodynamics_F00508	113	■1.000	`\Theta=0`	$\Theta=0$	for vanishing torsion, i.e., for $\Theta = 0$ in Einstein's standard GR.
mielke-geometrodynamics_F00509	113	■1.000	`\Gamma^{\alpha \beta}`	$\Gamma^{\alpha \beta}$	By varying (5.3.1) with respect to the linear connection $\Gamma^{\alpha\beta}$, we obtain the second
mielke-geometrodynamics_F00510	113	■0.839	`\tilde{\mathrm{T}}_{\mu \nu}`	$\tilde{\mathrm{T}}_{\mu \nu}$	$\tilde{\mathrm{T}}_{\mu\nu}$ contains additional terms depending quadratically on the spin angular momen-
mielke-geometrodynamics_F00511	113	■1.000	`\tau_{\mathrm{s}}=\mathrm{o}`	$\tau_{\mathrm{s}}=\mathrm{o}$	tum of matter; cf. HEHL et al. (1976). In the instance that $\tau_{\mathrm{s}} = \mathrm{o}$, the EC equation
mielke-geometrodynamics_F00512	113	■1.000	`\ell^{* 4}`	$\ell^{* 4}$	terms in (5.3.12) are proportional to ℓ^{*4} and could thus, with a modified Planck
mielke-geometrodynamics_F00513	114	■1.000	`\ell^{*}=10^{-33} \mathrm{~cm}`	$\ell^{*}=10^{-33} \mathrm{~cm}$	length $\ell^* = 10^{-33}$ cm, play a role on the macroscopic scale only in cosmology; cf.,
mielke-geometrodynamics_F00514	114	■1.000	`10^{80}`	10^{80}	be threatened by a final singularity, since the gravitational collapse of 10^{80} baryons
mielke-geometrodynamics_F00515	114	■0.528	`R^{\{\lambda \alpha \beta}`	$R^{\{\lambda \alpha \beta}$	into the Riemannian curvature $R^{\{\}\alpha\beta}$ plus contortion pieces.
mielke-geometrodynamics_F00516	114	■1.000	`{ }^{(i)} T_{\alpha}`	${ }^{(i)} T_{\alpha}$	allelism as a consequence. Here ${}^{(i)}T_\alpha$ are the three irreducible pieces of the torsion.
mielke-geometrodynamics_F00517	114	■1.000	`\mathscr{A}`	$\mathscr{A}$	When only this axial torsion $\mathscr{A}$ enters algebraically, the EC Lagrangian
mielke-geometrodynamics_F00518	114	■1.000	`L_{\mathrm{HE}}`	L_{HE}	generalizes[1] the metric Hilbert–Einstein Lagrangian L_{HE} to an RC spacetime with
mielke-geometrodynamics_F00519	114	■0.647	`T^{\alpha} \wedge{ }^{*}\left({ }^{(1)} T_{ \alpha}-2^{(2)} T_{\alpha}-\frac{1}{2}{ }^{(3)} T_{ \alpha}\right)`	$T^{\alpha} \wedge{ }^{*}\left({ }^{(1)} T_{\alpha}-2^{(2)} T_{\alpha}-\frac{1}{2}{ }^{(3)} T_{\alpha}\right)$	dure, since the particular combination $T^\alpha\wedge{}^*\left({}^{(1)}T_\alpha - 2\,{}^{(2)}T_\alpha - \frac{1}{2}\,{}^{(3)}T_\alpha\right)$ of irreducible pieces is
mielke-geometrodynamics_F00520	114	■0.558	`d C_{\mathrm{TT}}{ }^{*}`	$d C_{\mathrm{TT}}{ }^{*}$	the space of gravity theories, the term dC_{TT^*} interrelates GR with its teleparallelism equivalent
mielke-geometrodynamics_F00521	115	■1.000	`G_{\alpha}^{\{ \}}=G_{\alpha}{ }^{\beta} \eta_{\beta}`	$G_{\alpha}^{\{ \}}=G_{\alpha}{ }^{\beta} \eta_{\beta}$	can be decomposed into the Einstein three-form $G_\alpha^{\{\}} = G_\alpha{}^\beta\eta_\beta$ with respect to the
mielke-geometrodynamics_F00522	115	■1.000	`\ell=\sqrt{\kappa}`	$\ell=\sqrt{\kappa}$	coupled via the "bare" fundamental length $\ell = \sqrt{\kappa}$. Then "on shell," EC theory
mielke-geometrodynamics_F00523	115	■1.000	`k \rightarrow \infty`	$k \rightarrow \infty$	bounded in the ultraviolet limit $k \to \infty$. Quite generally, the renormalization flow is
mielke-geometrodynamics_F00524	115	■1.000	`\Gamma_{k}`	Γ_{k}	for the effective action Γ_k, i.e.,
mielke-geometrodynamics_F00525	116	■0.999	`k \partial_{k}`	$k \partial_{k}$	where $k\partial_k$ is a scale derivative returning the Euler degree of homogeneous functionals.
mielke-geometrodynamics_F00526	116	■1.000	`k`	k	where k is the renormalization group (RG) scale in momentum space and Λ the
mielke-geometrodynamics_F00527	116	■1.000	`\rho_{\Lambda}`	ρ_{Λ}	cosmological constant related to dark energy (DE) of density ρ_Λ; see also BJORKEN
mielke-geometrodynamics_F00528	116	■1.000	`d_{\mathrm{N}}`	d_{N}	where d_{N} is the anomalous dimension of the running Newton coupling g_{N}.

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mielke-geometrodynamics_F00529	116	1.000	`g_{\mathrm{N}}`	g_{N}	where d_{N} is the anomalous dimension of the running Newton coupling g_{N}.
mielke-geometrodynamics_F00530	116	1.000	`g_{\mathrm{N} *}`	$g_{\mathrm{N} *}$	stants (5.4.3) run into some nontrivial fixed points $g_{\mathrm{N}*}$ and λ_*, depending on the
mielke-geometrodynamics_F00531	116	1.000	`\lambda_{*}`	λ_{*}	stants (5.4.3) run into some nontrivial fixed points $g_{\mathrm{N}*}$ and λ_*, depending on the
mielke-geometrodynamics_F00532	116	1.000	`R^{n}`	R^{n}	high-order polynomials R^n of the Ricci scalar, similarly as in the classically bifur-
mielke-geometrodynamics_F00533	116	1.000	`f(R)`	$f(R)$	cating $f(R)$ models (SCHUNCK et al. 2005), but then the issue of physical ghosts or
mielke-geometrodynamics_F00534	117	1.000	`\kappa`	κ	also scales with the gravitational constant κ, but inversely compared to the Hilbert–
mielke-geometrodynamics_F00535	117	1.000	`1 / k^{4}`	$1 / k^{4}$	bilinears is mandatory, including the $1/k^4$ dependence in the functional integral.
mielke-geometrodynamics_F00536	117	1.000	`f_{*}^{2}`	f_{*}^{2}	and the corresponding dimensionless renormalized running coupling f_*^2 becomes
mielke-geometrodynamics_F00537	117	1.000	`R_{\alpha \beta} \equiv 0`	$R_{\alpha \beta} \equiv 0$	One way to avoid such anomalies is to employ curvature constraints like $R_{\alpha\beta} \equiv 0$,
mielke-geometrodynamics_F00538	117	1.000	`S L(5, \mathbb{R})`	$S L(5, \mathbb{R})$	from a "minimalist" $SL(5, \mathbb{R})$ gauge model that includes only a "bare" Pontryagin-
mielke-geometrodynamics_F00539	118	1.000	`\Delta c / c \simeq 10^{-17}`	$\Delta c / c \simeq 10^{-17}$	Lorentz-invariant, nowadays measured at the unprecedented $\Delta c/c \simeq 10^{-17}$ level
mielke-geometrodynamics_F00540	118	1.000	`10^{-13}`	10^{-13}	et al. 2012) at the 10^{-13} level. Moreover, the upper bound for a violation of the
mielke-geometrodynamics_F00541	118	1.000	`10^{-8}`	10^{-8}	quarks, is about 10^{-8}; cf. HUGHES (1993).
mielke-geometrodynamics_F00542	118	0.979	`\bar{v}_{e}`	$\bar{v}_{e}$	Assuming that not only the dominant delayed antineutrinos $\overline{\nu}_e$ but at least one prompt
mielke-geometrodynamics_F00543	118	0.813	`v_{e}`	v_{e}	neutrino ν_e was in the pulse, one could infer that the equivalence principle is valid
mielke-geometrodynamics_F00544	118	1.000	`10^{-6}`	10^{-6}	(GUZZO et al. 2002) for electron–neutrinos ν_e up to 10^{-6}.
mielke-geometrodynamics_F00545	118	0.996	`C P T`	$C P T$	Moreover, a test of the CPT theorem for the neutral kaon system indirectly
mielke-geometrodynamics_F00546	118	1.000	`K^{0}`	K^{0}	yielded an upper bound for the relative difference of the *inertial masses* of the K^0
mielke-geometrodynamics_F00547	118	1.000	`\bar{K}^{0}`	$\bar{K}^{0}$	meson and its antiparticle $\overline{K}^0$ of less than 10^{-18}, when employing (MURAYAMA
mielke-geometrodynamics_F00548	118	1.000	`10^{-18}`	10^{-18}	meson and its antiparticle $\overline{K}^0$ of less than 10^{-18}, when employing (MURAYAMA
mielke-geometrodynamics_F00549	120	0.761	`(5, \mathbb{R})`	$(5, \mathbb{R})$	Mielke EW (2012) Einstein-Weyl gravity from a topological gauge SL $(5, \mathbb{R})$ invariant action. Adv
mielke-geometrodynamics_F00550	122	1.000	`(1986,1991)`	$(1986,1991)$	On the other hand, Ashtekar's reformulation ASHTEKAR (1986, 1991) of general
mielke-geometrodynamics_F00551	122	0.996	`d C_{\mathrm{TT}}`	$d C_{\mathrm{TT}}$	ary term dC_{TT} constructed from a *translational Chern–Simons term* (CS) multiplied
mielke-geometrodynamics_F00552	122	0.987	`\left(\mathrm{GR}_{|}\right)`	$\left(\mathrm{GR}_{\|}\right)$	as was already suggested by EINSTEIN (1928), a theory with *teleparallelism* $(\mathrm{GR}_{\|})$.
mielke-geometrodynamics_F00553	123	1.000	`\ell:=\sqrt{8 \pi \hbar G_{\mathrm{N}} / c^{3}} \simeq 10^{-33} \mathrm{~cm}`	$\ell:=\sqrt{8 \pi \hbar G_{\mathrm{N}} / c^{3}} \simeq 10^{-33} \mathrm{~cm}$	scale $\ell := \sqrt{8\pi\hbar G_{\mathrm{N}}/c^3} \simeq 10^{-33}$ cm, a topologically rich spectrum of dislocations
mielke-geometrodynamics_F00554	123	1.000	`q`	q	In the one-dimensional harmonic oscillator model (MIELKE 1998) with q as gen-
mielke-geometrodynamics_F00555	123	0.994	`\mathscr{C}=q^{2} / 2`	$\mathscr{C}=q^{2} / 2$	term derived from the *Chern–Simons* (CS)-type term $\mathscr{C} = q^2/2$ as *generating func-*

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mielke-geometrodynamics_F00556	123	0.999	`\mathscr{S}=\exp (-\mathscr{C})=\exp \left(-q^{2} / 2 \right)`	$\mathscr{S}=\exp(-\mathscr{C})=\exp\left(-q^{2}/2\right)$	*tion*. After quantization, the corresponding operator $\mathscr{S}=\exp(-\mathscr{C})=\exp(-q^2/2)$
mielke-geometrodynamics_F00557	123	0.999	`\Psi=N \exp \left(i \theta \int \mathscr{C}\right)`	$\Psi=N\exp\left(i\theta\int\mathscr{C}\right)$	shown that $\Psi=N\exp(i\theta\int\mathscr{C})$ is, up to an overall factor, the unique solution of the
mielke-geometrodynamics_F00558	123	1.000	`P:=\mathbb{R}^{4} \otimes S O(1,3)`	$P:=\mathbb{R}^{4}\otimes SO(1,3)$	Since the Poincaré group $P:=\mathbb{R}^4\rtimes SO(1,3)$ is the semidirect product of trans-
mielke-geometrodynamics_F00559	124	1.000	`\mathscr{A}:={ }^{*}\left(\vartheta_{\alpha} \wedge T^{ \alpha}\right)=`	$\mathscr{A}:={}^{*}\left(\vartheta_{\alpha}\wedge T^{\alpha}\right)=$	They are *translational* and Lorentz-rotational CS terms, where $\mathscr{A}:={}^*(\vartheta_\alpha\wedge T^\alpha)=$
mielke-geometrodynamics_F00560	124	1.000	`\mathscr{A}_{i} d x^{i}`	$\mathscr{A}_{i}dx^{i}$	$\mathscr{A}_i dx^i$ is the *axial torsion* one-form. In the first *translational* CS three-form, there
mielke-geometrodynamics_F00561	124	1.000	`\ell`	ℓ	necessarily occurs a fundamental length ℓ for dimensional reasons.
mielke-geometrodynamics_F00562	124	0.907	`\mathrm{SO}(1,4)`	$\mathrm{SO}(1,4)$	the de Sitter group SO(1, 4) or SO(2, 3) as a subgroup, both CS three-forms are
mielke-geometrodynamics_F00563	124	0.907	`\mathrm{SO}(2,3)`	$\mathrm{SO}(2,3)$	the de Sitter group SO(1, 4) or SO(2, 3) as a subgroup, both CS three-forms are
mielke-geometrodynamics_F00564	124	1.000	`g_{\alpha \beta}`	$g_{\alpha\beta}$	free Pontryagin form, in the NY term (6.2.4), a metric $g_{\alpha\beta}$ is needed to raise and
mielke-geometrodynamics_F00565	124	1.000	`T_{\alpha}=g_{\alpha \beta} T^{\beta}`	$T_{\alpha}=g_{\alpha\beta}T^{\beta}$	lower the indices, for instance in $T_\alpha=g_{\alpha\beta}T^\beta$.
mielke-geometrodynamics_F00566	124	1.000	`\vartheta_{\alpha}`	ϑ_{α}	ϑ_α, on the first Bianchi identity
mielke-geometrodynamics_F00567	125	1.000	`R_{\alpha \beta}^{\{ \}}`	$R_{\alpha\beta}^{\{\}}$	of general relativity (GR), where $R^{\{\}}_{\alpha\beta}$ denotes the Riemannian curvature with respect
mielke-geometrodynamics_F00568	125	1.000	`\Gamma_{\alpha \beta}^{\{ \}}`	$\Gamma_{\alpha\beta}^{\{\}}$	to the Levi-Civita connection $\Gamma^{\{\}}_{\alpha\beta}$.
mielke-geometrodynamics_F00569	125	1.000	`K_{\alpha \beta}`	$K_{\alpha\beta}$	is dual to the contortion one-form $K_{\alpha\beta}$, which features in the decomposition $\Gamma_{\alpha\beta}=$
mielke-geometrodynamics_F00570	125	1.000	`\Gamma_{\alpha \beta}=`	$\Gamma_{\alpha\beta}=$	is dual to the contortion one-form $K_{\alpha\beta}$, which features in the decomposition $\Gamma_{\alpha\beta}=$
mielke-geometrodynamics_F00571	125	0.985	`\left.\left.\left.-\Gamma_{\beta \alpha}=\Gamma_{\alpha \beta}^{\{ \}}-K_{\alpha \beta}=\Gamma_{\alpha \beta}^{\{ \}}+e_{\alpha}\right\rfloor T_{\beta}+\left(e_{\alpha}\right\rfloor e_{\beta}\right\rfloor T_{\gamma}\right) \wedge \vartheta^{\gamma}`	$-\Gamma_{\beta\alpha}=\Gamma_{\alpha\beta}^{\{\}}-K_{\alpha\beta}=\Gamma_{\alpha\beta}^{\{\}}+e_{\alpha}\rfloor T_{\beta}+\left(e_{\alpha}\rfloor e_{\beta}\rfloor T_{\gamma}\right)\wedge\vartheta^{\gamma}$	$-\Gamma_{\beta\alpha}=\Gamma^{\{\}}_{\alpha\beta}-K_{\alpha\beta}=\Gamma^{\{\}}_{\alpha\beta}+e_\alpha\rfloor T_\beta+(e_\alpha\rfloor e_\beta\rfloor T_\gamma)\wedge\vartheta^\gamma$ of the RC connection.
mielke-geometrodynamics_F00572	125	0.971	`C_{\mathrm{TT} *}:=\vartheta^{\alpha} \wedge{ }^{*} T_{ \alpha} / 2 \ell^{2}`	$C_{\mathrm{TT}*}:=\vartheta^{\alpha}\wedge{}^{*}T_{\alpha}/2\ell^{2}$	where $C_{\mathrm{TT}*}:=\vartheta^\alpha\wedge{}^*T_\alpha/2\ell^2$ is a dual CS term, proper *teleparallelism* (GR$_{\Vert}$) spec-
mielke-geometrodynamics_F00573	125	1.000	`R^{\alpha \beta}=0`	$R^{\alpha\beta}=0$	the proper constraint $R^{\alpha\beta}=0$ has to be imposed by subtracting $R^{\alpha\beta}\wedge\lambda_{\alpha\beta}$ from
mielke-geometrodynamics_F00574	125	1.000	`R^{\alpha \beta} \wedge \lambda_{\alpha \beta}`	$R^{\alpha\beta}\wedge\lambda_{\alpha\beta}$	the proper constraint $R^{\alpha\beta}=0$ has to be imposed by subtracting $R^{\alpha\beta}\wedge\lambda_{\alpha\beta}$ from
mielke-geometrodynamics_F00575	125	1.000	`\lambda_{\alpha \beta}=-\lambda_{\beta \alpha}`	$\lambda_{\alpha\beta}=-\lambda_{\beta\alpha}$	(6.2.9), where the 2-form $\lambda_{\alpha\beta}=-\lambda_{\beta\alpha}$ is a Lagrange multiplier, see HEHL et al.
mielke-geometrodynamics_F00576	126	1.000	`\vartheta^{\alpha}, \Gamma^{\alpha \beta}`	$\vartheta^{\alpha},\Gamma^{\alpha\beta}$	By varying this Lagrangian independently with respect to ϑ^α, $\Gamma^{\alpha\beta}$, and the mul-
mielke-geometrodynamics_F00577	126	1.000	`\lambda_{\alpha \beta}`	$\lambda_{\alpha\beta}$	tiplier $\lambda_{\alpha\beta}$, we obtain as field equations
mielke-geometrodynamics_F00578	126	0.856	`L_{|}`	$L_{\Vert}$	first field Eq. (6.2.13) is the same as that of the unconstrained Lagrangian $L_{\Vert}$. The
mielke-geometrodynamics_F00579	126	1.000	`\lambda_{\alpha \beta}=\left(\gamma / \ell^{2}\right) \eta_{\alpha \beta}`	$\lambda_{\alpha\beta}=\left(\gamma/\ell^{2}\right)\eta_{\alpha\beta}$	*spinless* matter, a possible solution is $\lambda_{\alpha\beta}=(\gamma/\ell^2)\eta_{\alpha\beta}$, as will be shown below.
mielke-geometrodynamics_F00580	126	0.982	`\boldsymbol{G R}_{|}`	$\boldsymbol{GR}_{\Vert}$	***6.2.2 Yang–Mills-Type Formulation of Complex GR$_{\Vert}$***

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mielke-geometrodynamics_F00581	127	1.000	`\sigma= \pm i \gamma`	$\sigma= \pm i \gamma$	of BAEKLER et al. (1982); if $\sigma = \pm i\gamma$ is chosen for one of the free parameters there.
mielke-geometrodynamics_F00582	127	1.000	`\theta_{\mathrm{T}}= \pm i`	$\theta_{\mathrm{T}}= \pm i$	version (6.2.9) of Einstein's GR for the choice $\theta_{\mathrm{T}} = \pm i$ into a *chiral gauge theory*
mielke-geometrodynamics_F00583	127	0.984	`\stackrel{( \pm)}{T}{ }^{\alpha}:=\frac{1}{2}\left(T^{\alpha} \pm i^{*} T^{\alpha}\right)`	$\stackrel{(\pm)}{T}{ }^{\alpha}:=\frac{1}{2}\left(T^{\alpha} \pm i^{*} T^{\alpha}\right)$	where $\stackrel{(\pm)}{T}{}^{\alpha} := \frac{1}{2}(T^{\alpha} \pm i\,{}^{*}T^{\alpha})$ denotes the self-dual or anti-self-dual torsion. The
mielke-geometrodynamics_F00584	128	1.000	`{ }^{(\tau)} \lambda_{\alpha \beta}`	${ }^{(\tau)} \lambda_{\alpha \beta}$	et al. 1989) an additional piece ${}^{(\tau)}\lambda_{\alpha\beta}$ in the ansatz for the Lagrangian multiplier,
mielke-geometrodynamics_F00585	128	1.000	`D^{(\tau)} \lambda_{\alpha \beta}=\tau_{\alpha \beta}`	$D^{(\tau)} \lambda_{\alpha \beta}=\tau_{\alpha \beta}$	provided it satisfies $D\,{}^{(\tau)}\lambda_{\alpha\beta} = \tau_{\alpha\beta}$.
mielke-geometrodynamics_F00586	128	0.584	`\stackrel{( \pm)}{\Pi^{\alpha}}`	$\stackrel{(\pm)}{\Pi^{\alpha}}$	The complex translational field momenta $\stackrel{(\pm)}{\Pi}{}^{\alpha}$ replace the role of the torsion if we
mielke-geometrodynamics_F00587	128	0.731	`\delta \stackrel{( \pm)}{L}_{|} / \delta \vartheta^{\alpha}`	$\delta \stackrel{(\pm)}{L}_{\Vert} / \delta \vartheta^{\alpha}$	allow us to rederive the field Eq. (6.2.24) via $\delta \stackrel{(\pm)}{L}_{\Vert}/\delta\vartheta^{\alpha}$ from a variational principle.
mielke-geometrodynamics_F00588	128	1.000	`R_{\alpha \beta}=0`	$R_{\alpha \beta}=0$	Cartan spacetime with $R_{\alpha\beta} = 0$ and in view of the first Noether identity (6.8.11),
mielke-geometrodynamics_F00589	128	0.999	`\stackrel{( \pm)}{\Pi}_{\alpha}`	$\stackrel{(\pm)}{\Pi}_{\alpha}$	terms of the momenta $\stackrel{(\pm)}{\Pi}_{\alpha}$, it would be convenient to have a *relocalized* energy–momentum current
mielke-geometrodynamics_F00590	128	0.121	`\stackrel{( \pm)}{\Sigma}{ }_{\alpha}`	$\stackrel{(\pm)}{\Sigma}{ }_{\alpha}$	$\stackrel{(\pm)}{\Sigma}_{\alpha}$ for which $\stackrel{(\pm)(\pm)}{D\,\Sigma}_{\alpha} \cong 0$ holds.
mielke-geometrodynamics_F00591	128	0.121	`\stackrel{( \pm)( \pm)}{D}`	$\stackrel{(\pm)(\pm)}{D}$	$\stackrel{(\pm)}{\Sigma}_{\alpha}$ for which $\stackrel{(\pm)(\pm)}{D\,\Sigma}_{\alpha} \cong 0$ holds.
mielke-geometrodynamics_F00592	129	0.821	`\stackrel{(\mp)}{A_{\alpha}}{ }^{\beta}`	$\stackrel{(\mp)}{A_{\alpha}}{ }^{\beta}$	valued 1-form $\stackrel{(\mp)}{A}_{\alpha}{}^{\beta}$ deforms our original Riemann–Cartan connection $\Gamma_{\alpha}{}^{\beta}$ similarly
mielke-geometrodynamics_F00593	129	0.946	`\underline{\Gamma}_{|}^{\alpha \beta} \rightarrow 0`	$\underline{\Gamma}_{\Vert}^{\alpha \beta} \rightarrow 0$	(1973), THIRRING (1978), HEHL et al. (1980). In the teleparallelism limit $\underline{\Gamma}_{\Vert}^{\alpha\beta} \to 0$,
mielke-geometrodynamics_F00594	129	1.000	`H_{\alpha}`	H_{α}	the “superpotential” H_{α} is the *Møller* or *Freud complex* (FREUD 1939)
mielke-geometrodynamics_F00595	129	0.700	`\overparen{D} \eta_{\alpha}:=D \eta_{\alpha}-`	$\overparen{D} \eta_{\alpha}:=D \eta_{\alpha}-$	[3]The *transposed* connection, which has the property (MIELKE et al. 1989) that $\overparen{D}\eta_{\alpha} := D\eta_{\alpha} -$
mielke-geometrodynamics_F00596	129	0.521	`\left(e_{\alpha} \downharpoonleft T^{\beta}\right) \wedge \eta_{\beta} \equiv 0`	$\left(e_{\alpha} \downharpoonleft T^{\beta}\right) \wedge \eta_{\beta} \equiv 0$	$(e_{\alpha}\rfloor T^{\beta}) \wedge \eta_{\beta} \equiv 0$, may be regarded as a special real version of our Sen-type connection, for
mielke-geometrodynamics_F00597	129	0.593	`\stackrel{( \pm)}{D} \eta_{\alpha}= \pm(i / 2) \ell^{2} \eta_{\alpha \beta} \wedge \stackrel{( \pm)}{\Pi}^{\beta}`	$\stackrel{(\pm)}{D} \eta_{\alpha}= \pm(i / 2) \ell^{2} \eta_{\alpha \beta} \wedge \stackrel{(\pm)}{\Pi}^{\beta}$	which $\stackrel{(\pm)}{D}\eta_{\alpha} = \pm(i/2)\ell^{2}\eta_{\alpha\beta} \wedge \stackrel{(\pm)}{\Pi}{}^{\beta}$ holds.
mielke-geometrodynamics_F00598	130	1.000	`E_{\alpha}`	E_{α}	et al. (2000) that E_{α} is a covariant conserved gravitational energy, i.e., that $DE_{\alpha} =$
mielke-geometrodynamics_F00599	130	1.000	`D E_{\alpha}=`	$D E_{\alpha}=$	et al. (2000) that E_{α} is a covariant conserved gravitational energy, i.e., that $DE_{\alpha} =$
mielke-geometrodynamics_F00600	130	1.000	`D D H_{\alpha} \equiv-R_{\alpha}{ }^{\beta} \wedge H_{\beta}=0`	$D D H_{\alpha} \equiv-R_{\alpha}{ }^{\beta} \wedge H_{\beta}=0$	$DDH_{\alpha} \equiv -R_{\alpha}{}^{\beta} \wedge H_{\beta} = 0$ due to the vanishing of the Ricci identity in a teleparallel
mielke-geometrodynamics_F00601	130	0.984	`\mathbf{G R}_{\boldsymbol{|}}`	$\mathbf{G R}_{\boldsymbol{\Vert}}$	**6.4 Hamiltonian Formulation of Complex GR$_{\Vert}$**
mielke-geometrodynamics_F00602	130	1.000	`\Sigma_{t}`	Σ_{t}	into a family of spacelike three-dimensional hypersurfaces Σ_{t} that are parametrized
mielke-geometrodynamics_F00603	130	1.000	`x^{0}=t`	$x^{0}=t$	by the coordinate time $x^{0} = t$. Then there exists the future-directed *timelike* vector
mielke-geometrodynamics_F00604	130	0.496	`u:=-N^{2} d t, \Longleftrightarrow`	$u:=-N^{2} d t, \Longleftrightarrow$	which is uniquely linked to the hypersurface orthogonal 1-form $u := -N^{2}\,dt$, $\Longleftrightarrow$
mielke-geometrodynamics_F00605	130	0.963	`d u \wedge u=0`	$d u \wedge u=0$	$du \wedge u = 0$ via $u = g(n,\ .)$. The lapse function N and the shift vector N^{A}, which

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mielke-geometrodynamics_F00606	130	0.963	`u=g(n,`	$u=g(n,$	$du \wedge u=0$ via $u=g(n,\ .)$. The lapse function N and the shift vector N^{A}, which
mielke-geometrodynamics_F00607	130	0.963	`) . The lapse function N`	$) . The lapse function N$	$du \wedge u=0$ via $u=g(n,\ .)$. The lapse function N and the shift vector N^{A}, which
mielke-geometrodynamics_F00608	130	0.963	`N^{A}`	N^{A}	$du \wedge u=0$ via $u=g(n,\ .)$. The lapse function N and the shift vector N^{A}, which
mielke-geometrodynamics_F00609	130	0.974	`p`	p	The *normal* part of a p-form Ψ is defined by
mielke-geometrodynamics_F00610	130	0.997	`(3+1)`	$(3+1)$	operators. (For more details on this very concise (3+1) decomposition of exterior
mielke-geometrodynamics_F00611	131	1.000	`L=L\left(\vartheta^{\alpha}, \Psi, D \Psi\right)`	$L=L\left(\vartheta^{\alpha}, \Psi, D \Psi\right)$	Let us consider a Lagrangian $L=L(\vartheta^{\alpha},\ \Psi,\ D\Psi)$ which is of first differential
mielke-geometrodynamics_F00612	131	0.973	`\left.\mathscr{K}=L_{\perp}:=n\right\rfloor L`	$\left.\mathscr{K}=L_{\perp}:=n\right\rfloor L$	where $\mathscr{K}=L_{\perp}:=n\rfloor L$ denotes the normal part of the Lagrangian. Since the “time
mielke-geometrodynamics_F00613	131	0.677	`{ }^{\bullet}`	${ }^{\bullet}$	derivative” ${}^{\bullet}$ is more precisely given by the Lie derivative $\ell_{n}:=dn\rfloor+n\rfloor d$ along
mielke-geometrodynamics_F00614	131	0.677	`\left.\ell_{n}:=d n\right\rfloor+n \downarrow d`	$\left.\ell_{n}:=d n\right\rfloor+n \downarrow d$	derivative” ${}^{\bullet}$ is more precisely given by the Lie derivative $\ell_{n}:=dn\rfloor+n\rfloor d$ along
mielke-geometrodynamics_F00615	131	1.000	`n`	n	the timelike vector field n, we have
mielke-geometrodynamics_F00616	131	0.965	`\left.n^{\alpha}:=n\right\rfloor \vartheta^{\alpha}`	$\left.n^{\alpha}:=n\right\rfloor \vartheta^{\alpha}$	and transvect the resulting expression with the four-vector $n^{\alpha}:=n\rfloor\vartheta^{\alpha}$ of *lapse and*
mielke-geometrodynamics_F00617	131	0.605	`\left.n^{\alpha} e_{\alpha}\right\rfloor \Psi=n \downarrow \Psi`	$\left.n^{\alpha} e_{\alpha}\right\rfloor \Psi=n \downarrow \Psi$	*shift*, for which the useful relation $n^{\alpha}\, e_{\alpha}\rfloor\Psi=n\rfloor\Psi$ can be derived. For the projection
mielke-geometrodynamics_F00618	132	0.919	`\mathrm{L}_{n}=D n \downarrow+n \downarrow D`	$\mathrm{L}_{n}=D n \downarrow+n \downarrow D$	In introducing the gauge-covariant Lie derivative $\text{Ł}_{n}=Dn\rfloor+n\rfloor D$, separating
mielke-geometrodynamics_F00619	132	0.651	`n\rfloor \Psi=n \downharpoonleft(d t \wedge n\rfloor \Psi)=n\rfloor \Psi`	$n\rfloor \Psi=n \downharpoonleft(d t \wedge n\rfloor \Psi)=n\rfloor \Psi$	sion is related to the Hamiltonian (6.4.5). Indeed, using $n\rfloor\Psi=n\rfloor(dt\wedge n\rfloor\Psi)=n\rfloor\Psi$
mielke-geometrodynamics_F00620	132	0.479	`\left.\Gamma_{\perp}{ }^{\alpha \beta}:=n\right\rfloor \Gamma^{\alpha \beta}`	$\left.\Gamma_{\perp}{ }^{\alpha \beta}:=n\right\rfloor \Gamma^{\alpha \beta}$	where the normal vector $n^{\alpha}:=n\rfloor\vartheta^{\alpha}$ comprises *lapse and shift* and $\Gamma_{\perp}{}^{\alpha\beta}:=n\rfloor\Gamma^{\alpha\beta}$
mielke-geometrodynamics_F00621	132	1.000	`\mathscr{G}_{\alpha}`	$\mathscr{G}_{\alpha}$	is the normal part of the Lorentz connection and $\mathscr{G}_{\alpha}$ and $\mathscr{G}_{\alpha\beta}$ represent the tangential
mielke-geometrodynamics_F00622	132	1.000	`\mathscr{G}_{\alpha \beta}`	$\mathscr{G}_{\alpha \beta}$	is the normal part of the Lorentz connection and $\mathscr{G}_{\alpha}$ and $\mathscr{G}_{\alpha\beta}$ represent the tangential
mielke-geometrodynamics_F00623	133	0.709	`\left.D_{\alpha}:=e_{\alpha}\right\rfloor D`	$\left.D_{\alpha}:=e_{\alpha}\right\rfloor D$	where $D_{\alpha}:=e_{\alpha}\rfloor D$ are the anholonomic components of the exterior covariant
mielke-geometrodynamics_F00624	133	0.927	`D \lambda_{\alpha \beta}`	$D \lambda_{\alpha \beta}$	arise the extra tangential term $\underline{D\lambda}_{\alpha\beta}$ in the *Lorentz* generator $\mathscr{G}_{\alpha\beta}$, where
mielke-geometrodynamics_F00625	133	1.000	`\underline{\Psi}:=\Psi-d t \wedge n\rfloor \Psi`	$\underline{\Psi}:=\Psi-d t \wedge n\rfloor \Psi$	$\underline{\Psi}:=\Psi-dt\wedge n\rfloor\Psi$ in the notation of differential forms. However, as suggested
mielke-geometrodynamics_F00626	133	0.893	`\underline{\lambda}_{\alpha \beta}=\left(1 / 2 \ell^{2}\right) *\left(\vartheta_{\alpha} \wedge \vartheta_{\beta}\right)=-\left(1 / N \ell^{2}\right) n_{[\alpha}{ }^{*} \underline{\vartheta}_{\beta]}`	$\underline{\lambda}_{\alpha \beta}=\left(1 / 2 \ell^{2}\right) *\left(\vartheta_{\alpha} \wedge \vartheta_{\beta}\right)=-\left(1 / N \ell^{2}\right) n_{[\alpha}{ }^{*} \underline{\vartheta}_{\beta]}$	by $\underline{\lambda}_{\alpha\beta}=(1/2\ell^{2})\,{}^{*}(\vartheta_{\alpha}\wedge\vartheta_{\beta})=-(1/N\ell^{2})n_{[\alpha}{}^{\underline{*}}\underline{\vartheta}_{\beta]}$. This is another manifestation of
mielke-geometrodynamics_F00627	133	0.925	`d \stackrel{(\mp)}{C}_{\text {TT }}`	$d \stackrel{(\mp)}{C}_{\text {TT }}$	by the boundary term $d\stackrel{(\mp)}{C}_{\mathrm{TT}}$.
mielke-geometrodynamics_F00628	133	0.989	`(\alpha=\hat{0})`	$(\alpha=\hat{0})$	After this trivialization of the Lorentz constraints, the Hamiltonian $(\alpha=\hat{0})$ and
mielke-geometrodynamics_F00629	133	1.000	`(\alpha=B)`	$(\alpha=B)$	diffeomorphism constraints $(\alpha=B)$ of complexified GR$_{\parallel}$, in terms of a complex
mielke-geometrodynamics_F00630	133	0.464	`\stackrel{( \pm)}{\underline{\mathscr{A}}_{\alpha}}:=\stackrel{*}{\underline{\Pi}}_{\alpha}`	$\stackrel{(\pm)}{\underline{\mathscr{A}}_{\alpha}}:=\stackrel{*}{\underline{\Pi}}_{\alpha}$	(1994). Here the tangential Ashtekar-type connection $\stackrel{(\pm)}{\underline{\mathscr{A}}}_{\alpha}:={}^{\underline{*}}\stackrel{(\pm)}{\underline{\Pi}}_{\alpha}$ is three-dual
mielke-geometrodynamics_F00631	133	0.130	`\stackrel{*}{-}_{\alpha}`	$\stackrel{*}{-}_{\alpha}$	ties ${}^{\underline{*}}\underline{\vartheta}_{\alpha}$, where we note that the spatial Hodge dual is involutive, i.e., ${}^{\underline{*}}{}^{\underline{*}}=+1$.

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mielke-geometrodynamics_F00632	133	0.130	`\stackrel{* *}{=}=+1`	$\stackrel{* *}{=}=+1$	ties $\underline{*}\vartheta_\alpha$, where we note that the spatial Hodge dual is involutive, i.e., $\underline{*}\,\underline{*}=+1$.
mielke-geometrodynamics_F00633	133	1.000	`\underline{\vartheta}^{\hat{0}}=0`	$\underline{\vartheta}^{\hat{0}}=0$	Moreover, by adopting tetrads in the temporal gauge $\underline{\vartheta}^{\hat{0}}=0$ of SCHWINGER (1963)
mielke-geometrodynamics_F00634	133	0.858	`\stackrel{( \pm)}{\mathscr{A}_{\hat{0}}}=d^{( \pm)}`	$\stackrel{(\pm)}{\mathscr{A}_{\hat{0}}}=d^{(\pm)}$	and the gauge $\stackrel{(\pm)}{\mathscr{A}}_{\hat{0}}=d\stackrel{(\pm)}{\phi}$ of NESTER (1989), the Hamiltonian constraint
mielke-geometrodynamics_F00635	134	1.000	`\alpha=\hat{0}`	$\alpha=\hat{0}$	i.e., (6.4.19) for $\alpha=\hat{0}$, vanishes identically.
mielke-geometrodynamics_F00636	134	0.561	`\stackrel{*}{-\vartheta}_{\alpha}`	$\stackrel{*}{-\vartheta}_{\alpha}$	parts of the basic one-forms, i.e., the *triad densities* $\underline{*}\underline{\vartheta}_\alpha$ and the tangential part of
mielke-geometrodynamics_F00637	134	0.677	`\stackrel{( \pm)}{\mathscr{A}_{\alpha}}:=\stackrel{*}{ \underline{\Pi}}_{\alpha}`	$\stackrel{(\pm)}{\mathscr{A}_{\alpha}}:=\stackrel{*}{ \underline{\Pi}}_{\alpha}$	the self-dual or anti-self-dual connection $\stackrel{(\pm)}{\mathscr{A}}_\alpha := \underline{*}\stackrel{(\pm)}{\underline{\Pi}}_\alpha$, the three-dual of the chiral
mielke-geometrodynamics_F00638	134	0.995	`\stackrel{ \pm}{p}`	$\stackrel{ \pm}{p}$	momenta $\stackrel{(\pm)}{\Pi}_\alpha$, become the generalized coordinates q and momenta $\stackrel{\pm}{p}$ of the bosonic
mielke-geometrodynamics_F00639	134	0.337	`\stackrel{*( \pm)}{\underline{\Pi}_{\alpha}}`	$\stackrel{*(\pm)}{\underline{\Pi}_{\alpha}}$	prevails (MIELKE 2002), where the complex field "momenta" $\underline{*}\stackrel{(\pm)}{\underline{\Pi}}_\alpha$ become differ-
mielke-geometrodynamics_F00640	134	0.316	`\underline{*}^{\vartheta^{\beta}}`	$\underline{*}^{\vartheta^{\beta}}$	ential operators, whereas the triad densities $\underline{*}\underline{\vartheta}^\beta$ remain generalized coordinates q.
mielke-geometrodynamics_F00641	134	1.000	`\alpha=B`	$\alpha=B$	For the remaining Gauss constraint (6.4.8) with $\alpha=B$, let us try the state vector
mielke-geometrodynamics_F00642	134	0.731	`\underline{C}_{\mathrm{TT}}`	$\underline{C}_{\mathrm{TT}}$	Chern–Simons term $\underline{C}_{\mathrm{TT}}$ in terms of the triads or triad densities and the tangential
mielke-geometrodynamics_F00643	134	0.862	`\underline{*}^{\underline{\theta^{B}}}`	$\underline{*}^{\underline{\theta^{B}}}$	rewritten in terms of the triad densities $\underline{*}\underline{\vartheta}^B$ with $B=1,2,3$ and the tangential part
mielke-geometrodynamics_F00644	134	0.862	`B=1,2,3`	$B=1,2,3$	rewritten in terms of the triad densities $\underline{*}\underline{\vartheta}^B$ with $B=1,2,3$ and the tangential part
mielke-geometrodynamics_F00645	134	0.637	`\stackrel{( \pm)}{\underline{T}}^{B}`	$\stackrel{(\pm)}{\underline{T}}^{B}$	momentum operator (6.4.22) returns the chiral torsion $\stackrel{(\pm)}{\underline{T}}{}^B$ as a prefactor:
mielke-geometrodynamics_F00646	134	1.000	`\underline{\Gamma}_{|}^{\alpha \beta} \stackrel{*}{=} 0`	$\underline{\Gamma}_{\|}^{\alpha \beta} \stackrel{*}{=} 0$	Since the corresponding connection $\underline{\Gamma}_{\|}^{\alpha\beta} \stackrel{*}{=} 0$ is of pure gauge type, the state vector
mielke-geometrodynamics_F00647	135	0.999	`\Psi \wedge{ }^{*} \Phi=\Phi \wedge{ }^{*} \Psi`	$\Psi \wedge{ }^{*} \Phi=\Phi \wedge{ }^{*} \Psi$	where the rule $\Psi \wedge {}^*\Phi = \Phi \wedge {}^*\Psi$ for differential forms of the same degree has
mielke-geometrodynamics_F00648	135	0.994	`S U(2)`	$S U(2)$	borrowed from $SU(2)$ Chern–Simons field theory (WITTEN 1989; GUADAGNINI
mielke-geometrodynamics_F00649	135	1.000	`\mathscr{H}_{\Lambda} \Psi_{\Lambda}(A)=0`	$\mathscr{H}_{\Lambda} \Psi_{\Lambda}(A)=0$	straint $\mathscr{H}_\Lambda\, \Psi_\Lambda(A)=0$ of GR with cosmological constant in the nonperturbative
mielke-geometrodynamics_F00650	135	1.000	`\ell \rightarrow 0`	$\ell \rightarrow 0$	ℓ, it is dimensionless and nonsingular even for $\ell \rightarrow 0$, due to $dim\, \underline{\vartheta}^B=[\ell]$.
mielke-geometrodynamics_F00651	135	1.000	`\operatorname{dim} \underline{\vartheta}^{B}=[\ell]`	$\operatorname{dim} \underline{\vartheta}^{B}=[\ell]$	ℓ, it is dimensionless and nonsingular even for $\ell \rightarrow 0$, due to $dim\, \underline{\vartheta}^B=[\ell]$.
mielke-geometrodynamics_F00652	135	0.937	`\underline{T}^{(\mp)}{ }^{B}=0`	$\underline{T}^{(\mp)}{ }^{B}=0$	configurations with $\stackrel{(\mp)}{\underline{T}}{}^B=0$, i.e., *self-dual or anti-self dual* tangential torsion. Con-
mielke-geometrodynamics_F00653	135	0.802	`\stackrel{( \pm)}{\underline{T}}=0`	$\stackrel{(\pm)}{\underline{T}}=0$	straint are dominated by *self-dual* torsion solutions satisfying $\stackrel{(\pm)}{\underline{T}}=0$.
mielke-geometrodynamics_F00654	135	1.000	`\mathbb{R}^{4} \cup \infty=S^{4}`	$\mathbb{R}^{4} \cup \infty=S^{4}$	$\mathbb{R}^4 \cup \infty = S^4$ with the spherically symmetric metric
mielke-geometrodynamics_F00655	136	0.997	`(r, \psi, \theta, \phi)`	(r, ψ, θ, ϕ)	where (r, ψ, θ, ϕ) are the standard hyperspherical coordinates that parameterize the
mielke-geometrodynamics_F00656	136	1.000	`S^{3}`	S^{3}	unit three-sphere S^3. The function f carries physical dimension [length]. This space

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mielke-geometrodynamics_F00657	136	1.000	f	f	unit three-sphere S^3. The function f carries physical dimension [length]. This space
mielke-geometrodynamics_F00658	136	1.000	d f= \pm 2 h d r	$df = \pm 2hdr$	which is self-dual or anti-self-dual, provided that $df = \pm 2\,hdr$. Here the self-dual
mielke-geometrodynamics_F00659	136	0.998	\eta_{\mu \nu}^{A}:=\eta_{0}{ }^{A}{ }_{\mu \nu}+\delta_{\mu}^{A} \delta_{\nu}^{0}-\delta_{\nu}^{A} \delta_{\mu}^{0}	$\eta_{\mu \nu}^{A}:=\eta_{0}{ }^{A}{ }_{\mu \nu}+\delta_{\mu}^{A} \delta_{\nu}^{0}-\delta_{\nu}^{A} \delta_{\mu}^{0}$	generalized Levi-Civita symbol $\eta_{\mu\nu}^{A} := \eta_0{}^{A}{}_{\mu\nu} + \delta_\mu^A\delta_\nu^0 - \delta_\nu^A\delta_\mu^0$ of 'T HOOFT (1991),
mielke-geometrodynamics_F00660	136	1.000	r \rightarrow \infty	$r \rightarrow \infty$	radial infinity $r \rightarrow \infty$ yields
mielke-geometrodynamics_F00661	136	0.954	\stackrel{(\mp)}{T}^{B}=0	$\stackrel{(\mp)}{T}^{B}=0$	self-dual, i.e., $\stackrel{(\mp)}{\underline{T}}{}^{B} = 0$, then integration over the chiral CS term $\stackrel{(\pm)}{\underline{C}}_{\mathrm{TT}}$ yields the
mielke-geometrodynamics_F00662	136	0.954	\stackrel{(\pm)}{C}_{\text {TT }}	$\stackrel{(\pm)}{C}_{\text {TT }}$	self-dual, i.e., $\stackrel{(\mp)}{\underline{T}}{}^{B} = 0$, then integration over the chiral CS term $\stackrel{(\pm)}{\underline{C}}_{\mathrm{TT}}$ yields the
mielke-geometrodynamics_F00663	136	1.000	\Psi_{\\|}(\vartheta) \propto e^{-C}	$\Psi_{\Vert}(\vartheta) \propto e^{-C}$	Thus one can surmise that the state vector $\Psi_{\Vert}(\vartheta) \propto e^{-C}$ as a solution of the
mielke-geometrodynamics_F00664	136	1.000	\underline{\vartheta}^{\hat{0}}=3 \kappa d \theta_{\mathrm{L}}=h d r= \pm d f / 2	$\underline{\vartheta}^{\hat{0}}=3 \kappa d \theta_{\mathrm{L}}=h d r= \pm d f / 2$	[4]Interestingly, in the gauge $\underline{\vartheta}^{\hat{0}} = 3\kappa d\theta_{\mathrm{L}} = hdr = \pm df/2$, such instantons are solutions to the
mielke-geometrodynamics_F00665	136	0.991	T^{\hat{0}}=0	$T^{\hat{0}}=0$	topological Eq. (6.7.12), due to $T^{\hat{0}} = 0$.
mielke-geometrodynamics_F00666	137	1.000	A=A_{\mu}^{I} \lambda_{I}	$A=A_{\mu}^{I} \lambda_{I}$	From the *holonomy* of a Lie-algebra-valued connection one-form $A = A_\mu^I\lambda_I$
mielke-geometrodynamics_F00667	137	0.967	d x^{\mu} \in \mathscr{C}	$d x^{\mu} \in \mathscr{C}$	$dx^\mu \in \mathscr{C}$, one can construct a *gauge-invariant* operator, or character, along the loop
mielke-geometrodynamics_F00668	137	1.000	\gamma^{\mu}(s)	$\gamma^{\mu}(s)$	with coordinates $\gamma^\mu(s)$. Then the Wilson loop can be written as
mielke-geometrodynamics_F00669	137	0.826	\Psi(A)	$\Psi(A)$	The state vector $\Psi(A)$ for the gauge potential can be transformed via the Wilson
mielke-geometrodynamics_F00670	137	1.000	\mathscr{C} / \mathscr{G}	$\mathscr{C} / \mathscr{G}$	The integration is performed over a coset space $\mathscr{C}/\mathscr{G}$ in order to factor out gauge-
mielke-geometrodynamics_F00671	137	1.000	G(\gamma)	$G(\gamma)$	of the Wilson loop is the *self-linking number* $G(\gamma)$ of Gauss. It is interesting to see
mielke-geometrodynamics_F00672	137	1.000	V(\gamma)	$V(\gamma)$	of the Jones polynomial $V(\gamma)$. If we could formally transform these results of lattice
mielke-geometrodynamics_F00673	138	1.000	|\Psi\rangle	$\vert\Psi\rangle$	state vector $\vert\Psi\rangle$ of the universe would be characterized by knot invariants, such as
mielke-geometrodynamics_F00674	138	1.000	\mathbb{R}^{4}	$\mathbb{R}^{4}$	group $\mathbb{R}^4$ as structure group. Then, instead of Wilson-type loops, we encounter
mielke-geometrodynamics_F00675	138	0.979	\left.n^{\alpha}=n\right\rfloor \vartheta^{\alpha}	$\left.n^{\alpha}=n\right\rfloor \vartheta^{\alpha}$	GUADAGNINI 1993) via one of the orthonormal legs $n^\alpha = n\rfloor\vartheta^\alpha$. In view of (6.4.25),
mielke-geometrodynamics_F00676	138	1.000	\left(1 / 16 \pi^{2}\right)	$\left(1 / 16 \pi^{2}\right)$	It is well known from differential geometry that the Euler number $(1/16\pi^2)$
mielke-geometrodynamics_F00677	138	1.000	\int R^{\alpha \beta} \wedge R_{\alpha \beta}^{(\star)}	$\int R^{\alpha \beta} \wedge R_{\alpha \beta}^{(\star)}$	$\int R^{\alpha\beta} \wedge R_{\alpha\beta}^{(\star)}$ and the Pontryagin integral $(1/8\pi^2)\int R_\alpha{}^\beta \wedge R_\beta{}^\alpha$ bear an intimate
mielke-geometrodynamics_F00678	138	1.000	\left(1 / 8 \pi^{2}\right) \int R_{\alpha}{ }^{\beta} \wedge R_{\beta}{ }^{\alpha}	$\left(1 / 8 \pi^{2}\right) \int R_{\alpha}{ }^{\beta} \wedge R_{\beta}{ }^{\alpha}$	$\int R^{\alpha\beta} \wedge R_{\alpha\beta}^{(\star)}$ and the Pontryagin integral $(1/8\pi^2)\int R_\alpha{}^\beta \wedge R_\beta{}^\alpha$ bear an intimate
mielke-geometrodynamics_F00679	138	1.000	\epsilon	ϵ	to the following argument: For a real constant ϵ with $0 \leq \epsilon \leq 1$, the RC connec-
mielke-geometrodynamics_F00680	138	1.000	0 \leq \epsilon \leq 1	$0 \leq \epsilon \leq 1$	to the following argument: For a real constant ϵ with $0 \leq \epsilon \leq 1$, the RC connec-
mielke-geometrodynamics_F00681	138	0.973	\Gamma_{\alpha \beta}(\epsilon)=\Gamma_{\alpha \beta}^{\{ \}}-\epsilon K_{\alpha \beta}	$\Gamma_{\alpha \beta}(\epsilon)=\Gamma_{\alpha \beta}^{\{ \}}-\epsilon K_{\alpha \beta}$	to $\Gamma_{\alpha\beta}(\epsilon) = \Gamma_{\alpha\beta}^{\{\}} - \epsilon K_{\alpha\beta}$ such that $\Gamma_{\alpha\beta}(0) = \Gamma_{\alpha\beta}^{\{\}}$. Provided a topological quantity
mielke-geometrodynamics_F00682	138	0.973	\Gamma_{\alpha \beta}(0)=\Gamma_{\alpha \beta}^{\{ \}}	$\Gamma_{\alpha \beta}(0)=\Gamma_{\alpha \beta}^{\{ \}}$	to $\Gamma_{\alpha\beta}(\epsilon) = \Gamma_{\alpha\beta}^{\{\}} - \epsilon K_{\alpha\beta}$ such that $\Gamma_{\alpha\beta}(0) = \Gamma_{\alpha\beta}^{\{\}}$. Provided a topological quantity

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mielke-geometrodynamics_F00683	138	1.000	`\Gamma_{\alpha \beta}`	$\Gamma_{\alpha \beta}$	depends on the connection $\Gamma_{\alpha\beta}$, it must be invariant under this continuous defor-
mielke-geometrodynamics_F00684	138	0.679	`\left.T:=e_{\alpha}\right\rfloor T^{\alpha}=d \sigma`	$\left.T:=e_{\alpha}\right\rfloor T^{\alpha}=d \sigma$	torsion $T := e_\alpha \rfloor T^\alpha = d\sigma$ derived from a scalar field via a gradient (GREGORASH &
mielke-geometrodynamics_F00685	138	1.000	`N`	N	manifold, which may enclose the singularity line of the dislocation N times, yields
mielke-geometrodynamics_F00686	139	1.000	`\sigma=`	$\sigma=$	provided the scalar field is not single-valued, as, for instance, in the case of $\sigma =$
mielke-geometrodynamics_F00687	139	0.997	`\arcsin (|x|)`	$\arcsin (\lvert x\rvert)$	$\arcsin(\lvert x\rvert)$.
mielke-geometrodynamics_F00688	139	1.000	`\pi_{3}(S O(4))=`	$\pi_{3}(S O(4))=$	nontrivial winding number will occur as a result of the homotopy group $\pi_3(SO(4)) =$
mielke-geometrodynamics_F00689	139	1.000	`\mathbb{Z} \oplus \mathbb{Z}`	$\mathbb{Z} \oplus \mathbb{Z}$	$\mathbb{Z} \oplus \mathbb{Z}$. On the other hand, the 3D hypersurface admits the intriguing mapping $\ell\underline{\vartheta}^A \rightarrow$
mielke-geometrodynamics_F00690	139	1.000	`\ell \underline{\vartheta}^{A} \rightarrow`	$\ell \underline{\vartheta}^{A} \rightarrow$	$\mathbb{Z} \oplus \mathbb{Z}$. On the other hand, the 3D hypersurface admits the intriguing mapping $\ell\underline{\vartheta}^A \rightarrow$
mielke-geometrodynamics_F00691	139	0.999	`\underline{\Gamma}^{\star A}:=\frac{1}{2} \eta^{A B C} \underline{\Gamma}_{B C}`	$\underline{\Gamma}^{\star A}:=\frac{1}{2} \eta^{A B C} \underline{\Gamma}_{B C}$	$\underline{\Gamma}^{\star A} := \frac{1}{2}\eta^{ABC}\underline{\Gamma}_{BC}$ of the triads to the corresponding three-dual of the $\mathfrak{su}(2)$-valued
mielke-geometrodynamics_F00692	139	0.999	`\mathfrak{s u}(2)`	$\mathfrak{s u}(2)$	$\underline{\Gamma}^{\star A} := \frac{1}{2}\eta^{ABC}\underline{\Gamma}_{BC}$ of the triads to the corresponding three-dual of the $\mathfrak{su}(2)$-valued
mielke-geometrodynamics_F00693	139	0.718	`\underline{C}_{\mathrm{TT}} \rightarrow \underline{C}_{\mathrm{RR}}+\Lambda^{\prime} \underline{\eta}`	$\underline{C}_{\mathrm{TT}} \rightarrow \underline{C}_{\mathrm{RR}}+\Lambda^{\prime} \underline{\eta}$	connection (BAEKLER et al. 1992). Since this implies a mapping $\underline{C}_{\mathrm{TT}} \rightarrow \underline{C}_{\mathrm{RR}} + \Lambda'\underline{\eta}$
mielke-geometrodynamics_F00694	139	1.000	`\mathbb{R} \times W^{3}`	$\mathbb{R} \times W^{3}$	small scale of "quantum gravity," the underlying spacetime manifold $\mathbb{R} \times W^3$ may
mielke-geometrodynamics_F00695	139	1.000	`d j_{5}`	$d j_{5}$	In classical EC theory, the net axial current production dj_5 seems (CHANG &
mielke-geometrodynamics_F00696	139	1.000	`d \mathscr{A} \wedge d \mathscr{A}`	$d \mathscr{A} \wedge d \mathscr{A}$	$d\mathscr{A} \wedge d\mathscr{A}$ involving the axial torsion. Since torsion instantons are characterized
mielke-geometrodynamics_F00697	140	0.685	`\theta_{\mathrm{T}}=2 / \gamma`	$\theta_{\mathrm{T}}=2 / \gamma$	[5]The translational angle $\theta_{\mathrm{T}} = 2/\gamma$ is at times identified (FREIDEL et al. 2005) with the inverse
mielke-geometrodynamics_F00698	140	0.985	`d C_{\text {TT }}`	$d C_{\text {TT }}$	lational Chern–Simons term dC_{TT} were considered earlier by MIELKE (1992).
mielke-geometrodynamics_F00699	141	1.000	`\vartheta_{[\beta} \wedge H_{\alpha]}`	$\vartheta_{[\beta} \wedge H_{\alpha]}$	Note that the angular momentum part $\vartheta_{[\beta} \wedge H_{\alpha]}$ of the gauge fields induced by
mielke-geometrodynamics_F00700	141	1.000	`D H_{\alpha \beta}`	$D H_{\alpha \beta}$	the NY term cancels identically against one part of $DH_{\alpha\beta}$. In the case of Dirac fields
mielke-geometrodynamics_F00701	141	1.000	`\mu_{\alpha}`	μ_{α}	(MIELKE 2004), the *spin–energy potential* μ_α, a two-form, is related to the axial
mielke-geometrodynamics_F00702	141	1.000	`j_{5}=\bar{\psi}^{*} \gamma \gamma_{5} \psi`	$j_{5}=\bar{\psi}^{*} \gamma \gamma_{5} \psi$	current three-form $j_5 = \overline{\psi}^*\gamma\gamma_5\psi$ via
mielke-geometrodynamics_F00703	141	1.000	`D \theta_{\mathrm{L}} \neq 0`	$D \theta_{\mathrm{L}} \neq 0$	that (6.7.6) for $D\theta_{\mathrm{L}} \neq 0$ provides torsion with a dynamical coupling to RC curva-
mielke-geometrodynamics_F00704	141	1.000	`\mu_{\alpha} \wedge \vartheta^{\alpha}=0`	$\mu_{\alpha} \wedge \vartheta^{\alpha}=0$	i.e., $\mu_\alpha \wedge \vartheta^\alpha = 0$, in view of (6.7.8).
mielke-geometrodynamics_F00705	141	0.726	`\left.T=e_{\alpha}\right\rfloor T^{\alpha}`	$\left.T=e_{\alpha}\right\rfloor T^{\alpha}$	where the vector torsion one-form $T = e_\alpha \rfloor T^\alpha$ enters as an intermediate source. Due
mielke-geometrodynamics_F00706	141	0.815	`D D \theta=d d \theta \equiv 0`	$D D \theta=d d \theta \equiv 0$	to the Poincaré lemma $DD\theta = dd\theta \equiv 0$ for a (pseudo) scalar field, Eq. (6.7.10) has
mielke-geometrodynamics_F00707	141	1.000	`\vartheta^{\beta}`	ϑ^{β}	as a first integral. After a contraction with ϑ^β, the dual of the axial torsion one-form
mielke-geometrodynamics_F00708	141	0.996	`\mathscr{A}:={ }^{*}\left(\vartheta^{\alpha} \wedge T_{\alpha}\right)`	$\mathscr{A}:={ }^{*}\left(\vartheta^{\alpha} \wedge T_{\alpha}\right)$	$\mathscr{A} := {}^*(\vartheta^\alpha \wedge T_\alpha)$, i.e., the translational CS term, turns out to be related to the volume
mielke-geometrodynamics_F00709	141	0.999	`\theta_{\mathrm{L}}`	θ_{L}	a coupling to a kinetic term arising from the pseudoscalar field θ_{L} rescaling the

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mielke-geometrodynamics_F00710	141	1.000	`T^{[\gamma} \wedge T^{\mu]} \eta_{\alpha \beta \gamma \mu}=0`	$T^{[\gamma} \wedge T^{\mu]} \eta_{\alpha \beta \gamma \mu}=0$	$T^{[\gamma} \wedge T^{\mu]} \eta_{\alpha\beta\gamma\mu}=0$. Again we end up with a first-order equation for torsion, where,
mielke-geometrodynamics_F00711	142	1.000	`D H`	$D H$	Part of the sources for the Yang–Mills-type divergence terms "DH" are the mate-
mielke-geometrodynamics_F00712	143	1.000	`L_{g}`	L_{g}	aid of the coframe ϑ^{α}, in general gives us back the gauge Lagrangian L_g amended
mielke-geometrodynamics_F00713	143	1.000	`{ }^{(i)} T^{\alpha}`	${ }^{(i)} T^{\alpha}$	Lagrangian, which is at most *quadratic* in the irreducible pieces ${}^{(i)}T^{\alpha}$ and ${}^{(j)}R^{\alpha\beta}$
mielke-geometrodynamics_F00714	143	1.000	`{ }^{(j)} R^{\alpha \beta}`	${ }^{(j)} R^{\alpha \beta}$	Lagrangian, which is at most *quadratic* in the irreducible pieces ${}^{(i)}T^{\alpha}$ and ${}^{(j)}R^{\alpha\beta}$
mielke-geometrodynamics_F00715	143	1.000	`\ell \simeq 10^{-32} \mathrm{~cm}`	$\ell \simeq 10^{-32} \mathrm{~cm}$	The gravitational coupling constant is absorbed in the Planck length $\ell \simeq 10^{-32}$ cm,
mielke-geometrodynamics_F00716	143	1.000	`a_{0}, a_{i}(i=1,2,3), \kappa`	$a_{0}, a_{i}(i=1,2,3), \kappa$	whereas a_0, a_i $(i=1,2,3)$, κ, and b_j $(j=1,\ldots,6)$ are dimensionless coupling
mielke-geometrodynamics_F00717	143	1.000	`b_{j}(j=1, \ldots, 6)`	$b_{j}(j=1, \ldots, 6)$	whereas a_0, a_i $(i=1,2,3)$, κ, and b_j $(j=1,\ldots,6)$ are dimensionless coupling
mielke-geometrodynamics_F00718	144	1.000	`C: Q \rightarrow Q^{\prime}=-Q`	$C: Q \rightarrow Q^{\prime}=-Q$	a charge conjugation $C: Q \rightarrow Q' = -Q$, one obtains the same gravitational field.
mielke-geometrodynamics_F00719	144	1.000	`C`	C	Maxwell equations are invariant under CPT, charge conjugation C has to be accom-
mielke-geometrodynamics_F00720	144	0.963	`T`	T	panied by an additional time reversal T (particles moving backward in time $\stackrel{\wedge}{=}$ antipar-
mielke-geometrodynamics_F00721	144	0.963	`\stackrel{\wedge}{=}`	$\stackrel{\wedge}{=}$	panied by an additional time reversal T (particles moving backward in time $\stackrel{\wedge}{=}$ antipar-
mielke-geometrodynamics_F00722	144	1.000	`a_{0}=0, a_{i}=1`	$a_{0}=0, a_{i}=1$	by the choice $a_0=0, a_i=1$, and $b_j=1$. In order to obtain its complexified version,
mielke-geometrodynamics_F00723	144	1.000	`b_{j}=1`	$b_{j}=1$	by the choice $a_0=0, a_i=1$, and $b_j=1$. In order to obtain its complexified version,
mielke-geometrodynamics_F00724	144	1.000	`\theta_{\mathrm{T}}`	θ_{T}	parameters θ_{T} and θ_{L}:
mielke-geometrodynamics_F00725	145	1.000	`P: \vartheta^{\prime A}=-\vartheta^{A}`	$P: \vartheta^{\prime A}=-\vartheta^{A}$	are invariant under space reflections $P: \vartheta'^{A} = -\vartheta^{A}$, where $A=1,2,3$, due to the
mielke-geometrodynamics_F00726	145	1.000	`A=1,2,3`	$A=1,2,3$	are invariant under space reflections $P: \vartheta'^{A} = -\vartheta^{A}$, where $A=1,2,3$, due to the
mielke-geometrodynamics_F00727	145	1.000	`\eta_{\alpha \beta \gamma \delta}`	$\eta_{\alpha \beta \gamma \delta}$	occurrence of the determinant of the metric in $\eta_{\alpha\beta\gamma\delta}$. In contradistinction to these
mielke-geometrodynamics_F00728	145	1.000	`R_{\alpha}{ }^{\beta} \wedge R_{\beta}{ }^{\alpha}`	$R_{\alpha}{ }^{\beta} \wedge R_{\beta}{ }^{\alpha}$	The term $R_{\alpha}{}^{\beta} \wedge R_{\beta}{}^{\alpha}$ is known to yield, after integration over spacetime, a topo-
mielke-geometrodynamics_F00729	145	1.000	`\langle 0| R^{\alpha \beta}|0\rangle=0`	$\langle 0\| R^{\alpha \beta}\|0\rangle=0$	Even above the Planck scale, we can at most require $\langle 0\|R^{\alpha\beta}\|0\rangle = 0$, but we could
mielke-geometrodynamics_F00730	151	0.938	`\Omega_{\mathrm{C}}`	Ω_{C}	Let Ω_{C} be the 2-form of a generalized curvature tensor (NOMIZU 1972) that is
mielke-geometrodynamics_F00731	151	0.929	`\operatorname{SL}(2, \mathbb{C}) \approx \widetilde{\mathrm{SO}}(3,1)`	$\operatorname{SL}(2, \mathbb{C}) \approx \widetilde{\mathrm{SO}}(3,1)$	[1]The bispinor representation of the covering group $\mathrm{SL}(2,\mathbb{C}) \approx \widetilde{\mathrm{SO}}(3,1)$ is inapt for this purpose,
mielke-geometrodynamics_F00732	151	1.000	`\sigma_{\alpha \beta}=\frac{1}{2}\left[\gamma_{\alpha}, \gamma_{\beta}\right]`	$\sigma_{\alpha \beta}=\frac{1}{2}\left[\gamma_{\alpha}, \gamma_{\beta}\right]$	since its infinitesimal generators $\sigma_{\alpha\beta} = \frac{1}{2}[\gamma_{\alpha}, \gamma_{\beta}]$ are neither self-dual nor anti-self-dual. As a direct
mielke-geometrodynamics_F00733	151	1.000	`\sigma_{\alpha \beta}^{(\star)}=\gamma^{5} \sigma_{\alpha \beta}`	$\sigma_{\alpha \beta}^{(\star)}=\gamma^{5} \sigma_{\alpha \beta}$	sum of the spinor and its conjugate representation, they rather satisfy the relation $\sigma^{(\star)}_{\alpha\beta} = \gamma^5 \sigma_{\alpha\beta}$,
mielke-geometrodynamics_F00734	151	0.829	`\gamma^{5}:=\frac{1}{4!} \varepsilon_{\mathrm{abcd}} \gamma^{\mathrm{a}} \gamma^{\mathrm{b}} \gamma^{ \mathrm{c}} \gamma^{\mathrm{d}}`	$\gamma^{5}:=\frac{1}{4!} \varepsilon_{\mathrm{abcd}} \gamma^{\mathrm{a}} \gamma^{\mathrm{b}} \gamma^{\mathrm{c}} \gamma^{\mathrm{d}}$	whereby the matrix $\gamma^5 := \frac{1}{4!}\varepsilon_{\mathrm{abcd}}\gamma^{\mathrm{a}}\gamma^{\mathrm{b}}\gamma^{\mathrm{c}}\gamma^{\mathrm{d}}$ is, within the Clifford algebra, an axial Lorentz scalar.
mielke-geometrodynamics_F00735	152	1.000	`s=0`	$s=0$	In a four-dimensional Riemannian manifold with the Euclidean signature $s=0$,

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mielke-geometrodynamics_F00736	153	1.000	`R_{\alpha \beta}`	$R_{\alpha\beta}$	where $R_{\alpha\beta}$ is the curvature two-form. In WEYL (1919), in the concluding chapter
mielke-geometrodynamics_F00737	153	0.813	`\Gamma^{\alpha \beta}:=\Gamma_{i}^{\alpha \beta} d x^{i}`	$\Gamma^{\alpha\beta}:=\Gamma_i^{\alpha\beta}dx^i$	$\Gamma^{\alpha\beta} := \Gamma_i^{\alpha\beta} dx^i$. In his purely affine approach, Yang did not consider the canonical
mielke-geometrodynamics_F00738	153	0.977	`\vartheta^{\alpha}=E_{j}{ }^{\alpha} d x^{j}`	$\vartheta^\alpha = E_j{}^\alpha dx^j$	with respect to the coframe $\vartheta^\alpha = E_j{}^\alpha\, dx^j$ dual to the local anholonomic frame e_α.
mielke-geometrodynamics_F00739	153	0.977	`e_{\alpha}`	e_α	with respect to the coframe $\vartheta^\alpha = E_j{}^\alpha\, dx^j$ dual to the local anholonomic frame e_α.
mielke-geometrodynamics_F00740	154	1.000	`G_{i j}`	G_{ij}	which is dual to the usual Einstein tensor G_{ij} in Riemannian spacetime. In exterior
mielke-geometrodynamics_F00741	154	0.494	`\operatorname{Ric}_{\alpha \beta}:=(-1)^{s} *\left(R_{(\alpha}{ }^{\delta} \wedge \eta_{\delta \mid \beta)}\right)`	$\operatorname{Ric}_{\alpha\beta}:=(-1)^s*\left(R_{(\alpha}{}^{\delta}\wedge\eta_{\delta\mid\beta)}\right)$	$\mathrm{Ric}_{\alpha\beta} := (-1)^s\,{}^*(R_{(\alpha}{}^{\delta}\wedge\eta_{\delta\mid\beta)})$.
mielke-geometrodynamics_F00742	154	1.000	`\tau_{\alpha \beta}=\vartheta_{[\alpha} \wedge`	$\tau_{\alpha\beta}=\vartheta_{[\alpha}\wedge$	Macroscopically, the matter coupling[2] to the spin current three-form $\tau_{\alpha\beta} = \vartheta_{[\alpha} \wedge$
mielke-geometrodynamics_F00743	154	0.998	`\mu_{\beta]}`	$\mu_{\beta]}$	$\mu_{\beta]}$ seems to disfavor such models, a criticism which has already raised by FAIRCHILD
mielke-geometrodynamics_F00744	154	0.589	`\Gamma^{\{\beta}{ }^{\alpha \beta} \propto \partial g`	$\Gamma^{\{\beta\alpha\beta}\propto\partial g$	In Riemannian spacetime, with its Levi–Civita connection $\Gamma^{\{\}\alpha\beta} \propto \partial g$, these
mielke-geometrodynamics_F00745	154	1.000	`{ }^{(\star)}`	${}^{(\star)}$	and the Lie dual ${}^{(\star)}$ of the curvature $R_{\alpha\beta}$ in a space(time) of signature s.
mielke-geometrodynamics_F00746	154	1.000	`L_{\text {SKY }}^{(\star)}=2 L_{\text {SKY }}`	$L_{\text{SKY}}^{(\star)}=2L_{\text{SKY}}$	i.e., they lead to $L^{(\star)}_{\mathrm{SKY}} = 2L_{\mathrm{SKY}}$. Both second-order duality conditions satisfy Yang's
mielke-geometrodynamics_F00747	154	1.000	`j_{5}:=\bar{\psi}^{*} \gamma \gamma_{5} \psi`	$j_5:=\bar{\psi}^*\gamma\gamma_5\psi$	current $j_5 := \overline{\psi}\,{}^*\gamma\gamma_5\psi$.
mielke-geometrodynamics_F00748	155	1.000	`D \eta_{\alpha \beta}=0`	$D\eta_{\alpha\beta}=0$	its Lie dual, due to $D\eta_{\alpha\beta}=0$ in a Riemannian spacetime. More on this “vacuum
mielke-geometrodynamics_F00749	155	1.000	`\zeta= \pm 1`	$\zeta=\pm1$	whereby only $\zeta=\pm1$ (and zero) are admitted.
mielke-geometrodynamics_F00750	155	1.000	`\zeta=+1`	$\zeta=+1$	$\zeta=+1$: *self-double-dual curvature*
mielke-geometrodynamics_F00751	155	1.000	`\zeta=-1`	$\zeta=-1$	$\zeta=-1$: *anti-self-double-dual curvature*
mielke-geometrodynamics_F00752	155	0.999	`\mathscr{A}:={ }^{*}\left(\vartheta_{\alpha} \wedge T^{\alpha}\right)`	$\mathscr{A}:={}^*\left(\vartheta_\alpha\wedge T^\alpha\right)$	Riemannian curvature $R^{\{\}}_{\alpha\beta}$ and the axial torsion one-form $\mathscr{A} := {}^*(\vartheta_\alpha\wedge T^\alpha)$ is moti-
mielke-geometrodynamics_F00753	155	0.638	`\left\langle d j_{5}\right\rangle=2 i^{*} m\left\langle \bar{\psi} \gamma_{5} \psi\right\rangle-\left(R_{\alpha \beta}^{\{ \}} \wedge R^{\{j \alpha \beta}+\frac{1}{2} d \mathscr{A} \wedge d \mathscr{A}\right) /`	$\left\langle dj_5\right\rangle=2i^*m\left\langle\bar{\psi}\gamma_5\psi\right\rangle-\left(R_{\alpha\beta}^{\{\}}\wedge R^{\{j\alpha\beta}+\frac{1}{2}d\mathscr{A}\wedge d\mathscr{A}\right)/$	vated by the *axial anomaly* $\langle dj_5\rangle = 2i\,{}^*m\langle\overline{\psi}\gamma_5\psi\rangle - \left(R^{\{\}}_{\alpha\beta}\wedge R^{\{\}\alpha\beta}+\frac12 d\mathscr{A}\wedge d\mathscr{A}\right)/$
mielke-geometrodynamics_F00754	155	1.000	`48 \pi^{2}`	$48\pi^2$	$48\pi^2$ in the coupling to Dirac fields ψ.
mielke-geometrodynamics_F00755	156	1.000	`d^{4} x`	d^4x	arises, where the four-dimensional integration over d^4x is already implied in our
mielke-geometrodynamics_F00756	156	0.996	`{ }^{*} R_{\alpha \beta}=\theta_{\mathrm{L}}^{\star} R_{\alpha \beta}^{(\star)}+\left(\theta_{\mathrm{T}}^{*} \Lambda / 6\right) \eta_{\alpha \beta}`	${}^*R_{\alpha\beta}=\theta_{\mathrm{L}}^\star R_{\alpha\beta}^{(\star)}+\left(\theta_{\mathrm{T}}^*\Lambda/6\right)\eta_{\alpha\beta}$	1994) via a modified duality ${}^*R_{\alpha\beta} = \theta^\star_{\mathrm L}R^{(\star)}_{\alpha\beta} + (\theta^*_{\mathrm T}\Lambda/6)\,\eta_{\alpha\beta}$, which, however, explic-
mielke-geometrodynamics_F00757	156	0.998	`\sigma_{\alpha}:=\Sigma_{\alpha}-D^{\{ \}} \mu_{\alpha}`	$\sigma_\alpha:=\Sigma_\alpha-D^{\{\}}\mu_\alpha$	& MAGGIOLO 2005) symmetrized energy–momentum current $\sigma_\alpha := \Sigma_\alpha - D^{\{\}}\mu_\alpha$.)
mielke-geometrodynamics_F00758	156	0.682	`\left.\left.R:=e_{\alpha}\right\rfloor e_{\beta}\right\rfloor R^{\alpha \beta}`	$\left.\left.R:=e_\alpha\right\rfloor e_\beta\right\rfloor R^{\alpha\beta}$	trace with $R := e_\alpha\rfloor e_\beta\rfloor R^{\alpha\beta}$ as irreducible pieces. Then the double-dual curvature
mielke-geometrodynamics_F00759	157	1.000	`o_{i j}^{\text {Mink }}`	o_{ij}^{Mink}	with o^{Mink}_{ij} denoting the (pseudo)orthogonal Minkowski metric. Earlier, EINSTEIN &
mielke-geometrodynamics_F00760	157	0.529	`D^{\{ \}} G_{\alpha}^{\{ \}} \equiv 0`	$D^{\{\}}G_\alpha^{\{\}}\equiv0$	Due to the contracted Bianchi identity $D^{\{\}}G^{\{\}}_\alpha \equiv 0$ in Riemannian spacetime, the

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mielke-geometrodynamics_F00761	157	0.999	`R=-4 \Lambda`	$R=-4 \Lambda$	curvature scalar is constant, i.e., more precisely, $R=-4\Lambda$. Thus, (7.4.6) is equivalent
mielke-geometrodynamics_F00762	157	1.000	`\Lambda \simeq 1 / \kappa`	$\Lambda \simeq 1 / \kappa$	with cosmological constant $\Lambda \simeq 1/\kappa$. It is crucial to note that the latter carries phys-
mielke-geometrodynamics_F00763	157	1.000	`R=0`	$R=0$	sical Einstein–Fokker background due to $R=0$ in double-dual Thompson spaces.
mielke-geometrodynamics_F00764	157	1.000	`h_{i j}:=g_{i j}-g_{i j}^{\mathrm{dS}}`	$h_{i j}:=g_{i j}-g_{i j}^{\mathrm{dS}}$	On the other hand, the linearized Einstein equations for $h_{ij} := g_{ij} - g_{ij}^{\mathrm{dS}}$ in a de
mielke-geometrodynamics_F00765	157	1.000	`g_{i j}^{\mathrm{dS}}`	$g_{i j}^{\mathrm{dS}}$	Sitter background g_{ij}^{dS} with a positive cosmological constant are
mielke-geometrodynamics_F00766	158	1.000	`m / H`	m / H	de Sitter space if m/H is kept finite in the massless limit. For gravitons, the mass
mielke-geometrodynamics_F00767	158	1.000	`m \leq 7 \times 10^{-32} \mathrm{eV}`	$m \leq 7 \times 10^{-32} \mathrm{eV}$	(GOLDHABER & NIETO 2010) of $m \leq 7 \times 10^{-32}$ eV, but larger than the Hubble
mielke-geometrodynamics_F00768	158	1.000	`H \simeq 10^{-33} \mathrm{eV}`	$H \simeq 10^{-33} \mathrm{eV}$	scale of $H \simeq 10^{-33}$ eV.
mielke-geometrodynamics_F00769	158	0.994	`\Lambda_{\text {obs }}>0`	$\Lambda_{\text {obs }}>0$	is induced via the tiny cosmological constant $\Lambda_{\mathrm{obs}} > 0$ of our expanding universe.
mielke-geometrodynamics_F00770	158	0.994	`1 / g^{2}`	$1 / g^{2}$	constant $1/g^2$ is 'traded" for a dimensional one. This "dimensional transmutation" à
mielke-geometrodynamics_F00771	159	0.662	`\operatorname{SU}(2)`	$\operatorname{SU}(2)$	Originally, Yang and Mills formulated their model as an $SU(2)$ gauge theory with
mielke-geometrodynamics_F00772	159	0.968	`1974)`	1974)	1974) is an $SO(1,3)$ gauge theory, whose covering gauge group, after a Wick rotation
mielke-geometrodynamics_F00773	159	1.000	`\overline{S O}(4) \cong`	$\overline{S O}(4) \cong$	to Euclidean space in the path integral quantization, is locally isomorphic to $\overline{SO}(4) \cong$
mielke-geometrodynamics_F00774	159	1.000	`S U(2) \otimes S U(2)`	$S U(2) \otimes S U(2)$	$SU(2) \otimes SU(2)$. This doubling of the (internal) unitary structure group $SU(2)$ was
mielke-geometrodynamics_F00775	159	0.719	`\kappa \Lambda_{\text {obs }} \simeq 10^{-123}`	$\kappa \Lambda_{\text {obs }} \simeq 10^{-123}$	occur at the tiny *dimensionless scale* of $\kappa\Lambda_{\mathrm{obs}} \simeq 10^{-123}$, as suggested by observa-
mielke-geometrodynamics_F00776	159	0.999	`\Delta L`	ΔL	In perturbative quantum gravity (ALVAREZ 1989), there arise counterterms ΔL of
mielke-geometrodynamics_F00777	159	0.995	`\operatorname{Ric}_{\alpha \beta}:={ }^{*}\left(R_{(\alpha}{ }^{\delta} \wedge \eta_{\delta \mid \beta)}\right)`	$\operatorname{Ric}_{\alpha \beta}:={ }^{*}\left(R_{(\alpha}{ }^{\delta} \wedge \eta_{\delta \mid \beta)}\right)$	version of the zero-form $\mathrm{Ric}_{\alpha\beta} := {}^*(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta\mid\beta)})$.
mielke-geometrodynamics_F00778	160	1.000	`H_{\alpha \beta}`	$H_{\alpha \beta}$	Here the field momenta H_α and $H_{\alpha\beta}$ are understood as arising from a generating
mielke-geometrodynamics_F00779	160	1.000	`L_{\text {eff }}`	$L_{\text {eff }}$	n-form G as part of some effective gauge Lagrangian L_{eff} that includes the countert-
mielke-geometrodynamics_F00780	160	0.637	`n-2`	$n-2$	In our dynamical approach, the $(n-2)$-forms H_α and $H_{\alpha\beta}$ will be gauge field
mielke-geometrodynamics_F00781	160	1.000	`P:=\mathbb{R}^{4} \otimes`	$P:=\mathbb{R}^{4} \otimes$	tively. Due to the *semidirect product* structure of the Poincaré group $P := \mathbb{R}^4 \rtimes$
mielke-geometrodynamics_F00782	160	0.945	`H^{\alpha \beta} \rightarrow \vartheta^{\alpha}`	$H^{\alpha \beta} \rightarrow \vartheta^{\alpha}$	$SO(1,3)$, the gauge field momenta contribute to the gauge potentials via $H^{\alpha\beta} \rightarrow \vartheta^\alpha$
mielke-geometrodynamics_F00783	160	1.000	`H^{\alpha} \rightarrow \Gamma^{\alpha \beta}`	$H^{\alpha} \rightarrow \Gamma^{\alpha \beta}$	and $H^\alpha \rightarrow \Gamma^{\alpha\beta}$ in the FR (7.7.2, 7.7.3) just in an *intertwined* manner.
mielke-geometrodynamics_F00784	160	0.991	`{ }^{*} D H_{\alpha}{ }^{\beta}`	${ }^{*} D H_{\alpha}{ }^{\beta}$	similarly as for Yang–Mills fields, a term proportional to ${}^*DH_\alpha{}^\beta$. However, "on
mielke-geometrodynamics_F00785	160	0.388	`{ }^{*}\left(H^{\beta} \wedge \vartheta_{\alpha}\right) \equiv e_{\alpha}`	${ }^{*}\left(H^{\beta} \wedge \vartheta_{\alpha}\right) \equiv e_{\alpha}$	${}^*(H^\beta \wedge \vartheta_\alpha) \equiv e_\alpha \rfloor {}^*H^\beta$ due to an algebraic identity. In the FR (7.7.2) of the coframe,
mielke-geometrodynamics_F00786	160	0.388	`H^{\beta}`	H^{β}	${}^*(H^\beta \wedge \vartheta_\alpha) \equiv e_\alpha \rfloor {}^*H^\beta$ due to an algebraic identity. In the FR (7.7.2) of the coframe,

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mielke-geometrodynamics_F00787	160	0.644	`{ }^{*} D H^{\alpha} \cong{ }^{*} E^{\alpha}`	${ }^{*} D H^{\alpha} \cong{ }^{*} E^{\alpha}$	the same situation arises, with the modification that the "on shell" term ${}^*DH^\alpha \cong {}^*E^\alpha$
mielke-geometrodynamics_F00788	160	1.000	`\widetilde{G}`	$\widetilde{G}$	order generation functional $\widetilde{G}$. When coupling to matter, FRs have to be handled with
mielke-geometrodynamics_F00789	161	1.000	`\widetilde{H}_{\alpha \beta}:=-\partial \widetilde{L} / \partial R^{\alpha \beta}`	$\widetilde{H}_{\alpha \beta}:=-\partial \widetilde{L} / \partial R^{\alpha \beta}$	momenta $\widetilde{H}_{\alpha\beta} := -\partial\widetilde{L}/\partial R^{\alpha\beta}$ can be expanded as $\widetilde{H}_{\alpha\beta} = \stackrel{(2)}{H}_{\alpha\beta} + \stackrel{(4)}{H}_{\alpha\beta} + \cdots$. (For the
mielke-geometrodynamics_F00790	161	1.000	`\widetilde{H}_{\alpha \beta}=\stackrel{(2)}{H}_{\alpha \beta}+\stackrel{(4)}{H}_{\alpha \beta}+\cdots`	$\widetilde{H}_{\alpha \beta}=\stackrel{(2)}{H}_{\alpha \beta}+\stackrel{(4)}{H}_{\alpha \beta}+\cdots$	momenta $\widetilde{H}_{\alpha\beta} := -\partial\widetilde{L}/\partial R^{\alpha\beta}$ can be expanded as $\widetilde{H}_{\alpha\beta} = \stackrel{(2)}{H}_{\alpha\beta} + \stackrel{(4)}{H}_{\alpha\beta} + \cdots$. (For the
mielke-geometrodynamics_F00791	161	1.000	`\widetilde{H}_{\alpha}:=-\partial \widetilde{L} / \partial T^{\alpha}`	$\widetilde{H}_{\alpha}:=-\partial \widetilde{L} / \partial T^{\alpha}$	time being, the field momentum $\widetilde{H}_\alpha := -\partial\widetilde{L}/\partial T^\alpha$ conjugate to the torsion $T^\alpha :=$
mielke-geometrodynamics_F00792	161	1.000	`T^{\alpha}:=`	$T^{\alpha}:=$	time being, the field momentum $\widetilde{H}_\alpha := -\partial\widetilde{L}/\partial T^\alpha$ conjugate to the torsion $T^\alpha :=$
mielke-geometrodynamics_F00793	161	1.000	`D \vartheta^{\alpha}`	$D \vartheta^{\alpha}$	$D\vartheta^\alpha$ will be set to zero.)
mielke-geometrodynamics_F00794	161	1.000	`\widetilde{H}_{\alpha \beta \mu \nu}`	$\widetilde{H}_{\alpha \beta \mu \nu}$	For a *fixed point* of the transformation (7.7.9), the Hessian $\widetilde{H}_{\alpha\beta\mu\nu}$ obviously has
mielke-geometrodynamics_F00795	161	1.000	`\partial \widetilde{H}_{\mu \nu} / \partial R^{\alpha \beta}=0`	$\partial \widetilde{H}_{\mu \nu} / \partial R^{\alpha \beta}=0$	to vanish. This condition, i.e., $\partial\widetilde{H}_{\mu\nu}/\partial R^{\alpha\beta} = 0$, can be readily solved.
mielke-geometrodynamics_F00796	161	1.000	`\theta_{\mathrm{T}} R^{\alpha \beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta}`	$\theta_{\mathrm{T}} R^{\alpha \beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta}$	If parity-violating terms such as $\theta_{\mathrm{T}} R^{\alpha\beta} \wedge \vartheta_\alpha \wedge \vartheta_\beta$ arising from the Nieh–Yan
mielke-geometrodynamics_F00797	162	1.000	`\widetilde{L}`	$\widetilde{L}$	Lagrangian $\widetilde{L}$.
mielke-geometrodynamics_F00798	162	1.000	`\eta_{\alpha \beta}`	$\eta_{\alpha \beta}$	Accordingly, $\eta_{\alpha\beta}$ and $\widetilde{H}_{\alpha\beta}$ interchange their roles as generalized coordinates and
mielke-geometrodynamics_F00799	162	1.000	`\widetilde{H}_{\alpha \beta}`	$\widetilde{H}_{\alpha \beta}$	Accordingly, $\eta_{\alpha\beta}$ and $\widetilde{H}_{\alpha\beta}$ interchange their roles as generalized coordinates and
mielke-geometrodynamics_F00800	162	1.000	`\widetilde{L}=L_{\mathrm{EC}}`	$\widetilde{L}=L_{\mathrm{EC}}$	momenta, respectively. If we had started from $\widetilde{L} = L_{\mathrm{EC}}$, then we would be led back
mielke-geometrodynamics_F00801	162	1.000	`L=L_{\mathrm{EC}}`	$L=L_{\mathrm{EC}}$	to $L = L_{\mathrm{EC}}$ for the choice $\theta^*_{\mathrm{T}} = 1$ and $\theta_{\mathrm{T}} = 0$. In the case $\theta^*_{\mathrm{T}} = 1$ and $\theta_{\mathrm{T}} = i$, this
mielke-geometrodynamics_F00802	162	1.000	`\theta_{\mathrm{T}}^{*}=1`	$\theta_{\mathrm{T}}^{*}=1$	to $L = L_{\mathrm{EC}}$ for the choice $\theta^*_{\mathrm{T}} = 1$ and $\theta_{\mathrm{T}} = 0$. In the case $\theta^*_{\mathrm{T}} = 1$ and $\theta_{\mathrm{T}} = i$, this
mielke-geometrodynamics_F00803	162	1.000	`\theta_{\mathrm{T}}=0`	$\theta_{\mathrm{T}}=0$	to $L = L_{\mathrm{EC}}$ for the choice $\theta^*_{\mathrm{T}} = 1$ and $\theta_{\mathrm{T}} = 0$. In the case $\theta^*_{\mathrm{T}} = 1$ and $\theta_{\mathrm{T}} = i$, this
mielke-geometrodynamics_F00804	162	1.000	`\theta_{\mathrm{T}}=i`	$\theta_{\mathrm{T}}=i$	to $L = L_{\mathrm{EC}}$ for the choice $\theta^*_{\mathrm{T}} = 1$ and $\theta_{\mathrm{T}} = 0$. In the case $\theta^*_{\mathrm{T}} = 1$ and $\theta_{\mathrm{T}} = i$, this
mielke-geometrodynamics_F00805	162	0.436	`\left.\Gamma_{\alpha}{ }^{\beta} \rightarrow \stackrel{(\pm)}{\Gamma}_{\alpha}{ }^{\beta}:=\Gamma_{\alpha}{ }^{\beta} \mp\left(i \ell^{2} / 2\right) e_{\alpha}\right\rfloor^{(\mp)}{ }^{\beta}`	$\left.\Gamma_{\alpha}{ }^{\beta} \rightarrow \stackrel{(\pm)}{\Gamma}_{\alpha}{ }^{\beta}:=\Gamma_{\alpha}{ }^{\beta} \mp\left(i \ell^{2} / 2\right) e_{\alpha}\right\rfloor^{(\mp)}{ }^{\beta}$	ition $\Gamma_\alpha{}^\beta \rightarrow \stackrel{(\pm)}{\Gamma}_\alpha{}^\beta := \Gamma_\alpha{}^\beta \mp (i\ell^2/2)\, e_\alpha \rfloor \stackrel{(\mp)}{H}{}^\beta$, induced by the translational Chern–
mielke-geometrodynamics_F00806	162	0.999	`i d C_{\mathrm{TT}}`	$i d C_{\mathrm{TT}}$	Simons term idC_{TT}. The resulting Sen-type connection still couples *minimally* to the
mielke-geometrodynamics_F00807	162	0.999	`{ }^{* *} \Phi=(-1)^{p(4-p)+s} \Phi`	${ }^{* *} \Phi=(-1)^{p(4-p)+s} \Phi$	holds. Applied to p-forms, it is almost involutive, i.e., ${}^{**}\Phi = (-1)^{p(4-p)+s}\Phi$. For
mielke-geometrodynamics_F00808	162	0.997	`s:=\operatorname{sig}=1`	$s:=\operatorname{sig}=1$	spacetimes in which $s := \operatorname{sig} = 1$ holds, it induces an *almost complex structure*, and
mielke-geometrodynamics_F00809	163	1.000	`\left\{\eta, \eta_{\alpha}, \eta_{\alpha \beta}, \eta_{\alpha \beta \gamma} \eta_{\alpha \beta \gamma \delta}\right\}`	$\left\{\eta, \eta_{\alpha}, \eta_{\alpha \beta}, \eta_{\alpha \beta \gamma} \eta_{\alpha \beta \gamma \delta}\right\}$	makes it possible to generate the so-called η, or dual, basis $\{\eta,\ \eta_\alpha,\ \eta_{\alpha\beta},\ \eta_{\alpha\beta\gamma}\ \eta_{\alpha\beta\gamma\delta}\}$
mielke-geometrodynamics_F00810	163	0.925	`\left.\eta_{\alpha}:=e_{\alpha}\right\rfloor \eta={ }^{*} \vartheta_{\alpha}, \eta_{\alpha \beta}:=`	$\left.\eta_{\alpha}:=e_{\alpha}\right\rfloor \eta={ }^{*} \vartheta_{\alpha}, \eta_{\alpha \beta}:=$	of exterior forms via consecutive interior products: $\eta_\alpha := e_\alpha \rfloor \eta = {}^*\vartheta_\alpha,\ \eta_{\alpha\beta} :=$

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mielke-geometrodynamics_F00811	163	0.920	`\left.\left.\left.e_{\beta}\right\rfloor \eta_{\alpha}=\eta_{\alpha \beta \gamma} \vartheta^{\gamma}=e_{\beta}\right\rfloor e_{\alpha}\right\rfloor \eta={ }^{*}\left(\vartheta_{\alpha} \wedge \vartheta_{\beta}\right)=\frac{1}{2} \eta_{\alpha \beta \gamma \delta} \vartheta^{\gamma} \wedge \vartheta^{\delta}`	$e_{\beta}\rfloor \eta_{\alpha}=\eta_{\alpha \beta \gamma} \vartheta^{\gamma}=e_{\beta}\rfloor e_{\alpha}\rfloor \eta={ }^{*}\left(\vartheta_{\alpha} \wedge \vartheta_{\beta}\right)=\frac{1}{2} \eta_{\alpha \beta \gamma \delta} \vartheta^{\gamma} \wedge \vartheta^{\delta}$	$e_\beta\rfloor\eta_\alpha = \eta_{\alpha\beta\gamma}\,\vartheta^\gamma = e_\beta\rfloor e_\alpha\rfloor\eta = {}^*(\vartheta_\alpha\wedge\vartheta_\beta) = \frac{1}{2}\eta_{\alpha\beta\gamma\delta}\,\vartheta^\gamma\wedge\vartheta^\delta$, and $\eta_{\alpha\beta\gamma} := e_\gamma$
mielke-geometrodynamics_F00812	163	0.920	`\eta_{\alpha \beta \gamma}:=e_{\gamma}`	$\eta_{\alpha \beta \gamma}:=e_{\gamma}$	$e_\beta\rfloor\eta_\alpha = \eta_{\alpha\beta\gamma}\,\vartheta^\gamma = e_\beta\rfloor e_\alpha\rfloor\eta = {}^*(\vartheta_\alpha\wedge\vartheta_\beta) = \frac{1}{2}\eta_{\alpha\beta\gamma\delta}\,\vartheta^\gamma\wedge\vartheta^\delta$, and $\eta_{\alpha\beta\gamma} := e_\gamma$
mielke-geometrodynamics_F00813	163	0.862	`\rfloor \eta_{\alpha \beta}={ }^{*}\left(\vartheta_{\alpha} \wedge \vartheta_{\beta} \wedge \vartheta_{\gamma}\right)`	$\rfloor \eta_{\alpha \beta}={ }^{*}\left(\vartheta_{\alpha} \wedge \vartheta_{\beta} \wedge \vartheta_{\gamma}\right)$	$\rfloor\eta_{\alpha\beta} = {}^*(\vartheta_\alpha\wedge\vartheta_\beta\wedge\vartheta_\gamma)$. Anholonomic indices are lowered by $o_{\alpha\beta} = e^i{}_\alpha\, e^j{}_\beta\, g_{ij}$,
mielke-geometrodynamics_F00814	163	0.862	`o_{\alpha \beta}=e^{i}{ }_{\alpha} e^{j}{ }_{\beta} g_{i j}`	$o_{\alpha \beta}=e^{i}{ }_{\alpha} e^{j}{ }_{\beta} g_{i j}$	$\rfloor\eta_{\alpha\beta} = {}^*(\vartheta_\alpha\wedge\vartheta_\beta\wedge\vartheta_\gamma)$. Anholonomic indices are lowered by $o_{\alpha\beta} = e^i{}_\alpha\, e^j{}_\beta\, g_{ij}$,
mielke-geometrodynamics_F00815	163	1.000	`o_{\alpha \beta}`	$o_{\alpha \beta}$	where $o_{\alpha\beta}$ denotes the Minkowski metric.
mielke-geometrodynamics_F00816	163	1.000	`\Phi=\frac{1}{p!} \Phi_{i_{1} \cdots i_{p}} d x^{i_{1}} \wedge \cdots \wedge d x^{i_{p}}`	$\Phi=\frac{1}{p!} \Phi_{i_{1} \cdots i_{p}} d x^{i_{1}} \wedge \cdots \wedge d x^{i_{p}}$	dimensional manifold, the Hodge dual of a p-form $\Phi = \frac{1}{p!}\Phi_{i_1\cdots i_p}\, dx^{i_1}\wedge\cdots\wedge dx^{i_p}$
mielke-geometrodynamics_F00817	163	0.769	`{ }^{* *} \Phi=(-1)^{p(4-p)+\operatorname{sig}} \Phi`	${ }^{* *} \Phi=(-1)^{p(4-p)+\operatorname{sig}} \Phi$	so that it is almost involutive, i.e., ${}^{**}\Phi = (-1)^{p(4-p)+\mathrm{sig}}\Phi$.
mielke-geometrodynamics_F00818	163	0.998	`\left\{\mathbf{1}_{4}, \gamma_{\alpha}, \sigma_{\alpha \beta}, \gamma_{5}, \gamma_{5} \gamma_{\alpha}\right\}`	$\left\{\mathbf{1}_{4}, \gamma_{\alpha}, \sigma_{\alpha \beta}, \gamma_{5}, \gamma_{5} \gamma_{\alpha}\right\}$	In the familiar Pauli representation, the 16 matrices $\{\mathbf{1}_4, \gamma_\alpha, \sigma_{\alpha\beta}, \gamma_5, \gamma_5\gamma_\alpha\}$, where
mielke-geometrodynamics_F00819	163	0.994	`\sigma_{\alpha \beta}:=\frac{i}{2}\left(\gamma_{\alpha} \gamma_{\beta}-\gamma_{\beta} \gamma_{\alpha}\right)`	$\sigma_{\alpha \beta}:=\frac{i}{2}\left(\gamma_{\alpha} \gamma_{\beta}-\gamma_{\beta} \gamma_{\alpha}\right)$	$\sigma_{\alpha\beta} := \frac{i}{2}(\gamma_\alpha\gamma_\beta - \gamma_\beta\gamma_\alpha)$ are the Lorentz generators, and $\gamma_5 = -i\,\gamma_{\hat 0}\,\gamma_{\hat 1}\,\gamma_{\hat 2}\,\gamma_{\hat 3}$ with
mielke-geometrodynamics_F00820	163	0.994	`\gamma_{5}=-i \gamma_{\hat{0}} \gamma_{\hat{1}} \gamma_{\hat{2}} \gamma_{\hat{3}}`	$\gamma_{5}=-i \gamma_{\hat{0}} \gamma_{\hat{1}} \gamma_{\hat{2}} \gamma_{\hat{3}}$	$\sigma_{\alpha\beta} := \frac{i}{2}(\gamma_\alpha\gamma_\beta - \gamma_\beta\gamma_\alpha)$ are the Lorentz generators, and $\gamma_5 = -i\,\gamma_{\hat 0}\,\gamma_{\hat 1}\,\gamma_{\hat 2}\,\gamma_{\hat 3}$ with
mielke-geometrodynamics_F00821	163	0.736	`\gamma_{5}^{2}=(-1)^{\mathrm{sig}+1} \mathbf{1}_{4}`	$\gamma_{5}^{2}=(-1)^{\mathrm{sig}+1} \mathbf{1}_{4}$	$\gamma_5^2 = (-1)^{\mathrm{sig}+1}\mathbf{1}_4$ constitute a basis of the *Clifford algebra* in four dimensions with
mielke-geometrodynamics_F00822	163	1.000	`\operatorname{Tr}\left(\gamma_{\alpha} \gamma_{\beta}\right)=4 o_{\alpha \beta}`	$\operatorname{Tr}\left(\gamma_{\alpha} \gamma_{\beta}\right)=4 o_{\alpha \beta}$	They are normalized by the traces $\mathrm{Tr}(\gamma_\alpha\,\gamma_\beta) = 4\,o_{\alpha\beta}$ and $\mathrm{Tr}(\sigma_{\alpha\beta}\,\sigma^{\gamma\delta}) = 8\,\delta^\gamma_{[\alpha}\,\delta^\delta_{\beta]}$,
mielke-geometrodynamics_F00823	163	1.000	`\operatorname{Tr}\left(\sigma_{\alpha \beta} \sigma^{\gamma \delta}\right)=8 \delta_{[\alpha}^{\gamma} \delta_{\beta]}^{\delta}`	$\operatorname{Tr}\left(\sigma_{\alpha \beta} \sigma^{\gamma \delta}\right)=8 \delta_{[\alpha}^{\gamma} \delta_{\beta]}^{\delta}$	They are normalized by the traces $\mathrm{Tr}(\gamma_\alpha\,\gamma_\beta) = 4\,o_{\alpha\beta}$ and $\mathrm{Tr}(\sigma_{\alpha\beta}\,\sigma^{\gamma\delta}) = 8\,\delta^\gamma_{[\alpha}\,\delta^\delta_{\beta]}$,
mielke-geometrodynamics_F00824	163	1.000	`[\alpha \beta]=\frac{1}{2}(\alpha \beta-\beta \alpha)`	$[\alpha \beta]=\frac{1}{2}(\alpha \beta-\beta \alpha)$	where $[\alpha\,\beta] = \frac{1}{2}(\alpha\beta - \beta\alpha)$ denotes the antisymmetrization of indices. For the unit
mielke-geometrodynamics_F00825	164	0.988	`\operatorname{Lie}`	Lie	whereas for the *Lie* (or right) dual $\sigma^{(\star)}_{\alpha\beta} := \frac{1}{2}\sigma^{\gamma\delta}\,\eta_{\alpha\beta\gamma\delta} = i\gamma_5\,\sigma_{\alpha\beta}$, we obtain the asso-
mielke-geometrodynamics_F00826	164	0.988	`\sigma_{\alpha \beta}^{(\star)}:=\frac{1}{2} \sigma^{\gamma \delta} \eta_{\alpha \beta \gamma \delta}=i \gamma_{5} \sigma_{\alpha \beta}`	$\sigma_{\alpha \beta}^{(\star)}:=\frac{1}{2} \sigma^{\gamma \delta} \eta_{\alpha \beta \gamma \delta}=i \gamma_{5} \sigma_{\alpha \beta}$	whereas for the *Lie* (or right) dual $\sigma^{(\star)}_{\alpha\beta} := \frac{1}{2}\sigma^{\gamma\delta}\,\eta_{\alpha\beta\gamma\delta} = i\gamma_5\,\sigma_{\alpha\beta}$, we obtain the asso-
mielke-geometrodynamics_F00827	164	0.998	`\sigma^{(\star)}`	$\sigma^{(\star)}$	$\sigma^{(\star)}$ solely by multiplication by γ_5, which here is regarded as just an anticommuting
mielke-geometrodynamics_F00828	164	0.998	`\gamma_{5}`	γ_{5}	$\sigma^{(\star)}$ solely by multiplication by γ_5, which here is regarded as just an anticommuting
mielke-geometrodynamics_F00829	164	0.995	`12 i^{*} \gamma`	$12 i^{*} \gamma$	relates this to $12i\,{}^*\gamma$, the Hodge dual[3] for the basis of Cliffords has been recovered.
mielke-geometrodynamics_F00830	164	1.000	`P_{ \pm}=\frac{1}{2}\left(1 \pm \gamma_{5}\right)`	$P_{ \pm}=\frac{1}{2}\left(1 \pm \gamma_{5}\right)$	representation involves explicitly the *chirality projector* $P_\pm = \frac{1}{2}(1\pm\gamma_5)$, obeying
mielke-geometrodynamics_F00831	164	1.000	`P_{ \pm} P_{ \pm}=P_{ \pm}`	$P_{ \pm} P_{ \pm}=P_{ \pm}$	$P_\pm\,P_\pm = P_\pm$.
mielke-geometrodynamics_F00832	164	0.981	`\gamma \rightarrow \widetilde{\gamma}=e^{\varphi} \gamma`	$\gamma \rightarrow \widetilde{\gamma}=e^{\varphi} \gamma$	(ATIYAH et al. 1978) under the Weyl rescaling $\gamma\rightarrow\widetilde{\gamma} = e^\varphi\gamma$, where φ can be viewed
mielke-geometrodynamics_F00833	164	1.000	`{ }^{\#}`	${ }^{\#}$	${}^\#$ on *arbitrary* two-forms allows us to *reconstruct* a spacetime metric h, which is,

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mielke-geometrodynamics_F00834	164	■ 1.000	`h`	h	$^{\#}$ on *arbitrary* two-forms allows us to *reconstruct* a spacetime metric h, which is,
mielke-geometrodynamics_F00835	164	■ 0.524	`{ }^{* *}=-1`	${}^{**}=-1$	[3]For Minkowski signature (sign = 1), the Hodge dual satisfies ${}^{**}=-1$, and therefore, i^* is used
mielke-geometrodynamics_F00836	164	■ 0.524	`i^{*}`	i^{*}	[3]For Minkowski signature (sign = 1), the Hodge dual satisfies ${}^{**}=-1$, and therefore, i^* is used
mielke-geometrodynamics_F00837	165	■ 1.000	`\Gamma:=\frac{i}{4} \Gamma^{\alpha \beta} \sigma_{\alpha \beta}`	$\Gamma:=\frac{i}{4} \Gamma^{\alpha \beta} \sigma_{\alpha \beta}$	of gravity. In terms of the Clifford-algebra-valued *connection* $\Gamma := \frac{i}{4}\Gamma^{\alpha\beta}\sigma_{\alpha\beta}$, the
mielke-geometrodynamics_F00838	165	■ 1.000	`\overline{S O}_{\circ}(1,3) \cong S L(2, C)`	$\overline{S O}_{\circ}(1,3) \cong S L(2, C)$	$\overline{SO}_\circ(1,3) \cong SL(2,C)$-covariant exterior derivative
mielke-geometrodynamics_F00839	165	■ 0.975	`[\Psi, \Phi]:=\Psi \wedge \Phi-(-1)^{p q}`	$[\Psi, \Phi]:=\Psi \wedge \Phi-(-1)^{p q}$	of p-forms employs the algebra-valued *form commutator* $[\Psi,\Phi]:=\Psi\wedge\Phi-(-1)^{pq}$
mielke-geometrodynamics_F00840	165	■ 0.755	`\Phi \wedge \Psi`	$\Phi \wedge \Psi$	$\Phi \wedge \Psi$.
mielke-geometrodynamics_F00841	165	■ 1.000	`K:=\frac{i}{4} K^{\alpha \beta} \sigma_{\alpha \beta}`	$K:=\frac{i}{4} K^{\alpha \beta} \sigma_{\alpha \beta}$	involves $K := \frac{i}{4}K^{\alpha\beta}\sigma_{\alpha\beta}$, the contortion one-form.
mielke-geometrodynamics_F00842	166	■ 1.000	`{ }^{(2)} \Theta:=`	${}^{(2)} \Theta:=$	The torsion two-form can be irreducibly decomposed into the trace part ${}^{(2)}\Theta :=$
mielke-geometrodynamics_F00843	166	■ 1.000	`\frac{1}{3} \gamma \wedge T`	$\frac{1}{3} \gamma \wedge T$	$\frac{1}{3}\gamma\wedge T$, the axial torsion ${}^{(3)}\Theta := -\frac{1}{3}{}^*(\gamma\wedge\mathscr{A})$, and the tensor torsion ${}^{(1)}\Theta :=$
mielke-geometrodynamics_F00844	166	■ 1.000	`{ }^{(3)} \Theta:=-\frac{1}{3} *(\gamma \wedge \mathscr{A})`	${}^{(3)} \Theta:=-\frac{1}{3} *(\gamma \wedge \mathscr{A})$	$\frac{1}{3}\gamma\wedge T$, the axial torsion ${}^{(3)}\Theta := -\frac{1}{3}{}^*(\gamma\wedge\mathscr{A})$, and the tensor torsion ${}^{(1)}\Theta :=$
mielke-geometrodynamics_F00845	166	■ 1.000	`{ }^{(1)} \Theta:=`	${}^{(1)} \Theta:=$	$\frac{1}{3}\gamma\wedge T$, the axial torsion ${}^{(3)}\Theta := -\frac{1}{3}{}^*(\gamma\wedge\mathscr{A})$, and the tensor torsion ${}^{(1)}\Theta :=$
mielke-geometrodynamics_F00846	166	■ 1.000	`\Theta-{ }^{(2)} \Theta-{ }^{(3)} \Theta`	$\Theta-{}^{(2)} \Theta-{}^{(3)} \Theta$	$\Theta - {}^{(2)}\Theta - {}^{(3)}\Theta$, where the one-forms of the trace and axial vector torsion, respec-
mielke-geometrodynamics_F00847	166	■ 0.984	`\widetilde{\mathscr{A}}=\mathscr{A}`	$\widetilde{\mathscr{A}}=\mathscr{A}$	Under a Weyl rescaling, the axial torsion remains invariant, i.e., $\widetilde{\mathscr{A}}=\mathscr{A}$. Both
mielke-geometrodynamics_F00848	167	■ 1.000	`D \Omega^{g(\star)} \equiv 0`	$D \Omega^{g(\star)} \equiv 0$	i.e., $D\Omega^{g(\star)}\equiv 0$, provided that $D\eta_{\alpha\beta\gamma\delta}=0$ holds as in spaces with vanishing Weyl
mielke-geometrodynamics_F00849	167	■ 1.000	`D \eta_{\alpha \beta \gamma \delta}=0`	$D \eta_{\alpha \beta \gamma \delta}=0$	i.e., $D\Omega^{g(\star)}\equiv 0$, provided that $D\eta_{\alpha\beta\gamma\delta}=0$ holds as in spaces with vanishing Weyl
mielke-geometrodynamics_F00850	167	■ 0.995	`\mathscr{A}:={ }^{*}\left(\vartheta_{\alpha} \wedge T^{\alpha}\right)=\mathscr{A}_{i} d x^{i}`	$\mathscr{A}:={}^{*}\left(\vartheta_{\alpha} \wedge T^{\alpha}\right)=\mathscr{A}_{i} d x^{i}$	where $\mathscr{A} := {}^*(\vartheta_\alpha\wedge T^\alpha) = \mathscr{A}_i dx^i$ is the *axial torsion* one-form. The translational
mielke-geometrodynamics_F00851	168	■ 0.471	`R^{\{ \}}:=(-1)^{\text {sig }+1 ~ *}\left(R^{\{j\} \beta} \wedge \eta_{\beta \alpha}\right)`	$R^{\{ \}}:=(-1)^{\text {sig }+1 ~ *}\left(R^{\{j\} \beta} \wedge \eta_{\beta \alpha}\right)$	vature scalar $R^{\{\}} := (-1)^{\text{sig}+1}{}^*(R^{\{\}\alpha\beta}\wedge\eta_{\beta\alpha})$ and the axial torsion piece $d\mathscr{A}\wedge d\mathscr{A}$
mielke-geometrodynamics_F00852	168	■ 0.995	`L_{\text {SKY }}`	$L_{\text {SKY }}$	Lagrangian L_{SKY} as well as a Ricci-squared term and a scalar-curvature-squared
mielke-geometrodynamics_F00853	168	■ 0.827	`\operatorname{Ric}_{\alpha \beta}:=(-1)^{\text {sig } *}\left(R_{(\alpha}{ }^{\delta} \wedge \eta_{\delta \mid \beta)}\right)`	$\operatorname{Ric}_{\alpha \beta}:=(-1)^{\text {sig } *}\left(R_{(\alpha}{}^{\delta} \wedge \eta_{\delta \mid \beta)}\right)$	$\mathrm{Ric}_{\alpha\beta} := (-1)^{\text{sig}}{}^*(R_{(\alpha}{}^{\delta}\wedge\eta_{\delta\mid\beta)})$, is also known as the Gauss–Bonnet term.
mielke-geometrodynamics_F00854	169	■ 1.000	`G_{\alpha}:=R^{\beta \gamma} \wedge \eta_{\alpha \beta \gamma} / 2`	$G_{\alpha}:=R^{\beta \gamma} \wedge \eta_{\alpha \beta \gamma} / 2$	$G_\alpha := R^{\beta\gamma}\wedge\eta_{\alpha\beta\gamma}/2$ and the curvature scalar R. The latter is the zero-form
mielke-geometrodynamics_F00855	169	■ 0.894	`G^{\{ \}}=G_{\alpha}^{\{ \} \beta} \eta_{\beta} \gamma^{\alpha}`	$G^{\{ \}}=G_{\alpha}^{\{ \} \beta} \eta_{\beta} \gamma^{\alpha}$	decomposes into the Einstein three-form $G^{\{\}} = G_\alpha^{\{\}\beta}\eta_\beta\gamma^\alpha$ with respect to the
mielke-geometrodynamics_F00856	170	■ 0.983	`2+1`	$2+1$	Feynman RP (1981) The qualitative behavior of Yang-Mills theory in 2+1 dimensions. Nuclear
mielke-geometrodynamics_F00857	171	■ 0.998	`\mathrm{m} \rightarrow 0`	$\mathrm{m} \rightarrow 0$	Kogan II, Mouslopoulos S, Papazoglou A (2001) The m→0 limit for massive graviton in dS_4 and
mielke-geometrodynamics_F00858	171	■ 0.998	`\mathrm{dS}_{4}`	dS_{4}	Kogan II, Mouslopoulos S, Papazoglou A (2001) The m→0 limit for massive graviton in dS_4 and
mielke-geometrodynamics_F00859	171	■ 0.999	`\mathrm{AdS}_{4}`	AdS_{4}	AdS_4. How to circumvent the van Dam-Veltman-Zakharov discontinuity. Phys Lett B 503(1):173–

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mielke-geometrodynamics_F00860	171	0.878	`5, \mathbb{R}`	$5, \mathbb{R}$	Mielke EW (2012) Einstein-Weyl gravity from a topological SL(5, $\mathbb{R}$) gauge invariant action. Adv
mielke-geometrodynamics_F00861	172	1.000	`R+R^{2}+Q^{2}`	$R+R^{2}+Q^{2}$	Zhytnikov V (1994) Wavelike exact solutions of $R+R^2+Q^2$ gravity. J Math Phys 35(11):6001–
mielke-geometrodynamics_F00862	173	0.999	`(2014,2006)`	$(2014,2006)$	"with flying colors"; see WILL (2014, 2006) for recent reviews. However, Einstein's
mielke-geometrodynamics_F00863	173	0.999	`\lambda_{\mathrm{DE}}=\left(\hbar c / \rho_{\mathrm{DE}}\right)^{1 / 4} \simeq 85 \mu \mathrm{~m}`	$\lambda_{\mathrm{DE}}=\left(\hbar c / \rho_{\mathrm{DE}}\right)^{1 / 4} \simeq 85 \mu \mathrm{~m}$	cal dark energy (DE) scale of $\lambda_{\mathrm{DE}}=(\hbar c/\rho_{\mathrm{DE}})^{1/4} \simeq 85$ µm. Thus this "window" of
mielke-geometrodynamics_F00864	174	1.000	`[\hbar]=[p][q]`	$[\hbar]=[p][q]$	dimensional transmutation due to the physical dimension $[\hbar]=[p][q]$ of Planck's
mielke-geometrodynamics_F00865	174	1.000	`\hbar`	$\hbar$	(reduced) constant $\hbar$.
mielke-geometrodynamics_F00866	174	1.000	`C:=\frac{i}{4} C^{\alpha \beta} \sigma_{\alpha \beta}`	$C:=\frac{i}{4} C^{\alpha \beta} \sigma_{\alpha \beta}$	ghosts, into operator transformations. Let $C:=\frac{i}{4}C^{\alpha\beta}\,\sigma_{\alpha\beta}$ denote the zero-form of the
mielke-geometrodynamics_F00867	174	0.998	`\Psi:=\frac{i}{4} \Psi_{j}^{\alpha \beta} \sigma_{\alpha \beta} d x^{j}`	$\Psi:=\frac{i}{4} \Psi_{j}^{\alpha \beta} \sigma_{\alpha \beta} d x^{j}$	$\Psi:=\frac{i}{4}\Psi_i^{\alpha\beta}\sigma_{\alpha\beta}dx^j$ the topological ghost one-form, and $\Phi:=\frac{i}{4}\Phi^{\alpha\beta}\sigma_{\alpha\beta}$ the correspond-
mielke-geometrodynamics_F00868	174	0.998	`\Phi:=\frac{i}{4} \Phi^{\alpha \beta} \sigma_{\alpha \beta}`	$\Phi:=\frac{i}{4} \Phi^{\alpha \beta} \sigma_{\alpha \beta}$	$\Psi:=\frac{i}{4}\Psi_i^{\alpha\beta}\sigma_{\alpha\beta}dx^j$ the topological ghost one-form, and $\Phi:=\frac{i}{4}\Phi^{\alpha\beta}\sigma_{\alpha\beta}$ the correspond-
mielke-geometrodynamics_F00869	174	1.000	`\sigma_{\alpha \beta}`	$\sigma_{\alpha \beta}$	of the generator $\sigma_{\alpha\beta}$ of the linear (or Lorentz) group.
mielke-geometrodynamics_F00870	175	1.000	`\Gamma`	Γ	This is consistent with the interpretation of Γ and Ω^g as connection one-form and
mielke-geometrodynamics_F00871	175	1.000	`\Omega^{g}`	Ω^{g}	This is consistent with the interpretation of Γ and Ω^g as connection one-form and
mielke-geometrodynamics_F00872	175	1.000	`C, \Psi`	C, Ψ	number of degrees of freedom carried by the ghosts C, Ψ, and Φ is equal to that of
mielke-geometrodynamics_F00873	175	1.000	`s^{2}=0`	$s^{2}=0$	variables, i.e., $s^2=0$.
mielke-geometrodynamics_F00874	175	0.948	`^{2}`	^{2}	(1988) by introducing the *graded*[2] connection and curvature forms
mielke-geometrodynamics_F00875	175	1.000	`s s=0`	$s s=0$	Moreover, a straightforward proof of the nilpotency $ss=0$ of the BRST operator
mielke-geometrodynamics_F00876	175	1.000	`(d \oplus s)(d \oplus s) \equiv 0`	$(d \oplus s)(d \oplus s) \equiv 0$	s follows simply from $(d\oplus s)(d\oplus s)\equiv 0$ as a result of the graded Bianchi identity
mielke-geometrodynamics_F00877	175	1.000	`[s, d]:=s d+d s=0`	$[s, d]:=s d+d s=0$	(8.2.3), the anticommutation of the graded commutator $[s,d]:=sd+ds=0$, and
mielke-geometrodynamics_F00878	175	0.615	`\bar{C}, \bar{\chi}`	$\bar{C}, \bar{\chi}$	ism, where the Lorentz-algebra-valued antighosts $\overline{C}$, $\overline{\chi}$, and $\overline{\Phi}$ obey the following
mielke-geometrodynamics_F00879	175	0.615	`\bar{\Phi}`	$\bar{\Phi}$	ism, where the Lorentz-algebra-valued antighosts $\overline{C}$, $\overline{\chi}$, and $\overline{\Phi}$ obey the following
mielke-geometrodynamics_F00880	175	1.000	`[\Psi, \Phi]:=\Psi \wedge \Phi-`	$[\Psi, \Phi]:=\Psi \wedge \Phi-$	ghost number g, such that the graded commutator is more generally defined by $[\Psi,\Phi]:=\Psi\wedge\Phi-$
mielke-geometrodynamics_F00881	175	0.510	`(-1)^{\left(p_{1}+g_{1}\right)\left(p_{2}+g_{2}\right)} \Phi \wedge \Psi`	$(-1)^{\left(p_{1}+g_{1}\right)\left(p_{2}+g_{2}\right)} \Phi \wedge \Psi$	$(-1)^{(p_1+g_1)(p_2+g_2)}\Phi\wedge\Psi$, (CHANG & SOO 1992).
mielke-geometrodynamics_F00882	176	1.000	`\bar{\eta}`	$\bar{\eta}$	By construction, s is nilpotent for the antighosts, since the Lagrange multiplier $\overline{\eta}$,
mielke-geometrodynamics_F00883	176	1.000	`\mathbb{B}`	$\mathbb{B}$	the self-dual two-forms $\mathbb{B}$, and β are auxiliary fields introduced as trivial pairs.
mielke-geometrodynamics_F00884	176	0.998	`\alpha=\alpha_{i} d x^{i}`	$\alpha=\alpha_{i} d x^{i}$	By introducing a BRST gauge field $\alpha=\alpha_i dx^i$ with ghost number -1 and a
mielke-geometrodynamics_F00885	176	0.999	`\lambda`	λ	commuting ghost λ of α, one can promote (DE CARVALHO & BAULIEU 1992) the
mielke-geometrodynamics_F00886	176	0.999	`\alpha`	α	commuting ghost λ of α, one can promote (DE CARVALHO & BAULIEU 1992) the

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mielke-geometrodynamics_F00887	176	1.000	`C \rightarrow(\alpha+`	$C \rightarrow(\alpha+$	promotion, which likewise, can be generated via the field redefinitions $C \rightarrow (\alpha +$
mielke-geometrodynamics_F00888	176	0.712	`\lambda) C, \Psi \rightarrow(\alpha+\lambda) \Psi`	$\lambda) C, \Psi \rightarrow(\alpha+\lambda) \Psi$	$\lambda)C$, $\Psi \rightarrow (\alpha+\lambda)\Psi$ as well as $\Phi \rightarrow (\alpha+\lambda)^2\Phi$ of the ghosts. Thus, local BRST
mielke-geometrodynamics_F00889	176	0.712	`\Phi \rightarrow(\alpha+\lambda)^{2} \Phi`	$\Phi \rightarrow(\alpha+\lambda)^{2} \Phi$	$\lambda)C$, $\Psi \rightarrow (\alpha+\lambda)\Psi$ as well as $\Phi \rightarrow (\alpha+\lambda)^2\Phi$ of the ghosts. Thus, local BRST
mielke-geometrodynamics_F00890	176	1.000	`\gamma=\vartheta^{\alpha} \gamma_{\alpha}`	$\gamma=\vartheta^{\alpha} \gamma_{\alpha}$	In the affine or (broken) Poincaré gauge theory, the coframe $\gamma = \vartheta^\alpha\gamma_\alpha$ (locally
mielke-geometrodynamics_F00891	176	1.000	`\widetilde{\Theta}:=\Theta \oplus \psi`	$\widetilde{\Theta}:=\Theta \oplus \psi$	respectively. In the graded torsion two-form $\widetilde{\Theta} := \Theta \oplus \psi$, there occurs the transla-
mielke-geometrodynamics_F00892	176	0.999	`\psi:=\psi_{j}^{\alpha} \gamma_{\alpha} d x^{j}`	$\psi:=\psi_{j}^{\alpha} \gamma_{\alpha} d x^{j}$	tional “Clifform” $\psi := \psi^\alpha_j\gamma_\alpha dx^j$ of the corresponding topological ghost.
mielke-geometrodynamics_F00893	177	0.885	`s \rightarrow \widetilde{s}=s+£_{\zeta}`	$s \rightarrow \widetilde{s}=s+£_{\zeta}$	alizing the BRST transformations s via $s \rightarrow \widetilde{s} = s + Ł_\zeta$ involving the covariant Lie
mielke-geometrodynamics_F00894	177	0.566	`\left.\left.\mathrm{L}_{\zeta}:=\zeta\right\lrcorner D-D \zeta\right\rfloor`	$\mathrm{L}_{\zeta}:=\zeta \lrcorner D-D \zeta \rfloor$	derivative $Ł_\zeta := \zeta\rfloor D - D\zeta\rfloor$ along an anticommuting ghost vector field $\zeta = \zeta^i\partial_i$.
mielke-geometrodynamics_F00895	177	0.566	`\zeta=\zeta^{i} \partial_{i}`	$\zeta=\zeta^{i} \partial_{i}$	derivative $Ł_\zeta := \zeta\rfloor D - D\zeta\rfloor$ along an anticommuting ghost vector field $\zeta = \zeta^i\partial_i$.
mielke-geometrodynamics_F00896	177	0.339	`\mathfrak{L}_{\zeta}`	$\mathfrak{L}_{\zeta}$	of $Ł_\zeta$, since ζ has ghost number one. Its BRST transformation is $s\zeta = \phi + Ł_\zeta\zeta$, such
mielke-geometrodynamics_F00897	177	0.339	`\zeta`	ζ	of $Ł_\zeta$, since ζ has ghost number one. Its BRST transformation is $s\zeta = \phi + Ł_\zeta\zeta$, such
mielke-geometrodynamics_F00898	177	0.339	`s \zeta=\phi+\mathfrak{L}_{\zeta} \zeta`	$s \zeta=\phi+\mathfrak{L}_{\zeta} \zeta$	of $Ł_\zeta$, since ζ has ghost number one. Its BRST transformation is $s\zeta = \phi + Ł_\zeta\zeta$, such
mielke-geometrodynamics_F00899	177	1.000	`p+1`	$p+1$	Its action on a p-form, reduces to a finite series with up to $p+1$ terms. In effect, the
mielke-geometrodynamics_F00900	177	0.497	`\exp (\zeta\rfloor) \widetilde{\Omega}^{g}`	$\exp (\zeta\rfloor) \widetilde{\Omega}^{g}$	by $\exp(\zeta\rfloor)\widetilde{\Omega}^g$ and $\exp(\zeta\rfloor)(\widetilde{\Theta} \oplus \phi)$, respectively, where $\phi := \phi^\alpha\gamma_\alpha$ are the ghosts of
mielke-geometrodynamics_F00901	177	0.497	`\left.\exp (\zeta\rfloor\right)(\widetilde{\Theta} \oplus \phi)`	$\exp (\zeta\rfloor)(\widetilde{\Theta} \oplus \phi)$	by $\exp(\zeta\rfloor)\widetilde{\Omega}^g$ and $\exp(\zeta\rfloor)(\widetilde{\Theta} \oplus \phi)$, respectively, where $\phi := \phi^\alpha\gamma_\alpha$ are the ghosts of
mielke-geometrodynamics_F00902	177	0.497	`\phi:=\phi^{\alpha} \gamma_{\alpha}`	$\phi:=\phi^{\alpha} \gamma_{\alpha}$	by $\exp(\zeta\rfloor)\widetilde{\Omega}^g$ and $\exp(\zeta\rfloor)(\widetilde{\Theta} \oplus \phi)$, respectively, where $\phi := \phi^\alpha\gamma_\alpha$ are the ghosts of
mielke-geometrodynamics_F00903	177	1.000	`d`	d	(BAULIEU 1987), the exterior derivative d suffers from the similarity transformation
mielke-geometrodynamics_F00904	177	1.000	`\Gamma^{(\mathrm{T})}=(\gamma-D \xi) / \ell`	$\Gamma^{(\mathrm{T})}=(\gamma-D \xi) / \ell$	*translational* connection $\Gamma^{(T)} = (\gamma - D\xi)/\ell$ and corresponding curvature $\Omega^{(T)} =$
mielke-geometrodynamics_F00905	177	1.000	`\Omega^{(\mathrm{T})}=`	$\Omega^{(\mathrm{T})}=$	*translational* connection $\Gamma^{(T)} = (\gamma - D\xi)/\ell$ and corresponding curvature $\Omega^{(T)} =$
mielke-geometrodynamics_F00906	177	1.000	`D \Gamma^{(\mathrm{T})}=(\Theta-D D \xi) / \ell`	$D \Gamma^{(\mathrm{T})}=(\Theta-D D \xi) / \ell$	$D\Gamma^{(T)} = (\Theta - DD\xi)/\ell$. Then the translational connection $\Gamma^{(T)}$ should be graded
mielke-geometrodynamics_F00907	177	1.000	`\Gamma^{(\mathrm{T})}`	$\Gamma^{(\mathrm{T})}$	$D\Gamma^{(T)} = (\Theta - DD\xi)/\ell$. Then the translational connection $\Gamma^{(T)}$ should be graded
mielke-geometrodynamics_F00908	177	1.000	`\gamma \rightarrow \widetilde{\gamma}:=\ell\left( \Gamma^{(\mathrm{T})} \oplus c\right)`	$\gamma \rightarrow \widetilde{\gamma}:=\ell\left(\Gamma^{(\mathrm{T})} \oplus c\right)$	as well, e.g., by the substitution $\gamma \rightarrow \widetilde{\gamma} := \ell(\Gamma^{(T)} \oplus c)$ in the first structure equation.
mielke-geometrodynamics_F00909	177	1.000	`\xi:=\xi^{\alpha} \gamma_{\alpha}`	$\xi:=\xi^{\alpha} \gamma_{\alpha}$	The “quartet” of scalars $\xi := \xi^\alpha\gamma_\alpha$ corresponds to the “generalized radius vectors”
mielke-geometrodynamics_F00910	177	1.000	`A(4, \mathbb{R}) / G L(4, \mathbb{R}) \approx \mathbb{R}^{4}`	$A(4, \mathbb{R}) / G L(4, \mathbb{R}) \approx \mathbb{R}^{4}$	of Cartan and lives on the coset space $A(4,\mathbb{R})/GL(4,\mathbb{R}) \approx \mathbb{R}^4$ of the affine group.
mielke-geometrodynamics_F00911	178	1.000	`\Gamma^{(\mathrm{T})} \stackrel{*}{=} 0`	$\Gamma^{(\mathrm{T})} \stackrel{*}{=} 0$	naturally arises in the affine gauge theory after, locally, the gauge $\Gamma^{(T)} \stackrel{*}{=} 0$ is imposed
mielke-geometrodynamics_F00912	178	0.995	`S U(N)`	$S U(N)$	and replace the internal $SU(N)$ group by the linear group $SL(4,\mathbb{R})$ of the tangent
mielke-geometrodynamics_F00913	178	0.995	`S L(4, \mathbb{R})`	$S L(4, \mathbb{R})$	and replace the internal $SU(N)$ group by the linear group $SL(4,\mathbb{R})$ of the tangent

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mielke-geometrodynamics_F00914	178	0.899	`\mathrm{SO}(1,3)`	$\mathrm{SO}(1,3)$	space embracing the Lorentz group $\mathrm{SO}(1,3)$ (or $\mathrm{SO}(4)$ in Euclidean space with
mielke-geometrodynamics_F00915	178	0.467	`=0`	$=0$	signature sig $= 0$) as subgroup. Then one starts from the gravitational Pontryagin
mielke-geometrodynamics_F00916	178	0.999	`s\{\cdots\}`	$s\{\cdots\}$	This purely topological action can be amended by any s-exact four-form $s\{\cdots\}$,
mielke-geometrodynamics_F00917	178	1.000	`d^{*} \Gamma=0, D^{*} \Psi=0`	$d^{*} \Gamma=0, D^{*} \Psi=0$	(1988), we may choose the Lorentz-type conditions $d^{*}\Gamma = 0$, $D^{*}\Psi = 0$ on the
mielke-geometrodynamics_F00918	178	1.000	`\Omega^{g( \pm)}=0`	$\Omega^{g(\pm)}=0$	self-duality condition $\Omega^{g(\pm)} = 0$ on the curvature as *gauge constraints* consistently
mielke-geometrodynamics_F00919	178	1.000	`C P`	$C P$	[3]For the CP-invariant Euler term, a similar result would hold, i.e., $sL_{\mathrm{Euler}} =$
mielke-geometrodynamics_F00920	178	1.000	`s L_{\text {Euler }}=`	$s L_{\text {Euler }}=$	[3]For the CP-invariant Euler term, a similar result would hold, i.e., $sL_{\mathrm{Euler}} =$
mielke-geometrodynamics_F00921	178	0.998	`(-1)^{\text {sig }+1} d\left(\Psi^{\alpha \beta} \wedge R_{\alpha \beta}^{(\star)}\right)`	$(-1)^{\text {sig }+1} d\left(\Psi^{\alpha \beta} \wedge R_{\alpha \beta}^{(\star)}\right)$	$(-1)^{\mathrm{sig}+1} d\left(\Psi^{\alpha\beta} \wedge R_{\alpha\beta}^{(\star)}\right)$. However, the latter is only partially metric-free, since it involves
mielke-geometrodynamics_F00922	179	1.000	`\bar{\chi}`	$\bar{\chi}$	with the four-form character of a Lagrangian, the auxiliary fields $\overline{\chi}$ and β have
mielke-geometrodynamics_F00923	179	0.993	`\widetilde{\Psi}_{\mathrm{F}}:=\bar{C}_{\alpha \beta} \left(d^{*} \Gamma^{\alpha \beta}+\frac{1}{2}{ }^{*} \mathbb{B}^{\alpha \beta}\right)`	$\widetilde{\Psi}_{\mathrm{F}}:=\bar{C}_{\alpha \beta} \left(d^{*} \Gamma^{\alpha \beta}+\frac{1}{2}{ }^{*} \mathbb{B}^{\alpha \beta}\right)$	$\widetilde{\Psi}_{\mathrm{F}}:=\overline{C}_{\alpha\beta}(d^{*}\Gamma^{\alpha\beta} + \frac{1}{2}{}^{*}\mathbb{B}^{\alpha\beta})$.
mielke-geometrodynamics_F00924	179	0.835	`\{\cdots\}`	$\{\cdots\}$	four-form $\{\cdots\}$ is able to induce a dependence on the metric that in is not present in
mielke-geometrodynamics_F00925	179	0.998	`d^{*} \Gamma={ }^{*} \mathbb{B}`	$d^{*} \Gamma={ }^{*} \mathbb{B}$	$d^{*}\Gamma = {}^{*}\mathbb{B}$ on the linear connection. Moreover, for vanishing real *gauge parameter*
mielke-geometrodynamics_F00926	179	1.000	`\rho=0`	$\rho=0$	$\rho = 0$, the equation of motion for the auxiliary Lagrange-multiplier-type two-form
mielke-geometrodynamics_F00927	180	1.000	`\rho=1`	$\rho=1$	In the case of the choice $\rho = 1$ of the real gauge parameter, the two-form β is present
mielke-geometrodynamics_F00928	180	1.000	`D \Omega^{g} \equiv 0`	$D \Omega^{g} \equiv 0$	Due to the second Bianchi identity $D\Omega^{g} \equiv 0$, the dual field equation
mielke-geometrodynamics_F00929	180	0.960	`T_{\mu \nu}:=2^{*}\left(\partial L / \partial g_{\mu \nu} \right)=0`	$T_{\mu \nu}:=2^{*}\left(\partial L / \partial g_{\mu \nu} \right)=0$	i.e., $T_{\mu\nu}:=2^{*}(\partial L/\partial g_{\mu\nu}) = 0$. In turn, this implies that the BRST quantization is
mielke-geometrodynamics_F00930	181	0.999	`\mathscr{A}:={ }^{*}\left(\vartheta_{\alpha} \wedge \right.`	$\mathscr{A}:={ }^{*}\left(\vartheta_{\alpha} \wedge \right.$	respect to the Riemannian curvature $R_{\alpha\beta}^{\{\}}$ and the *axial torsion* one-form $\mathscr{A}:={}^{*}(\vartheta_{\alpha} \wedge$
mielke-geometrodynamics_F00931	181	0.999	`T^{\alpha}`	T^{α}	T^{α}) is rather well motivated by the *axial anomaly* in RC spacetime
mielke-geometrodynamics_F00932	181	1.000	`\gamma:=\vartheta^{\alpha} \gamma_{\alpha}`	$\gamma:=\vartheta^{\alpha} \gamma_{\alpha}$	In addition, the coframe $\gamma:=\vartheta^{\alpha}\gamma_{\alpha}$ is constrained by the Hilbert–de Donder gauge
mielke-geometrodynamics_F00933	181	1.000	`d^{*} \gamma=0`	$d^{*} \gamma=0$	condition $d^{*}\gamma = 0$ complying with the BRST algebra of translations.
mielke-geometrodynamics_F00934	181	1.000	`\widetilde{L}_{\mathrm{FP}}:=L_{\mathrm{FP}}+\Delta L_{ \mathrm{FP}}`	$\widetilde{L}_{\mathrm{FP}}:=L_{\mathrm{FP}}+\Delta L_{ \mathrm{FP}}$	Then for $\rho = 0$ there arises from $\widetilde{L}_{\mathrm{FP}}:=L_{\mathrm{FP}} + \Delta L_{\mathrm{FP}}$ the *modified double duality*
mielke-geometrodynamics_F00935	181	1.000	`i`	i	on the curvature two-form Ω^{g}, which is real due to the extra imaginary unit i.
mielke-geometrodynamics_F00936	181	0.692	`\stackrel{ \pm}{\Omega^{g}}`	$\stackrel{ \pm}{\Omega^{g}}$	only one sign for the dual curvature $\stackrel{\pm}{\Omega}{}^{g}$ leads to a self-consistent constraint. This
mielke-geometrodynamics_F00937	182	0.811	`\theta=\pi / 2`	$\theta=\pi / 2$	of the curvature, which for $\theta = \pi/2$ specializes to the self-dual curvature $\stackrel{+}{\Omega}{}^{g}$, and for
mielke-geometrodynamics_F00938	182	0.811	`\stackrel{+}{\Omega}{ }^{g}`	$\stackrel{+}{\Omega}{ }^{g}$	of the curvature, which for $\theta = \pi/2$ specializes to the self-dual curvature $\stackrel{+}{\Omega}{}^{g}$, and for
mielke-geometrodynamics_F00939	182	1.000	`\theta=3 \pi / 2`	$\theta=3 \pi / 2$	$\theta = 3\pi/2$ to the anti-self-dual curvature $\stackrel{-}{\Omega}{}^{g}$. Since the "unit"-curvature two-form σ

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mielke-geometrodynamics_F00940	182	1.000	`\bar{\Omega}^{g}`	$\bar{\Omega}^{g}$	$\theta=3\pi/2$ to the anti-self-dual curvature $\bar{\Omega}^{g}$. Since the "unit"-curvature two-form σ
mielke-geometrodynamics_F00941	182	1.000	`\theta_{\mathrm{L}}=\theta_{\mathrm{T}}=0`	$\theta_{\mathrm{L}}=\theta_{\mathrm{T}}=0$	a special case of the Ansatz (8.6.4) for the choice $\theta_{\mathrm{L}}=\theta_{\mathrm{T}}=0$ and $\theta_{\mathrm{L}}^{\star}=\mp(-1)^{\text{sig}}$. Interestingly
mielke-geometrodynamics_F00942	182	1.000	`\theta_{\mathrm{L}}^{\star}=\mp(-1)^{\text {sig }}`	$\theta_{\mathrm{L}}^{\star}=\mp(-1)^{\text {sig }}$	a special case of the Ansatz (8.6.4) for the choice $\theta_{\mathrm{L}}=\theta_{\mathrm{T}}=0$ and $\theta_{\mathrm{L}}^{\star}=\mp(-1)^{\text{sig}}$. Interestingly
mielke-geometrodynamics_F00943	183	0.983	`L_{\mathrm{qPG}}`	L_{qPG}	tion parts of the gravitational Lagrangian L_{qPG}. In the field momenta, each of the
mielke-geometrodynamics_F00944	183	1.000	`a_{(M)}`	$a_{(M)}$	the Lagrangian with individual dimensionless weights $a_{(M)}$ and $b_{(N)}$, respectively.
mielke-geometrodynamics_F00945	183	1.000	`b_{(N)}`	$b_{(N)}$	the Lagrangian with individual dimensionless weights $a_{(M)}$ and $b_{(N)}$, respectively.
mielke-geometrodynamics_F00946	183	1.000	`\theta_{\mathrm{T}}, \theta_{\mathrm{T}}^{*}`	$\theta_{\mathrm{T}}, \theta_{\mathrm{T}}^{*}$	for the rotational field momenta (MIELKE 1992; ZHYTNIKOV 1994), where θ_{T}, θ_{T}^{*},
mielke-geometrodynamics_F00947	183	1.000	`\theta_{\mathrm{L}}^{\star}`	$\theta_{\mathrm{L}}^{\star}$	θ_{L}, and $\theta_{\mathrm{L}}^{\star}$ are dimensionless constants. The constraint (8.5.3) corresponds to the
mielke-geometrodynamics_F00948	183	1.000	`a_{0}=0, b_{(N)}=1`	$a_{0}=0, b_{(N)}=1$	SKY Lagrangian with $a_0=0$, $b_{(N)}=1$, and the choice $\theta_{\mathrm{L}}^{\star}=\mp(-1)^{\text{sig}}$ as well as
mielke-geometrodynamics_F00949	183	0.359	`d C_{\mathrm{TT}}{ }^{*}:=d\left(\vartheta^{\alpha} \wedge{ }^{*} T_{\alpha}\right) / 2 \ell^{2}`	$d C_{\mathrm{TT}}{ }^{*}:=d\left(\vartheta^{\alpha} \wedge{ }^{*} T_{\alpha}\right) / 2 \ell^{2}$	$dC_{\mathrm{TT}^*}:=d(\vartheta^{\alpha}\wedge{}^*T_{\alpha})/2\ell^2$ inducing the teleparallelism equivalence, as well as the
mielke-geometrodynamics_F00950	183	1.000	`\frac{1}{2} s \bar{\chi}^{\alpha \beta} \wedge\left(H_{\alpha \beta}+\partial L_{\theta} / \partial R^{\alpha \beta}\right)`	$\frac{1}{2} s \bar{\chi}^{\alpha \beta} \wedge\left(H_{\alpha \beta}+\partial L_{\theta} / \partial R^{\alpha \beta}\right)$	BRST gauge constraint $\frac{1}{2}s\overline{\chi}^{\alpha\beta}\wedge(H_{\alpha\beta}+\partial L_{\theta}/\partial R^{\alpha\beta})$ generalizing the corresponding
mielke-geometrodynamics_F00951	183	0.680	`\widetilde{L}_{\mathrm{FP}}`	$\widetilde{L}_{\mathrm{FP}}$	terms in the Faddeev–Popov Lagrangian $\widetilde{L}_{\mathrm{FP}}$ modified by (8.5.1).
mielke-geometrodynamics_F00952	184	1.000	`D`	D	The covariant derivative D of the "unit"-curvature pieces proportional to $\eta_{\alpha\beta}$ and
mielke-geometrodynamics_F00953	184	1.000	`\vartheta_{\alpha} \wedge \vartheta_{\beta}`	$\vartheta_{\alpha} \wedge \vartheta_{\beta}$	$\vartheta_{\alpha}\wedge\vartheta_{\beta}$ in the DD Ansatz is responsible for the two additional torsion terms. For
mielke-geometrodynamics_F00954	184	0.999	`\tau_{\alpha \beta}=0`	$\tau_{\alpha \beta}=0$	spinless matter, i.e., $\tau_{\alpha\beta}=0$, one can algebraically resolve (8.6.9) using Eq. (A.1.26)
mielke-geometrodynamics_F00955	184	1.000	`H_{\alpha}(* *)`	$H_{\alpha}(* *)$	of HEHL et al. (1995) for the translational momenta $H_{\alpha}(**)$ constrained by double
mielke-geometrodynamics_F00956	185	1.000	`H_{\alpha \beta}(* *) \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta}`	$H_{\alpha \beta}(* *) \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta}$	The contraction $H_{\alpha\beta}(**)\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}$ of the DD Ansatz (8.6.4) together with the linear
mielke-geometrodynamics_F00957	185	1.000	`\theta_{\mathrm{L}} \neq 0`	$\theta_{\mathrm{L}} \neq 0$	expansion (8.6.3) implies, for $\theta_{\mathrm{L}}\neq 0$, that the pseudoscalar curvature vanishes for
mielke-geometrodynamics_F00958	185	1.000	`R_{\alpha \beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta}=0`	$R_{\alpha \beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta}=0$	consistency, i.e., $R_{\alpha\beta}\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}=0$. Then the scalar curvature
mielke-geometrodynamics_F00959	185	1.000	`\theta_{\mathrm{L}}=0`	$\theta_{\mathrm{L}}=0$	[5]There occurs an interesting modification in the case that $\theta_{\mathrm{L}}=0$, since then the pseudoscalar cur-
mielke-geometrodynamics_F00960	185	1.000	`\theta_{\mathrm{T}} R^{\alpha \beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta} / 4 \ell^{2}`	$\theta_{\mathrm{T}} R^{\alpha \beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta} / 4 \ell^{2}$	vature four-form $\theta_{\mathrm{T}}R^{\alpha\beta}\wedge\vartheta_{\alpha}\wedge\vartheta_{\beta}/4\ell^2$ needs to be subtracted from (8.6.12), which would induce
mielke-geometrodynamics_F00961	186	1.000	`T^{\alpha} \simeq 0`	$T^{\alpha} \simeq 0$	vanishing torsion $T^{\alpha}\simeq 0$ via Lagrange multipliers in the gravitational gauge field
mielke-geometrodynamics_F00962	186	1.000	`T^{\alpha}=0`	$T^{\alpha}=0$	Then obviously, $T^{\alpha}=0$ will emerge by varying the Lagrange multiplier, and the
mielke-geometrodynamics_F00963	186	1.000	`\lambda_{\alpha}`	λ_{α}	second field equation (8.6.7) amounts to an algebraic equation for the two-form λ_{α}.
mielke-geometrodynamics_F00964	186	0.382	`\Gamma^{\{j \alpha \beta}`	$\Gamma^{\{j \alpha \beta}$	which, generically, is of third order in the Levi-Civita connection $\Gamma^{\{\}\alpha\beta}$ of Rie-
mielke-geometrodynamics_F00965	186	0.803	`\mu_{\alpha}=\frac{1}{4} \vartheta_{\alpha} \wedge{ }^{*} j_{5}`	$\mu_{\alpha}=\frac{1}{4} \vartheta_{\alpha} \wedge{ }^{*} j_{5}$	In the case of Dirac spinors, the *spin-energy potential* $\mu_{\alpha}=\frac{1}{4}\vartheta_{\alpha}\wedge{}^*j_5$ is dual to

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mielke-geometrodynamics_F00966	186	1.000	`\mu_{\alpha} \wedge \delta T^{\alpha}`	$\mu_{\alpha} \wedge \delta T^{\alpha}$	can likewise be obtained from the variation $\mu_\alpha \wedge \delta T^\alpha$.
mielke-geometrodynamics_F00967	186	1.000	`D R_{\alpha \beta} \equiv 0, D R_{\alpha \beta}^{(\star)} \equiv`	$D R_{\alpha \beta} \equiv 0, D R_{\alpha \beta}^{(\star)} \equiv$	coupling constants in the θ-type boundary term (8.6.5): since $DR_{\alpha\beta} \equiv 0$, $DR_{\alpha\beta}^{(\star)} \equiv$
mielke-geometrodynamics_F00968	186	0.420	`0, D \eta_{\alpha \beta}=0`	$0, D \eta_{\alpha \beta}=0$	0, $D\eta_{\alpha\beta} = 0$, and $D(\vartheta_\alpha \wedge \vartheta_\beta) = 2T_{[\alpha} \wedge \vartheta_{\beta]} = 0$ in a Riemannian spacetime, the
mielke-geometrodynamics_F00969	186	0.420	`D\left(\vartheta_{\alpha} \wedge \vartheta_{\beta} \right)=2 T_{[\alpha} \wedge \vartheta_{\beta]}=0`	$D\left(\vartheta_{\alpha} \wedge \vartheta_{\beta}\right)=2 T_{[\alpha} \wedge \vartheta_{\beta]}=0$	0, $D\eta_{\alpha\beta} = 0$, and $D(\vartheta_\alpha \wedge \vartheta_\beta) = 2T_{[\alpha} \wedge \vartheta_{\beta]} = 0$ in a Riemannian spacetime, the
mielke-geometrodynamics_F00970	186	1.000	`R_{\alpha \beta}^{(\star)}`	$R_{\alpha \beta}^{(\star)}$	the Lie dual $R_{\alpha\beta}^{(\star)}$ of the curvature does not contribute in (8.6.8), due to the Bach–
mielke-geometrodynamics_F00971	186	0.428	`G^{\{ \}}:=R^{\{1\} \beta \gamma} \wedge \eta_{\alpha \beta \gamma} \gamma^{\alpha} / 2`	$G^{\{ \}}:=R^{\{1\} \beta \gamma} \wedge \eta_{\alpha \beta \gamma} \gamma^{\alpha} / 2$	Here $G^{\{\}}{:=}R^{\{\}\beta\gamma} \wedge \eta_{\alpha\beta\gamma}\gamma^\alpha/2$ is the Einstein three-form dual to the usual Einstein
mielke-geometrodynamics_F00972	186	0.997	`G_{i j}:=\operatorname{Ric}_{i j}^{\{ \}}-\frac{1}{2} g_{i j}`	$G_{i j}:=\operatorname{Ric}_{i j}^{\{ \}}-\frac{1}{2} g_{i j}$	tensor $G_{ij}{:=}\mathrm{Ric}_{ij}^{\{\}} - \frac{1}{2}g_{ij}$ written in terms of the Riemannian connection $\Gamma^{\{\}}$. Matter
mielke-geometrodynamics_F00973	186	0.993	`\kappa_{\text {eff }}=\ell^{2} / \theta_{\mathrm{T}}^{*}`	$\kappa_{\text {eff }}=\ell^{2} / \theta_{\mathrm{T}}^{*}$	whereas the effective gravitational coupling constant $\kappa_{\text{eff}} = \ell^2/\theta_{\mathrm{T}}^*$ depends on a
mielke-geometrodynamics_F00974	186	1.000	`\theta_{\mathrm{T}}^{*}`	θ_{T}^{*}	"bare" length scale ℓ "renormalized" by the vacuum angle θ_{T}^*.
mielke-geometrodynamics_F00975	187	1.000	`\widehat{\Sigma}_{\alpha}`	$\widehat{\Sigma}_{\alpha}$	*Einstein equation* (8.6.18) to the symmetrized energy–momentum current $\widehat{\Sigma}_\alpha$ of
mielke-geometrodynamics_F00976	187	0.987	`\kappa_{\text {eff }}=8 \pi G_{\mathrm{N}} / c^{4}`	$\kappa_{\text {eff }}=8 \pi G_{\mathrm{N}} / c^{4}$	matter requires, at least locally, the macroscopic Newtonian value $\kappa_{\text{eff}} = 8\pi G_{\mathrm{N}}/c^4$
mielke-geometrodynamics_F00977	187	0.894	`L_{\text {qPG }}`	$L_{\text {qPG }}$	most quadratic Lagrangian L_{qPG} with ten[6] parameters $a_{(M)}$ and $b_{(N)}$.
mielke-geometrodynamics_F00978	187	1.000	`\langle\varphi\rangle \propto 1 / \ell`	$\langle\varphi\rangle \propto 1 / \ell$	tive to associate this with a (spontaneous) *symmetry breaking* $\langle\varphi\rangle \propto 1/\ell$ of the scale
mielke-geometrodynamics_F00979	188	1.000	`a`	a	When these covectors are induced by an axion a and dilaton φ as respective potentials,
mielke-geometrodynamics_F00980	188	1.000	`\Phi=a+i f_{\varphi} \exp \left(-\varphi / f_{\varphi} \right)`	$\Phi=a+i f_{\varphi} \exp \left(-\varphi / f_{\varphi}\right)$	even combine into a single complex scalar, the *axidilaton* $\Phi = a + if_\varphi \exp(-\varphi/f_\varphi)$;
mielke-geometrodynamics_F00981	192	0.954	`\Lambda_{\text {eff }}>0`	$\Lambda_{\text {eff }}>0$	(here generalized to an n-dimensional manifold) and second for $\Lambda_{\text{eff}} > 0$ to the Nariai
mielke-geometrodynamics_F00982	193	0.989	`\mathrm{N}=\mathrm{N}(\mathrm{r} \pm \mathrm{t})`	$\mathrm{N}=\mathrm{N}(\mathrm{r} \pm \mathrm{t})$	with $\mathrm{N} = \mathrm{N}(\mathrm{r} \pm \mathrm{t})$ denoting an arbitrary harmonic function of r and t; cf. Ni (1975).
mielke-geometrodynamics_F00983	193	0.989	`r`	r	with $\mathrm{N} = \mathrm{N}(\mathrm{r} \pm \mathrm{t})$ denoting an arbitrary harmonic function of r and t; cf. Ni (1975).
mielke-geometrodynamics_F00984	193	0.998	`\Lambda_{\text {eff }}=0`	$\Lambda_{\text {eff }}=0$	spacetime, however, the Schwarzschild solution (9.1.1, 2) with $\Lambda_{\text{eff}} = 0$ is of para-
mielke-geometrodynamics_F00985	194	0.358	`{ }^{*}`	${ }^{*}$	nonsingular one, we first pass to a different coordinate system. Let r^* be the "tortoise"
mielke-geometrodynamics_F00986	194	0.996	`\mathrm{u}^{2}-v^{2}=1`	$\mathrm{u}^{2}-v^{2}=1$	$\mathrm{u}^2 - \nu^2 = 1$, that is, for $\mathrm{r} = \mathrm{r}^* = \mathrm{o}$, this geometry becomes singular in real space-
mielke-geometrodynamics_F00987	194	0.996	`\mathrm{r}=\mathrm{r}^{*}=\mathrm{o}`	$\mathrm{r}=\mathrm{r}^{*}=\mathrm{o}$	$\mathrm{u}^2 - \nu^2 = 1$, that is, for $\mathrm{r} = \mathrm{r}^* = \mathrm{o}$, this geometry becomes singular in real space-
mielke-geometrodynamics_F00988	194	1.000	`\tilde{\mathrm{v}}:=\mathrm{iv}`	$\tilde{\mathrm{v}}:=\mathrm{iv}$	be avoided by making use of a purely "imaginary" coordinate $\tilde{\mathrm{v}} := \mathrm{iv}$. The sphere
mielke-geometrodynamics_F00989	194	0.531	`\mathrm{S}^{2}`	S^{2}	S^2 given by $\mathrm{e}^\nu = 0$ deteriorates to a rotational axis, while the imaginary time axis
mielke-geometrodynamics_F00990	194	0.531	`\mathrm{e}^{v}=0`	$\mathrm{e}^{v}=0$	S^2 given by $\mathrm{e}^\nu = 0$ deteriorates to a rotational axis, while the imaginary time axis
mielke-geometrodynamics_F00991	194	1.000	`\tau:=`	$\tau:=$	$\tau :=$ it represents an angle variable about this axis in Euclidean spacetime. (Corre-
mielke-geometrodynamics_F00992	194	1.000	`\mathrm{K}_{3}`	K_{3}	K_3-surface represents the only compact manifold that is allowed by a metric with

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mielke-geometrodynamics_F00993	195	0.629	`\omega^{g}`	ω^{g}	connection ω^{g} into a Christoffel-type connection $\omega^{\{\}}$ and the remaining contortion
mielke-geometrodynamics_F00994	195	0.975	`R C`	RC	2-form in an RC spacetime. Its Riemannian part may be decomposed further into
mielke-geometrodynamics_F00995	195	0.917	`\Omega_{\mathrm{C}}^{\{ \}}`	$\Omega_{\mathrm{C}}^{\{ \}}$	where $\Omega_{\mathrm{C}}^{\{\}}$ denotes Weyl's conformal curvature 2-form in a Riemannian spacetime.
mielke-geometrodynamics_F00996	195	1.000	`\mathrm{f}_{\mathrm{i}}=\mathrm{o}`	$\mathrm{f}_{\mathrm{i}}=\mathrm{o}$	choice $\mathrm{f}_{\mathrm{i}}=\mathrm{o}$ and $\zeta=-1$, as discussed by BAEKLER et al. (1982), the duality ansatz
mielke-geometrodynamics_F00997	196	1.000	`K \wedge K`	$K \wedge K$	fact that $K \wedge K$ is already self-double-dual.
mielke-geometrodynamics_F00998	196	1.000	`\mu=0`	$\mu=0$	cosmos, vanishes without Schwarzschild's source term, i.e., for $\mu=0$. This is in
mielke-geometrodynamics_F00999	196	1.000	`m u`	$m u$	parameter mu vanishes in the metric ground form. Here the radial functions occurring
mielke-geometrodynamics_F01000	197	1.000	`\kappa<0`	$\kappa<0$	For a real solution, $\kappa<0$ is necessary. In this case, torsion can exist only in an anti-de
mielke-geometrodynamics_F01001	197	1.000	`\mathrm{t} \rightarrow-\mathrm{i} \tau`	$\mathrm{t} \rightarrow-\mathrm{i} \tau$	Weyl's trick, i.e., the formal substitution $\mathrm{t} \rightarrow -\mathrm{i}\tau$ of the timelike variable. It follows
mielke-geometrodynamics_F01002	197	0.687	`\mathrm{f}=\mathrm{T}_{\hat{0} \hat{1}}^{. . \hat{0}}`	$\mathrm{f}=\mathrm{T}_{\hat{0} \hat{1}}^{\cdot\cdot \hat{0}}$	from the matrix representation (9.1.15) that the anholonomic components $\mathrm{f}=\mathrm{T}_{\hat{0}\hat{1}}^{\cdot\cdot\hat{0}}$
mielke-geometrodynamics_F01003	197	0.790	`\mathrm{g}=\mathrm{T}_{\hat{1} \hat{2}}^{\cdot \hat{2}}= \mathrm{T}_{\hat{1} \hat{3}}^{\cdot \hat{3}}`	$\mathrm{g}=\mathrm{T}_{\hat{1} \hat{2}}^{\cdot \hat{2}}=\mathrm{T}_{\hat{1} \hat{3}}^{\cdot \hat{3}}$	and $\mathrm{g}=\mathrm{T}_{\hat{1}\hat{2}}^{\cdot\cdot\hat{2}}=\mathrm{T}_{\hat{1}\hat{3}}^{\cdot\cdot\hat{3}}$ are the only ones that are then allowed to be nonzero. A solution
mielke-geometrodynamics_F01004	198	1.000	`\mathrm{n}=4`	$\mathrm{n}=4$	ourselves to a manifold of dimension $\mathrm{n}=4$. The previous classification of the source-
mielke-geometrodynamics_F01005	198	0.998	`\mathrm{R}^{6}`	R^{6}	space R^{6}. For then
mielke-geometrodynamics_F01006	199	0.995	`\gamma_{2}^{g}`	γ_{2}^{g}	the form γ_{2}^{g} over the basis manifold, which provides for a topological invariant. It is
mielke-geometrodynamics_F01007	199	1.000	`\mathrm{H}_{\mathrm{P}}(\mathrm{M}, \mathbb{R})`	$\mathrm{H}_{\mathrm{P}}(\mathrm{M}, \mathbb{R})$	rank of the pth homology group $\mathrm{H}_{\mathrm{P}}(\mathrm{M}, \mathbb{R})$, i.e., from the so-called Betti numbers
mielke-geometrodynamics_F01008	199	1.000	`\mathrm{b}:=\operatorname{dim} \mathrm{H}_{\mathrm{p}}( \mathrm{M}, \mathbb{R})`	$\mathrm{b}:=\operatorname{dim} \mathrm{H}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$	$\mathrm{b}:=\dim \mathrm{H}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$, which were introduced by Riemann. Similarly as in the case of
mielke-geometrodynamics_F01009	199	1.000	`\chi(\mathrm{M})`	$\chi(\mathrm{M})$	the vanishing of $\chi(\mathrm{M})$. In particular, it is known that a compact even-dimensional
mielke-geometrodynamics_F01010	199	1.000	`\mathrm{s}=1`	$\mathrm{s}=1$	manifold admits a pseudo-Riemannian metric of signature $\mathrm{s}=1$ ("Lorentz metric")
mielke-geometrodynamics_F01011	199	0.996	`\alpha: \omega^{\mathrm{g}}-\omega^{\mathrm{g}}`	$\alpha: \omega^{\mathrm{g}}-\omega^{\mathrm{g}}$	$\alpha: \omega^{\mathrm{g}}-\overset{\infty}{\omega}{}^{\mathrm{g}}$ of the embedding of the boundary $\partial\mathrm{M}$ in M. Although it is exactly
mielke-geometrodynamics_F01012	200	1.000	`\mathrm{B}^{\mathrm{n}}`	B^{n}	an open ball B^{n} is to be cut out of M and N in order to identify the then respectively
mielke-geometrodynamics_F01013	200	1.000	`\partial \mathrm{B}^{\mathrm{n}}=\mathrm{S}^{ \mathrm{n}-1}`	$\partial \mathrm{B}^{\mathrm{n}}=\mathrm{S}^{\mathrm{n}-1}$	existing boundary spheres $\partial \mathrm{B}^{\mathrm{n}}=\mathrm{S}^{\mathrm{n}-1}$ with each other via a diffeomorphism that
mielke-geometrodynamics_F01014	200	0.736	`\mathrm{W}^{\mathrm{n}}:=\mathrm{S}^{1} \times \mathrm{S}^{\mathrm{n}-1}`	$\mathrm{W}^{\mathrm{n}}:=\mathrm{S}^{1} \times \mathrm{S}^{\mathrm{n}-1}$	$\mathrm{W}^{\mathrm{n}}:=\mathrm{S}^{1} \times \mathrm{S}^{\mathrm{n}-1}$ (WHEELER 1955), the manifold $\mathrm{M}\#(\mathrm{S}^{1} \times \mathrm{S}^{\mathrm{n}-1})$ comes into exis-
mielke-geometrodynamics_F01015	200	0.736	`\left(\mathrm{S}^{1} \times \mathrm{S}^{\mathrm{n}-1} \right)`	$\left(\mathrm{S}^{1} \times \mathrm{S}^{\mathrm{n}-1}\right)$	$\mathrm{W}^{\mathrm{n}}:=\mathrm{S}^{1} \times \mathrm{S}^{\mathrm{n}-1}$ (WHEELER 1955), the manifold $\mathrm{M}\#(\mathrm{S}^{1} \times \mathrm{S}^{\mathrm{n}-1})$ comes into exis-
mielke-geometrodynamics_F01016	200	0.980	`\mathbb{C P}^{2}`	$\mathbb{C P}^{2}$	number" of a surface (cf. SEIFERT & THREFALL 1934), and $\mathbb{C}\mathrm{P}^{2}$ denotes the complex
mielke-geometrodynamics_F01017	200	1.000	`\mathrm{E}^{2}`	E^{2}	projective two-space in E^{2}. For spaces with uneven dimensions,
mielke-geometrodynamics_F01018	201	1.000	`\mathrm{T}^{4}=\mathrm{S}^{1} \times \mathrm{S}^{1} \times \mathrm{S}^{1} \times \mathrm{S}^{1}`	$\mathrm{T}^{4}=\mathrm{S}^{1} \times \mathrm{S}^{1} \times \mathrm{S}^{1} \times \mathrm{S}^{1}$	by the flat torus $\mathrm{T}^{4}=\mathrm{S}^{1} \times \mathrm{S}^{1} \times \mathrm{S}^{1} \times \mathrm{S}^{1}$. Solutions with anti-self-double-dual cur-
mielke-geometrodynamics_F01019	202	0.982	`\mathscr{S}_{*}(\mathrm{M}):=\mathscr{M} / \tilde{ \mathscr{D}}`	$\mathscr{S}_{*}(\mathrm{M}):=\mathscr{M} / \tilde{\mathscr{D}}$	space" $\mathscr{S}_{*}(\mathrm{M}):=\mathscr{M} / \tilde{\mathscr{D}}$ of all Riemannian metrics modulo the diffeomorphisms

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mielke-geometrodynamics_F01020	203	1.000	`S^{2}`	S^{2}	Euler–Poincaré topological invariant (9.3.1). The two-sphere part S^2 of these $\mathbb{R} \times$
mielke-geometrodynamics_F01021	203	1.000	`\mathbb{R} \times`	$\mathbb{R} \times$	Euler–Poincaré topological invariant (9.3.1). The two-sphere part S^2 of these $\mathbb{R} \times$
mielke-geometrodynamics_F01022	203	1.000	`S^{1} \times S^{2}`	$S^{1} \times S^{2}$	$S^1 \times S^2$ topological configurations may even be knotted (MIELKE 1977a), and within
mielke-geometrodynamics_F01023	203	1.000	`\mathbf{R}^{4}`	$\mathbf{R}^{4}$	piecewise linear structures (MILNOR 2011) on a noncompact $\mathbf{R}^4$. However, mani-
mielke-geometrodynamics_F01024	203	1.000	`U_{4}`	U_{4}	Baekler P (1980) The unique spherically symmetric solution of the U_4-theory of gravity in the
mielke-geometrodynamics_F01025	205	0.999	`\mathrm{R}+\mathrm{R}^{2}`	$\mathrm{R}+\mathrm{R}^{2}$	Neville DE (1980) Birkhoff theorems for $\mathrm{R}+\mathrm{R}^2$ gravity theories with torsion. Phys Rev D
mielke-geometrodynamics_F01026	208	0.837	`C_{\mathrm{TL}}`	C_{TL}	term C_{TL}, which simulates to some extent Einstein's theory in 3D, allowing us to
mielke-geometrodynamics_F01027	208	0.994	`\varepsilon \rightarrow 1 / \varepsilon`	$\varepsilon \rightarrow 1 / \varepsilon$	$\varepsilon \rightarrow 1/\varepsilon$.
mielke-geometrodynamics_F01028	208	0.674	`\Gamma^{\beta \gamma}=\Gamma_{j}^{\beta \gamma} d x^{j}`	$\Gamma^{\beta \gamma}=\Gamma_{j}^{\beta \gamma} d x^{j}$	connection $\Gamma^{\beta\gamma} = \Gamma_j^{\beta\gamma} dx^j$, i.e.,
mielke-geometrodynamics_F01029	209	1.000	`\Gamma_{\alpha}^{\star}`	$\Gamma_{\alpha}^{\star}$	Since the coframe ϑ_α has physical dimensions [length] and the connection $\Gamma_\alpha^\star$ is
mielke-geometrodynamics_F01030	209	1.000	`\vartheta_{\alpha} / \ell`	$\vartheta_{\alpha} / \ell$	dimensionless, one can use ϑ_α/ℓ and $\Gamma_\alpha^\star$ as *dimensionless* gauge potentials of local
mielke-geometrodynamics_F01031	209	1.000	`C=\operatorname{Tr}\{A \wedge F\}`	$C=\operatorname{Tr}\{A \wedge F\}$	The corresponding Chern–Simons three-forms of gauge type $C = \mathrm{Tr}\{A \wedge F\}$ are the
mielke-geometrodynamics_F01032	209	1.000	`T_{\alpha}`	T_{α}	are uniquely related to the torsion T_α, the curvature $R_\alpha^\star$, and a cosmological term η_α,
mielke-geometrodynamics_F01033	209	1.000	`R_{\alpha}^{\star}`	$R_{\alpha}^{\star}$	are uniquely related to the torsion T_α, the curvature $R_\alpha^\star$, and a cosmological term η_α,
mielke-geometrodynamics_F01034	209	1.000	`\eta_{\alpha}`	η_{α}	are uniquely related to the torsion T_α, the curvature $R_\alpha^\star$, and a cosmological term η_α,
mielke-geometrodynamics_F01035	210	1.000	`n^{2}`	n^{2}	(table below)
mielke-geometrodynamics_F01036	210	1.000	`n^{2}(n-1) / 2`	$n^{2}(n-1)/2$	(table below)

	Objects	p-form	Components	n = 4	3	2
ϑ^α	Vector	1	n^2	16	9	1
$\Gamma_\alpha^\star$	Vector	1	n^2	16	9	1
T^α	Vector	2	$n^2(n-1)/2$	24	9	2
$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1
Σ_α	Vector	$n-1$	n^2	16	9	4
$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2
η_α	Vector	$n-1$	n^2	16	9	4

	Objects	p-form	Components	n = 4	3	2
ϑ^α	Vector	1	n^2	16	9	1
$\Gamma_\alpha^\star$	Vector	1	n^2	16	9	1
T^α	Vector	2	$n^2(n-1)/2$	24	9	2
$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1
Σ_α	Vector	$n-1$	n^2	16	9	4
$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2
η_α	Vector	$n-1$	n^2	16	9	4

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<tr><td>mielke-geometrodynamics_F01037</td><td>210</td><td>■1.000</td><td>R^{\alpha \beta}</td><td>$R^{\alpha \beta}$</td><td><table><tr><th></th><th>Objects</th><th>p-form</th><th>Components</th><th>$n=4$</th><th>3</th><th>2</th></tr><tr><td>ϑ^α</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>$\Gamma^\star_\alpha$</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>T^α</td><td>Vector</td><td>2</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>$R^{\alpha\beta}$</td><td>Bivector</td><td>2</td><td>$n^2(n-1)^2/4$</td><td>36</td><td>9</td><td>1</td></tr><tr><td>Σ_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr><tr><td>$\tau_{\alpha\beta}$</td><td>Bivector</td><td>$n-1$</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>η_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr></table></td></tr>
<tr><td>mielke-geometrodynamics_F01038</td><td>210</td><td>■1.000</td><td>n^{2}(n-1)^{2} / 4</td><td>$n^{2}(n-1)^{2} / 4$</td><td><table><tr><th></th><th>Objects</th><th>p-form</th><th>Components</th><th>$n=4$</th><th>3</th><th>2</th></tr><tr><td>ϑ^α</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>$\Gamma^\star_\alpha$</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>T^α</td><td>Vector</td><td>2</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>$R^{\alpha\beta}$</td><td>Bivector</td><td>2</td><td>$n^2(n-1)^2/4$</td><td>36</td><td>9</td><td>1</td></tr><tr><td>Σ_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr><tr><td>$\tau_{\alpha\beta}$</td><td>Bivector</td><td>$n-1$</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>η_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr></table></td></tr>
<tr><td>mielke-geometrodynamics_F01039</td><td>210</td><td>■1.000</td><td>\tau_{\alpha \beta}</td><td>$\tau_{\alpha \beta}$</td><td><table><tr><th></th><th>Objects</th><th>p-form</th><th>Components</th><th>$n=4$</th><th>3</th><th>2</th></tr><tr><td>ϑ^α</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>$\Gamma^\star_\alpha$</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>T^α</td><td>Vector</td><td>2</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>$R^{\alpha\beta}$</td><td>Bivector</td><td>2</td><td>$n^2(n-1)^2/4$</td><td>36</td><td>9</td><td>1</td></tr><tr><td>Σ_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr><tr><td>$\tau_{\alpha\beta}$</td><td>Bivector</td><td>$n-1$</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>η_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr></table></td></tr>
<tr><td>mielke-geometrodynamics_F01040</td><td>210</td><td>■1.000</td><td>n=3</td><td>$n=3$</td><td>components in various dimensions: Observe that only for $n = 3$ do all fields have</td></tr>
<tr><td>mielke-geometrodynamics_F01041</td><td>210</td><td>■0.872</td><td>\mathrm{SO}(1,2)</td><td>$\mathrm{SO}(1,2)$</td><td>dimensional Lorentz group SO(1, 2) as a $9 = 5 \otimes 3 \otimes 1$ multiplet after bivectors are</td></tr>
<tr><td>mielke-geometrodynamics_F01042</td><td>210</td><td>■0.872</td><td>9=5 \otimes 3 \otimes 1</td><td>$9=5 \otimes 3 \otimes 1$</td><td>dimensional Lorentz group SO(1, 2) as a $9 = 5 \otimes 3 \otimes 1$ multiplet after bivectors are</td></tr>
<tr><td>mielke-geometrodynamics_F01043</td><td>211</td><td>■1.000</td><td>H=-\partial L / \partial F</td><td>$H=-\partial L / \partial F$</td><td>are the translational and rotational field momenta (or the excitation $H = -\partial L/\partial F$</td></tr>
<tr><td>mielke-geometrodynamics_F01044</td><td>211</td><td>■1.000</td><td>H_{\alpha}=H_{[\alpha \beta]} \vartheta^{\beta}</td><td>$H_{\alpha}=H_{[\alpha \beta]} \vartheta^{\beta}$</td><td>(In the particular case of an antisymmetric field $H_\alpha = H_{[\alpha\beta]}\vartheta^\beta$, this is dual to the</td></tr>
<tr><td>mielke-geometrodynamics_F01045</td><td>211</td><td>■1.000</td><td>S_{\alpha}=-{ }^{*} H_{\alpha}</td><td>$S_{\alpha}=-{ }^{*} H_{\alpha}$</td><td>translational field strength, i.e., $S_\alpha = -{}^*H_\alpha$.) The sources for the gravitational field</td></tr>
<tr><td>mielke-geometrodynamics_F01046</td><td>211</td><td>■0.957</td><td>\mathbb{R}^{3} \otimes S O(1,2)</td><td>$\mathbb{R}^{3} \otimes S O(1,2)$</td><td>$\mathbb{R}^3 \ltimes SO(1, 2)$, we arrive at the MIELKE & BAEKLER (1991) (MB) model, which is</td></tr>
<tr><td>mielke-geometrodynamics_F01047</td><td>211</td><td>■1.000</td><td>\theta_{\mathrm{T}}, \theta_{\mathrm{L}}</td><td>$\theta_{\mathrm{T}}, \theta_{\mathrm{L}}$</td><td>Allowing for arbitrary “vacuum angles” θ_{T}, θ_{L}, and $\theta_{\mathrm{TL}} = -\chi$, the most general</td></tr>
<tr><td>mielke-geometrodynamics_F01048</td><td>211</td><td>■1.000</td><td>\theta_{\mathrm{TL}}=-\chi</td><td>$\theta_{\mathrm{TL}}=-\chi$</td><td>Allowing for arbitrary “vacuum angles” θ_{T}, θ_{L}, and $\theta_{\mathrm{TL}} = -\chi$, the most general</td></tr>
<tr><td>mielke-geometrodynamics_F01049</td><td>212</td><td>■0.997</td><td>C_{\mathrm{T}}, C_{\mathrm{L}}</td><td>$C_{\mathrm{T}}, C_{\mathrm{L}}$</td><td>where C_{T}, C_{L}, and C_{TL} are respectively the translational, rotational, and “mixed” CS</td></tr>
<tr><td>mielke-geometrodynamics_F01050</td><td>212</td><td>■1.000</td><td>C_{\mathrm{T}}</td><td>C_{T}</td><td>tional term C_{T} is covariant, it appears that the MB model is only semi-topological,</td></tr>
<tr><td>mielke-geometrodynamics_F01051</td><td>212</td><td>■1.000</td><td>\Gamma^{\star \alpha}</td><td>$\Gamma^{\star \alpha}$</td><td>Consequently, varying the Lagrangian (10.2.2) with respect to ϑ^α and $\Gamma^{\star\alpha}$ yields</td></tr>
<tr><td>mielke-geometrodynamics_F01052</td><td>212</td><td>■1.000</td><td>\theta_{\mathrm{T}} / 2 \ell</td><td>$\theta_{\mathrm{T}} / 2 \ell$</td><td>$\theta_{\mathrm{T}}/2\ell$, resembling a cosmological constant familiar from 4D gravity.</td></tr>
<tr><td>mielke-geometrodynamics_F01053</td><td>212</td><td>■1.000</td><td>\varepsilon=\theta_{\mathrm{TL}} \theta_{\mathrm{T}} / A</td><td>$\varepsilon=\theta_{\mathrm{TL}} \theta_{\mathrm{T}} / A$</td><td>where the contortional constants $\varepsilon = \theta_{\mathrm{TL}}\theta_{\mathrm{T}}/A$ and $\rho = -\theta_{\mathrm{T}}^2/A$ are related to the</td></tr>
</table>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
mielke-geometrodynamics_ F01054	212	1.000	\rho=-\theta_{\mathrm{T}}^{2} / A	$\rho=-\theta_{\mathrm{T}}^{2} / A$	where the contortional constants $\varepsilon = \theta_{\mathrm{TL}}\theta_{\mathrm{T}}/A$ and $\rho = -\theta_{\mathrm{T}}^2/A$ are related to the
mielke-geometrodynamics_ F01055	212	1.000	A:=-(-1)^{s} \theta_{\mathrm{TL}}^{2}+2 \theta_{ \mathrm{T}} \theta_{\mathrm{L}} \neq 0	$A:=-(-1)^{s} \theta_{\mathrm{TL}}^{2}+2 \theta_{\mathrm{T}} \theta_{\mathrm{L}} \neq 0$	$A := -(-1)^s\theta_{\mathrm{TL}}^2 + 2\theta_{\mathrm{T}}\theta_{\mathrm{L}} \neq 0$ is nonvanishing.
mielke-geometrodynamics_ F01056	213	0.994	K_{\alpha}^{\star}=-{ }^{*} T_{\alpha} / 2=-(-1)^{s}(\varepsilon / 2 \ell) \vartheta_{\alpha}	$K_{\alpha}^{\star}=-{ }^{*} T_{\alpha} / 2=-(-1)^{s}(\varepsilon / 2 \ell) \vartheta_{\alpha}$	for the RC curvature. From the relation $K_{\alpha}^{\star} = -{}^{*}T_{\alpha}/2 = -(-1)^s(\varepsilon/2\ell)\vartheta_{\alpha}$ for the
mielke-geometrodynamics_ F01057	213	1.000	\varepsilon=0	$\varepsilon=0$	$\varepsilon = 0$, i.e., in a purely Riemannian spacetime. Alternatively, for $\rho = 0$, i.e., in the
mielke-geometrodynamics_ F01058	213	1.000	\Lambda_{\text {eff }}	Λ_{eff}	cosmological term Λ_{eff} has the anti-de Sitter (AdS) metric
mielke-geometrodynamics_ F01059	214	1.000	J	J	respectively. Observe that the shift is proportional to the angular momentum J of
mielke-geometrodynamics_ F01060	214	1.000	J^{2} \leq M^{2}	$J^{2} \leq M^{2}$	the solution, which allows for $J^2 \leq M^2$ to interpret this configuration as a *rotating*
mielke-geometrodynamics_ F01061	214	1.000	M=-1	$M=-1$	normalization $M = -1$ and $J = 0$, it reduces to the AdS metric.
mielke-geometrodynamics_ F01062	214	1.000	J=0	$J=0$	normalization $M = -1$ and $J = 0$, it reduces to the AdS metric.
mielke-geometrodynamics_ F01063 [conf 0.080]	214	0.080	\Gamma_{\alpha}^{\left\{{ }^{\}} \star\right.} \rightarrow \Gamma_{\alpha}^{\star}=\Gamma_{\alpha}^{ \left\{{ }^{\{\star}\right.}-K_{\alpha}^{\star}	$\Gamma_{\alpha}^{\{\}\star} \rightarrow \Gamma_{\alpha}^{\star}=\Gamma_{\alpha}^{\{\{\star}-K_{\alpha}^{\star}$	rest essentially on a "prolongation" $\Gamma_{\alpha}^{\{\}\star} \to \Gamma_{\alpha}^{\star} = \Gamma_{\alpha}^{\{\}\star} - K_{\alpha}^{\star}$ of the Riemannian to
mielke-geometrodynamics_ F01064	214	1.000	\varepsilon	ε	effective cosmological constant Λ_{eff} is related to ε and ρ via (10.2.10), the black hole
mielke-geometrodynamics_ F01065	214	1.000	\Gamma_{\alpha}^{\star}=\frac{1}{2} \eta_{\alpha \beta \gamma} \Gamma^{\beta \gamma}	$\Gamma_{\alpha}^{\star}=\frac{1}{2} \eta_{\alpha \beta \gamma} \Gamma^{\beta \gamma}$	where ε is a continuous parameter, involves the Lie dual $\Gamma_{\alpha}^{\star} = \frac{1}{2}\eta_{\alpha\beta\gamma}\Gamma^{\beta\gamma}$ of the
mielke-geometrodynamics_ F01066	214	1.000	\widetilde{T}_{\alpha}=0	$\widetilde{T}_{\alpha}=0$	Riemannian spacetime with $\widetilde{T}_{\alpha} = 0$ and deformed *teleparallelism* in the local gauge
mielke-geometrodynamics_ F01067	214	0.999	\widetilde{\Gamma}_{\alpha}^{\star} \stackrel{*}{=} 0	$\widetilde{\Gamma}_{\alpha}^{\star} \stackrel{*}{=} 0$	$\widetilde{\Gamma}_{\alpha}^{\star} \stackrel{*}{=} 0$, equivalent to the covariant constraint of vanishing modified RC curvature,
mielke-geometrodynamics_ F01068	214	1.000	\widetilde{R}_{\alpha}^{\star}=0	$\widetilde{R}_{\alpha}^{\star}=0$	i.e., $\widetilde{R}_{\alpha}^{\star} = 0$. In the latter case, coframe and connection are locally *Lie dual* to each
mielke-geometrodynamics_ F01069	215	1.000	1 / \varepsilon	$1/\varepsilon$	large rotational momentum proportional to $1/\varepsilon$ and conversely, resembling, to some
mielke-geometrodynamics_ F01070	215	0.915	\rho=(-1)^{\text {sign }} \varepsilon^{2} / 4	$\rho=(-1)^{\text{sign}} \varepsilon^{2} / 4$	as originally envisioned by CARTAN (1924). Here $\rho = (-1)^{\text{sign}}\varepsilon^2/4$ depends quadrat-
mielke-geometrodynamics_ F01071	215	0.518	R_{\alpha}^{\star}=(-1)^{\text {sign }} \varepsilon T_{ \alpha} / 4 \ell	$R_{\alpha}^{\star}=(-1)^{\text{sign}} \varepsilon T_{\alpha} / 4 \ell$	should not ignore that $R_{\alpha}^{\star} = (-1)^{\text{sign}}\varepsilon\, T_{\alpha}/4\ell$ induces, for $\varepsilon \neq 0$, a constant-curvature
mielke-geometrodynamics_ F01072	215	0.518	\varepsilon \neq 0	$\varepsilon \neq 0$	should not ignore that $R_{\alpha}^{\star} = (-1)^{\text{sign}}\varepsilon\, T_{\alpha}/4\ell$ induces, for $\varepsilon \neq 0$, a constant-curvature
mielke-geometrodynamics_ F01073	216	1.000	\varepsilon \rightarrow-\varepsilon	$\varepsilon \rightarrow-\varepsilon$	symmetry" $\varepsilon \to -\varepsilon$ in (10.3.9) and (10.3.10), we see that the limit $\gamma = (1-\delta\varepsilon) \to$
mielke-geometrodynamics_ F01074	216	1.000	\gamma=(1-\delta \varepsilon) \rightarrow	$\gamma=(1-\delta \varepsilon) \rightarrow$	symmetry" $\varepsilon \to -\varepsilon$ in (10.3.9) and (10.3.10), we see that the limit $\gamma = (1-\delta\varepsilon) \to$
mielke-geometrodynamics_ F01075	216	1.000	\gamma \neq 0	$\gamma \neq 0$	(10.2.1). For $\gamma \neq 0$, the vacuum solution (10.3.9) yields
mielke-geometrodynamics_ F01076	216	1.000	\varepsilon \rightarrow 0	$\varepsilon \rightarrow 0$	in the limit $\varepsilon \to 0$ of *weak* axial torsion coupling, the translational field momenta
mielke-geometrodynamics_ F01077	216	1.000	H_{\alpha}^{\star}=\gamma \vartheta_{\alpha} / \ell	$H_{\alpha}^{\star}=\gamma \vartheta_{\alpha} / \ell$	i.e., $H_{\alpha}^{\star} = \gamma\vartheta_{\alpha}/\ell$, similar to the case of the EC action. Conversely, in the limit
mielke-geometrodynamics_ F01078	216	1.000	\varepsilon \rightarrow \infty	$\varepsilon \rightarrow \infty$	$\varepsilon \to \infty$ of *strong* axial torsion coupling, the translational momenta (10.3.9) become
mielke-geometrodynamics_ F01079	216	1.000	H_{\alpha}=-\gamma \vartheta_{\alpha} / \delta \ell^{2}	$H_{\alpha}=-\gamma \vartheta_{\alpha} / \delta \ell^{2}$	$H_{\alpha} = -\gamma\vartheta_{\alpha}/\delta\ell^2$, and the rotational momenta turn out to be $H_{\alpha}^{\star} = 0$. This implies

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mielke-geometrodynamics_F01080	216	1.000	`H_{\alpha}^{\star}=0`	$H_{\alpha}^{\star}=0$	$H_\alpha = -\gamma\vartheta_\alpha/\delta\ell^2$, and the rotational momenta turn out to be $H_\alpha^\star = 0$. This implies
mielke-geometrodynamics_F01081	217	1.000	`\{H, d t, d \mathscr{A}, d h\}`	$\{H, d t, d \mathscr{A}, d h\}$	a basis $\{H, dt, d\mathscr{A}, dh\}$ of one-forms can be constructed that, for nonzero torsion
mielke-geometrodynamics_F01082	217	1.000	`H=0`	$H=0$	For $H=0$, the three one-forms on the right-hand side become linearly dependent,
mielke-geometrodynamics_F01083	217	0.998	`H_{\alpha} \neq 0`	$H_{\alpha} \neq 0$	thereby sending us back to the exact solution (10.3.9). However, for $H_\alpha \neq 0$, an exact
mielke-geometrodynamics_F01084	217	1.000	`d \mathscr{A}`	$d \mathscr{A}$	can be proposed in which one coordinate “leg,” for instance $d\mathscr{A}$, needs to be con-
mielke-geometrodynamics_F01085	218	1.000	`d^{*} \vartheta_{\alpha}=0`	$d^{*} \vartheta_{\alpha}=0$	generalizes the Hilbert–de Donder or *transversality condition* $d^*\vartheta_\alpha = 0$ for the
mielke-geometrodynamics_F01086	218	1.000	`\eta_{\alpha}={ }^{*} \vartheta_{\alpha}`	$\eta_{\alpha}={ }^{*} \vartheta_{\alpha}$	To this end, we will employ the relation $\eta_\alpha = {}^*\vartheta_\alpha$ for the η_α basis and iterate the
mielke-geometrodynamics_F01087	219	1.000	`m=\varepsilon / \ell`	$m=\varepsilon / \ell$	where $m=\varepsilon/\ell$, results in vacuum and the gauge-covariant Hilbert–de Donder or
mielke-geometrodynamics_F01088	219	0.782	`\vartheta^{\alpha}=E_{i}{ }^{\alpha} d x^{i}`	$\vartheta^{\alpha}=E_{i}{ }^{\alpha} d x^{i}$	is as follows: In 3D, the coframe $\vartheta^\alpha = E_i{}^\alpha dx^i$ has $3\times 3 = 9$ components, of which
mielke-geometrodynamics_F01089	219	0.782	`3 \times 3=9`	$3 \times 3=9$	is as follows: In 3D, the coframe $\vartheta^\alpha = E_i{}^\alpha dx^i$ has $3\times 3 = 9$ components, of which
mielke-geometrodynamics_F01090	219	0.924	`S O(1,2)`	$S O(1,2)$	3 are pure gauge due to the $SO(1,2)$ Lorentz transformation $\vartheta'^\alpha = \Lambda_\beta{}^\alpha(x)\vartheta^\beta$. The
mielke-geometrodynamics_F01091	219	0.924	`\vartheta^{\prime \alpha}=\Lambda_{\beta}{ }^{\alpha}(x) \vartheta^{\beta}`	$\vartheta^{\prime \alpha}=\Lambda_{\beta}{ }^{\alpha}(x) \vartheta^{\beta}$	3 are pure gauge due to the $SO(1,2)$ Lorentz transformation $\vartheta'^\alpha = \Lambda_\beta{}^\alpha(x)\vartheta^\beta$. The
mielke-geometrodynamics_F01092	220	0.992	`2 / \theta_{\mathrm{L}}`	$2 / \theta_{\mathrm{L}}$	contributes to the mass with an “anyon”-type g-factor of $2/\theta_{\mathrm{L}}$; cf. JACKIW & NAIR
mielke-geometrodynamics_F01093	220	1.000	`\theta_{\mathrm{T}} \theta_{\mathrm{L}} / 2 \chi=1`	$\theta_{\mathrm{T}} \theta_{\mathrm{L}} / 2 \chi=1$	to be constrained to $\theta_{\mathrm{T}}\theta_{\mathrm{L}}/2\chi = 1$, or at least to be positive in order to avoid nega-
mielke-geometrodynamics_F01094	220	1.000	`\theta_{\mathrm{T}} \neq 0`	$\theta_{\mathrm{T}} \neq 0$	$\theta_{\mathrm{T}} \neq 0$, the vacuum angle of the translational Chern–Simons term C_{T}. In any case,
mielke-geometrodynamics_F01095	221	1.000	`\varepsilon \neq \tilde{\varepsilon}`	$\varepsilon \neq \tilde{\varepsilon}$	Then, for *different* continuous parameters $\varepsilon \neq \tilde\varepsilon$, we obtain
mielke-geometrodynamics_F01096	221	1.000	`\tilde{\theta}_{\mathrm{L}}`	$\tilde{\theta}_{\mathrm{L}}$	parent. Consequently, the constants θ_{L} and $\tilde\theta_{\mathrm{L}}$ give rise to the two central charges
mielke-geometrodynamics_F01097	222	1.000	`C^{k l}=C^{l k}`	$C^{k l}=C^{l k}$	is the one-form associated with the symmetric Cotton tensor $C^{kl}=C^{lk}$. In general,
mielke-geometrodynamics_F01098	222	0.940	`\Gamma^{\{\lambda \alpha \beta}`	$\Gamma^{\{\lambda \alpha \beta}$	this is a third-order equation in the Levi-Civita connection $\Gamma^{\{\}\alpha\beta}$, i.e., of fourth order
mielke-geometrodynamics_F01099	222	1.000	`T_{\alpha}=D \vartheta_{\alpha}=0`	$T_{\alpha}=D \vartheta_{\alpha}=0$	Due to $T_\alpha = D\vartheta_\alpha = 0$ in Riemannian spacetime, implying $H_\alpha = 0$ in 3D, the mod-
mielke-geometrodynamics_F01100	222	1.000	`H_{\alpha}=0`	$H_{\alpha}=0$	Due to $T_\alpha = D\vartheta_\alpha = 0$ in Riemannian spacetime, implying $H_\alpha = 0$ in 3D, the mod-
mielke-geometrodynamics_F01101	222	0.891	`D H_{\alpha}^{\star}=\delta \ell D H_{\alpha}+(\gamma / \ell) D \vartheta_{\alpha} \equiv 0`	$D H_{\alpha}^{\star}=\delta \ell D H_{\alpha}+(\gamma / \ell) D \vartheta_{\alpha} \equiv 0$	ified S-duality leads to $DH_\alpha^\star = \delta\ell DH_\alpha + (\gamma/\ell)D\vartheta_\alpha \equiv 0$, and the two-form $D^{\{\}}H_{[\alpha\beta]}$
mielke-geometrodynamics_F01102	222	0.891	`D^{[4]} H_{[\alpha \beta]}`	$D^{[4]} H_{[\alpha \beta]}$	ified S-duality leads to $DH_\alpha^\star = \delta\ell DH_\alpha + (\gamma/\ell)D\vartheta_\alpha \equiv 0$, and the two-form $D^{\{\}}H_{[\alpha\beta]}$
mielke-geometrodynamics_F01103	222	0.539	`G_{\alpha}=R_{\alpha}^{\left\{{ }^{\}} \star\right.}`	$G_{\alpha}=R_{\alpha}^{\left\{{ }^{\}} \star\right.}$	in Riemannian spacetime. Since $G_\alpha = R_\alpha^{\{\}\star}$ is the Einstein current two-form in 3D,
mielke-geometrodynamics_F01104	222	0.990	`\widetilde{\Sigma}_{\alpha}:=\Sigma_{\alpha}-D^{\{ \}} \mu_{\alpha}`	$\widetilde{\Sigma}_{\alpha}:=\Sigma_{\alpha}-D^{\{ \}} \mu_{\alpha}$	*symmetric* Belinfante–Rosenfeld two-form $\widetilde\Sigma_\alpha := \Sigma_\alpha - D^{\{\}}\mu_\alpha$ as source.
mielke-geometrodynamics_F01105	223	1.000	`E=p c_{\mathrm{F}}`	$E=p c_{\mathrm{F}}$	spectrum is $E = pc_{\mathrm{F}}$, where c_{F} is the effective speed of fermions in the 2D carbon
mielke-geometrodynamics_F01106	223	1.000	`c_{\mathrm{F}}`	c_{F}	spectrum is $E = pc_{\mathrm{F}}$, where c_{F} is the effective speed of fermions in the 2D carbon

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mielke-geometrodynamics_F01107	223	0.825	`\mathrm{SO}(3)`	$\mathrm{SO}(3)$	space(time): in the Euclidean case, the covering group of the rotation group SO(3)
mielke-geometrodynamics_F01108	224	0.902	`s=1`	$s=1$	On the other hand, for Lorentzian signature $s=1$, the covering group of SO(1,2) is
mielke-geometrodynamics_F01109	224	0.942	`S L(2, \mathbb{R})`	$S L(2, \mathbb{R})$	isomorphic to the real group $SL(2,\mathbb{R})$, or as well to the symplectic algebra $\mathfrak{sp}(2,\mathbb{R})$.
mielke-geometrodynamics_F01110	224	0.942	`\mathfrak{s p}(2, \mathbb{R})`	$\mathfrak{s p}(2, \mathbb{R})$	isomorphic to the real group $SL(2,\mathbb{R})$, or as well to the symplectic algebra $\mathfrak{sp}(2,\mathbb{R})$.
mielke-geometrodynamics_F01111	224	1.000	`\gamma_{\alpha} \gamma_{\beta}+\gamma_{\beta} \gamma_{\alpha}=2 g_{\alpha \beta}`	$\gamma_{\alpha} \gamma_{\beta}+\gamma_{\beta} \gamma_{\alpha}=2 g_{\alpha \beta}$	thus providing a realization of the Clifford algebra $\gamma_\alpha\gamma_\beta+\gamma_\beta\gamma_\alpha=2g_{\alpha\beta}$ in 3D.
mielke-geometrodynamics_F01112	224	1.000	`D \psi:=d \psi-\frac{1}{2} \gamma_{\alpha} \wedge \Gamma^{\star \alpha} \psi`	$D \psi:=d \psi-\frac{1}{2} \gamma_{\alpha} \wedge \Gamma^{\star \alpha} \psi$	where $D\psi := d\psi - \frac{1}{2}\gamma_\alpha \wedge \Gamma^{\star\alpha}\psi$ denotes the covariant spinor derivative in 3D.
mielke-geometrodynamics_F01113	224	1.000	`L_{\mathrm{D}}`	L_{D}	is obtained by varying L_{D} with respect to $\overline{\Psi}$. Since
mielke-geometrodynamics_F01114	224	1.000	`\bar{\Psi}`	$\bar{\Psi}$	is obtained by varying L_{D} with respect to $\overline{\Psi}$. Since
mielke-geometrodynamics_F01115	224	1.000	`\hat{D}`	$\hat{D}$	The curvature associated with the deformed covariant derivative $\hat{D}$ is
mielke-geometrodynamics_F01116	225	0.999	`F:=d A`	$F:=d A$	field strength reads $F:=dA$, i.e., it remains an exact two-form. In 3D, the Maxwell
mielke-geometrodynamics_F01117	225	0.539	`d\left(A \wedge^{*} F\right) / 2`	$d\left(A \wedge^{*} F\right) / 2$	However, this three-form is gauge-invariant only modulo an exact form $d(A\wedge {}^*F)/2$.
mielke-geometrodynamics_F01118	225	0.528	`\square:=d^{*} d^{*}-{ }^{*} d^{*} d`	$\square:=d^{*} d^{*}-{ }^{*} d^{*} d$	where $\square := d^*d^* - {}^*d^*d$ is the d'Alembertian in $2+1$ dimensions. Since (10.6.3) is a
mielke-geometrodynamics_F01119	225	1.000	`\lambda=\hbar / c m_{\text {photon }}`	$\lambda=\hbar / c m_{\text {photon }}$	reasons, via the Compton wavelength $\lambda=\hbar/cm_{\text{photon}}$ for massive photons.
mielke-geometrodynamics_F01120	226	0.987	`x^{\alpha}`	x^{α}	vention that x^α and y^α are spacelike orthogonal vectors that span the (x,y)-plane
mielke-geometrodynamics_F01121	226	0.987	`y^{\alpha}`	y^{α}	vention that x^α and y^α are spacelike orthogonal vectors that span the (x,y)-plane
mielke-geometrodynamics_F01122	226	0.987	`(x, y)`	(x, y)	vention that x^α and y^α are spacelike orthogonal vectors that span the (x,y)-plane
mielke-geometrodynamics_F01123	226	1.000	`n^{\alpha}`	n^{α}	Moreover, the vector n^α is a timelike unit vector normal to the hypersurface with
mielke-geometrodynamics_F01124	226	1.000	`n^{\alpha} n_{\alpha}=s`	$n^{\alpha} n_{\alpha}=s$	$n^\alpha n_\alpha = s$, the signature s of our 3D spacetime. Following SOLENG (1992), ANANDAN
mielke-geometrodynamics_F01125	226	1.000	`\tau_{\alpha}^{\star}`	$\tau_{\alpha}^{\star}$	(1994), and BAKKE et al. (2009), we assume that the two-forms Σ_α and $\tau_\alpha^\star$ of the
mielke-geometrodynamics_F01126	227	1.000	`X`	X	of the defect; cf. Fig. 10.2. From the specification (10.7.1) of the one-forms X and
mielke-geometrodynamics_F01127	227	1.000	`Y`	Y	Y, it can easily be inferred that the only nonzero components are $\Sigma_{\hat{0}}\neq 0$ and $\tau_{\hat{1}\hat{2}}=$
mielke-geometrodynamics_F01128	227	1.000	`\Sigma_{\hat{0}} \neq 0`	$\Sigma_{\hat{0}} \neq 0$	Y, it can easily be inferred that the only nonzero components are $\Sigma_{\hat{0}}\neq 0$ and $\tau_{\hat{1}\hat{2}}=$
mielke-geometrodynamics_F01129	227	1.000	`\tau_{\hat{1} \hat{2}}=`	$\tau_{\hat{1} \hat{2}}=$	Y, it can easily be inferred that the only nonzero components are $\Sigma_{\hat{0}}\neq 0$ and $\tau_{\hat{1}\hat{2}}=$
mielke-geometrodynamics_F01130	227	0.975	`-\tau_{\hat{2} \hat{1}} \neq 0`	$-\tau_{\hat{2} \hat{1}} \neq 0$	$-\tau_{\hat{2}\hat{1}}\neq 0$.
mielke-geometrodynamics_F01131	227	1.000	`x^{[\alpha} y^{\beta]} R_{\alpha \beta}=R_{\hat{1} \hat{2}}=-R_{\hat{2} \hat{1}} \neq 0`	$x^{[\alpha} y^{\beta]} R_{\alpha \beta}=R_{\hat{1} \hat{2}}=-R_{\hat{2} \hat{1}} \neq 0$	with x^α and y^α reveal that $x^{[\alpha}y^{\beta]}R_{\alpha\beta}=R_{\hat{1}\hat{2}}=-R_{\hat{2}\hat{1}}\neq 0$ are the only nonvanishing
mielke-geometrodynamics_F01132	227	0.511	`\left.N^{\alpha}=n\right\rfloor \vartheta^{\alpha}`	$\left.N^{\alpha}=n\right\rfloor \vartheta^{\alpha}$	We recall that $N^\alpha = n\rfloor\vartheta^\alpha$ is the lapse and shift vector in the (2+1) decomposition à
mielke-geometrodynamics_F01133	228	1.000	`d(\mathscr{A} \wedge d \mathscr{A})`	$d(\mathscr{A} \wedge d \mathscr{A})$	Pointryagin-type term $d(\mathscr{A}\wedge d\mathscr{A})$ from the axial torsion. Moreover, the Nieh–Yan

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mielke-geometrodynamics_F01134	228	1.000	d C_{\mathrm{T}}	dC_{T}	term dC_{T} proportional to $d^{*}\mathscr{A}$ *vanishes identically* for this example of a spinning
mielke-geometrodynamics_F01135	228	1.000	d^{*} \mathscr{A}	$d^{*}\mathscr{A}$	term dC_{T} proportional to $d^{*}\mathscr{A}$ *vanishes identically* for this example of a spinning
mielke-geometrodynamics_F01136	228	1.000	\mathscr{N}=1	$\mathscr{N}=1$	try generator, i.e., $\mathscr{N}=1$, represents the simplest consistent coupling of a Rarita–
mielke-geometrodynamics_F01137	228	0.961	\left.\Psi_{\alpha}:=e_{\alpha}\right\rfloor \Psi	$\left.\Psi_{\alpha}:=e_{\alpha}\right\rfloor \Psi$	as $\Psi_{\alpha}:=e_{\alpha}\rfloor\Psi$ involves the tetrad that is inverse to the frame in the anholonomic
mielke-geometrodynamics_F01138	228	1.000	i, j, k, \ldots=0,1,2	$i, j, k, \ldots=0,1,2$	run over $i, j, k, \ldots=0, 1, 2$, whereas $\alpha, \beta, \ldots=\hat{0}, \hat{1}, \hat{2}$ for the anholonomic indices.
mielke-geometrodynamics_F01139	228	1.000	\alpha, \beta, \ldots=\hat{0}, \hat{1}, \hat{2}	$\alpha, \beta, \ldots=\hat{0}, \hat{1}, \hat{2}$	run over $i, j, k, \ldots=0, 1, 2$, whereas $\alpha, \beta, \ldots=\hat{0}, \hat{1}, \hat{2}$ for the anholonomic indices.
mielke-geometrodynamics_F01140	228	1.000	\bar{\Psi}:=\Psi^{\dagger} \gamma^{0}	$\bar{\Psi}:=\Psi^{\dagger} \gamma^{0}$	$\overline{\Psi}:=\Psi^{\dagger}\gamma^{0}$.
mielke-geometrodynamics_F01141	228	1.000	\Psi:=\Psi_{\alpha} \vartheta^{\alpha}	$\Psi:=\Psi_{\alpha} \vartheta^{\alpha}$	[3]In four dimensions (4D), the RS field $\Psi:=\Psi_{\alpha}\vartheta^{\alpha}$ entering (10.8.1) is a *Majorana-spinor*-valued
mielke-geometrodynamics_F01142	228	0.700	\Psi=C \bar{\Psi}^{\mathrm{T}}	$\Psi=C \bar{\Psi}^{\mathrm{T}}$	i.e., $\Psi=C\overline{\Psi}^{\mathrm{T}}$, where C is the charge conjugation matrix given by $C=-i\gamma_{0}$ satisfying $C^{\dagger}=$
mielke-geometrodynamics_F01143	228	0.700	C=-i \gamma_{0}	$C=-i \gamma_{0}$	i.e., $\Psi=C\overline{\Psi}^{\mathrm{T}}$, where C is the charge conjugation matrix given by $C=-i\gamma_{0}$ satisfying $C^{\dagger}=$
mielke-geometrodynamics_F01144	228	0.700	C^{\dagger}=	$C^{\dagger}=$	i.e., $\Psi=C\overline{\Psi}^{\mathrm{T}}$, where C is the charge conjugation matrix given by $C=-i\gamma_{0}$ satisfying $C^{\dagger}=$
mielke-geometrodynamics_F01145	228	1.000	C^{-1}, \quad C^{\mathrm{T}}=-C	$C^{-1}, \quad C^{\mathrm{T}}=-C$	$C^{-1}, \quad C^{\mathrm{T}}=-C$ and $C^{-1}\gamma^{\alpha}C=-(\gamma^{\alpha})^{\mathrm{T}}$. Consequently,
mielke-geometrodynamics_F01146	228	1.000	C^{-1} \gamma^{\alpha} C=-\left(\gamma^{\alpha}\right)^{\mathrm{T}}	$C^{-1} \gamma^{\alpha} C=-\left(\gamma^{\alpha}\right)^{\mathrm{T}}$	$C^{-1}, \quad C^{\mathrm{T}}=-C$ and $C^{-1}\gamma^{\alpha}C=-(\gamma^{\alpha})^{\mathrm{T}}$. Consequently,
mielke-geometrodynamics_F01147	228	1.000	\gamma^{\alpha}	γ^{α}	For the real Majorana representation, all γ^{α} are purely imaginary, and the components of the
mielke-geometrodynamics_F01148	229	0.803	L_{\mathrm{RS}}	L_{RS}	The generalization of L_{RS} analyzed by MIELKE & MAGGIOLO (2012) exists only
mielke-geometrodynamics_F01149	230	0.995	\vartheta^{\alpha}, \Gamma_{\alpha}^{\star}, \Psi	$\vartheta^{\alpha}, \Gamma_{\alpha}^{\star}, \Psi$	independent variables $(\vartheta^{\alpha}, \Gamma_{\alpha}^{\star}, \Psi)$ yields
mielke-geometrodynamics_F01150	230	1.000	c	c	where σ stands in for a spinor-valued zero-form and c is a real constant. Let us probe
mielke-geometrodynamics_F01151	230	1.000	\overline{c \gamma \sigma}=c \bar{\sigma} \gamma	$\overline{c \gamma \sigma}=c \bar{\sigma} \gamma$	where $\overline{c\gamma\sigma}=c\overline{\sigma}\gamma$ holds for the Dirac adjoint.
mielke-geometrodynamics_F01152	230	1.000	\delta L / \delta \Gamma_{\alpha}^{\star} \cong 0	$\delta L / \delta \Gamma_{\alpha}^{\star} \cong 0$	In the following, let us assume that the second field equation $\delta L/\delta\Gamma_{\alpha}^{\star}\cong 0$ satisfies,
mielke-geometrodynamics_F01153	231	1.000	c=-m	$c=-m$	$\tau_{\alpha}^{\star}$, vanish. Moreover, we require $c=-m$, in order to eliminate kinetic terms like
mielke-geometrodynamics_F01154	231	1.000	\gamma \wedge D \Psi	$\gamma \wedge D \Psi$	$\gamma\wedge D\Psi$. Then, using the 3D formula
mielke-geometrodynamics_F01155	232	0.999	L_{\infty}	L_{∞}	for the coupling constants of the bosonic part of L_{∞}. Consequently, massless RS
mielke-geometrodynamics_F01156	232	1.000	\mathscr{N}=2	$\mathscr{N}=2$	More recently, ALVAREZ et al. (2015) considered an $\mathscr{N}=2$ supersymmetry with-
mielke-geometrodynamics_F01157	233	0.988	\mathrm{D}=3	$\mathrm{D}=3$	Deser S, McCarthy J (1990) Self-dual formulations of $\mathrm{D}=3$ gravity theories. Phys Lett B
mielke-geometrodynamics_F01158	233	0.666	(2+1)	$(2+1)$	Garcia-Compean H, Obregon O, Ramirez C, Sabido M (2001) On S duality in (2+1) Chern–Simons
mielke-geometrodynamics_F01159	237	0.772	1 / 2	$1/2$	1/2 are described by bispinor representations $D^{(-\frac{1}{2},\frac{3}{2})}\otimes D^{(\frac{1}{2},\frac{3}{2})}(\widetilde{SO}_{\circ}(1,3))$ of the
mielke-geometrodynamics_F01160	237	0.772	D^{\left(-\frac{1}{2}, \frac{3}{2}\right)} \otimes D^{\left(\frac{1}{2}, \frac{3}{2}\right)}\left(\widetilde{SO}_{\circ}(1,3)\right)	$D^{\left(-\frac{1}{2}, \frac{3}{2}\right)} \otimes D^{\left(\frac{1}{2}, \frac{3}{2}\right)}\left(\widetilde{SO}_{\circ}(1,3)\right)$	1/2 are described by bispinor representations $D^{(-\frac{1}{2},\frac{3}{2})}\otimes D^{(\frac{1}{2},\frac{3}{2})}(\widetilde{SO}_{\circ}(1,3))$ of the

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mielke-geometrodynamics_F01161	237	1.000	`S O_{\circ}(1,3)`	$S O_{\circ}(1,3)$	proper Lorentz group $SO_{\circ}(1,3)$ (GEL'FAND et al. 1963).[2] These representations
mielke-geometrodynamics_F01162	237	0.983	`P_{\circ}:=\mathbb{R}^{4} \otimes S O_{\circ}(1,3)`	$P_{\circ}:=\mathbb{R}^{4} \otimes S O_{\circ}(1,3)$	resentations of the (restricted) Poincaré group $P_{\circ} := \mathbb{R}^4 \ltimes SO_{\circ}(1,3)$ of flat
mielke-geometrodynamics_F01163	237	1.000	`\psi_{\infty}:=\psi_{1 / 2} \otimes \psi_{5 / 2} \otimes \psi_{9 / 2} \otimes \ldots`	$\psi_{\infty}:=\psi_{1 / 2} \otimes \psi_{5 / 2} \otimes \psi_{9 / 2} \otimes \ldots$	$\psi_{\infty} := \psi_{1/2} \otimes \psi_{5/2} \otimes \psi_{9/2} \otimes \ldots$
mielke-geometrodynamics_F01164	237	0.706	`\mathrm{SO}(s, n-s)`	$\mathrm{SO}(s, n-s)$	(pseudo-orthogonal) structure group $\mathrm{SO}(s, n-s)$ of the tangent bundle of an
mielke-geometrodynamics_F01165	237	1.000	`s \neq 0`	$s \neq 0$	n-dimensional manifold[3] of signature s. Provided $s \neq 0$, this topological space
mielke-geometrodynamics_F01166	237	1.000	`S O_{\circ}(s, n-s)`	$S O_{\circ}(s, n-s)$	ing to homotopy theory. Concerning the connected component $SO_{\circ}(s, n-s)$ of the
mielke-geometrodynamics_F01167	237	0.929	`\widetilde{S O}_{\circ}(s, n-s)(=: \operatorname{Spin}(n)`	$\widetilde{S O}_{\circ}(s, n-s)(=: \operatorname{Spin}(n)$	$\widetilde{SO}_{\circ}(s, n-s)$ (=: Spin(n) for $s = 0$), whose irreducible representations provide
mielke-geometrodynamics_F01168	237	0.929	`s=0)`	$s=0)$	$\widetilde{SO}_{\circ}(s, n-s)$ (=: Spin(n) for $s = 0$), whose irreducible representations provide
mielke-geometrodynamics_F01169	238	1.000	`\Lambda: \tilde{\mathrm{G}} \rightarrow \mathrm{G}`	$\Lambda: \tilde{\mathrm{G}} \rightarrow \mathrm{G}$	This transition is achieved by a group homomorphism $\Lambda : \tilde{\mathrm{G}} \rightarrow \mathrm{G}$, which is
mielke-geometrodynamics_F01170	238	1.000	`C(s, n-s)`	$C(s, n-s)$	based on the following explicit construction: Let $C(s, n-s)$ denote the so-called
mielke-geometrodynamics_F01171	238	1.000	`\gamma_{\alpha}`	γ_{α}	Clifford algebra, which is spanned by *generalized* complex Dirac matrices γ_{α}. These
mielke-geometrodynamics_F01172	238	1.000	`N \times N`	$N \times N$	and admit realizations by $N \times N$ matrices for which $N = 2^{[n/2]}$ corresponds to the
mielke-geometrodynamics_F01173	238	1.000	`N=2^{[n / 2]}`	$N=2^{[n / 2]}$	and admit realizations by $N \times N$ matrices for which $N = 2^{[n/2]}$ corresponds to the
mielke-geometrodynamics_F01174	238	1.000	`\mathbf{E}^{n}`	$\mathbf{E}^{n}$	Let the transformations of the Euclidean vector space $\mathbf{E}^n$ with coordinates x^{α} be
mielke-geometrodynamics_F01175	238	0.963	`\mathrm{g} \in S O(s, n-s)`	$\mathrm{g} \in S O(s, n-s)$	determined by the (Lorentz) rotations $\mathrm{g} \in SO(s, n-s)$ on behalf of the group action
mielke-geometrodynamics_F01176	238	0.983	`2^{N}`	2^{N}	to a 2^N-dimensional bundle, the so-called Clifford bundle CV. For details, see ATIYAH
mielke-geometrodynamics_F01177	239	0.976	`\widetilde{S O}_{\circ}(1,3) \approx S L(2, \mathbb{C})`	$\widetilde{S O}_{\circ}(1,3) \approx S L(2, \mathbb{C})$	orthochronous Lorentz group $SO_{\circ}(1,3)$ is represented by $\widetilde{SO}_{\circ}(1,3) \approx SL(2, \mathbb{C})$
mielke-geometrodynamics_F01178	239	1.000	`\widetilde{S O}_{\circ}(s, n-s)`	$\widetilde{S O}_{\circ}(s, n-s)$	infinitesimal generators of the group $\widetilde{SO}_{\circ}(s, n-s)$ are
mielke-geometrodynamics_F01179	239	0.588	`\mathrm{P}_{0}:=\mathbb{R}^{\mathrm{n}} \otimes \widetilde{S O}`	$\mathrm{P}_{0}:=\mathbb{R}^{\mathrm{n}} \otimes \widetilde{S O}$	as induced representations of the "covering" Poincaré group $\mathrm{P}_0 := \mathbb{R}^{\mathrm{n}} \ltimes \widetilde{SO}_{\circ}$
mielke-geometrodynamics_F01180	239	1.000	`(s, n-s)`	$(s, n-s)$	$(s, n-s)$ in a curved spacetime. It is necessary to obtain from a more general
mielke-geometrodynamics_F01181	239	1.000	`L^{\mathrm{g}}(M)`	$L^{\mathrm{g}}(M)$	"geometric arena" the bundle $L^{\mathrm{g}}(M)$ of orthogonal frames concerning the foundation
mielke-geometrodynamics_F01182	239	0.974	`L_{\circ}^{g}(M)`	$L_{\circ}^{g}(M)$	a *spin structure* is imposed on a base space M as follows: Besides $L_{\circ}^{\mathrm{g}}(M)$, the
mielke-geometrodynamics_F01183	239	0.998	`\mathrm{f}: \tilde{L}(M) \rightarrow L_{\circ}^{g}(M)`	$\mathrm{f}: \tilde{L}(M) \rightarrow L_{\circ}^{g}(M)$	is introduced (MILNOR 1963) together with the bundle map $\mathrm{f} : \tilde{L}(M) \rightarrow L_{\circ}^{\mathrm{g}}(M)$,
mielke-geometrodynamics_F01184	239	0.950	`L_{o}^{g}(M)`	$L_{o}^{g}(M)$	A spin structure over M is equivalent to a double covering of $L_{\circ}^{\mathrm{g}}(M)$. This spin
mielke-geometrodynamics_F01185	239	0.949	`\tilde{L}^{\prime}(M), f^{\prime}`	$\tilde{L}^{\prime}(M), f^{\prime}$	structure is identical to a second one that is represented by $(\tilde{L}'(M), f')$ if the iso-
mielke-geometrodynamics_F01186	239	1.000	`f^{\prime}`	$f^{\prime}$	bundle mapping f'. Additionally, a *spin manifold* is understood as a Riemannian
mielke-geometrodynamics_F01187	239	1.000	`\mathrm{w}_{2} \in \mathrm{H}^{2}\left(M, \mathbb{Z}_{2}\right)`	$\mathrm{w}_{2} \in \mathrm{H}^{2}\left(M, \mathbb{Z}_{2}\right)$	it is necessary and sufficient that the Stiefel–Whitney class $\mathrm{w}_2 \in \mathrm{H}^2(M, \mathbb{Z}_2)$ of

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mielke-geometrodynamics_F01188	239	■ 1.000	`\mathrm{g}^{\text {th }}`	$\mathrm{g}^{\text {th }}$	the base space M be zero. Here the g^{th} Stiefel–Whitney class is considered as the
mielke-geometrodynamics_F01189	240	■ 1.000	`\mathrm{H}^{\mathrm{g}}\left(M, \mathbb{Z}_{2}\right)`	$\mathrm{H}^{\mathrm{g}}\left(M, \mathbb{Z}_{2}\right)$	characteristic class of the g^{th} relative cohomology group $\mathrm{H}^{\mathrm{g}}(M, \mathbb{Z}_2)$ of the manifold
mielke-geometrodynamics_F01190	240	■ 1.000	`\tilde{s}=\operatorname{dim} \mathrm{H}^{1}\left(M, \mathbb{Z}_{2}\right)`	$\tilde{s}=\operatorname{dim} \mathrm{H}^{1}\left(M, \mathbb{Z}_{2}\right)$	defined by the dimension $\tilde{s} = \dim \mathrm{H}^1(M, \mathbb{Z}_2)$ of the first relative cohomology group
mielke-geometrodynamics_F01191	240	■ 0.813	`\tilde{s}=2^{v}`	$\tilde{s}=2^{v}$	(MILNOR 1963). Accordingly, there are exactly $\tilde{s} = 2^{\nu}$ spin structures possible
mielke-geometrodynamics_F01192	240	■ 0.997	`\nu`	ν	hypersurface consists of the connected sum of ν handlebodies (“wormholes”) $W^3 :=$
mielke-geometrodynamics_F01193	240	■ 0.997	`W^{3}:=`	$W^{3}:=$	hypersurface consists of the connected sum of ν handlebodies (“wormholes”) $W^3 :=$
mielke-geometrodynamics_F01194	240	■ 0.999	`\mathrm{w}_{1}=0`	$\mathrm{w}_{1}=0$	s) bundles. The last bundle can be oriented if $\mathrm{w}_1 = 0$ is valid, while the number of
mielke-geometrodynamics_F01195	240	■ 0.993	`\mathrm{H}^{\circ}\left(M, \mathbb{Z}_{2}\right)`	$\mathrm{H}^{\circ}\left(M, \mathbb{Z}_{2}\right)$	distinguishable orientations has to be processed out of the dimension of $\mathrm{H}^{\circ}(M, \mathbb{Z}_2)$
mielke-geometrodynamics_F01196	240	■ 1.000	`S^{7}`	S^{7}	dimensional sphere S^7 (MILNOR 1965). Attention can be drawn to other topologically
mielke-geometrodynamics_F01197	240	■ 0.998	`(1968,1970)`	$(1968,1970)$	However, it was GEROCH (1968, 1970) who pointed out an intuitive and in the
mielke-geometrodynamics_F01198	240	■ 0.969	`T(M)`	$T(M)$	structures on $T(M)$ is dependent only on the mapping f (ISHAM 1978).
mielke-geometrodynamics_F01199	240	■ 0.925	`S U(3)`	$S U(3)$	invariance or Gell-Mann’s $SU(3)$-group as well into these global constructions, the
mielke-geometrodynamics_F01200	241	■ 1.000	`\tilde{\mathrm{L}}(\mathrm{M})`	$\tilde{\mathrm{L}}(\mathrm{M})$	*associated* to the spin bundle $\tilde{\mathrm{L}}(\mathrm{M})$. For this reason, it is also called an *associ-*
mielke-geometrodynamics_F01201	241	■ 0.614	`D^{\mathrm{F}}`	D^{F}	plex) representation D^{F} of dimension $N = 2^{[n/2]}$ of the structure group $\widetilde{SO}_{\circ}$
mielke-geometrodynamics_F01202	241	■ 0.614	`\widetilde{S O}_{\circ}`	$\widetilde{S O}_{\circ}$	plex) representation D^{F} of dimension $N = 2^{[n/2]}$ of the structure group $\widetilde{SO}_{\circ}$
mielke-geometrodynamics_F01203	241	■ 0.986	`\widetilde{S O}_{\circ}(1,3)`	$\widetilde{S O}_{\circ}(1,3)$	$(s, n-s)$, which contains the physical Lorentz group $\widetilde{SO}_{\circ}(1,3)$ as a subgroup.
mielke-geometrodynamics_F01204	241	■ 0.996	`S L(2, \mathbb{C}) \approx \widetilde{S O}_{\circ}(1,3)`	$S L(2, \mathbb{C}) \approx \widetilde{S O}_{\circ}(1,3)$	denotes the bispinor representation of the covering group $SL(2, \mathbb{C}) \approx \widetilde{SO}_{\circ}(1,3)$ of
mielke-geometrodynamics_F01205	241	■ 1.000	`\left(-\frac{1}{2}, \frac{3}{2}\right) \oplus\left( \frac{1}{2}, \frac{3}{2}\right)`	$\left(-\frac{1}{2}, \frac{3}{2}\right) \oplus\left(\frac{1}{2}, \frac{3}{2}\right)$	the proper Lorentz group, which itself is the direct sum $\left(-\frac{1}{2}, \frac{3}{2}\right) \oplus \left(\frac{1}{2}, \frac{3}{2}\right)$ of spin-
mielke-geometrodynamics_F01206	241	■ 0.999	`\frac{1}{2}`	$\frac{1}{2}$	$\frac{1}{2}$ representations labeled according to GEL’FAND et al. (1963). This formulation
mielke-geometrodynamics_F01207	241	■ 1.000	`\tilde{O}(1,3)`	$\tilde{O}(1,3)$	group $\tilde{O}(1,3)$ if required. Concerning the spinor basis $\tilde{b}_{(1)}, i = 1, \ldots, N$, their cross
mielke-geometrodynamics_F01208	241	■ 1.000	`\tilde{b}_{(1)}, i=1, \ldots, N`	$\tilde{b}_{(1)}, i=1, \ldots, N$	group $\tilde{O}(1,3)$ if required. Concerning the spinor basis $\tilde{b}_{(1)}, i = 1, \ldots, N$, their cross
mielke-geometrodynamics_F01209	242	■ 0.999	`\mathscr{G}_{S}`	$\mathscr{G}_{S}$	of the group $\mathscr{G}_S$ of (local) spin gauge transformations. This group results from $\mathscr{G}_p$, a
mielke-geometrodynamics_F01210	242	■ 0.999	`\mathscr{G}_{p}`	$\mathscr{G}_{p}$	of the group $\mathscr{G}_S$ of (local) spin gauge transformations. This group results from $\mathscr{G}_p$, a
mielke-geometrodynamics_F01211	242	■ 1.000	`\tilde{L}(M)`	$\tilde{L}(M)$	The spin bundle $\tilde{L}(M)$ can, analogously to the bundle $L^{\mathrm{g}}(M)$ of the orthonormal
mielke-geometrodynamics_F01212	242	■ 1.000	`\gamma^{\mathrm{n}+1}`	$\gamma^{\mathrm{n}+1}$	The matrix γ^{n+1} that is thereby implicitly defined is an axial Lorentz scalar and can
mielke-geometrodynamics_F01213	243	■ 1.000	`\gamma^{n+1}`	γ^{n+1}	If one realizes the Dirac matrices in the “Cartan basis,” then γ^{n+1} can be regarded as

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mielke-geometrodynamics_F01214	243	0.999	`\tilde{\omega}`	$\tilde{\omega}$	The introduction of a $\widetilde{SO}_{\circ}(s, n-s)$-valued 1-form $\tilde{\omega}$ of the connection in the asso-
mielke-geometrodynamics_F01215	243	0.790	`\Gamma_{i}{ }^{\alpha \beta}`	$\Gamma_{i}{ }^{\alpha \beta}$	The local components of the spin connection, i.e., Ricci's rotation coefficients $\Gamma_{i}{}^{\cdot\alpha\beta}$,
mielke-geometrodynamics_F01216	243	0.958	`\Gamma_{i j}{ }^{\cdot k}`	$\Gamma_{i j}{ }^{\cdot k}$	can be generated from the holonomic connection coefficients $\Gamma_{ij}{}^{\cdot k}$ by means of
mielke-geometrodynamics_F01217	244	0.995	`\mathbb{R}^{n} \otimes S O`	$\mathbb{R}^{n} \otimes S O$	If we had started from a spinor representation of the Poincaré group $\mathbb{R}^n \rtimes SO_{\circ}$
mielke-geometrodynamics_F01218	244	0.996	`\mathrm{e}_{\alpha}^{-i}(m)`	$\mathrm{e}_{\alpha}^{-i}(m)$	strived for an extension by inserting the tetrad fields $\mathrm{e}_{\alpha}^{\;i}(m)$, i.e., the local bundle coor-
mielke-geometrodynamics_F01219	245	1.000	`\mathcal{D}(\mathrm{M})`	$\mathcal{D}(\mathrm{M})$	with respect to the group $\mathcal{D}(\mathrm{M})$ of differentiable coordinate transformations is again
mielke-geometrodynamics_F01220	245	1.000	`\delta L_{\mathrm{D}} / \delta \bar{\psi}`	$\delta L_{\mathrm{D}} / \delta \bar{\psi}$	By variation of (11.2.1) for $\delta L_{\mathrm{D}}/\delta\overline{\psi}$ or for $\delta L_{\mathrm{D}}/\delta\psi$, respectively, the generally
mielke-geometrodynamics_F01221	245	1.000	`\delta L_{\mathrm{D}} / \delta \psi`	$\delta L_{\mathrm{D}} / \delta \psi$	By variation of (11.2.1) for $\delta L_{\mathrm{D}}/\delta\overline{\psi}$ or for $\delta L_{\mathrm{D}}/\delta\psi$, respectively, the generally
mielke-geometrodynamics_F01222	246	1.000	`\mathrm{D} \gamma=0`	$\mathrm{D} \gamma=0$	for a covariant constant γ, i.e., for $\mathrm{D}\gamma = 0$. However, the latter condition is globally
mielke-geometrodynamics_F01223	246	1.000	`\tau:=\delta \mathrm{L} / \delta \omega`	$\tau:=\delta \mathrm{L} / \delta \omega$	using the rules concerning exterior forms. According to the definitions $\tau := \delta\mathrm{L}/\delta\omega$
mielke-geometrodynamics_F01224	246	1.000	`\tau=i \bar{\psi} \gamma \psi`	$\tau=i \bar{\psi} \gamma \psi$	$\tau = i\overline{\psi}\gamma\psi$ of its charge current, but also on the (Lorentz) rotational gauge fields of
mielke-geometrodynamics_F01225	246	0.992	`\tau_{s}`	τ_{s}	respect to the current τ or τ_s, and this according to whether the "internal" Yang–
mielke-geometrodynamics_F01226	246	0.999	`j_{n+1}:=i \bar{\psi}^{*} \gamma \gamma^{n+1} \psi`	$j_{n+1}:=i \bar{\psi}^{*} \gamma \gamma^{n+1} \psi$	[6]A corresponding conservation law is also valid for the axial vector current $j_{n+1} := i\overline{\psi}^{*}\gamma\gamma^{n+1}\psi$
mielke-geometrodynamics_F01227	247	1.000	`\mathrm{H}^{\mathrm{n}-1}`	$\mathrm{H}^{\mathrm{n}-1}$	hypersurfaces H^{n-1}, this is not only independent of time, but even *positive definite*,
mielke-geometrodynamics_F01228	247	1.000	`\tau_{s}^{0}`	τ_{s}^{0}	Nevertheless, one could not interpret τ_s^0 as the probability density in a non quantized
mielke-geometrodynamics_F01229	247	0.999	`\tilde{\omega}^{\{ \}}`	$\tilde{\omega}^{\{ \}}$	into the Christoffel-like connection $\tilde{\omega}^{\{\}}$ and the spin contortion that is projected by
mielke-geometrodynamics_F01230	247	0.826	`\mathrm{D}^{\{ \}}`	$\mathrm{D}^{\{ \}}$	Here $\mathrm{D}^{\{\}}$ denotes—similarly as in HEHL et al. (1976)—the covariant derivative with
mielke-geometrodynamics_F01231	247	0.993	`\Theta=[K, \vartheta]`	$\Theta=[K, \vartheta]$	occurs in RC spacetime. Considering the resolution of the relation $\Theta = [K, \vartheta]$ for
mielke-geometrodynamics_F01232	248	1.000	`[\Theta, \vartheta]`	$[\Theta, \vartheta]$	it follows that the modified torsion $[\Theta, \vartheta]$ couples *algebraically* to the canonical
mielke-geometrodynamics_F01233	248	1.000	`\delta L_{D} / \delta \bar{\psi}`	$\delta L_{D} / \delta \bar{\psi}$	$\delta L_D/\delta\overline{\psi}$. This equation will be referred to as the *nonlinear Heisenberg–Pauli–Weyl*
mielke-geometrodynamics_F01234	249	1.000	`D=\vartheta^{\alpha} \nabla_{\alpha}`	$D=\vartheta^{\alpha} \nabla_{\alpha}$	with $D = \vartheta^{\alpha}\nabla_{\alpha}$ taken into considerations, as follows:
mielke-geometrodynamics_F01235	249	1.000	`(1938,1957)`	$(1938,1957)$	was IVANENKO (1938, 1957) who, for quantum-theoretic reasons, pointed out the
mielke-geometrodynamics_F01236	249	1.000	`P C`	$P C$	reflection and charge conjugation PC, and the time reflection T. To be added is the
mielke-geometrodynamics_F01237	249	0.520	`\psi:=\binom{\psi_{p}}{\psi_{N}}`	$\psi:=\binom{\psi_{p}}{\psi_{N}}$	fundamental equation for the isotopic doublet $\psi := \begin{pmatrix}\psi_P \\ \psi_N\end{pmatrix}$, this equation had to
mielke-geometrodynamics_F01238	249	1.000	`S L(2, \mathbb{C})`	$S L(2, \mathbb{C})$	group $SL(2,\mathbb{C})$ of Weyl to a gauge theory with $\overline{\mathrm{G}} = SL(2\mathrm{f}, \mathbb{C})$ as a structure group,
mielke-geometrodynamics_F01239	249	1.000	`\overline{\mathrm{G}}=S L(2 \mathrm{f}, \mathbb{C})`	$\overline{\mathrm{G}}=S L(2 \mathrm{f}, \mathbb{C})$	group $SL(2,\mathbb{C})$ of Weyl to a gauge theory with $\overline{\mathrm{G}} = SL(2\mathrm{f}, \mathbb{C})$ as a structure group,

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mielke-geometrodynamics_F01240	249	0.994	`\overline{\mathrm{G}}`	$\overline{\mathrm{G}}$	results in a $\overline{\mathrm{G}}$-invariant nonlinear Heisenberg–Pauli–Weyl spinor equation (MIELKE
mielke-geometrodynamics_F01241	251	1.000	`1 / r^{2}`	$1/r^{2}$	and thus replaces Coulomb's $1/r^2$-dependence on the distance $r := \vert\overrightarrow{\mathbf{x}}\vert$ from the cen-
mielke-geometrodynamics_F01242	251	1.000	`r:=|\overrightarrow{\mathbf{x}}|`	$r:=\vert\overrightarrow{\mathbf{x}}\vert$	and thus replaces Coulomb's $1/r^2$-dependence on the distance $r := \vert\overrightarrow{\mathbf{x}}\vert$ from the cen-
mielke-geometrodynamics_F01243	251	1.000	`(1934,1935)`	$(1934,1935)$	context. Theoretically, it was coined by BORN & INFELD (1934, 1935) and, as far as a quantum-
mielke-geometrodynamics_F01244	252	0.979	`n=4`	$n=4$	that is given by (11.3.7) exclusively in the physically(!) relevant case of $n = 4$ dimen-
mielke-geometrodynamics_F01245	252	1.000	`\varepsilon= \pm 1, \ell`	$\varepsilon= \pm 1, \ell$	the following analysis. On the other hand, the constants $\varepsilon = \pm 1$, ℓ, and a are to be
mielke-geometrodynamics_F01246	253	0.999	`v=v(\rho)`	$v=v(\rho)$	Concerning *static* spinor fields, the functions $\nu = \nu(\rho)$ and $\mu = \mu(\rho)$ depend only
mielke-geometrodynamics_F01247	253	0.999	`\mu=\mu(\rho)`	$\mu=\mu(\rho)$	Concerning *static* spinor fields, the functions $\nu = \nu(\rho)$ and $\mu = \mu(\rho)$ depend only
mielke-geometrodynamics_F01248	253	1.000	`\mathrm{r}:=|\overrightarrow{\mathbf{x}}|, \vartheta`	$\mathrm{r}:=\vert\overrightarrow{\mathbf{x}}\vert, \vartheta$	and the metric becomes static itself. Thereby the spherical coordinates $\mathrm{r} := \vert\overrightarrow{\mathbf{x}}\vert$, ϑ,
mielke-geometrodynamics_F01249	253	1.000	`\overrightarrow{\mathbf{x}}`	$\overrightarrow{\mathbf{x}}$	and ϕ are related to the Cartesian coordinates $\overrightarrow{\mathbf{x}}$ in the familiar manner. The search for
mielke-geometrodynamics_F01250	253	1.000	`g_{k l}`	$g_{k l}$	can be simplified by conformally relating the components g_{kl} of the metric with those
mielke-geometrodynamics_F01251	253	0.919	`\Gamma_{\kappa}^{\{f \cdot \alpha \beta}`	$\Gamma_{\kappa}^{\{f \cdot \alpha \beta}$	between the Christoffel symbols and the torsion-free rotation coefficients $\Gamma_{\kappa}^{\{\}\cdot\alpha\beta}$, the
mielke-geometrodynamics_F01252	254	0.742	`\mathrm{R}_{\circ}`	$\mathrm{R}_{\circ}$	$\mathrm{R}_{\circ}$. For this purpose, the de Sitter space is to be covered with the so-called horospher-
mielke-geometrodynamics_F01253	254	0.990	`\lambda, \mathbf{y}`	$\lambda, \mathbf{y}$	ical coordinate system $(\lambda, \mathbf{y})$. Concerning these coordinates, the line element of the
mielke-geometrodynamics_F01254	254	0.984	`S O_{\circ}(1,4)`	$S O_{\circ}(1,4)$	de Sitter group $SO_{\circ}(1,4)$ (see MIELKE (1977c), Eqs. (1.23) and (4.5) in the case of
mielke-geometrodynamics_F01255	254	1.000	`\bar{\Gamma}_{k}`	$\bar{\Gamma}_{k}$	For the sake of completeness, the conformally related component $\overline{\Gamma}_k$ of the spinor
mielke-geometrodynamics_F01256	254	1.000	`i \vec{\gamma} \cdot \vec{\partial}`	$i \vec{\gamma} \cdot \vec{\partial}$	spacelike Dirac operator $i\overrightarrow{\gamma} \cdot \overrightarrow{\partial}$ as being written again in Cartesian coordinates.
mielke-geometrodynamics_F01257	255	0.999	`\exp (-\mu / 2-\nu / 4)`	$\exp (-\mu / 2-\nu / 4)$	$\exp(-\mu/2 - \nu/4)$, as stated, into the separation ansatz
mielke-geometrodynamics_F01258	255	0.643	`\chi_{k}^{m}`	χ_{k}^{m}	Following the notation of ROSE (1961), the spin-weighted spherical harmonics χ_{κ}^{m}
mielke-geometrodynamics_F01259	255	1.000	`\mathrm{P}=(-1)^{l}`	$\mathrm{P}=(-1)^{l}$	of parity $\mathrm{P} = (-1)^l$ are given by
mielke-geometrodynamics_F01260	255	1.000	`l`	l	Here the parameter κ is related to the quantum numbers j and l of the total angular
mielke-geometrodynamics_F01261	255	1.000	`\mathrm{C}\left(l \frac{1}{2} \mathrm{j} ; \mathrm{m}-\overline{\mathrm{m}}, \overline{\mathrm{m}}\right)`	$\mathrm{C}\left(l \frac{1}{2} \mathrm{j} ; \mathrm{m}-\overline{\mathrm{m}}, \overline{\mathrm{m}}\right)$	Here $\mathrm{C}(l\frac{1}{2}\mathrm{j}; \mathrm{m}\text{-}\overline{\mathrm{m}}, \overline{\mathrm{m}})$ denote the Clebsch–Gordan coefficients of the rotation group
mielke-geometrodynamics_F01262	255	0.996	`\mathrm{Y}_{l}^{m}(\vartheta, \phi)`	$\mathrm{Y}_{l}^{m}(\vartheta, \phi)$	SU(2), and $\mathrm{Y}_l^m(\vartheta, \phi)$ are the spherical harmonics. Concerning the operators $\overrightarrow{J}$ and
mielke-geometrodynamics_F01263	255	0.996	`\vec{J}`	$\vec{J}$	SU(2), and $\mathrm{Y}_l^m(\vartheta, \phi)$ are the spherical harmonics. Concerning the operators $\overrightarrow{J}$ and
mielke-geometrodynamics_F01264	255	1.000	`\vec{L}`	$\vec{L}$	$\overrightarrow{L}$ of the total spin and the orbital angular momentum, these spinors satisfy the
mielke-geometrodynamics_F01265	256	1.000	`|\kappa|=1`	$\vert\kappa\vert=1$	follows from the properties of the spherical harmonics that this is the case for $\vert\kappa\vert = 1$

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mielke-geometrodynamics_F01266	256	1.000	`j=\frac{1}{2}, l=0,1`	$j=\frac{1}{2}, l=0,1$	to the values $j=\frac{1}{2}$, $l=0$, 1, and $\mathrm{m}=\pm\frac{1}{2}$, respectively. States of higher quantum
mielke-geometrodynamics_F01267	256	1.000	`\mathrm{m}= \pm \frac{1}{2}`	$\mathrm{m}= \pm \frac{1}{2}$	to the values $j=\frac{1}{2}$, $l=0$, 1, and $\mathrm{m}=\pm\frac{1}{2}$, respectively. States of higher quantum
mielke-geometrodynamics_F01268	256	1.000	`\int L_{\mathrm{D}^{\prime}}`	$\int L_{\mathrm{D}^{\prime}}$	functional $\int L_{\mathrm{D}'}$ belonging to (11.4.3). After integrating over the angular variables
mielke-geometrodynamics_F01269	256	1.000	`\rho^{*}`	ρ^{*}	schild solution, an utmost slow tending of ρ^* to the "coordinate singularity" is
mielke-geometrodynamics_F01270	257	1.000	`M^{*}`	M^{*}	mass M^* of the (possibly "strong") gravitational field is absorbed in the coefficients
mielke-geometrodynamics_F01271	257	0.999	`j=\frac{1}{2}`	$j=\frac{1}{2}$	*localized* and regular solutions. For a soliton with total spin $j=\frac{1}{2}$, i.e., for $\kappa=-1$,
mielke-geometrodynamics_F01272	257	0.999	`\kappa=-1`	$\kappa=-1$	*localized* and regular solutions. For a soliton with total spin $j=\frac{1}{2}$, i.e., for $\kappa=-1$,
mielke-geometrodynamics_F01273	257	1.000	`H(\rho)`	$H(\rho)$	For a soliton with total spin $j=\frac{1}{2}$ and $\kappa=-1$, the radial functions $H(\rho)$ in the
mielke-geometrodynamics_F01274	257	1.000	`l=0`	$l=0$	ansatz (11.4.15) account for a spherically symmetric distribution with $l=0$, while
mielke-geometrodynamics_F01275	257	1.000	`l=1`	$l=1$	according to (11.4.17), the quantum number $l=1$ of angular momentum can be
mielke-geometrodynamics_F01276	257	1.000	`F(\rho)`	$F(\rho)$	ascribed to $F(\rho)$. Let us compare this with the situation in a nonlocal version of the
mielke-geometrodynamics_F01277	258	0.943	`|\varphi|^{6}`	$\|\varphi\|^{6}$	$\|\varphi\|^6$-model of ROSEN (1965). With reference to an expansion in terms of spherical
mielke-geometrodynamics_F01278	258	1.000	`\mathrm{Y}_{l}^{\mathrm{m}}(\vartheta, \phi)`	$\mathrm{Y}_{l}^{\mathrm{m}}(\vartheta, \phi)$	harmonics $\mathrm{Y}_l^{\mathrm{m}}(\vartheta,\phi)$, the occurring *exact* radial solutions
mielke-geometrodynamics_F01279	259	0.726	`|\omega|<1`	$\|\omega\|<1$	ansatz (11.4.15) satisfies $\|\omega\|<1$. On the other hand, it is known (VÁZQUEZ 1977)
mielke-geometrodynamics_F01280	259	1.000	`|\omega|>1`	$\|\omega\|>1$	that bounded and square-integrable radial solutions *do not* exist for $\|\omega\|>1$.
mielke-geometrodynamics_F01281	259	1.000	`|\psi| \sim e^{-m_{\pi} r}`	$\|\psi\| \sim e^{-m_{\pi} r}$	$\|\psi\| \sim e^{-m_\pi r}$ between the centers of charge of the solitons. This is the case if the
mielke-geometrodynamics_F01282	259	1.000	`\mathrm{m}_{\mathrm{s}}`	m_{s}	The total mass m_{s} of a spinor soliton is obtained by the integration
mielke-geometrodynamics_F01283	259	1.000	`\Sigma_{\mathrm{D}}`	Σ_{D}	of the canonical energy momentum current Σ_{D} over a spacelike hypersurface H^3.
mielke-geometrodynamics_F01284	259	1.000	`m_{s}`	m_{s}	(11.4.40) is inversely proportional to the "fundamental" length ℓ^* and that m_s would
mielke-geometrodynamics_F01285	260	0.979	`(1951,1956)`	$(1951,1956)$	Following the preceding studies of FINKELSTEIN et al. (1951, 1956) and SOLER
mielke-geometrodynamics_F01286	260	0.356	`a=0`	$a=0$	(1970), which were mainly concerned with a model (11.4.3) with $a=0$, RAÑADA
mielke-geometrodynamics_F01287	260	0.988	`a=1`	$a=1$	equation (11.3.7), i.e., on the Eq. (11.4.3) with $a=1$ but with the opposite
mielke-geometrodynamics_F01288	260	1.000	`\varepsilon=+1`	$\varepsilon=+1$	sign $\varepsilon=+1$. For solitons corresponding to an $S_{1/2}$-bound state with normalized
mielke-geometrodynamics_F01289	260	1.000	`S_{1 / 2}`	$S_{1 / 2}$	sign $\varepsilon=+1$. For solitons corresponding to an $S_{1/2}$-bound state with normalized
mielke-geometrodynamics_F01290	260	1.000	`Q=e`	$Q=e$	charge $Q=e$, only nodeless radial solutions whose total rest mass shows an
mielke-geometrodynamics_F01291	263	1.000	`D_{\mathrm{F}}(x)`	$D_{\mathrm{F}}(x)$	tudes are not yet subjected to field ("second") quantization. Here $D_{\mathrm{F}}(x)$ denotes
mielke-geometrodynamics_F01292	263	1.000	`(1957,1958)`	$(1957,1958)$	A further reason for IVANENKO (1957, 1958) to consider the necessity of nonlin-

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mielke-geometrodynamics_F01293	263	1.000	`\ell^{*}=8.1 \times 10^{-33} \mathrm{~cm}`	$\ell^{*}=8.1 \times 10^{-33} \mathrm{~cm}$	$\ell^{*} = 8.1 \times 10^{-33}$ cm to the far larger amount of the Compton wavelength $\ell = 10^{-13}$
mielke-geometrodynamics_F01294	263	1.000	`\ell=10^{-13}`	$\ell=10^{-13}$	$\ell^{*} = 8.1 \times 10^{-33}$ cm to the far larger amount of the Compton wavelength $\ell = 10^{-13}$
mielke-geometrodynamics_F01295	263	1.000	`\ell=10^{-13} \mathrm{~cm}`	$\ell=10^{-13} \mathrm{~cm}$	field theory (1967), which starts from $\ell = 10^{-13}$ cm, can refrain from using a “bare”
mielke-geometrodynamics_F01296	269	1.000	`\langle d j\rangle=0`	$\langle d j\rangle=0$	charge current j can be conserved, i.e., $\langle dj\rangle = 0$, whereas the conservation of the
mielke-geometrodynamics_F01297	269	1.000	`\left\langle d j_{5}\right\rangle \neq 0`	$\left\langle d j_{5}\right\rangle \neq 0$	axial current j_5 is broken, $\langle dj_5\rangle \neq 0$, in the most common convention.
mielke-geometrodynamics_F01298	269	0.999	`j_{5}(x ; \varepsilon):=\bar{\psi}(x) \gamma_{5}{ }^{*} \gamma \psi(x+\varepsilon)`	$j_{5}(x ; \varepsilon):=\bar{\psi}(x) \gamma_{5}{ }^{*} \gamma \psi(x+\varepsilon)$	sidering the point-split current $j_5(x;\varepsilon) := \overline{\psi}(x)\gamma_5{}^{*}\gamma\psi(x+\varepsilon)$, where ε is an infin-
mielke-geometrodynamics_F01299	269	1.000	`\delta / \delta A`	$\delta / \delta A$	The variation $\delta/\delta A$ of the current $j_5(x;\varepsilon)$ is compensated by the variation of the
mielke-geometrodynamics_F01300	269	1.000	`j_{5}(x ; \varepsilon)`	$j_{5}(x ; \varepsilon)$	The variation $\delta/\delta A$ of the current $j_5(x;\varepsilon)$ is compensated by the variation of the
mielke-geometrodynamics_F01301	269	0.999	`x^{i} \rightarrow x^{i}+\varepsilon^{i}`	$x^{i} \rightarrow x^{i}+\varepsilon^{i}$	exponential. Since the parallel transport from $x^i \rightarrow x^i + \varepsilon^i$ along the infinitesimal
mielke-geometrodynamics_F01302	270	1.000	`P=\bar{\psi} \gamma_{5} \psi`	$P=\bar{\psi} \gamma_{5} \psi$	for massive fermions, where $P = \overline{\psi}\gamma_5\psi$ is the pseudoscalar current and $F := dA$
mielke-geometrodynamics_F01303	270	1.000	`C:=A \wedge d A`	$C:=A \wedge d A$	Simons (CS) term $C := A \wedge dA$ corresponds to the spin or helicity of the photon,
mielke-geometrodynamics_F01304	270	0.760	`\mathbf{A} \cdot \mathbf{B}`	$\mathbf{A} \cdot \mathbf{B}$	with its spacelike part $\mathbf{A} \cdot \mathbf{B}$ known as magnetic helicity (Jackiw & Pi 2000). Since
mielke-geometrodynamics_F01305	270	1.000	`\mathscr{D}:=D+i A \wedge`	$\mathscr{D}:=D+i A \wedge$	potential $A = A_i dx^i$ is accounted for via $\mathscr{D} := D + iA\wedge$, where $D\psi := d\psi + \Gamma\wedge$
mielke-geometrodynamics_F01306	270	1.000	`D \psi:=d \psi+\Gamma \wedge`	$D \psi:=d \psi+\Gamma \wedge$	potential $A = A_i dx^i$ is accounted for via $\mathscr{D} := D + iA\wedge$, where $D\psi := d\psi + \Gamma\wedge$
mielke-geometrodynamics_F01307	270	0.566	`\Gamma^{\alpha \beta}=\Gamma_{i}{ }^{\alpha \beta} d x^{i}`	$\Gamma^{\alpha \beta}=\Gamma_{i}{ }^{\alpha \beta} d x^{i}$	nection one-form $\Gamma^{\alpha\beta} = \Gamma_i{}^{\alpha\beta}dx^i$.
mielke-geometrodynamics_F01308	270	1.000	`\Theta:=D \gamma`	$\Theta:=D \gamma$	with respect to $\overline{\psi}$ and ψ. Making use of the torsion $\Theta := D\gamma$ and of the properties
mielke-geometrodynamics_F01309	271	0.652	`\left.\left.T:=\frac{1}{4} \operatorname{Tr}(\check{\gamma}\right\rfloor \Theta\right)=e_{\alpha}\right\lrcorner T^{\alpha}`	$T:=\frac{1}{4} \operatorname{Tr}(\check{\gamma}\rfloor \Theta\Big)=e_{\alpha}\lrcorner T^{\alpha}$	where $T := \frac{1}{4}Tr\left(\check{\gamma}\rfloor\Theta\right) = e_\alpha\rfloor T^\alpha$ is the one-form of the trace (or vector) torsion.
mielke-geometrodynamics_F01310	271	1.000	`K=\frac{i}{4} K^{\alpha \beta} \sigma_{\alpha \beta}`	$K=\frac{i}{4} K^{\alpha \beta} \sigma_{\alpha \beta}$	(or Christoffel) connection $\Gamma^{\{\}}$ and the *contortion* one-form $K = \frac{i}{4}K^{\alpha\beta}\sigma_{\alpha\beta}$, sat-
mielke-geometrodynamics_F01311	271	1.000	`D \gamma=[\gamma, K]=\gamma_{\alpha} T^{\alpha}`	$D \gamma=[\gamma, K]=\gamma_{\alpha} T^{\alpha}$	isfying $D\gamma = [\gamma, K] = \gamma_\alpha T^\alpha$. Accordingly, the Dirac Lagrangian (12.2.1) splits
mielke-geometrodynamics_F01312	271	0.509	`D^{(t)} \gamma=0`	$D^{(t)} \gamma=0$	$D^{\{\}}\gamma = 0$. Hence, in an RC spacetime, a Dirac spinor feels only the *axial torsion*,
mielke-geometrodynamics_F01313	271	1.000	`\gamma \rightarrow \gamma^{\beta}=`	$\gamma \rightarrow \gamma^{\beta}=$	which is invariant under Weyl rescalings and *chiral transformations* $\gamma \rightarrow \gamma^\beta =$
mielke-geometrodynamics_F01314	271	1.000	`e^{i \gamma^{5} \beta} \gamma e^{-i \gamma^{5} \beta}`	$e^{i \gamma^{5} \beta} \gamma e^{-i \gamma^{5} \beta}$	$e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$ of the coframe, but odd under parity $P : \vartheta^B \rightarrow -\vartheta^B$, where $B =$
mielke-geometrodynamics_F01315	271	1.000	`P: \vartheta^{B} \rightarrow-\vartheta^{B}`	$P: \vartheta^{B} \rightarrow-\vartheta^{B}$	$e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$ of the coframe, but odd under parity $P : \vartheta^B \rightarrow -\vartheta^B$, where $B =$
mielke-geometrodynamics_F01316	271	1.000	`B=`	$B=$	$e^{i\gamma^5\beta}\gamma e^{-i\gamma^5\beta}$ of the coframe, but odd under parity $P : \vartheta^B \rightarrow -\vartheta^B$, where $B =$
mielke-geometrodynamics_F01317	271	0.992	`1,2,3`	$1,2,3$	1, 2, 3.
mielke-geometrodynamics_F01318	271	1.000	`d j \simeq 0`	$d j \simeq 0$	From the Dirac equation (12.2.2) and its adjoint one can readily deduce that $dj \simeq 0$

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mielke-geometrodynamics_F01319	271	1.000	`m \rightarrow 0`	$m \rightarrow 0$	spacetime. If we restore chiral symmetry in the limit $m \rightarrow 0$, this leads to the classical
mielke-geometrodynamics_F01320	272	1.000	`d j_{5}=0`	$d j_{5}=0$	conservation law $dj_5 = 0$ of the axial current for massless Weyl spinors, or since
mielke-geometrodynamics_F01321	272	1.000	`\mathscr{A}=2 d \theta`	$\mathscr{A}=2 d \theta$	torsion via $\mathscr{A} = 2d\theta$. Then there arises in (12.2.3) a derivative coupling of the
mielke-geometrodynamics_F01322	272	1.000	`a=\theta f_{a}`	$a=\theta f_{a}$	would-be axion $a = \theta f_a$ to two fermions via the CPT-invariant term
mielke-geometrodynamics_F01323	272	0.972	`U(1)_{\mathrm{PQ}}`	$U(1)_{\mathrm{PQ}}$	associated with a spontaneously broken Peccei–Quinn symmetry $U(1)_{\mathrm{PQ}}$; cf. KIM &
mielke-geometrodynamics_F01324	273	1.000	`\eta^{\alpha \beta}:={ }^{*}\left(\vartheta^{\alpha} \wedge \vartheta^{\beta}\right)`	$\eta^{\alpha \beta}:={ }^{*}\left(\vartheta^{\alpha} \wedge \vartheta^{\beta}\right)$	where $\eta^{\alpha\beta} := {}^*(\vartheta^\alpha \wedge \vartheta^\beta)$ is dual to the unit two-form.
mielke-geometrodynamics_F01325	273	1.000	`\left\langle d j_{5}\right\rangle`	$\left\langle d j_{5}\right\rangle$	quantization, where the vacuum expectation value $\langle dj_5\rangle$ of the axial current picks up
mielke-geometrodynamics_F01326	273	0.999	`d C_{\mathrm{TT}} \cong(1 / 4) d j_{5} \rightarrow 0`	$d C_{\mathrm{TT}} \cong(1 / 4) d j_{5} \rightarrow 0$	NY four-form tends to zero "on shell," i.e., $dC_{\mathrm{TT}} \cong (1/4)dj_5 \rightarrow 0$. This is consistent
mielke-geometrodynamics_F01327	273	0.997	`{ }^{*} j_{5}`	${ }^{*} j_{5}$	Here we implicitly assume that the light-cone structure of the axial covector ${}^*j_5$
mielke-geometrodynamics_F01328	273	1.000	`n=4 k+2`	$n=4 k+2$	$n = 4k + 2$ dimensions (LEUTWYLER 1986).
mielke-geometrodynamics_F01329	274	0.999	`d^{*} \mathscr{A} \sim d C_{\mathrm{TT}}`	$d^{*} \mathscr{A} \sim d C_{\mathrm{TT}}$	arises in the axial anomaly, but *not* the NY-type term $d\,{}^*\mathscr{A} \sim dC_{\mathrm{TT}}$ as was claimed
mielke-geometrodynamics_F01330	274	1.000	`d \mathscr{A} \wedge d \mathscr{A}=-2 \mathscr{E} \cdot \mathscr{B} \eta`	$d \mathscr{A} \wedge d \mathscr{A}=-2 \mathscr{E} \cdot \mathscr{B} \eta$	arises in the anomaly. Thus only the Weyl invariant term $d\mathscr{A} \wedge d\mathscr{A} = -2\mathscr{E} \cdot \mathscr{B}\eta$
mielke-geometrodynamics_F01331	274	1.000	`F \wedge F=d A \wedge d A`	$F \wedge F=d A \wedge d A$	$F \wedge F = dA \wedge dA$ of the Pontryagin term (12.4.8). Torsion terms like $d\mathscr{A} \wedge d\mathscr{A}$
mielke-geometrodynamics_F01332	274	0.835	`d^{*} \mathscr{A} \wedge{ }^{*}\left(d^{*} \mathscr{A}\right)=4 \ell^{4} V_{\mathrm{NY}} \wedge{ }^{*} V_{\mathrm{NY}}`	$d^{*} \mathscr{A} \wedge{ }^{*}\left(d^{*} \mathscr{A}\right)=4 \ell^{4} V_{\mathrm{NY}} \wedge{ }^{*} V_{\mathrm{NY}}$	and $d\,{}^*\mathscr{A} \wedge {}^*(d\,{}^*\mathscr{A}) = 4\ell^4 V_{\mathrm{NY}} \wedge {}^*V_{\mathrm{NY}}$ have been considered previously, as part of
mielke-geometrodynamics_F01333	274	0.984	`d^{*} \mathscr{A}=2 \ell^{2} d C_{\mathrm{TT}}=2 \ell^{2} V_{\mathrm{NY}}`	$d^{*} \mathscr{A}=2 \ell^{2} d C_{\mathrm{TT}}=2 \ell^{2} V_{\mathrm{NY}}$	identity (12.4.10) for the NY term $d\,{}^*\mathscr{A} = 2\ell^2 dC_{\mathrm{TT}} = 2\ell^2 V_{\mathrm{NY}}$, the second term is
mielke-geometrodynamics_F01334	274	1.000	`Z`	Z	Z-factor, which would creep into the definition of the NY term. This is in sharp con-
mielke-geometrodynamics_F01335	274	0.588	`\Psi=\Psi_{i} d x^{i}`	$\Psi=\Psi_{i} d x^{i}$	Schwinger spinor-valued one-form $\Psi = \Psi_i dx^i$; cf. URRUTIA & VERGARA (1991),
mielke-geometrodynamics_F01336	275	0.999	`J_{5}:=i \bar{\Psi} \wedge \gamma \wedge \Psi`	$J_{5}:=i \bar{\Psi} \wedge \gamma \wedge \Psi$	The anomaly for the corresponding *axial current* $J_5 := i\bar{\Psi} \wedge \gamma \wedge \Psi$ is -21 times
mielke-geometrodynamics_F01337	275	1.000	`t \rightarrow+0`	$t \rightarrow+0$	In the heat kernel approach, there exists for small $t \rightarrow +0$ the asymptotic expansion
mielke-geometrodynamics_F01338	275	0.988	`\mathbb{A}:=`	$\mathbb{A}:=$	of the kernel in n dimensions, where the usual Feynman "dagger" convention $\not{A} :=$
mielke-geometrodynamics_F01339	275	0.794	`\check{\gamma}\rfloor A=\gamma^{\alpha} e_{\alpha} \downharpoonleft A=(-1)^{s+1}{ }^{*}\left[{ }^{*} \gamma \wedge A\right]`	$\check{\gamma}\rfloor A=\gamma^{\alpha} e_{\alpha} \downharpoonleft A=(-1)^{s+1}{ }^{*}\left[{ }^{*} \gamma \wedge A\right]$	$\check{\gamma}\rfloor A = \gamma^\alpha e_\alpha \rfloor A = (-1)^{s+1}\,{}^*\left[{}^*\gamma \wedge A\right]$ for one-forms is used.
mielke-geometrodynamics_F01340	275	0.996	`\square:=`	$\square:=$	terms additional to the generally covariant Riemannian d'Alembertian operator $\square :=$
mielke-geometrodynamics_F01341	275	0.980	`\partial_{i}\left(\sqrt{|g| g^{i j}} \partial_{j}\right) / \sqrt{|g|}`	$\partial_{i}\left(\sqrt{\vert g\vert g^{i j}} \partial_{j}\right) / \sqrt{\vert g\vert}$	$\partial_i\left(\sqrt{\vert g\vert}g^{ij}\partial_j\right)/\sqrt{\vert g\vert}$ can be explicitly identified (MIELKE & KREIMER 1998). Not
mielke-geometrodynamics_F01342	275	0.998	`K_{k}\left(x, \not D^{2}\right)`	$K_{k}\left(x, \not D^{2}\right)$	The coefficients $K_k(x, \not{D}^2)$ are completely determined by the form of the second-
mielke-geometrodynamics_F01343	275	0.952	`\not D^{2}`	$\not D^{2}$	order differential operator $\not{D}^2$, which is positive for Euclidean signature diag $\mathrm{o}_{\alpha\beta} =$
mielke-geometrodynamics_F01344	275	0.952	`\mathrm{o}_{\alpha \beta}=`	$\mathrm{o}_{\alpha \beta}=$	order differential operator $\not{D}^2$, which is positive for Euclidean signature diag $\mathrm{o}_{\alpha\beta} =$

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mielke-geometrodynamics_F01345	276	0.950	`(-1, \ldots,-1)`	$(-1, \ldots,-1)$	$(-1,\ldots,-1)$. For odd $k=1,3,\ldots$, these coefficients are zero, while the first non-
mielke-geometrodynamics_F01346	276	0.950	`k=1,3, \ldots`	$k=1,3, \ldots$	$(-1,\ldots,-1)$. For odd $k=1,3,\ldots$, these coefficients are zero, while the first non-
mielke-geometrodynamics_F01347	276	0.483	`d \mathscr{K}=d^{*} \stackrel{\smile}{D} \wedge^{*} \breve{D}^{*} \mathscr{A}`	$d \mathscr{K}=d^{*} \stackrel{\smile}{D} \wedge^{*} \breve{D}^{*} \mathscr{A}$	where the higher-order term $d\mathscr{K}=d^{*}\breve{D}\wedge^{*}\breve{D}\,^{*}\mathscr{A}$ involves the covariant derivative
mielke-geometrodynamics_F01348	276	0.507	`D=D^{\{ \}}+i \mathscr{A} \gamma_{5} / 4`	$D=D^{\{ \}}+i \mathscr{A} \gamma_{5} / 4$	$\breve{D}=D^{\{\}}+i\mathscr{A}\gamma_5/4$ modified by the axial torsion.
mielke-geometrodynamics_F01349	276	1.000	`K_{2}`	K_{2}	terms K_2 and K_4. Whereas in $n=4$ dimensions, the Pontryagin-type term K_4 is
mielke-geometrodynamics_F01350	276	1.000	`K_{4}`	K_{4}	terms K_2 and K_4. Whereas in $n=4$ dimensions, the Pontryagin-type term K_4 is
mielke-geometrodynamics_F01351	276	1.000	`k=4`	$k=4$	dimensionless and thus for $k=4$ gets multiplied by $t^{(k-4)/2}=1$, the term $K_2\sim$
mielke-geometrodynamics_F01352	276	1.000	`t^{(k-4) / 2}=1`	$t^{(k-4) / 2}=1$	dimensionless and thus for $k=4$ gets multiplied by $t^{(k-4)/2}=1$, the term $K_2\sim$
mielke-geometrodynamics_F01353	276	1.000	`K_{2} \sim`	$K_{2} \sim$	dimensionless and thus for $k=4$ gets multiplied by $t^{(k-4)/2}=1$, the term $K_2\sim$
mielke-geometrodynamics_F01354	276	0.960	`d^{*} \mathscr{A}=2 \ell^{2} d C_{\mathrm{TT}}`	$d^{*} \mathscr{A}=2 \ell^{2} d C_{\mathrm{TT}}$	$d\,^{*}\mathscr{A}=2\ell^2 dC_{\mathrm{TT}}$ carries dimensions. Since a *massive* Dirac spinor has canonical
mielke-geometrodynamics_F01355	276	0.986	`[\text { length }]^{-3 / 2}`	$[\text { length }]^{-3 / 2}$	dimension $[length]^{-3/2}$, it scales as $\psi\sim m^{3/2}$. Moreover, the term $t=1/M^2$ is
mielke-geometrodynamics_F01356	276	0.986	`\psi \sim m^{3 / 2}`	$\psi \sim m^{3 / 2}$	dimension $[length]^{-3/2}$, it scales as $\psi\sim m^{3/2}$. Moreover, the term $t=1/M^2$ is
mielke-geometrodynamics_F01357	276	0.986	`t=1 / M^{2}`	$t=1 / M^{2}$	dimension $[length]^{-3/2}$, it scales as $\psi\sim m^{3/2}$. Moreover, the term $t=1/M^2$ is
mielke-geometrodynamics_F01358	276	1.000	`M \rightarrow \infty`	$M \rightarrow \infty$	related to the regulator mass $M\rightarrow\infty$ in the FUJIKAWA (1979) method. Then the
mielke-geometrodynamics_F01359	276	1.000	`m \neq 0`	$m \neq 0$	heat kernel expansion will tend to zero in the chiral limit $m\rightarrow 0$. In the case $m\neq 0$,
mielke-geometrodynamics_F01360	276	1.000	`\vartheta^{\alpha} \rightarrow \widetilde{\vartheta}^{\alpha}=M \vartheta^{\alpha}`	$\vartheta^{\alpha} \rightarrow \widetilde{\vartheta}^{\alpha}=M \vartheta^{\alpha}$	To rescale the coframe by $\vartheta^{\alpha}\rightarrow\widetilde{\vartheta}^{\alpha}=M\vartheta^{\alpha}$ does not help, since that would also
mielke-geometrodynamics_F01361	276	1.000	`[\hbar]`	$[\hbar]$	change the dimension of the Dirac field in order to retain the physical dimension $[\hbar]$
mielke-geometrodynamics_F01362	276	1.000	`(n=2)`	$(n=2)$	sions, neither classically nor in QFT. On would surmise that in $(n=2)$-dimensional
mielke-geometrodynamics_F01363	276	1.000	`\sigma_{\alpha}:=\Sigma_{\alpha}-D \mu_{\alpha}`	$\sigma_{\alpha}:=\Sigma_{\alpha}-D \mu_{\alpha}$	where $\sigma_\alpha:=\Sigma_\alpha-D\mu_\alpha$ is the energy–momentum current Belinfante symmetrized
mielke-geometrodynamics_F01364	277	0.996	`\stackrel{( \pm)}{\Gamma}`	$\stackrel{(\pm)}{\Gamma}$	anti-self-dual variables $\stackrel{(\pm)}{\Gamma}$ in EC theory as well as in simple supergravity (MIELKE
mielke-geometrodynamics_F01365	277	0.839	`d C_{\mathrm{RR}}^{\{ \}}`	$d C_{\mathrm{RR}}^{\{ \}}$	The appearance of the Riemannian Pontryagin term $dC_{\mathrm{RR}}^{\{\}}$ in the anomaly (12.3.1)
mielke-geometrodynamics_F01366	277	1.000	`\Lambda \neq 0`	$\Lambda \neq 0$	essary condition $\Lambda\neq 0$ could even be the dominating configurations, one gets a net
mielke-geometrodynamics_F01367	277	0.570	`\underline{C}_{\mathrm{RR}}^{\{ \}}`	$\underline{C}_{\mathrm{RR}}^{\{ \}}$	to gravity, the tangential complexified CS term $\underline{C}_{\mathrm{RR}}^{\{\}}$ is known (KODAMA 1990) to
mielke-geometrodynamics_F01368	277	0.936	`\stackrel{( \pm)}{\underline{\Gamma}}`	$\stackrel{(\pm)}{\underline{\Gamma}}$	of gravity with cosmological constant, where the complex Ashtekar variable $\stackrel{(\pm)}{\underline{\Gamma}}$
mielke-geometrodynamics_F01369	277	1.000	`\mathscr{C}:=\Gamma / \mathscr{G}`	$\mathscr{C}:=\Gamma / \mathscr{G}$	theory on a three-dimensional hypersurface, with $\mathscr{C}:=\Gamma/\mathscr{G}$ of inequivalent gauge
mielke-geometrodynamics_F01370	278	0.423	`\mathrm{Ł}_{n}:=n \downarrow D+D n`	$Ł_{n}:=n \downarrow D+D n$	where $Ł_n:=n\rfloor D+Dn\rfloor$ is the gauge-covariant Lie derivative along the normal

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mielke-geometrodynamics_F01371	278	0.523	`\tau^{A}`	τ^{A}	for the matter current τ^A on the spacelike hypersurface (JIANG 1991). Only when
mielke-geometrodynamics_F01372	278	1.000	`\tau^{A}:=(1 / 2) \eta^{0 A \beta \gamma} \tau_{\beta \gamma}`	$\tau^{A}:=(1 / 2) \eta^{0 A \beta \gamma} \tau_{\beta \gamma}$	field, it is the canonical spin $\tau^A := (1/2)\eta^{0A\beta\gamma}\tau_{\beta\gamma}$ that appears on the right-hand
mielke-geometrodynamics_F01373	278	1.000	`\gamma_{\alpha} \gamma_{\beta}+\gamma_{\beta} \gamma_{\alpha}=2 \mathrm{o}_{\alpha \beta}`	$\gamma_{\alpha} \gamma_{\beta}+\gamma_{\beta} \gamma_{\alpha}=2 \mathrm{o}_{\alpha \beta}$	When the constant Dirac matrices γ_α satisfying $\gamma_\alpha\gamma_\beta+\gamma_\beta\gamma_\alpha = 2o_{\alpha\beta}$ saturate the
mielke-geometrodynamics_F01374	278	1.000	`\eta^{\alpha}:=`	$\eta^{\alpha}:=$	index of the orthonormal coframe one-form $\vartheta^\alpha = E_j{}^\alpha dx^j$ and its Hodge dual $\eta^\alpha :=$
mielke-geometrodynamics_F01375	278	0.996	`{ }^{*} \vartheta^{\alpha}`	${ }^{*} \vartheta^{\alpha}$	${}^*\vartheta^\alpha$, we obtain a basis of Clifford-algebra-valued exterior forms via
mielke-geometrodynamics_F01376	278	0.500	`\Gamma:=\frac{i}{4} \Gamma_{i}^{\alpha \beta} \sigma_{\alpha \beta} d x^{i}`	$\Gamma:=\frac{i}{4} \Gamma_{i}^{\alpha \beta} \sigma_{\alpha \beta} d x^{i}$	In terms of the Clifford-algebra-valued *connection* $\Gamma := \frac{i}{4}\Gamma_i{}^{\alpha\beta}\sigma_{\alpha\beta}dx^i$, the $SL(2,\mathbb{C})$-
mielke-geometrodynamics_F01377	278	1.000	`D=d+\Gamma \wedge`	$D=d+\Gamma \wedge$	covariant exterior derivative is given by $D = d+\Gamma\wedge$, where $\sigma_{\alpha\beta} = \frac{i}{2}(\gamma_\alpha\gamma_\beta-\gamma_\beta\gamma_\alpha)$
mielke-geometrodynamics_F01378	278	1.000	`\sigma_{\alpha \beta}=\frac{i}{2}\left(\gamma_{\alpha} \gamma_{\beta}-\gamma_{\beta} \gamma_{\alpha}\right)`	$\sigma_{\alpha \beta}=\frac{i}{2}\left(\gamma_{\alpha} \gamma_{\beta}-\gamma_{\beta} \gamma_{\alpha}\right)$	covariant exterior derivative is given by $D = d+\Gamma\wedge$, where $\sigma_{\alpha\beta} = \frac{i}{2}(\gamma_\alpha\gamma_\beta-\gamma_\beta\gamma_\alpha)$
mielke-geometrodynamics_F01379	278	1.000	`\sigma:=\frac{i}{2} \gamma \wedge \gamma=\frac{1}{2} \sigma_{\alpha \beta} \vartheta^{\alpha} \wedge \vartheta^{\beta}`	$\sigma:=\frac{i}{2} \gamma \wedge \gamma=\frac{1}{2} \sigma_{\alpha \beta} \vartheta^{\alpha} \wedge \vartheta^{\beta}$	are the Lorentz generators entering in the two-form $\sigma := \frac{i}{2}\gamma\wedge\gamma = \frac{1}{2}\sigma_{\alpha\beta}\vartheta^\alpha\wedge\vartheta^\beta$.
mielke-geometrodynamics_F01380	279	1.000	`R:={ }^{*}\left(R^{\alpha \beta} \wedge \eta_{\beta \alpha}\right)`	$R:={ }^{*}\left(R^{\alpha \beta} \wedge \eta_{\beta \alpha}\right)$	to the curvature scalar $R := {}^*(R^{\alpha\beta}\wedge\eta_{\beta\alpha})$ and the axial torsion piece $d\mathscr{A}\wedge d\mathscr{A}$ of
mielke-geometrodynamics_F01381	279	1.000	`\mathfrak{s l}(5, \mathbb{R})`	$\mathfrak{s l}(5, \mathbb{R})$	the $\mathfrak{sl}(5,\mathbb{R})$-valued connection $\hat\Gamma = \Gamma + (\vartheta^\alpha L^4{}_\alpha + \vartheta_\beta L^\beta{}_4)/\ell$ is expanded into the
mielke-geometrodynamics_F01382	279	1.000	`\hat{\Gamma}=\Gamma+\left(\vartheta^{\alpha} L^{4}{ }_{\alpha}+\vartheta_{\beta} L^{\beta}{ }_{4}\right) / \ell`	$\hat{\Gamma}=\Gamma+\left(\vartheta^{\alpha} L^{4}{ }_{\alpha}+\vartheta_{\beta} L^{\beta}{ }_{4}\right) / \ell$	the $\mathfrak{sl}(5,\mathbb{R})$-valued connection $\hat\Gamma = \Gamma + (\vartheta^\alpha L^4{}_\alpha + \vartheta_\beta L^\beta{}_4)/\ell$ is expanded into the
mielke-geometrodynamics_F01383	279	0.763	`\hat{C}_{\mathrm{RR}}`	$\hat{C}_{\mathrm{RR}}$	canonical dimension [*length*]. The corresponding Pontryagin term $\hat C_{\mathrm{RR}}$ splits via
mielke-geometrodynamics_F01384	280	1.000	`\pi^{0} \rightarrow 2 \gamma`	$\pi^{0} \rightarrow 2 \gamma$	Bell JS, Jackiw R (1969) A PCAC puzzle: $\pi^0 \to 2\gamma$ in the σ model. Il Nuovo Cimento A 60(1):
mielke-geometrodynamics_F01385	282	0.775	`\mathbb{R}^{4} \otimes \mathrm{SO}(1,3)`	$\mathbb{R}^{4} \otimes \mathrm{SO}(1,3)$	Lorentz or Poincaré group $\mathbb{R}^4 \rtimes \mathrm{SO}(1,3)$ to a metalinear group $\mathrm{SL}(5,\mathbb{R})$ as a gauge
mielke-geometrodynamics_F01386	283	1.000	`d B=0`	$d B=0$	Independent variations with respect to A and B lead to $dB=0$ and to the *constraint*
mielke-geometrodynamics_F01387	284	0.999	`\sigma=\sigma_{i} d x^{i}`	$\sigma=\sigma_{i} d x^{i}$	where $\sigma = \sigma_i dx^i$ is a one-form or a covector; cf. LUCCHESI et al. (1993). Since Bianchi
mielke-geometrodynamics_F01388	284	0.672	`\mathrm{SL}(5, \mathbb{R})`	$\mathrm{SL}(5, \mathbb{R})$	"semi-topological" BF scheme based on the metric-free structure group $\mathrm{SL}(5,\mathbb{R})$
mielke-geometrodynamics_F01389	284	1.000	`\operatorname{SL}(n+1, \mathbb{R})`	$\operatorname{SL}(n+1, \mathbb{R})$	In general, the metalinear group $\mathrm{SL}(n+1,\mathbb{R})$ with $\det g = 1$ is generated by
mielke-geometrodynamics_F01390	284	1.000	`\operatorname{det} g=1`	$\operatorname{det} g=1$	In general, the metalinear group $\mathrm{SL}(n+1,\mathbb{R})$ with $\det g = 1$ is generated by
mielke-geometrodynamics_F01391	284	1.000	`n(n+2)`	$n(n+2)$	$n(n+2)$ trace-free generators
mielke-geometrodynamics_F01392	284	1.000	`\mathfrak{g l}(n, \mathbb{R})`	$\mathfrak{g l}(n, \mathbb{R})$	comprise those of a $\mathfrak{gl}(n,\mathbb{R})$ subalgebra
mielke-geometrodynamics_F01393	284	1.000	`\alpha, \ldots, \delta=0, \ldots, n-1`	$\alpha, \ldots, \delta=0, \ldots, n-1$	where $\alpha,\ldots,\delta = 0,\ldots,n-1$.
mielke-geometrodynamics_F01394	284	0.367	`\ell P_{\alpha}:=Ł^{n+1}{ }_{\alpha}=L^{n+1}{ }_{\alpha}`	$\ell P_{\alpha}:=Ł^{n+1}{ }_{\alpha}=L^{n+1}{ }_{\alpha}$	The generators $\ell P_\alpha := \not{L}^{n+1}{}_\alpha = L^{n+1}{}_\alpha$ and $\ell P^\beta_* := \not{L}^{\beta}{}_{n+1} = L^\beta{}_{n+1}$ transform
mielke-geometrodynamics_F01395	284	0.367	`\ell P_{*}^{\beta}:=\swarrow^{\beta}{ }_{n+1}=L^{\beta}{ }_{n+1}`	$\ell P_{*}^{\beta}:=\swarrow^{\beta}{ }_{n+1}=L^{\beta}{ }_{n+1}$	The generators $\ell P_\alpha := \not{L}^{n+1}{}_\alpha = L^{n+1}{}_\alpha$ and $\ell P^\beta_* := \not{L}^{\beta}{}_{n+1} = L^\beta{}_{n+1}$ transform
mielke-geometrodynamics_F01396	285	0.998	`P_{*}^{\alpha}`	P_{*}^{α}	This reveals that P^α_* is a vector and P_β is a covector with respect to the $\mathrm{GL}(n,\mathbb{R})$

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mielke-geometrodynamics_F01397	285	0.998	`P_{\beta}`	P_{β}	This reveals that P_*^{α} is a vector and P_β is a covector with respect to the $\mathrm{GL}(n,\mathbb{R})$
mielke-geometrodynamics_F01398	285	0.998	`\operatorname{GL}(n, \mathbb{R})`	$\operatorname{GL}(n, \mathbb{R})$	This reveals that P_*^{α} is a vector and P_β is a covector with respect to the $\mathrm{GL}(n,\mathbb{R})$
mielke-geometrodynamics_F01399	285	0.996	`\mathrm{A}_{*}(n, \mathbb{R})`	$\mathrm{A}_{*}(n, \mathbb{R})$	to the semisimple *graded affine* group $\mathrm{A}_*(n,\mathbb{R})$ of the same rank:
mielke-geometrodynamics_F01400	285	0.998	`\mathrm{SO}(n+1)`	$\mathrm{SO}(n+1)$	(Pseudo)orthogonal groups like $\mathrm{SO}(n+1)$ involve a Cartan–Killing metric with
mielke-geometrodynamics_F01401	285	1.000	`\hat{g}_{A B}`	$\hat{g}_{A B}$	components $\hat{g}_{AB}$. Then a basis of the Lie generators
mielke-geometrodynamics_F01402	285	0.998	`n(n+1) / 2`	$n(n+1) / 2$	is only $n(n+1)/2$-dimensional. The same would apply to the respective connections
mielke-geometrodynamics_F01403	285	0.999	`A(x) \in \mathrm{A}_{*}(n, R)`	$A(x) \in \mathrm{A}_{*}(n, R)$	where $A(x)\in\mathrm{A}_*(n,\ R)$ and the $\Lambda(x)\in\mathrm{GL}(n,\mathbb{R})$ are linear gauge transformations,
mielke-geometrodynamics_F01404	285	0.999	`\Lambda(x) \in \mathrm{GL}(n, \mathbb{R})`	$\Lambda(x) \in \mathrm{GL}(n, \mathbb{R})$	where $A(x)\in\mathrm{A}_*(n,\ R)$ and the $\Lambda(x)\in\mathrm{GL}(n,\mathbb{R})$ are linear gauge transformations,
mielke-geometrodynamics_F01405	285	1.000	`\tau(x):=\exp \left[\xi^{\alpha} P_{\alpha}\right] \in \mathscr{T}(n, R)`	$\tau(x):=\exp \left[\xi^{\alpha} P_{\alpha}\right] \in \mathscr{T}(n, R)$	while $\tau(x):=\exp[\xi^\alpha P_\alpha]\in\mathscr{T}(n,R)$ and $\tau_*(x):=\exp[\xi_\beta^* P_*^\beta]\in\mathscr{T}_*(n,R)$ represent
mielke-geometrodynamics_F01406	285	1.000	`\tau_{*}(x):=\exp \left[\xi_{\beta}^{*} P_{*}^{\beta} \right] \in \mathscr{T}_{*}(n, R)`	$\tau_{*}(x):=\exp \left[\xi_{\beta}^{*} P_{*}^{\beta} \right] \in \mathscr{T}_{*}(n, R)$	while $\tau(x):=\exp[\xi^\alpha P_\alpha]\in\mathscr{T}(n,R)$ and $\tau_*(x):=\exp[\xi_\beta^* P_*^\beta]\in\mathscr{T}_*(n,R)$ represent
mielke-geometrodynamics_F01407	286	1.000	`\ell \rightarrow \infty`	$\ell \rightarrow \infty$	For $\ell\rightarrow\infty$, this holds automatically due to the degeneracy of the commutation
mielke-geometrodynamics_F01408	286	1.000	`\operatorname{Ad} A(B)=[A, B]`	$\operatorname{Ad} A(B)=[A, B]$	the adjoint representation $\mathrm{Ad}A(B)=[A,B]$. The curvature
mielke-geometrodynamics_F01409	286	1.000	`\hat{\Gamma}`	$\hat{\Gamma}$	The inhomogeneous transformation law (13.4.3) for the graded affine connection $\hat{\Gamma}$
mielke-geometrodynamics_F01410	286	1.000	`\Gamma^{(L)}`	$\Gamma^{(L)}$	can be split into that of the linear connection $\Gamma^{(L)}$, which acquires the conventional
mielke-geometrodynamics_F01411	286	0.990	`\mathrm{GL}(n, \mathbb{R})`	$\mathrm{GL}(n, \mathbb{R})$	of a Yang–Mills-type connection for the gauge group $\mathrm{GL}(n,\mathbb{R})$. The remaining trans-
mielke-geometrodynamics_F01412	287	0.998	`D \tau(x):=d \tau(x)+\Gamma^{(L)} \tau(x)`	$D \tau(x):=d \tau(x)+\Gamma^{(L)} \tau(x)$	derivative $D\tau(x):=d\tau(x)+\Gamma^{(L)}\tau(x)$, the translational parts $\Gamma^{(T)}$ and $\Gamma_*^{(T)}$ do not
mielke-geometrodynamics_F01413	287	0.998	`\Gamma^{(T)}`	$\Gamma^{(T)}$	derivative $D\tau(x):=d\tau(x)+\Gamma^{(L)}\tau(x)$, the translational parts $\Gamma^{(T)}$ and $\Gamma_*^{(T)}$ do not
mielke-geometrodynamics_F01414	287	0.998	`\Gamma_{*}^{(T)}`	$\Gamma_{*}^{(T)}$	derivative $D\tau(x):=d\tau(x)+\Gamma^{(L)}\tau(x)$, the translational parts $\Gamma^{(T)}$ and $\Gamma_*^{(T)}$ do not
mielke-geometrodynamics_F01415	287	1.000	`\xi=\xi^{\alpha} P_{\alpha}`	$\xi=\xi^{\alpha} P_{\alpha}$	Since $\xi=\xi^\alpha P_\alpha$ and $\xi_*=\xi_\beta^* P_*^\beta$ acquire their values in the coset space
mielke-geometrodynamics_F01416	287	1.000	`\xi_{*}=\xi_{\beta}^{*} P_{*}^{\beta}`	$\xi_{*}=\xi_{\beta}^{*} P_{*}^{\beta}$	Since $\xi=\xi^\alpha P_\alpha$ and $\xi_*=\xi_\beta^* P_*^\beta$ acquire their values in the coset space
mielke-geometrodynamics_F01417	287	1.000	`\mathscr{T}(n, R) \otimes \mathscr{T}_{*}(n, R)`	$\mathscr{T}(n, R) \otimes \mathscr{T}_{*}(n, R)$	“hide” the action of the local pseudotranslations $\mathscr{T}(n,R)\otimes\mathscr{T}_*(n,R)$ on ϑ and θ.
mielke-geometrodynamics_F01418	287	1.000	`\xi=\xi^{*}=0`	$\xi=\xi^{*}=0$	If we postulate even stronger constraints $\xi=\xi^*=0$ of “zero section” vector
mielke-geometrodynamics_F01419	287	1.000	`\theta_{\alpha}=g_{\alpha \beta} \vartheta^{\beta}`	$\theta_{\alpha}=g_{\alpha \beta} \vartheta^{\beta}$	mutually transposed one-forms, i.e., $\theta_\alpha=g_{\alpha\beta}\vartheta^\beta$.
mielke-geometrodynamics_F01420	287	1.000	`D \xi^{\alpha} \stackrel{*}{=} \vartheta^{\alpha}`	$D \xi^{\alpha} \stackrel{*}{=} \vartheta^{\alpha}$	When $D\xi^\alpha\stackrel{*}{=}\vartheta^\alpha$, a Cartan transport of a radius vector ξ^α occurs via “rolling
mielke-geometrodynamics_F01421	287	1.000	`\xi^{\alpha}`	ξ^{α}	When $D\xi^\alpha\stackrel{*}{=}\vartheta^\alpha$, a Cartan transport of a radius vector ξ^α occurs via “rolling
mielke-geometrodynamics_F01422	288	0.593	`1 /[`	$1/[$	In view of the canonical dimension $1/[\text{length}]$ of the generators P_α and P_*^β of pseudo-

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mielke-geometrodynamics_F01423	288	0.593	`]`	]	In view of the canonical dimension 1/[length] of the generators P_α and P_*^β of pseudo-
mielke-geometrodynamics_F01424	288	0.593	`P_{*}^{\beta}`	P_{*}^{β}	In view of the canonical dimension 1/[length] of the generators P_α and P_*^β of pseudo-
mielke-geometrodynamics_F01425	288	0.647	`\vartheta^{\alpha}=\ell \Gamma_{\hat{4}}^{\alpha}`	$\vartheta^{\alpha}=\ell \Gamma_{\hat{4}}^{\alpha}$	i.e., it becomes a Cartan connection, where $\vartheta^\alpha = \ell\Gamma_{\hat{4}}{}^\alpha$ and $\theta_\beta = \ell\Gamma_\beta{}^{\hat{4}}$ are two inde-
mielke-geometrodynamics_F01426	288	0.647	`\theta_{\beta}=\ell \Gamma_{\beta}{ }^{\hat{4}}`	$\theta_{\beta}=\ell \Gamma_{\beta}{ }^{\hat{4}}$	i.e., it becomes a Cartan connection, where $\vartheta^\alpha = \ell\Gamma_{\hat{4}}{}^\alpha$ and $\theta_\beta = \ell\Gamma_\beta{}^{\hat{4}}$ are two inde-
mielke-geometrodynamics_F01427	288	1.000	`\Lambda=3 / \ell^{2}`	$\Lambda=3 / \ell^{2}$	where $\Lambda = 3/\ell^2$ will later be identified with the cosmological constant.
mielke-geometrodynamics_F01428	289	0.998	`d \hat{C}`	$d \hat{C}$	$d\hat{C}$ provides us, after variation with respect to $\Gamma_A{}^B$, with the Bianchi identity
mielke-geometrodynamics_F01429	289	0.998	`\Gamma_{A}{ }^{B}`	$\Gamma_{A}{ }^{B}$	$d\hat{C}$ provides us, after variation with respect to $\Gamma_A{}^B$, with the Bianchi identity
mielke-geometrodynamics_F01430	289	1.000	`\hat{B}`	$\hat{B}$	The variation with respect to $\hat{B}$ would lead to $\hat{R} = 0$, a physically too strong constraint.
mielke-geometrodynamics_F01431	289	1.000	`\hat{R}=0`	$\hat{R}=0$	The variation with respect to $\hat{B}$ would lead to $\hat{R} = 0$, a physically too strong constraint.
mielke-geometrodynamics_F01432	289	0.994	`\eta_{A B C D E}:=\sqrt{\hat{g}} \varepsilon_{A B C D E}`	$\eta_{A B C D E}:=\sqrt{\hat{g}} \varepsilon_{A B C D E}$	Here, the antisymmetric unit tensor $\eta_{ABCDE} := \sqrt{\hat{g}}\varepsilon_{ABCDE}$ is constructed from the
mielke-geometrodynamics_F01433	289	1.000	`\varepsilon_{A B C D E}`	$\varepsilon_{A B C D E}$	*metric-free* Levi-Civita symbol ε_{ABCDE}, being invariant under the five-dimensional
mielke-geometrodynamics_F01434	289	0.913	`\hat{g}`	$\hat{g}$	structure group SL(5, $\mathbb{R}$), together with the determinant $\hat{g}$ of the invertible group
mielke-geometrodynamics_F01435	289	1.000	`B^{A B}:=\hat{g}^{A C} B_{C}{ }^{B}`	$B^{A B}:=\hat{g}^{A C} B_{C}{ }^{B}$	$B^{AB} := \hat{g}^{AC}B_C{}^B$. Thus, the additional quadratic term provides a *dynamical symmetry-*
mielke-geometrodynamics_F01436	289	0.853	`\operatorname{SO}(2,3)`	$\operatorname{SO}(2,3)$	or SO(2, 3), respectively. Moreover, the five-index structure of η_{ABCDE} has forced us
mielke-geometrodynamics_F01437	289	0.853	`\eta_{A B C D E}`	$\eta_{A B C D E}$	or SO(2, 3), respectively. Moreover, the five-index structure of η_{ABCDE} has forced us
mielke-geometrodynamics_F01438	289	0.997	`\Phi^{A}=\left\{\phi^{\alpha}, \phi^{\hat{4}}\right\}`	$\Phi^{A}=\left\{\phi^{\alpha}, \phi^{\hat{4}}\right\}$	to complement the construction by a quintet of pseudoscalars $\Phi^A = \{\phi^\alpha, \phi^{\hat{4}}\}$, our
mielke-geometrodynamics_F01439	289	0.988	`U\left(\Phi^{B} \Phi_{B}\right)`	$U\left(\Phi^{B} \Phi_{B}\right)$	arise from a minimum of a Higgs-type potential $U(\Phi^B\Phi_B)$; cf. PAGELS (1984) and
mielke-geometrodynamics_F01440	290	1.000	`\Phi=\Re \Phi+i \Im \Phi`	$\Phi=\Re \Phi+i \Im \Phi$	$\Phi = \Re\Phi + i\Im\Phi$ is invariant
mielke-geometrodynamics_F01441	290	0.995	`d C_{\mathrm{RR}}`	$d C_{\mathrm{RR}}$	where the resulting Pontryagin term dC_{RR} serves merely to reproduce the Bianchi
mielke-geometrodynamics_F01442	290	0.989	`\stackrel{\circ}{\Gamma}^{\alpha \beta}=`	$\stackrel{\circ}{\Gamma}^{\alpha \beta}=$	identity (13.5.5) for the curvature, now in terms of the Lorentz connection $\stackrel{\circ}{\Gamma}^{\alpha\beta} =$
mielke-geometrodynamics_F01443	290	1.000	`-\stackrel{\circ}{\Gamma}^{\beta \alpha}`	$-\stackrel{\circ}{\Gamma}^{\beta \alpha}$	$-\stackrel{\circ}{\Gamma}^{\beta\alpha}$. The variation with respect to the Lorentz-valued two-form $B^{\alpha\beta}$ leads to
mielke-geometrodynamics_F01444	290	1.000	`B^{\alpha \beta}`	$B^{\alpha \beta}$	$-\stackrel{\circ}{\Gamma}^{\beta\alpha}$. The variation with respect to the Lorentz-valued two-form $B^{\alpha\beta}$ leads to
mielke-geometrodynamics_F01445	290	1.000	`\mu \neq 0`	$\mu \neq 0$	as an equation of motion. Thus, for $\mu \neq 0$, our Lagrangian is “on shell” equivalent
mielke-geometrodynamics_F01446	291	1.000	`\vartheta^{\alpha}=g^{\alpha \beta} \theta_{\beta}`	$\vartheta^{\alpha}=g^{\alpha \beta} \theta_{\beta}$	For a nondegenerate coframe $\vartheta^\alpha = g^{\alpha\beta}\theta_\beta$ and vanishing torsion, the covariant
mielke-geometrodynamics_F01447	291	1.000	`\kappa=`	$\kappa=$	Pontryagin four-forms (here suppressed). Einstein's gravitational constant $\kappa =$
mielke-geometrodynamics_F01448	291	1.000	`8 \pi G_{\mathrm{N}}=8 \pi \ell_{\text {Planck }}^{2}`	$8 \pi G_{\mathrm{N}}=8 \pi \ell_{\text {Planck }}^{2}$	$8\pi G_{\mathrm{N}} = 8\pi\ell^2_{\mathrm{Planck}}$ and the density $\rho_\Lambda = \Lambda/\kappa = \Omega_\Lambda\rho_{\mathrm{crit}} < \rho_{\mathrm{crit}}$ corresponding to

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mielke-geometrodynamics_F01449	291	1.000	\rho_{\Lambda}=\Lambda / \kappa=\Omega_{\Lambda} \rho_{ \text {crit }}<\rho_{\text {crit }}	$\rho_{\Lambda}=\Lambda / \kappa=\Omega_{\Lambda} \rho_{\text {crit }}<\rho_{\text {crit }}$	$8\pi G_{\mathrm{N}} = 8\pi \ell^2_{\text{Planck}}$ and the density $\rho_\Lambda = \Lambda/\kappa = \Omega_\Lambda \rho_{\text{crit}} < \rho_{\text{crit}}$ corresponding to
mielke-geometrodynamics_F01450	291	1.000	\kappa=\mu \ell^{2} / 4	$\kappa=\mu \ell^{2} / 4$	implies that these constants are related via $\kappa = \mu\ell^2/4$ and $\Lambda = 12\kappa/\mu\ell^4 = 3/\ell^2$ to
mielke-geometrodynamics_F01451	291	1.000	\Lambda=12 \kappa / \mu \ell^{4}=3 / \ell^{2}	$\Lambda=12 \kappa / \mu \ell^{4}=3 / \ell^{2}$	implies that these constants are related via $\kappa = \mu\ell^2/4$ and $\Lambda = 12\kappa/\mu\ell^4 = 3/\ell^2$ to
mielke-geometrodynamics_F01452	291	1.000	\mu \simeq 10^{-123}	$\mu \simeq 10^{-123}$	The symmetry-breaking parameter $\mu \simeq 10^{-123}$ is observationally an extremely small
mielke-geometrodynamics_F01453	291	1.000	\hat{B} \wedge \hat{B}	$\hat{B} \wedge \hat{B}$	breaking $\hat{B} \wedge \hat{B}$ term, as anticipated by SMOLIN (2000) in the restricted case of
mielke-geometrodynamics_F01454	291	1.000	G_{\mathrm{N}}	G_{N}	gravitational constant G_{N} has its origin in SSB, as suggested already by ENGLERT et
mielke-geometrodynamics_F01455	292	0.997	O(5)	$O(5)$	Since our $\mathrm{SL}(5,\mathbb{R})$ gauge model is broken into two steps from $O(5)$ down to the
mielke-geometrodynamics_F01456	292	0.466	\operatorname{SO}(1,3)	$\operatorname{SO}(1,3)$	Lorentz group $\mathrm{SO}(1,3)$ as an exact subgroup, a local holonomic spacetime metric
mielke-geometrodynamics_F01457	292	1.000	g_{i j}	$g_{i j}$	g_{ij} is then induced. In the last step, we assumed that the translational connections
mielke-geometrodynamics_F01458	292	0.999	\Gamma_{4}{ }^{\alpha} \otimes_{\mathrm{s}} \Gamma_{\alpha}{ }^{4}	$\Gamma_{4}{ }^{\alpha} \otimes_{\mathrm{s}} \Gamma_{\alpha}{ }^{4}$	$\Gamma_4{}^\alpha \otimes_{\mathrm{s}} \Gamma_\alpha{}^4$ is dimensionless, we had to multiply the vacuum expectation (13.6.1) of
mielke-geometrodynamics_F01459	292	1.000	\ell^{2} / \mu^{2}=\left(\kappa / 4 \Lambda_{\mathrm{H}}\right)	$\ell^{2} / \mu^{2}=\left(\kappa / 4 \Lambda_{\mathrm{H}}\right)$	the derivative of the Higgs field by a *huge* dimensional factor $\ell^2/\mu^2 = (\kappa/4\Lambda_{\mathrm{H}})$. This
mielke-geometrodynamics_F01460	294	1.000	W^{+}	W^{+}	W^+ and W^- gauge bosons or even "heavy photons" Z, again via pair creation. These
mielke-geometrodynamics_F01461	294	1.000	W^{-}	W^{-}	W^+ and W^- gauge bosons or even "heavy photons" Z, again via pair creation. These
mielke-geometrodynamics_F01462	294	0.633	\mu \rightarrow 0	$\mu \rightarrow 0$	In the limit $\mu \to 0$ of $\mathrm{SL}(5,\mathbb{R})$ symmetry restoration, concepts like macroscopic
mielke-geometrodynamics_F01463	294	1.000	d s	$d s$	logical field theory does a line element ds emerge as a necessary means for measuring
mielke-geometrodynamics_F01464	294	1.000	1 / \alpha:=	$1 / \alpha:=$	slowly decay (ZEE 2004). The inverse *dimensionless* coupling constants $1/\alpha :=$
mielke-geometrodynamics_F01465	295	1.000	\hbar c / e^{2} \simeq 137	$\hbar c / e^{2} \simeq 137$	$\hbar c/e^2 \simeq 137$ of Sommerfeld and $1/\mu \simeq 10^{123}$ of gravity seem to be on a par except
mielke-geometrodynamics_F01466	295	1.000	1 / \mu \simeq 10^{123}	$1 / \mu \simeq 10^{123}$	$\hbar c/e^2 \simeq 137$ of Sommerfeld and $1/\mu \simeq 10^{123}$ of gravity seem to be on a par except
mielke-geometrodynamics_F01467	295	0.991	C_{\mathrm{TT}}	C_{TT}	contains a translational CS three-form C_{TT} that gives rise to the *parity-violating*
mielke-geometrodynamics_F01468	295	1.000	\mathfrak{g l}(4, \mathbb{R})	$\mathfrak{g l}(4, \mathbb{R})$	of Nieh and Yan (NY) (NIEH 2007). On the other hand, the $\mathfrak{gl}(4,\mathbb{R})$-valued Pontryagin
mielke-geometrodynamics_F01469	296	0.865	\operatorname{Ric}_{\alpha \beta}:=(-1)^{\text {sig }} *	$\operatorname{Ric}_{\alpha \beta}:=(-1)^{\text {sig }} *$	sion involving the symmetric Ricci tensor, i.e., the zero-form $\mathrm{Ric}_{\alpha\beta} := (-1)^{\text{sig}} *$
mielke-geometrodynamics_F01470	296	0.998	\left(R_{(\alpha}{ }^{\delta} \wedge \eta_{\delta \mid \beta)}\right)	$\left(R_{(\alpha}{ }^{\delta} \wedge \eta_{\delta \mid \beta)}\right)$	$(R_{(\alpha}{}^\delta \wedge \eta_{\delta\mid\beta)})$, is known as the *Gauss-Bonnet identity*.
mielke-geometrodynamics_F01471	300	1.000	\mathscr{S}:=\mathscr{M} / \mathscr{D}	$\mathscr{S}:=\mathscr{M} / \mathscr{D}$	[2]The quotient space $\mathscr{S} := \mathscr{M}/\mathscr{D}$ of all Riemannian metrics on a 3-dimensional manifold modulo the
mielke-geometrodynamics_F01472	300	1.000	\mathscr{D}	$\mathscr{D}$	group $\mathscr{D}$ of the diffeomorphisms of M^3 is called the "superspace". Accordingly, it has a completely
mielke-geometrodynamics_F01473	300	1.000	\mathrm{M}^{3}	M^{3}	group $\mathscr{D}$ of the diffeomorphisms of M^3 is called the "superspace". Accordingly, it has a completely
mielke-geometrodynamics_F01474	301	1.000	\mathrm{F}_{i j}	$\mathrm{F}_{i j}$	ponents F_{ij} of the electromagnetic field strengths. Such a theory was developed not
mielke-geometrodynamics_F01475	301	1.000	g_{[i j]} g^{[i j]}	$g_{[i j]} g^{[i j]}$	term proportional to $g_{[ij]}g^{[ij]}$ occurs side by side with Hilbert's Lagrangian 4-form,

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mielke-geometrodynamics_F01476	301	1.000	`\mathrm{g}_{[i j]}:=\ell^{* 2} \mathrm{~F}_{i j}`	$\mathrm{g}_{[i j]}:=\ell^{* 2} \mathrm{~F}_{i j}$	FAT (1979a), in a more recent study, has given up the identification $\mathrm{g}_{[ij]} := \ell^{*2}\mathrm{F}_{ij}$.
mielke-geometrodynamics_F01477	301	1.000	`m \in M`	$m \in M$	be dependent not only on the point $m \in M$ but also on locally distinguished direc-
mielke-geometrodynamics_F01478	303	1.000	`\Sigma_{\omega}`	Σ_{ω}	mathematical structure of the canonical energy–momentum current Σ_ω of the Yang–
mielke-geometrodynamics_F01479	304	0.995	`\Sigma_{\text {mat }}=0`	$\Sigma_{\text {mat }}=0$	$\Sigma_{\mathrm{mat}} = 0$.
mielke-geometrodynamics_F01480	304	1.000	`\delta=\pi / 2`	$\delta=\pi / 2$	$\delta = \pi/2$ is exactly equal to the operation of taking the Hodge dual[3]
mielke-geometrodynamics_F01481	305	1.000	`\Sigma_{\mathrm{YM}}`	Σ_{YM}	transformation). The *invariance* of Σ_{YM} with respect to duality rotations can easily
mielke-geometrodynamics_F01482	305	1.000	`\cos ^{2} \delta+`	$\cos ^{2} \delta+$	be proved by writing out and applying the elementary functional relation $\cos^2\delta +$
mielke-geometrodynamics_F01483	305	1.000	`\sin ^{2} \delta=1`	$\sin ^{2} \delta=1$	$\sin^2\delta = 1$ for trigonometric functions.
mielke-geometrodynamics_F01484	307	0.999	`R_{i j}^{\{ \}}`	$R_{i j}^{\{ \}}$	for the holonomically written components $R_{ij}^{\{\}}$ or the Ricci tensor. Combined with
mielke-geometrodynamics_F01485	307	0.993	`\bar{\tau}_{e}`	$\bar{\tau}_{e}$	out of the matter currents $\bar{\tau}_e$ and $\bar{\tau}_p$ of a theory (SCHWINGER 1968; JULIA &
mielke-geometrodynamics_F01486	307	0.993	`\bar{\tau}_{p}`	$\bar{\tau}_{p}$	out of the matter currents $\bar{\tau}_e$ and $\bar{\tau}_p$ of a theory (SCHWINGER 1968; JULIA &
mielke-geometrodynamics_F01487	308	1.000	`\delta=0`	$\delta=0$	traditional electrodynamics, which is defined by the choice $\delta = 0$ of a “dual gauge”
mielke-geometrodynamics_F01488	308	0.562	`\xi \wedge \vartheta`	$\xi \wedge \vartheta$	After the exterior multiplication of (14.1.32) by $\xi \wedge \vartheta$ and (14.1.33) by ${}^*\xi \wedge \vartheta$ and
mielke-geometrodynamics_F01489	308	0.562	`{ }^{*} \xi \wedge \vartheta`	${ }^{*} \xi \wedge \vartheta$	After the exterior multiplication of (14.1.32) by $\xi \wedge \vartheta$ and (14.1.33) by ${}^*\xi \wedge \vartheta$ and
mielke-geometrodynamics_F01490	308	0.966	`d \delta`	$d \delta$	$d\delta$, Eq. (14.1.37) has to be contracted by ${}^*\Sigma_{\mathrm{YM}}(\xi)$. On account of the invariance
mielke-geometrodynamics_F01491	308	0.966	`{ }^{*} \Sigma_{\mathrm{YM}}(\xi)`	${ }^{*} \Sigma_{\mathrm{YM}}(\xi)$	$d\delta$, Eq. (14.1.37) has to be contracted by ${}^*\Sigma_{\mathrm{YM}}(\xi)$. On account of the invariance
mielke-geometrodynamics_F01492	309	1.000	`d d \delta \equiv 0`	$d d \delta \equiv 0$	Due to the identity $dd\delta \equiv 0$, this follows in an unequivocal way from the exactness
mielke-geometrodynamics_F01493	309	0.928	`b_{1}=\operatorname{dim} H^{1}(M)=\nu`	$b_{1}=\operatorname{dim} H^{1}(M)=\nu$	of the form $d\delta$. For manifolds with Betti number $b_1 = \dim H^1(M) = \nu$, e.g., for
mielke-geometrodynamics_F01494	309	1.000	`v`	v	those that exhibit ν-fold global connectivity, however,
mielke-geometrodynamics_F01495	310	1.000	`(1962,1969)`	$(1962,1969)$	Encouraged by this remarkable result, WHEELER (1962, 1969) was inclined not
mielke-geometrodynamics_F01496	311	1.000	`\mathrm{n}=4+\mathrm{K}`	$\mathrm{n}=4+\mathrm{K}$	(or supergravity) in $\mathrm{n} = 4 + \mathrm{K}$ dimensions, which is then shown or assumed to
mielke-geometrodynamics_F01497	311	0.844	`(1926,1928)`	$(1926,1928)$	(1926, 1928) proved that this is possible after proceeding to 5-dimensional[5] man-
mielke-geometrodynamics_F01498	312	1.000	`M^{4+K}`	M^{4+K}	M^{4+K} that is endowed with a bundle $\hat{L}(M^{4+K})$ of linear frames and is furthermore
mielke-geometrodynamics_F01499	312	1.000	`\hat{L}\left(M^{4+K}\right)`	$\hat{L}\left(M^{4+K}\right)$	M^{4+K} that is endowed with a bundle $\hat{L}(M^{4+K})$ of linear frames and is furthermore
mielke-geometrodynamics_F01500	312	0.998	`\hat{\vartheta}`	$\hat{\vartheta}$	equipped with a canonical 1-form $\hat{\vartheta}$ associated with $\hat{L}(M^{4+K})$. Only later does one
mielke-geometrodynamics_F01501	312	1.000	`\hat{E} \approx M^{4} \times B`	$\hat{E} \approx M^{4} \times B$	such that M^{4+K} takes on the form $\hat{E} \approx M^4 \times B$, where B is a (non)compact K-
mielke-geometrodynamics_F01502	312	1.000	`G / H`	G / H	dimensional homogeneous space, G/H. Here G denotes a semisimple Lie group of

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mielke-geometrodynamics_F01503	312	0.951	`\mathrm{K}=\operatorname{dim} \mathrm{G}`	$\mathrm{K}=\operatorname{dim} \mathrm{G}$	dimension K = dim G, and H an isotropic subgroup. As shown by COQUEREAUX &
mielke-geometrodynamics_F01504	312	0.600	`P\left(M^{4}, G, \pi, \delta\right)`	$P\left(M^{4}, G, \pi, \delta\right)$	JADCZYK (1983), the "geometric arena" is not the fiber bundle $P(M^4, G, \pi, \delta)$ but
mielke-geometrodynamics_F01505	312	0.983	`P\left(M^{4}, N / H, \pi, \delta\right)`	$P\left(M^{4}, N / H, \pi, \delta\right)$	$P(M^4, N/H, \pi, \delta)$, where N denotes the normalizer of H in G. The extended space
mielke-geometrodynamics_F01506	312	1.000	`\hat{E}`	$\hat{E}$	$\hat{E}$, which is locally isomorphic to $M^4 \times G/H$, is globally the associated bundle
mielke-geometrodynamics_F01507	312	1.000	`M^{4} \times G / H`	$M^{4} \times G / H$	$\hat{E}$, which is locally isomorphic to $M^4 \times G/H$, is globally the associated bundle
mielke-geometrodynamics_F01508	312	0.996	`\hat{E}\left(M^{4}, F=G / H, N / H, P\right)`	$\hat{E}\left(M^{4}, F=G / H, N / H, P\right)$	$\hat{E}(M^4, F = G/H, N/H, P)$ with N/H as structure group.
mielke-geometrodynamics_F01509	312	0.996	`N / H`	N / H	$\hat{E}(M^4, F = G/H, N/H, P)$ with N/H as structure group.
mielke-geometrodynamics_F01510	312	0.858	`\omega=\omega^{j}{ }_{\alpha} L_{j} \otimes \vartheta^{ \alpha}`	$\omega=\omega^{j}{ }_{\alpha} L_{j} \otimes \vartheta^{\alpha}$	and $\omega = \omega^j{}_\alpha L_j \otimes \vartheta^\alpha$ is a 1-form with values in the Lie algebra of G or N/H,
mielke-geometrodynamics_F01511	312	0.760	`\stackrel{\circ}{\vartheta}`	$\stackrel{\circ}{\vartheta}$	Mills-type gauge fields. The 1-form $\stackrel{\circ}{\vartheta}$ is defined on the cotangent bundle $T_e{}^*(G)$ of
mielke-geometrodynamics_F01512	312	0.760	`T_{e}{ }^{*}(G)`	$T_{e}{ }^{*}(G)$	Mills-type gauge fields. The 1-form $\stackrel{\circ}{\vartheta}$ is defined on the cotangent bundle $T_e{}^*(G)$ of
mielke-geometrodynamics_F01513	312	1.000	`\xi^{j}`	ξ^{j}	the group G, being parametrized by the coordinates ξ^j. In its expansion
mielke-geometrodynamics_F01514	312	1.000	`\varphi_{j}`	φ_{j}	the coefficients φ_j may be regarded as a multiplet of (Higgs-type) scalar fields trans-
mielke-geometrodynamics_F01515	312	1.000	`d \xi^{j}`	$d \xi^{j}$	(holonomic) basis $d\xi^j$ is "dual" to the infinitesimal generators I_j of G. Moreover,
mielke-geometrodynamics_F01516	312	1.000	`I_{j}`	I_{j}	(holonomic) basis $d\xi^j$ is "dual" to the infinitesimal generators I_j of G. Moreover,
mielke-geometrodynamics_F01517	312	0.881	`V=V^{j} I_{j}, \stackrel{\circ}{\vartheta}`	$V=V^{j} I_{j}, \stackrel{\circ}{\vartheta}$	as a frame "dual" to the Lie vector field $V = V^j I_j$, $\stackrel{\circ}{\vartheta}$ satisfies the Maurer–Cartan
mielke-geometrodynamics_F01518	312	1.000	`\stackrel{\circ}{\omega}`	$\stackrel{\circ}{\omega}$	A connection $\stackrel{\circ}{\omega}$ on G arises naturally via the Ansatz
mielke-geometrodynamics_F01519	313	1.000	`\stackrel{\circ}{\Theta}`	$\stackrel{\circ}{\Theta}$	The *internal* torsion $\stackrel{\circ}{\Theta}$ and curvature $\stackrel{\circ}{\Omega}$, respectively, are obtained from Cartan's
mielke-geometrodynamics_F01520	313	1.000	`\stackrel{\circ}{\Omega}`	$\stackrel{\circ}{\Omega}$	The *internal* torsion $\stackrel{\circ}{\Theta}$ and curvature $\stackrel{\circ}{\Omega}$, respectively, are obtained from Cartan's
mielke-geometrodynamics_F01521	313	0.811	`b=1 / 2`	$b=1 / 2$	$b = 1/2$: torsionless connection in a symmetric coset space B of constant
mielke-geometrodynamics_F01522	313	1.000	`\stackrel{\circ}{R}=1 / 4`	$\stackrel{\circ}{R}=1 / 4$	curvature $\stackrel{\circ}{R} = 1/4$.
mielke-geometrodynamics_F01523	313	1.000	`b=0,1`	$b=0,1$	$b = 0, 1$: Flat connection *with* parallelizing *torsion*.
mielke-geometrodynamics_F01524	313	1.000	`\operatorname{Tr}(\stackrel{\circ}{\vartheta} \otimes \stackrel{\circ}{\vartheta})`	$\operatorname{Tr}(\stackrel{\circ}{\vartheta} \otimes \stackrel{\circ}{\vartheta})$	G. The term $\mathrm{Tr}(\stackrel{\circ}{\vartheta} \otimes \stackrel{\circ}{\vartheta})$ just gives the internal line element
mielke-geometrodynamics_F01525	313	1.000	`\stackrel{\circ}{g}_{i j}`	$\stackrel{\circ}{g}_{i j}$	where $\stackrel{\circ}{g}_{ij}$ denotes the Cartan–Killing metric of G. As far as *simple* Lie groups are
mielke-geometrodynamics_F01526	313	0.997	`\operatorname{Tr}(\omega \otimes \stackrel{\circ}{ \vartheta})`	$\operatorname{Tr}(\omega \otimes \stackrel{\circ}{\vartheta})$	the terms $\mathrm{Tr}(\stackrel{\circ}{\vartheta} \otimes \stackrel{\circ}{\vartheta})$ and $\mathrm{Tr}(\omega \otimes \stackrel{\circ}{\vartheta})$ but also in $\mathrm{Tr}(\omega \otimes \omega)$, since ω takes values in
mielke-geometrodynamics_F01527	313	0.997	`\operatorname{Tr}(\omega \otimes \omega)`	$\operatorname{Tr}(\omega \otimes \omega)$	the terms $\mathrm{Tr}(\stackrel{\circ}{\vartheta} \otimes \stackrel{\circ}{\vartheta})$ and $\mathrm{Tr}(\omega \otimes \stackrel{\circ}{\vartheta})$ but also in $\mathrm{Tr}(\omega \otimes \omega)$, since ω takes values in

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mielke-geometrodynamics_F01528	313	1.000	`d \hat{s}^{2}`	$d \hat{s}^{2}$	classical, i.e., nonquantized models, the total signature of the line element $d\hat{s}^2$ has
mielke-geometrodynamics_F01529	313	0.983	`\mathrm{G} \times \mathrm{G} / \mathrm{G}_{\mathrm{D}}`	$\mathrm{G} \times \mathrm{G} / \mathrm{G}_{\mathrm{D}}$	[6]More precisely, the group manifold of G is the symmetric space $\mathrm{G} \times \mathrm{G}/\mathrm{G}_{\mathrm{D}}$, where G_{D} is the
mielke-geometrodynamics_F01530	313	0.983	`\mathrm{G}_{\mathrm{D}}`	G_{D}	[6]More precisely, the group manifold of G is the symmetric space $\mathrm{G} \times \mathrm{G}/\mathrm{G}_{\mathrm{D}}$, where G_{D} is the
mielke-geometrodynamics_F01531	314	1.000	`\hat{L}(\hat{E})`	$\hat{L}(\hat{E})$	may be imprinted on the bundle $\hat{L}(\hat{E})$ of the linear frames. Here ω^g denotes the
mielke-geometrodynamics_F01532	314	0.998	`L\left(M^{4}\right), \stackrel{\circ}{\omega}`	$L\left(M^{4}\right), \stackrel{\circ}{\omega}$	gravitational connection in $L(M^4)$, $\stackrel{\circ}{\omega}$ a connection in the group manifold, and $\hat{\alpha}$
mielke-geometrodynamics_F01533	314	0.998	`\hat{\alpha}`	$\hat{\alpha}$	gravitational connection in $L(M^4)$, $\stackrel{\circ}{\omega}$ a connection in the group manifold, and $\hat{\alpha}$
mielke-geometrodynamics_F01534	314	0.514	`4+\mathrm{K}`	$4+\mathrm{K}$	is valid. Connections possessing a nonvanishing $(4+\mathrm{K})$-dimensional torsion and
mielke-geometrodynamics_F01535	314	1.000	`\hat{\Omega}`	$\hat{\Omega}$	Accordingly, the higher-dimensional Riemann–Cartan curvature 2-form $\hat{\Omega}$, given
mielke-geometrodynamics_F01536	314	0.917	`(4+\mathrm{K})`	$(4+\mathrm{K})$	In our matrix notation, the $(4+\mathrm{K})$-dimensional exterior derivative $\hat{\mathrm{d}}$ is given by
mielke-geometrodynamics_F01537	314	0.917	`\hat{\mathrm{d}}`	$\hat{\mathrm{d}}$	In our matrix notation, the $(4+\mathrm{K})$-dimensional exterior derivative $\hat{\mathrm{d}}$ is given by
mielke-geometrodynamics_F01538	315	0.980	`M^{4}`	M^{4}	internal space B relative to the four-dimensional spacetime M^4 in the bundle $\hat{E}$.
mielke-geometrodynamics_F01539	315	0.999	`\Omega^{\prime}`	$\Omega^{\prime}$	$\hat{\alpha}$ and the 2-form Ω' of the Yang–Mills field strengths, we have obtained the decisive
mielke-geometrodynamics_F01540	316	0.988	`\alpha_{g}:=g^{2} / \hbar c`	$\alpha_{g}:=g^{2} / \hbar c$	modification by the gauge-coupling constant $\alpha_g := g^2/\hbar c$. The term $\hat{L}_{\mathrm{W}}^{G}$ depends
mielke-geometrodynamics_F01541	316	0.988	`\hat{L}_{\mathrm{W}}^{G}`	$\hat{L}_{\mathrm{W}}^{G}$	modification by the gauge-coupling constant $\alpha_g := g^2/\hbar c$. The term $\hat{L}_{\mathrm{W}}^{G}$ depends
mielke-geometrodynamics_F01542	316	0.928	`\stackrel{\circ}{R}`	$\stackrel{\circ}{R}$	whose value is determined by the scalar curvature $\stackrel{\circ}{R}$ and by the density $\sqrt{\det \stackrel{\circ}{g}}$ of
mielke-geometrodynamics_F01543	316	0.928	`\sqrt{\operatorname{det} \stackrel{\circ}{g}}`	$\sqrt{\operatorname{det} \stackrel{\circ}{g}}$	whose value is determined by the scalar curvature $\stackrel{\circ}{R}$ and by the density $\sqrt{\det \stackrel{\circ}{g}}$ of
mielke-geometrodynamics_F01544	316	1.000	`M^{4+K} \approx M^{4} \times B`	$M^{4+K} \approx M^{4} \times B$	(i) $M^{4+K} \approx M^4 \times B$ locally and
mielke-geometrodynamics_F01545	316	1.000	`R_{4} \gg R_{K}`	$R_{4} \gg R_{K}$	(ii) $R_4 \gg R_K$ for the corresponding characteristic radii.
mielke-geometrodynamics_F01546	316	0.611	`\mathscr{D}\left(M^{4+K}\right)`	$\mathscr{D}\left(M^{4+K}\right)$	"spontaneous" symmetry breaking of the general $(4+\mathrm{K})$-dimensional covariance group $\mathscr{D}(M^{4+K})$
mielke-geometrodynamics_F01547	316	0.686	`R_{\circ}`	$R_{\circ}$	which tends to diminish the characteristic radius $R_\circ$ of B to the order of the Planck length ℓ^*
mielke-geometrodynamics_F01548	317	0.998	`Y_{n}(B)`	$Y_{n}(B)$	into a complete set of harmonic functions $Y_n(B)$ on $B = G/H$; cf. SALAM &
mielke-geometrodynamics_F01549	317	0.998	`B=G / H`	$B=G / H$	into a complete set of harmonic functions $Y_n(B)$ on $B = G/H$; cf. SALAM &
mielke-geometrodynamics_F01550	317	1.000	`M^{4} \times S^{1}`	$M^{4} \times S^{1}$	closing up of the five-dimensional world to a cylinder $M^4 \times S^1$ of an exceedingly
mielke-geometrodynamics_F01551	317	0.725	`R_{\mathrm{o}}`	R_{o}	small radius $R_\circ$ is taken for granted, Eq. (14.2.21) degenerates to a Fourier expansion
mielke-geometrodynamics_F01552	317	1.000	`S^{1}`	S^{1}	on the circle S^1. The coefficients in the expansion (14.2.21) are fields of short range
mielke-geometrodynamics_F01553	318	1.000	`S U(2) \times U(1)`	$S U(2) \times U(1)$	of strong interactions and the $SU(2) \times U(1)$ gauge group of the Weinberg–Salam

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mielke-geometrodynamics_F01554	318	■0.948	`\mathbb{C} P^{2}=S U(3) / U(2)`	$\mathbb{C} P^{2}=S U(3) / U(2)$	where $\mathbb{C}P^2 = SU(3)/U(2)$ denotes the complex projective two-space and S^n the
mielke-geometrodynamics_F01555	318	■0.948	`S^{n}`	S^{n}	where $\mathbb{C}P^2 = SU(3)/U(2)$ denotes the complex projective two-space and S^n the
mielke-geometrodynamics_F01556	318	■0.994	`M^{4} \times B`	$M^{4} \times B$	difficulty arises due to the fact that the Cartesian product $M^4 \times B$ of the Minkowski
mielke-geometrodynamics_F01557	318	■1.000	`\mathrm{n}=11`	$\mathrm{n}=11$	Einstein space. In order to remedy this deficiency in, e.g., $\mathrm{n}=11$ dimensions, it is
mielke-geometrodynamics_F01558	318	■0.874	`S^{7}=S O(8) / S O(7)`	$S^{7}=S O(8) / S O(7)$	necessary to “squash” the (round) sphere $S^7 = SO(8)/SO(7)$ by partial rescalings
mielke-geometrodynamics_F01559	318	■0.999	`\mathrm{K}=4+2+1=7`	$\mathrm{K}=4+2+1=7$	As observed by WITTEN (1981), $\mathrm{K}=4+2+1=7$ of (14.2.25) represents the min-
mielke-geometrodynamics_F01560	319	■1.000	`\mathrm{n}=4+7=11`	$\mathrm{n}=4+7=11$	metric, belong to a single representation. Most remarkably, $\mathrm{n}=4+7=11$ is the
mielke-geometrodynamics_F01561	319	■1.000	`(\mathscr{N}=1)`	$(\mathscr{N}=1)$	*maximal* dimension of a theory of simple ($\mathscr{N}=1$) supergravity, cf. DEWITT &
mielke-geometrodynamics_F01562	319	■1.000	`\overline{\mathrm{s}} \leq 3 / 2`	$\overline{\mathrm{s}} \leq 3 / 2$	STORA (1984), that contains the necessary fermion fields of spin $\overline{\mathrm{s}} \leq 3/2$ without
mielke-geometrodynamics_F01563	319	■1.000	`\overline{\mathrm{s}}=2`	$\overline{\mathrm{s}}=2$	the bosonic fields exceeding the spin or helicity $\overline{\mathrm{s}}=2$ of the graviton. The bosonic
mielke-geometrodynamics_F01564	319	■0.928	`A_{\text {IJK }}`	$A_{\text {IJK }}$	metric tensor potential A_{IJK}, a remnant of the internal torsion, which may induce a
mielke-geometrodynamics_F01565	319	■0.835	`M^{11}=(\mathrm{AdS})^{4} \times S^{7}`	$M^{11}=(\mathrm{AdS})^{4} \times S^{7}$	tions, the underlying vacuum geometry factors into the product $M^{11}=(\mathrm{AdS})^4 \times S^7$
mielke-geometrodynamics_F01566	319	■1.000	`\tilde{\mathrm{S}}^{7}`	$\tilde{\mathrm{S}}^{7}$	ical model of the real world. For the “squashed” seven-sphere $\tilde{\mathrm{S}}^7$, there exists a
mielke-geometrodynamics_F01567	319	■1.000	`M^{5}=\mathbb{R}^{4} \times S^{1}`	$M^{5}=\mathbb{R}^{4} \times S^{1}$	BERGMANN (1938), who utilized the topology $M^5=\mathbb{R}^4\times S^1$ of a cylinder for the
mielke-geometrodynamics_F01568	320	■0.994	`\mathrm{E}_{2,4}`	$\mathrm{E}_{2,4}$	pseudo-Euclidean space $\mathrm{E}_{2,4}$ of conformal symmetries was pointed out by BUDINI
mielke-geometrodynamics_F01569	320	■0.998	`\hat{\omega}`	$\hat{\omega}$	point. Employing the dissociation (14.2.9) of the connection 1-form $\hat{\omega}$, one obtains
mielke-geometrodynamics_F01570	320	■1.000	`\hat{\psi}`	$\hat{\psi}$	field $\hat{\psi}$ with $2^{[2+\mathrm{K}/2]}$ components is adopted:
mielke-geometrodynamics_F01571	320	■1.000	`2^{[2+\mathrm{K} / 2]}`	$2^{[2+\mathrm{K} / 2]}$	field $\hat{\psi}$ with $2^{[2+\mathrm{K}/2]}$ components is adopted:
mielke-geometrodynamics_F01572	320	■1.000	`\chi_{\rho F}^{ \pm}(\xi)`	$\chi_{\rho F}^{ \pm}(\xi)$	Here the spinorial harmonics $\chi^{\pm}_{\rho F}(\xi)$ furnish a representation ρ of the symmetry
mielke-geometrodynamics_F01573	320	■0.989	`\stackrel{\circ}{\gamma}^{K+1}`	$\stackrel{\circ}{\gamma}^{K+1}$	of the internal helicity operator $\stackrel{\circ}{\gamma}{}^{K+1}$. After projection π to four-dimensional compo-
mielke-geometrodynamics_F01574	321	■1.000	`\tilde{\omega}^{\mathrm{g}}`	$\tilde{\omega}^{\mathrm{g}}$	the (spinor) connections $\tilde{\omega}^{\mathrm{g}}$ and ω, respectively, but that in addition, inherits a gauge
mielke-geometrodynamics_F01575	321	■1.000	`\gamma \wedge \gamma \wedge{ }^{*} \Omega \psi`	$\gamma \wedge \gamma \wedge{ }^{*} \Omega \psi$	covariant term $\gamma \wedge \gamma \wedge {}^*\Omega\psi$ of electric dipole type. This term in the Lagrangian
mielke-geometrodynamics_F01576	321	■0.527	`\hat{\mathrm{m}}=0`	$\hat{\mathrm{m}}=0$	which follows from the Lagrangian $(4+\mathrm{K})$-form in the case $\hat{\mathrm{m}}=0$. This equation
mielke-geometrodynamics_F01577	322	■1.000	`\stackrel{\circ}{D}`	$\stackrel{\circ}{D}$	depends via the covariant derivative $\stackrel{\circ}{D}$ on the spinor connection $\stackrel{\circ}{\tilde{\omega}}$ and therefore acts
mielke-geometrodynamics_F01578	322	■1.000	`\stackrel{\circ}{\tilde{\omega}}`	$\stackrel{\circ}{\tilde{\omega}}$	depends via the covariant derivative $\stackrel{\circ}{D}$ on the spinor connection $\stackrel{\circ}{\tilde{\omega}}$ and therefore acts
mielke-geometrodynamics_F01579	322	■0.978	`i \not D^{\text {(int) }}`	$i \not D^{\text {(int) }}$	only on the internal space B. Provided B is compact, $i \not D^{(\mathrm{int})}$ has a discrete spectrum

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mielke-geometrodynamics_F01580	322	1.000	`\gamma^{5} \psi=(\mp) \psi`	$\gamma^{5} \psi=(\mp) \psi$	obeying $\gamma^5\psi = (\mp)\psi$, transform according to inequivalent representations of the
mielke-geometrodynamics_F01581	322	1.000	`n=4+K`	$n=4+K$	dimension of the total space. For $n = 4 + K$, the spinor fields transform accord-
mielke-geometrodynamics_F01582	322	0.999	`\widetilde{S O}(1,3+K)`	$\widetilde{S O}(1,3+K)$	ing to the fundamental representation of the covering tangent group $\widetilde{SO}(1, 3 + K)$.
mielke-geometrodynamics_F01583	322	0.995	`\hat{\gamma}^{n+1}`	$\hat{\gamma}^{n+1}$	eigenvalues of the n-dimensional chirality operator $\hat{\gamma}^{n+1}$. Under the subgroup
mielke-geometrodynamics_F01584	322	0.615	`\overparen{S O}(1,3) \otimes \widetilde{S O}(K)`	$\overparen{S O}(1,3) \otimes \widetilde{S O}(K)$	$\widetilde{SO}(1, 3) \otimes \widetilde{SO}(K)$, such a representation branches according to
mielke-geometrodynamics_F01585	322	1.000	`n=2 p+1`	$n=2 p+1$	(i) $n = 2p + 1$: The transformation $\hat{\gamma}^A \rightarrow -\hat{\gamma}^A$ does not affect the infinitesimal
mielke-geometrodynamics_F01586	322	1.000	`\hat{\gamma}^{A} \rightarrow-\hat{\gamma}^{A}`	$\hat{\gamma}^{A} \rightarrow-\hat{\gamma}^{A}$	(i) $n = 2p + 1$: The transformation $\hat{\gamma}^A \rightarrow -\hat{\gamma}^A$ does not affect the infinitesimal
mielke-geometrodynamics_F01587	322	1.000	`\hat{\sigma}^{A B}`	$\hat{\sigma}^{A B}$	generators $\hat{\sigma}^{AB}$ but $\hat{\gamma}^{n+1} \rightarrow -\hat{\gamma}^{n+1}$. Since G-transformations are
mielke-geometrodynamics_F01588	322	1.000	`\hat{\gamma}^{n+1} \rightarrow-\hat{\gamma}^{n+1}`	$\hat{\gamma}^{n+1} \rightarrow-\hat{\gamma}^{n+1}$	generators $\hat{\sigma}^{AB}$ but $\hat{\gamma}^{n+1} \rightarrow -\hat{\gamma}^{n+1}$. Since G-transformations are
mielke-geometrodynamics_F01589	322	1.000	`n=4 p`	$n=4 p$	(ii) $n = 4p$: This implies that
mielke-geometrodynamics_F01590	322	1.000	`n=4 p+2`	$n=4 p+2$	(iii) $n = 4p + 2$: Then it follows that
mielke-geometrodynamics_F01591	323	0.543	`i \not D^{(\text {int })}`	$i \not D^{(\text {int })}$	Then it remains to reveal the conditions under which $i \not D^{(\text{int})}$ has zero eigenmodes.
mielke-geometrodynamics_F01592	323	1.000	`\mathrm{D}^{*}={ }^{*} \mathrm{D}`	$\mathrm{D}^{*}={ }^{*} \mathrm{D}$	Since $\mathrm{D}^* = {}^*\mathrm{D}$ holds for a metric-compatible (spinor) connection,
mielke-geometrodynamics_F01593	323	0.922	`\stackrel{\circ}{\Theta}=0`	$\stackrel{\circ}{\Theta}=0$	for vanishing torsion, i.e., $\stackrel{\circ}{\Theta} = 0$, Eq. (14.3.18) is the sum of a nonnegative and a
mielke-geometrodynamics_F01594	324	1.000	`n>2`	$n>2$	izable in $n > 2$ dimensions according to perturbation theory. For supergravity with
mielke-geometrodynamics_F01595	324	0.999	`n=11`	$n=11$	turbative expansion. Moreover, simple ($\mathcal{N} = 1$) supergravity in $n = 11$ dimensions
mielke-geometrodynamics_F01596	324	0.999	`G=\mathrm{SO}(32)`	$G=\mathrm{SO}(32)$	ical strings. These *superstring theories* with $G = \mathrm{SO}(32)$ or the exceptional group
mielke-geometrodynamics_F01597	324	1.000	`\mathrm{E}_{8} \times \mathrm{E}_{8}`	$\mathrm{E}_{8} \times \mathrm{E}_{8}$	$\mathrm{E}_8 \times \mathrm{E}_8$ as structure group seem to provide a unification of all physical interactions
mielke-geometrodynamics_F01598	324	1.000	`n=10+\operatorname{dim} G=506`	$n=10+\operatorname{dim} G=506$	in $n = 10 + \dim G = 506$ dimensions (DUFF et al. 1985) and a consistent and *finite*
mielke-geometrodynamics_F01599	324	0.993	`{ }^{10} J_{3} \hbar`	${ }^{10} J_{3} \hbar$	by the mass M, the z-component[10] $J_3\hbar$ of the angular momentum, and the total angu-
mielke-geometrodynamics_F01600	324	1.000	`\sqrt{J(J+1)} \hbar`	$\sqrt{J(J+1)} \hbar$	lar momentum $\sqrt{J(J+1)}\hbar$. Furthermore, it may carry the generalized "internal"
mielke-geometrodynamics_F01601	324	1.000	`Q^{(i)}`	$Q^{(i)}$	Here $Q^{(i)}$ and $P^{(i)}$ denote the "electric"- and "magnetic"-type charges of a "dually"
mielke-geometrodynamics_F01602	324	1.000	`P^{(i)}`	$P^{(i)}$	Here $Q^{(i)}$ and $P^{(i)}$ denote the "electric"- and "magnetic"-type charges of a "dually"
mielke-geometrodynamics_F01603	324	1.000	`J_{3} \hat{=} J`	$J_{3} \hat{=} J$	rotation group. As for our still purely *classical* considerations, $J_3 \hat{=} J$ remains valid.
mielke-geometrodynamics_F01604	325	0.798	`r^{2} e^{v}`	$r^{2} e^{v}$	substitutes the function r^2e^v in the Schwarzschild metric. The generalization of
mielke-geometrodynamics_F01605	326	0.976	`\bar{Q}^{(\mathrm{o})}`	$\bar{Q}^{(\mathrm{o})}$	mass M, generalized charge $\overline{Q}^{(\mathrm{o})}$, and (classical) angular momentum component $J_3\hbar$
mielke-geometrodynamics_F01606	326	0.976	`J_{3} \hbar`	$J_{3} \hbar$	mass M, generalized charge $\overline{Q}^{(\mathrm{o})}$, and (classical) angular momentum component $J_3\hbar$

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mielke-geometrodynamics_F01607	326	1.000	`\mu_{m}=\frac{Q_{e}}{M} J`	$\mu_{m}=\frac{Q_{e}}{M} J$	with respect to the axis of rotation, but it also has the magnetic moment $\mu_m = \frac{Q_e}{M} J$,
mielke-geometrodynamics_F01608	326	0.823	`\mathrm{M}_{\mathrm{ir}}`	M_{ir}	The total mass M of a black hole depends on its geometric remnant mass M_{ir},
mielke-geometrodynamics_F01609	326	0.938	`\mathrm{M}_{\mathrm{ir}}, \bar{Q}^{(\circ)}`	$\mathrm{M}_{\mathrm{ir}}, \bar{Q}^{(\circ)}$	dependence of M on the characteristic values M_{ir}, $\overline{Q}^{(\circ)}$, and J is nonlinear. More
mielke-geometrodynamics_F01610	326	0.811	`J \hat{=} J_{3}=0`	$J \hat{=} J_{3}=0$	formulated by REISSNER (1916) and NORDSTRÖM (1918) if $J \hat{=} J_3 = 0$ holds.
mielke-geometrodynamics_F01611	327	1.000	`\lambda_{3}`	λ_{3}	The generalized Gell-Mann matrix λ_3, which is identical to the Pauli matrix σ_3 for
mielke-geometrodynamics_F01612	327	1.000	`\sigma_{3}`	σ_{3}	The generalized Gell-Mann matrix λ_3, which is identical to the Pauli matrix σ_3 for
mielke-geometrodynamics_F01613	327	1.000	`\Delta`	Δ	Here $\varDelta$ is to be substituted by the term (14.4.5) with $J = 0$. Apart from the horizon
mielke-geometrodynamics_F01614	327	0.993	`r=r_{\mathrm{o}}`	$r=r_{\mathrm{o}}$	given by (14.4.6), this spacetime geometry has obviously still a further one at $r = r_{\circ}$,
mielke-geometrodynamics_F01615	327	1.000	`r=0`	$r=0$	away from the essential singularity at $r = 0$.
mielke-geometrodynamics_F01616	328	0.672	`Q=-\bar{Q} \cos \delta`	$Q=-\bar{Q} \cos \delta$	a particle of mass M with an electric charge $Q = -\overline{Q} \cos\delta$ and a magnetic monopole
mielke-geometrodynamics_F01617	328	1.000	`P=-\bar{Q} \sin \delta`	$P=-\bar{Q} \sin \delta$	strength $P = -\overline{Q} \sin\delta$. If the surface area $\mathrm{S_A}$ of the event horizon, which is defined
mielke-geometrodynamics_F01618	328	1.000	`\mathrm{S}_{\mathrm{A}}`	S_{A}	strength $P = -\overline{Q} \sin\delta$. If the surface area $\mathrm{S_A}$ of the event horizon, which is defined
mielke-geometrodynamics_F01619	328	0.546	`\bar{Q}`	$\bar{Q}$	and a large dual charge $\overline{Q}$, the charge–mass relation and the total mass M of the
mielke-geometrodynamics_F01620	328	0.990	`1 / \kappa`	$1 / \kappa$	necessarily fixed by the coupling constant $1/\kappa$ of the Lorentz "rotons".
mielke-geometrodynamics_F01621	330	1.000	`\mathrm{d}=10`	$\mathrm{d}=10$	Green MB, Schwarz JH (1984) Anomaly cancellations in supersymmetric $\mathrm{d} = 10$ gauge theory and
mielke-geometrodynamics_F01622	335	1.000	`1 / \alpha_{e}`	$1 / \alpha_{e}$	than electromagnetic forces by the factor $1/\alpha_e$. On the other hand, all experiments
mielke-geometrodynamics_F01623	335	0.967	`m_{\pi}=2 m_{e} / \alpha_{e}`	$m_{\pi}=2 m_{e} / \alpha_{e}$	(1964), is explained by a virtual exchange of scalar π-mesons of mass $m_\pi = 2\,m_e/\alpha_e$.
mielke-geometrodynamics_F01624	335	0.829	`S U(3)^{c}`	$S U(3)^{c}$	ory with an exact $SU(3)^c$ structure group. The introduction of these additional color
mielke-geometrodynamics_F01625	336	0.924	`f_{i j}`	$f_{i j}$	tensor fields f_{ij} and g_{ij} or via the canonical 1-forms ϑ^f and ϑ^g as follows:
mielke-geometrodynamics_F01626	336	0.924	`\vartheta^{f}`	ϑ^{f}	tensor fields f_{ij} and g_{ij} or via the canonical 1-forms ϑ^f and ϑ^g as follows:
mielke-geometrodynamics_F01627	336	0.924	`\vartheta^{g}`	ϑ^{g}	tensor fields f_{ij} and g_{ij} or via the canonical 1-forms ϑ^f and ϑ^g as follows:
mielke-geometrodynamics_F01628	336	1.000	`L_{f g}`	$L_{f g}$	necessitates the introduction of an additional mixing term L_{fg}. Were it not for this
mielke-geometrodynamics_F01629	336	1.000	`\vartheta^{\mathrm{f}}`	ϑ^{f}	mixing term is understood to have only a polynomial dependence on ϑ^{f} and ϑ^{g} of
mielke-geometrodynamics_F01630	336	1.000	`b-4`	$b-4$	degree a or $b-4$,
mielke-geometrodynamics_F01631	337	1.000	`\sum^{f}`	$\sum^{f}$	Let $\sum^f$ and $\sum^g$ be the canonical energy–momentum currents that are generated
mielke-geometrodynamics_F01632	337	1.000	`\sum^{g}`	$\sum^{g}$	Let $\sum^f$ and $\sum^g$ be the canonical energy–momentum currents that are generated

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mielke-geometrodynamics_F01633	338	0.921	`\mathrm{G}_{\mathrm{N}}`	G_{N}	of the (weak) Newtonian constant G_{N} of gravity to the coupling constant G_{S} of strong
mielke-geometrodynamics_F01634	338	0.921	`\mathrm{G}_{\mathrm{S}}`	G_{S}	of the (weak) Newtonian constant G_{N} of gravity to the coupling constant G_{S} of strong
mielke-geometrodynamics_F01635	338	1.000	`=10^{-13} \mathrm{~cm}`	$=10^{-13} \mathrm{~cm}$	spacetime structure beyond hadronic distances of the size of one fermi $= 10^{-13}$ cm.
mielke-geometrodynamics_F01636	338	1.000	`\mathrm{f}^{0}`	f^{0}	f^{0}-meson and the massless graviton in a first approximation.
mielke-geometrodynamics_F01637	339	1.000	`\mu^{g}, \mu^{f}`	μ^{g}, μ^{f}	is the result for the f-metric. The integration constants are μ^g, μ^f, and γ. For u = 1/2,
mielke-geometrodynamics_F01638	339	1.000	`\mathrm{u}=1 / 2`	$\mathrm{u}=1 / 2$	is the result for the f-metric. The integration constants are μ^g, μ^f, and γ. For u = 1/2,
mielke-geometrodynamics_F01639	339	0.992	`\mu^{g}=\Lambda=0`	$\mu^{g}=\Lambda=0$	$\mu^g = \Lambda = 0$, this solution reduces to the configuration that was given by SALAM &
mielke-geometrodynamics_F01640	339	1.000	`m_{f}=1-2 \mathrm{GeV} / c^{2}`	$m_{f}=1-2 \mathrm{GeV} / c^{2}$	f-mesons of mass $m_f = 1-2$ GeV/c^2. This additional property of the tensor dom-
mielke-geometrodynamics_F01641	340	1.000	`E_{j}^{\beta}`	E_{j}^{β}	the Poincaré gauge theory. These are given by the tetrads E_j^β and Ricci's rotation
mielke-geometrodynamics_F01642	340	1.000	`\Gamma_{i}^{\alpha \beta}`	$\Gamma_{i}^{\alpha \beta}$	coefficients $\Gamma_i^{\alpha\beta}$, respectively. With respect to anholonomic frame, these potentials
mielke-geometrodynamics_F01643	340	1.000	`\left(\frac{1}{2}, \frac{1}{2}\right)`	$\left(\frac{1}{2}, \frac{1}{2}\right)$	transform under the Lorentz group according to the representation $\left(\frac{1}{2}, \frac{1}{2}\right)$ or $(1,0) \oplus$
mielke-geometrodynamics_F01644	340	1.000	`(1,0) \oplus`	$(1,0) \oplus$	transform under the Lorentz group according to the representation $\left(\frac{1}{2}, \frac{1}{2}\right)$ or $(1,0) \oplus$
mielke-geometrodynamics_F01645	340	0.609	`(0,1)`	$(0,1)$	$(0,1)$, respectively, and are therefore have an holonomic spin $\bar{s}_a = 1$. After the tran-
mielke-geometrodynamics_F01646	340	0.609	`\operatorname{spin} \bar{s}_{a}=1`	$\operatorname{spin} \bar{s}_{a}=1$	$(0,1)$, respectively, and are therefore have an holonomic spin $\bar{s}_a = 1$. After the tran-
mielke-geometrodynamics_F01647	340	1.000	`\bar{s}=2`	$\bar{s}=2$	induces gauge potentials of maximal (reducible) spin or helicity $\bar{s} = 2$ (HEHL et al.
mielke-geometrodynamics_F01648	340	1.000	`\Lambda_{\text {eff }} \neq 0`	$\Lambda_{\text {eff }} \neq 0$	nonlinear theory with $\Lambda_{\text{eff}} \neq 0$. Other quantization procedures are needed in order to isolate the
mielke-geometrodynamics_F01649	341	1.000	`\ell=\ell^{*} / \sqrt{\kappa}`	$\ell=\ell^{*} / \sqrt{\kappa}$	proportional to $\ell = \ell^*/\sqrt{\kappa}$ is superposed in the metric background of the solutions.
mielke-geometrodynamics_F01650	341	1.000	`1 / \sqrt{\kappa}`	$1 / \sqrt{\kappa}$	ℓ^* scaled by $1/\sqrt{\kappa}$ is identical to the Compton wavelength of a proton according to
mielke-geometrodynamics_F01651	341	1.000	`10^{13} \mathrm{~cm}`	$10^{13} \mathrm{~cm}$	minimum radius would be increased to about 10^{13} cm as a result of the hypothesis of
mielke-geometrodynamics_F01652	341	0.999	`\ell^{2}`	ℓ^{2}	“strong” gravity. The relative strength of the coupling constants ℓ^2 and ℓ^{*2}, which
mielke-geometrodynamics_F01653	341	0.999	`\ell^{* 2}`	$\ell^{* 2}$	“strong” gravity. The relative strength of the coupling constants ℓ^2 and ℓ^{*2}, which
mielke-geometrodynamics_F01654	341	1.000	`\kappa=0,38 \times 10^{-38}`	$\kappa=0,38 \times 10^{-38}$	small number of order of magnitude $\kappa = 0,38 \times 10^{-38}$. It was again EINSTEIN in
mielke-geometrodynamics_F01655	342	1.000	`10^{20}`	10^{20}	there would be a fundamental length ℓ being enlarged by about 10^{20}. This hypothe-
mielke-geometrodynamics_F01656	342	1.000	`\mathrm{M}_{\mathrm{U}}`	M_{U}	between the total mass M_{U} of the universe and the mass M_{P} of the proton while
mielke-geometrodynamics_F01657	342	1.000	`\mathrm{M}_{\mathrm{P}}`	M_{P}	between the total mass M_{U} of the universe and the mass M_{P} of the proton while
mielke-geometrodynamics_F01658	342	1.000	`\ell_{\mathrm{H}}`	ℓ_{H}	relating it to the product of the relative Hubble radius ℓ_{H} and the relative Compton
mielke-geometrodynamics_F01659	342	1.000	`S U(f)`	$S U(f)$	resentations of the unitary groups $SU(f)$. Concerning this aspect, particular success

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mielke-geometrodynamics_F01660	342	0.995	`S U(4)`	$SU(4)$	to a far larger extent for the enlarged group $SU(4)$ that is postulated in the context of the charm
mielke-geometrodynamics_F01661	342	0.997	`S U(5)`	$SU(5)$	hypothesis, and also for the grand unified theories (GUTs) that embarks upon $SU(5)$ as internal
mielke-geometrodynamics_F01662	343	0.999	`S O(3) \approx S U(2)`	$SO(3) \approx SU(2)$	transform according to spinor representations of the group $SO(3) \approx SU(2)$ of the
mielke-geometrodynamics_F01663	343	1.000	`S U(4) \approx S O(6) \supset`	$SU(4) \approx SO(6) \supset$	nal spin within a common group-theoretic scheme, the group $SU(4) \approx SO(6) \supset$
mielke-geometrodynamics_F01664	343	1.000	`S O(4) \approx S U(2) \times S U(2)`	$SO(4) \approx SU(2) \times SU(2)$	$SO(4) \approx SU(2) \times SU(2)$ was proposed by Wigner. Later on, the group $SU(6)$ was
mielke-geometrodynamics_F01665	343	1.000	`S U(6)`	$SU(6)$	$SO(4) \approx SU(2) \times SU(2)$ was proposed by Wigner. Later on, the group $SU(6)$ was
mielke-geometrodynamics_F01666	343	1.000	`J=1 / 2`	$J=1/2$	$J = 1/2$ and $J = 3/2$ of total angular momentum are thought to fill out a 56-fold
mielke-geometrodynamics_F01667	343	1.000	`J=3 / 2`	$J=3/2$	$J = 1/2$ and $J = 3/2$ of total angular momentum are thought to fill out a 56-fold
mielke-geometrodynamics_F01668	343	1.000	`S L(6, \mathbb{C}) \supset S L(2, \mathbb{C}) \times S U(3)`	$SL(6,\mathbb{C}) \supset SL(2,\mathbb{C}) \times SU(3)$	this problem is to extend $SU(6)$ to the linear group $SL(6,\mathbb{C}) \supset SL(2,\mathbb{C}) \times SU(3)$.
mielke-geometrodynamics_F01669	343	0.962	`\mathrm{SU}(3)`	$\mathrm{SU}(3)$	Thereby a relativistic union of the internal symmetry group SU(3) with the (simply
mielke-geometrodynamics_F01670	343	1.000	`S L(6, \mathbb{C})`	$SL(6,\mathbb{C})$	spacetime, is achieved. If the symmetry is regarded only as a local one, an $SL(6,\mathbb{C})$
mielke-geometrodynamics_F01671	343	0.933	`\mathrm{P}(\mathrm{M}, \overline{\mathrm{G}}, \pi, \delta)`	$\mathrm{P}(\mathrm{M}, \overline{\mathrm{G}}, \pi, \delta)$	fiber bundle $\mathrm{P}(\mathrm{M}, \overline{\mathrm{G}}, \pi, \delta)$ as a geometric arena. But this time, it is
mielke-geometrodynamics_F01672	343	1.000	`[S U(4) \otimes]^{4}`	$[SU(4)\otimes]^{4}$	chiral $[SU(4)\otimes]^4$ model of ELIAS et al. (1978) ready to hand for the case $f = c = 4$.
mielke-geometrodynamics_F01673	343	1.000	`f=c=4`	$f=c=4$	chiral $[SU(4)\otimes]^4$ model of ELIAS et al. (1978) ready to hand for the case $f = c = 4$.
mielke-geometrodynamics_F01674	344	0.993	`\overline{\mathrm{L}}(\mathrm{M})`	$\overline{\mathrm{L}}(\mathrm{M})$	with $\overline{\mathrm{L}}(\mathrm{M})$. This 1-form is supposed to have holonomic components in the local
mielke-geometrodynamics_F01675	344	0.993	`\mathrm{A}_{2}`	A_{2}	fields—as is, for instance, realized in nature via the $SU(3)$-nonet f, f', A_2, and K*
mielke-geometrodynamics_F01676	344	0.995	`S L(2 N, \mathbb{C}) \supset S L(2, \mathbb{C}) \times U(N)`	$SL(2N,\mathbb{C}) \supset SL(2,\mathbb{C}) \times U(N)$	erators of $SL(2N,\mathbb{C}) \supset SL(2,\mathbb{C}) \times U(N)$ can be realized by the matrices
mielke-geometrodynamics_F01677	344	1.000	`\lambda_{i}`	λ_{i}	where the generalized Gell-Mann matrices λ_i are the N^2-1 vector operators of the
mielke-geometrodynamics_F01678	344	1.000	`\mathrm{N}^{2}-1`	$\mathrm{N}^{2}-1$	where the generalized Gell-Mann matrices λ_i are the N^2-1 vector operators of the
mielke-geometrodynamics_F01679	344	0.983	`\operatorname{SU}(N)`	$\operatorname{SU}(N)$	$SU(N)$ subgroup. If the gauge potentials of the unitary groups—which are contained
mielke-geometrodynamics_F01680	345	1.000	`\mathrm{A}_{i}`	A_{i}	The Yang–Mills-like gauge potentials A_i aside, this total connection 1-form com-
mielke-geometrodynamics_F01681	345	0.998	`\bar{\gamma}`	$\bar{\gamma}$	field strengths that belong to the 1-forms $\overline{\gamma}$ and $\overline{\omega}$ are the 2-form
mielke-geometrodynamics_F01682	345	1.000	`L_{\mathrm{W}}(\mathrm{f})`	$L_{\mathrm{W}}(\mathrm{f})$	freedom, this is achieved by replacing the Weyl term $L_{\mathrm{W}}(\mathrm{f})$ in the total Lagrangian
mielke-geometrodynamics_F01683	345	1.000	`f=c=0`	$f=c=0$	If the internal degrees of freedom are suppressed, i.e., for the case $f = c = 0$, the
mielke-geometrodynamics_F01684	345	1.000	`f=3`	$f=3$	as a structure group. In the framework of an $SL(6,\mathbb{C})$ gauge theory, i.e., for $f = 3$
mielke-geometrodynamics_F01685	345	0.987	`c=0`	$c=0$	and $c = 0$, they also proposed a mixing term $\overline{L}_{\mathrm{fg}}$ that is responsible for the medi-
mielke-geometrodynamics_F01686	345	0.987	`\bar{L}_{\mathrm{fg}}`	$\bar{L}_{\mathrm{fg}}$	and $c = 0$, they also proposed a mixing term $\overline{L}_{\mathrm{fg}}$ that is responsible for the medi-

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mielke-geometrodynamics_F01687	346	1.000	G=S U(2)	$G=SU(2)$	made allowance only for the isospin group, i.e., $G = SU(2)$. If we followed that
mielke-geometrodynamics_F01688	346	0.989	S U(4)^{f} \otimes S U(4)^{c}	$SU(4)^{f} \otimes SU(4)^{c}$	theories. Within the scheme A of this $SU(4)^f \otimes SU(4)^c$-gauge theory, the quarklike
mielke-geometrodynamics_F01689	346	1.000	S L(8, \mathbb{C})^{f} \otimes S U(8, \mathbb{C})^{c}	$SL(8, \mathbb{C})^{f} \otimes SU(8, \mathbb{C})^{c}$	$SL(8,\mathbb{C})^f \otimes SU(8,\mathbb{C})^c$-gauge theory, however, is rather more related—following
mielke-geometrodynamics_F01690	347	1.000	M=\mathbb{R} \times S^{1} \times S^{2}	$M=\mathbb{R} \times S^{1} \times S^{2}$	fields within the nontrivial topology of the space $M = \mathbb{R} \times S^1 \times S^2$. Its apparent
mielke-geometrodynamics_F01691	348	0.994	G L(8, \mathbb{C})^{f} \otimes	$GL(8, \mathbb{C})^{f} \otimes$	1980) as indicated previously; see also CASTRO (2012). Thereby, a $GL(8,\mathbb{C})^f \otimes$
mielke-geometrodynamics_F01692	348	1.000	G L(8, \mathbb{C})^{c}	$GL(8, \mathbb{C})^{c}$	$GL(8,\mathbb{C})^c$ gauge unification of all basic particle forces emerged into a rather spec-
mielke-geometrodynamics_F01693	348	1.000	U(5)	$U(5)$	Since $SU(4)$ and $U(5)$ together with the local Lorentz group can also be embedded
mielke-geometrodynamics_F01694	348	0.987	S O(1,9)	$SO(1,9)$	(FADDEEV 2011) such as $SO(1,9)$ and the embedding of the $SO(10)$ GUT into the
mielke-geometrodynamics_F01695	348	0.987	S O(10)	$SO(10)$	(FADDEEV 2011) such as $SO(1,9)$ and the embedding of the $SO(10)$ GUT into the
mielke-geometrodynamics_F01696	348	0.994	S O(1,13)	$SO(1,13)$	group $SO(1,13)$ have been proposed (PERCACCI 1991). Even "graviweak" groups
mielke-geometrodynamics_F01697	348	0.998	S O(4, \mathbb{C})	$SO(4, \mathbb{C})$	such as $SO(4,\mathbb{C})$ have been considered by NESTI & PERCACCI (2010). Again, in
mielke-geometrodynamics_F01698	348	0.995	E(8)	$E(8)$	such as $E(8)$ and hyperbolic Kac–Moody algebras (KLEINSCHMIDT & NICOLAI
mielke-geometrodynamics_F01699	348	0.995	E(10)	$E(10)$	2010) like $E(10)$ have been suggested to achieve this. However, these approaches
mielke-geometrodynamics_F01700	349	0.984	(5, \mathbb{C})	$(5, \mathbb{C})$	Castro C (2012) A Clifford Cl $(5,\mathbb{C})$ unified gauge field theory of conformal gravity, Maxwell and
mielke-geometrodynamics_F01701	349	0.992	\operatorname{SL}(2 \mathrm{n}, \mathbb{C}) \otimes \operatorname{SL}(2, \mathbb{C})	$\operatorname{SL}(2 \mathrm{n}, \mathbb{C}) \otimes \operatorname{SL}(2, \mathbb{C})$	Dennis PW, Huang JC (1977) Spontaneous symmetry breakdown in an $\mathrm{SL}(2n,\mathbb{C}) \otimes \mathrm{SL}(2,\mathbb{C})$-
mielke-geometrodynamics_F01702	349	0.998	1000 \mathrm{~m}_{\mathrm{W}}	$1000 \mathrm{~m}_{\mathrm{W}}$	$1000\ \mathrm{m_W}$. Phys Rev Lett 40(14):920
mielke-geometrodynamics_F01703	350	0.473	2 \mathrm{n}, \mathbb{C}	$2 \mathrm{n}, \mathbb{C}$	Huang JC, Dennis PW (1981a) Spin-1 and spin-2 gauge fields in unified $\mathrm{SL}(2n,\mathbb{C})$ model. Phys
mielke-geometrodynamics_F01704	350	0.805	2^{+}	2^{+}	Isham C, Salam A, Strathdee J (1973a) 2^+ nonet as gauge particles for $\mathrm{SL}(6,\mathbb{C})$ symmetry. Phys
mielke-geometrodynamics_F01705	354	0.952	\mathrm{R}-\mathrm{R}_{\mathrm{o}}	$\mathrm{R}-\mathrm{R}_{\mathrm{o}}$	$\mathrm{R}-\mathrm{R_o}$) whose gradient balances the electrodynamic force.[1]
mielke-geometrodynamics_F01706	354	1.000	(1966,1967)	$(1966,1967)$	for instance, PRASAD (1966, 1967) and ROMAN & KOH (1966). These notions have
mielke-geometrodynamics_F01707	354	1.000	\kappa>0	$\kappa>0$	responsible for the negative (for $\kappa > 0$) pressure term
mielke-geometrodynamics_F01708	354	0.983	2 \mu / \mathrm{r}	$2 \mu / \mathrm{r}$	sented by a Schwarzschild–de Sitter microuniverse, whereby the $2\mu/\mathrm{r}$-singularity
mielke-geometrodynamics_F01709	355	1.000	\ell_{m}	ℓ_{m}	coordinate. The (dimensionless) ratio of the Compton wavelength ℓ_m of the test
mielke-geometrodynamics_F01710	355	0.998	\mathrm{F}=\mathrm{F}_{\mathrm{nm}}^{\iota}\left(\rho^{*} \right)	$\mathrm{F}=\mathrm{F}_{\mathrm{nm}}^{\iota}\left(\rho^{*}\right)$	with reference to the still to be determined functions $\mathrm{F}=\mathrm{F}^{\iota}_{\mathrm{nm}}(\rho^*)$.
mielke-geometrodynamics_F01711	356	0.877	\iota=0	$\iota=0$	For the sake of illustration (Fig. 16.1), the simplest case $\iota = 0$ for $\kappa > 0$, i.e., the
mielke-geometrodynamics_F01712	357	1.000	\rho^{*}=\frac{\pi}{2}	$\rho^{*}=\frac{\pi}{2}$	At $\rho^* = \frac{\pi}{2}$ there arises for $\beta^2 < 2$ a steeply rising potential wall that can never be
mielke-geometrodynamics_F01713	357	1.000	\beta^{2}<2	$\beta^{2}<2$	At $\rho^* = \frac{\pi}{2}$ there arises for $\beta^2 < 2$ a steeply rising potential wall that can never be
mielke-geometrodynamics_F01714	357	1.000	\mathrm{V}(\mathrm{r})=2 \mu / \mathrm{r}+\mathrm{cr}	$\mathrm{V}(\mathrm{r})=2 \mu / \mathrm{r}+\mathrm{cr}$	In the latter, it is usually the potential $\mathrm{V(r)} = 2\mu/\mathrm{r} + \mathrm{cr}$ that is postulated for the

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mielke-geometrodynamics_F01715	358	0.997	`\overline{\mathrm{Q}}`	$\overline{\mathrm{Q}}$	quantum number J of total angular momentum, and generalized charge $\bar{\mathrm{Q}}$ is (almost)
mielke-geometrodynamics_F01716	358	1.000	`\mathrm{Q}_{\mathrm{e}}`	Q_{e}	while the electric charge Q_{e} is determined via the generalized Gell-Mann–Nishijima
mielke-geometrodynamics_F01717	358	0.522	`\mathrm{M}_{\text {ir }}`	$\mathrm{M}_{\text {ir }}$	irreducible mass M_{ir}; concerning the tensor dominance model, it is thus suggestive
mielke-geometrodynamics_F01718	358	0.630	`\mathrm{M}^{*}:=\hbar \sqrt{8 \pi} / \ell^{*} \mathrm{c}`	$\mathrm{M}^{*}:=\hbar \sqrt{8 \pi} / \ell^{*} \mathrm{c}$	to propose that M_{ir} as well as the modified Planck mass $\mathrm{M}^{*} := \hbar\sqrt{8\pi}/\ell^{*}\mathrm{c}$ are
mielke-geometrodynamics_F01719	358	0.966	`\mathrm{M}_{\text {ir }}=\mathrm{M}^{*}=1 \mathrm{GeV} / \mathrm{c}^{2}`	$\mathrm{M}_{\text {ir }}=\mathrm{M}^{*}=1 \mathrm{GeV} / \mathrm{c}^{2}$	approximately identical to the mass of a proton, i.e., $\mathrm{M}_{\mathrm{ir}} = \mathrm{M}^{*} = 1\mathrm{GeV}/\mathrm{c}^{2}$. For
mielke-geometrodynamics_F01720	359	0.999	`M, J`	M, J	M, J, and I are the characteristics of particles that are here defined *canonically* in
mielke-geometrodynamics_F01721	359	0.999	`I`	I	M, J, and I are the characteristics of particles that are here defined *canonically* in
mielke-geometrodynamics_F01722	359	1.000	`\mathrm{J}=0`	$\mathrm{J}=0$	(However, if $\mathrm{J}=0$, this leads precisely back to the linear Gell-Mann–Okubo formula!)
mielke-geometrodynamics_F01723	359	0.999	`\mathrm{b}=\frac{1}{5}`	$\mathrm{b}=\frac{1}{5}$	surprisingly well represented by (16.3.3). (In Fig. 16.2, $\mathrm{b} = \frac{1}{5}$ has been chosen, while
mielke-geometrodynamics_F01724	359	0.998	`\Lambda(1116)`	$\Lambda(1116)$	M* has been accommodated to the state $\mathit{\Lambda}(1116)$ of excited baryons.) A detailed
mielke-geometrodynamics_F01725	360	1.000	`k_{B}=8.6 \times 10^{-5} \mathrm{eV} / \mathrm{K}`	$k_{B}=8.6 \times 10^{-5} \mathrm{eV} / \mathrm{K}$	Here $k_B = 8.6 \times 10^{-5}$ eV/K denotes the Boltzmann constant. Under the assump-
mielke-geometrodynamics_F01726	362	0.877	`S U(8)`	$S U(8)$	Okubo S (1975) $SU(4)$ and $SU(8)$ mass formulas and weak interactions. Phys Rev D 11(11):3261
mielke-geometrodynamics_F01727	362	0.824	`(6, \mathbb{C})`	$(6, \mathbb{C})$	Salam A (1973). On SL $(6, \mathbb{C})$ gauge invariance, In: Fundamental Interactions in Physics, 1973 Coral
mielke-geometrodynamics_F01728	364	0.591	`\equiv`	$\equiv$	$:=$: Equal by definition $\equiv$: Identically equal $\approx$: Isomorphic $\simeq$: Approximately equal $\sim$: Asymptotic expansion $\times$: Topological product $\otimes$: Direct product $\oplus$: Direct sum $\ltimes$: Semidirect product $\mathrm{G}_{\circ}$: Connected component of the group $\tilde{\mathrm{G}}$: Simply connected covering group $[\mathrm{A}, \mathrm{B}] := \mathrm{AB} - \mathrm{BA}$: Commutator $\{\mathrm{A}, \mathrm{B}\} := \mathrm{AB} + \mathrm{BA}$: Anticommutator $\mathrm{A}^{*} = \mathrm{C}\,\mathrm{A}$: Complex-conjugate matrix $\mathrm{A}^{+} = \mathrm{A}^{*\mathrm{T}}$: Hermitian adjoint matrix $\overset{-1}{\pi}(\mathrm{N})$: Original domain of N with respect to the mapping $\pi : \mathrm{M} \to \mathrm{N}$ $\mathrm{T}_{(\mu\nu)} := \frac{1}{2}(\mathrm{T}_{\mu\nu} + \mathrm{T}_{\nu\mu})$: Symmetrized second-rank tensor $\mathrm{T}_{[\mu\nu]} := \frac{1}{2}(\mathrm{T}_{\mu\nu} - \mathrm{T}_{\nu\mu})$: Antisymmetrized second-rank tensor

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mielke-geometrodynamics_F01729	364	0.591	`\approx`	$\approx$	$:=$: Equal by definition $\equiv$: Identically equal $\approx$: Isomorphic $\simeq$: Approximately equal $\sim$: Asymptotic expansion $\times$: Topological product $\otimes$: Direct product $\oplus$: Direct sum $\ltimes$: Semidirect product $\mathrm{G}_{\circ}$: Connected component of the group $\tilde{\mathrm{G}}$: Simply connected covering group $[\mathrm{A}, \mathrm{B}] := \mathrm{AB} - \mathrm{BA}$: Commutator $\{\mathrm{A}, \mathrm{B}\} := \mathrm{AB} + \mathrm{BA}$: Anticommutator $\mathrm{A}^{*}= \mathrm{C\,A}$: Complex-conjugate matrix $\mathrm{A}^{+} = \mathrm{A}^{*\mathrm{T}}$: Hermitian adjoint matrix $\overset{-1}{\pi}(\mathrm{N})$: Original domain of N with respect to the mapping $\pi : \mathrm{M} \to \mathrm{N}$ $\mathrm{T}_{(\mu\nu)} := \frac{1}{2}(\mathrm{T}_{\mu\nu} + \mathrm{T}_{\nu\mu})$: Symmetrized second-rank tensor $\mathrm{T}_{[\mu\nu]} := \frac{1}{2}(\mathrm{T}_{\mu\nu} - \mathrm{T}_{\nu\mu})$: Antisymmetrized second-rank tensor
mielke-geometrodynamics_F01730	364	0.591	`\mathrm{G}_{\circ}`	$\mathrm{G}_{\circ}$	$:=$: Equal by definition $\equiv$: Identically equal $\approx$: Isomorphic $\simeq$: Approximately equal $\sim$: Asymptotic expansion $\times$: Topological product $\otimes$: Direct product $\oplus$: Direct sum $\ltimes$: Semidirect product $\mathrm{G}_{\circ}$: Connected component of the group $\tilde{\mathrm{G}}$: Simply connected covering group $[\mathrm{A}, \mathrm{B}] := \mathrm{AB} - \mathrm{BA}$: Commutator $\{\mathrm{A}, \mathrm{B}\} := \mathrm{AB} + \mathrm{BA}$: Anticommutator $\mathrm{A}^{*}= \mathrm{C\,A}$: Complex-conjugate matrix $\mathrm{A}^{+} = \mathrm{A}^{*\mathrm{T}}$: Hermitian adjoint matrix $\overset{-1}{\pi}(\mathrm{N})$: Original domain of N with respect to the mapping $\pi : \mathrm{M} \to \mathrm{N}$ $\mathrm{T}_{(\mu\nu)} := \frac{1}{2}(\mathrm{T}_{\mu\nu} + \mathrm{T}_{\nu\mu})$: Symmetrized second-rank tensor $\mathrm{T}_{[\mu\nu]} := \frac{1}{2}(\mathrm{T}_{\mu\nu} - \mathrm{T}_{\nu\mu})$: Antisymmetrized second-rank tensor
mielke-geometrodynamics_F01731	364	0.591	`[\mathrm{A}, \mathrm{B}]:=\mathrm{AB}-\mathrm{BA}`	$[\mathrm{A}, \mathrm{B}]:=\mathrm{AB}-\mathrm{BA}$	$:=$: Equal by definition $\equiv$: Identically equal $\approx$: Isomorphic $\simeq$: Approximately equal $\sim$: Asymptotic expansion $\times$: Topological product $\otimes$: Direct product $\oplus$: Direct sum $\ltimes$: Semidirect product $\mathrm{G}_{\circ}$: Connected component of the group $\tilde{\mathrm{G}}$: Simply connected covering group $[\mathrm{A}, \mathrm{B}] := \mathrm{AB} - \mathrm{BA}$: Commutator $\{\mathrm{A}, \mathrm{B}\} := \mathrm{AB} + \mathrm{BA}$: Anticommutator $\mathrm{A}^{*}= \mathrm{C\,A}$: Complex-conjugate matrix $\mathrm{A}^{+} = \mathrm{A}^{*\mathrm{T}}$: Hermitian adjoint matrix $\overset{-1}{\pi}(\mathrm{N})$: Original domain of N with respect to the mapping $\pi : \mathrm{M} \to \mathrm{N}$ $\mathrm{T}_{(\mu\nu)} := \frac{1}{2}(\mathrm{T}_{\mu\nu} + \mathrm{T}_{\nu\mu})$: Symmetrized second-rank tensor $\mathrm{T}_{[\mu\nu]} := \frac{1}{2}(\mathrm{T}_{\mu\nu} - \mathrm{T}_{\nu\mu})$: Antisymmetrized second-rank tensor

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mielke-geometrodynamics_F01732	364	■0.591	`\{\mathrm{A}, \mathrm{B}\}:=\mathrm{AB}+\mathrm{BA}`	$\{\mathrm{A}, \mathrm{B}\}:=\mathrm{AB}+\mathrm{BA}$	$:=$: Equal by definition $\equiv$: Identically equal $\approx$: Isomorphic $\simeq$: Approximately equal $\sim$: Asymptotic expansion $\times$: Topological product $\otimes$: Direct product $\oplus$: Direct sum $\ltimes$: Semidirect product $\mathrm{G}_\circ$: Connected component of the group $\tilde{\mathrm{G}}$: Simply connected covering group $[\mathrm{A}, \mathrm{B}] := \mathrm{AB} - \mathrm{BA}$: Commutator $\{\mathrm{A}, \mathrm{B}\} := \mathrm{AB} + \mathrm{BA}$: Anticommutator $\mathrm{A}^* = \mathrm{C\,A}$: Complex-conjugate matrix $\mathrm{A}^+ = \mathrm{A}^{*\mathrm{T}}$: Hermitian adjoint matrix $\overset{-1}{\pi}(\mathrm{N})$: Original domain of N with respect to the mapping $\pi : \mathrm{M} \to \mathrm{N}$ $\mathrm{T}_{(\mu\nu)} := \frac{1}{2}(\mathrm{T}_{\mu\nu} + \mathrm{T}_{\nu\mu})$: Symmetrized second-rank tensor $\mathrm{T}_{[\mu\nu]} := \frac{1}{2}(\mathrm{T}_{\mu\nu} - \mathrm{T}_{\nu\mu})$: Antisymmetrized second-rank tensor
mielke-geometrodynamics_F01733	364	■0.591	`\mathrm{A}^{+}=\mathrm{A}^{* \mathrm{~T}}`	$\mathrm{A}^{+}=\mathrm{A}^{* \mathrm{~T}}$	$:=$: Equal by definition $\equiv$: Identically equal $\approx$: Isomorphic $\simeq$: Approximately equal $\sim$: Asymptotic expansion $\times$: Topological product $\otimes$: Direct product $\oplus$: Direct sum $\ltimes$: Semidirect product $\mathrm{G}_\circ$: Connected component of the group $\tilde{\mathrm{G}}$: Simply connected covering group $[\mathrm{A}, \mathrm{B}] := \mathrm{AB} - \mathrm{BA}$: Commutator $\{\mathrm{A}, \mathrm{B}\} := \mathrm{AB} + \mathrm{BA}$: Anticommutator $\mathrm{A}^* = \mathrm{C\,A}$: Complex-conjugate matrix $\mathrm{A}^+ = \mathrm{A}^{*\mathrm{T}}$: Hermitian adjoint matrix $\overset{-1}{\pi}(\mathrm{N})$: Original domain of N with respect to the mapping $\pi : \mathrm{M} \to \mathrm{N}$ $\mathrm{T}_{(\mu\nu)} := \frac{1}{2}(\mathrm{T}_{\mu\nu} + \mathrm{T}_{\nu\mu})$: Symmetrized second-rank tensor $\mathrm{T}_{[\mu\nu]} := \frac{1}{2}(\mathrm{T}_{\mu\nu} - \mathrm{T}_{\nu\mu})$: Antisymmetrized second-rank tensor
mielke-geometrodynamics_F01734	364	■0.591	`\pi^{-1}(\mathrm{~N})`	$\pi^{-1}(\mathrm{~N})$	$:=$: Equal by definition $\equiv$: Identically equal $\approx$: Isomorphic $\simeq$: Approximately equal $\sim$: Asymptotic expansion $\times$: Topological product $\otimes$: Direct product $\oplus$: Direct sum $\ltimes$: Semidirect product $\mathrm{G}_\circ$: Connected component of the group $\tilde{\mathrm{G}}$: Simply connected covering group $[\mathrm{A}, \mathrm{B}] := \mathrm{AB} - \mathrm{BA}$: Commutator $\{\mathrm{A}, \mathrm{B}\} := \mathrm{AB} + \mathrm{BA}$: Anticommutator $\mathrm{A}^* = \mathrm{C\,A}$: Complex-conjugate matrix $\mathrm{A}^+ = \mathrm{A}^{*\mathrm{T}}$: Hermitian adjoint matrix $\overset{-1}{\pi}(\mathrm{N})$: Original domain of N with respect to the mapping $\pi : \mathrm{M} \to \mathrm{N}$ $\mathrm{T}_{(\mu\nu)} := \frac{1}{2}(\mathrm{T}_{\mu\nu} + \mathrm{T}_{\nu\mu})$: Symmetrized second-rank tensor $\mathrm{T}_{[\mu\nu]} := \frac{1}{2}(\mathrm{T}_{\mu\nu} - \mathrm{T}_{\nu\mu})$: Antisymmetrized second-rank tensor

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mielke-geometrodynamics_F01735	364	■ 0.591	`\pi: \mathrm{M} \rightarrow \mathrm{N}`	$\pi : \mathrm{M} \rightarrow \mathrm{N}$	$:=$: Equal by definition $\equiv$: Identically equal $\approx$: Isomorphic $\simeq$: Approximately equal $\sim$: Asymptotic expansion $\times$: Topological product $\otimes$: Direct product $\oplus$: Direct sum $\ltimes$: Semidirect product $\mathrm{G}_\circ$: Connected component of the group $\tilde{\mathrm{G}}$: Simply connected covering group $[\mathrm{A},\mathrm{B}] := \mathrm{AB}-\mathrm{BA}$: Commutator $\{\mathrm{A},\mathrm{B}\} := \mathrm{AB}+\mathrm{BA}$: Anticommutator $\mathrm{A}^{*}= \mathrm{C\,A}$: Complex-conjugate matrix $\mathrm{A}^{+} = \mathrm{A}^{*\mathrm{T}}$: Hermitian adjoint matrix $\overset{-1}{\pi}(\mathrm{N})$: Original domain of N with respect to the mapping $\pi : \mathrm{M} \to \mathrm{N}$ $\mathrm{T}_{(\mu\nu)} := \frac{1}{2}(\mathrm{T}_{\mu\nu}+\mathrm{T}_{\nu\mu})$: Symmetrized second-rank tensor $\mathrm{T}_{[\mu\nu]} := \frac{1}{2}(\mathrm{T}_{\mu\nu}-\mathrm{T}_{\nu\mu})$: Antisymmetrized second-rank tensor
mielke-geometrodynamics_F01736	364	■ 0.591	`\mathrm{T}_{(\mu \nu)}:=\frac{1}{2}\left(\mathrm{~T}_{\mu \nu}+\mathrm{T}_{\nu \mu}\right)`	$\mathrm{T}_{(\mu \nu)}:=\frac{1}{2}\left(\mathrm{~T}_{\mu \nu}+\mathrm{T}_{\nu \mu}\right)$	$:=$: Equal by definition $\equiv$: Identically equal $\approx$: Isomorphic $\simeq$: Approximately equal $\sim$: Asymptotic expansion $\times$: Topological product $\otimes$: Direct product $\oplus$: Direct sum $\ltimes$: Semidirect product $\mathrm{G}_\circ$: Connected component of the group $\tilde{\mathrm{G}}$: Simply connected covering group $[\mathrm{A},\mathrm{B}] := \mathrm{AB}-\mathrm{BA}$: Commutator $\{\mathrm{A},\mathrm{B}\} := \mathrm{AB}+\mathrm{BA}$: Anticommutator $\mathrm{A}^{*}= \mathrm{C\,A}$: Complex-conjugate matrix $\mathrm{A}^{+} = \mathrm{A}^{*\mathrm{T}}$: Hermitian adjoint matrix $\overset{-1}{\pi}(\mathrm{N})$: Original domain of N with respect to the mapping $\pi : \mathrm{M} \to \mathrm{N}$ $\mathrm{T}_{(\mu\nu)} := \frac{1}{2}(\mathrm{T}_{\mu\nu}+\mathrm{T}_{\nu\mu})$: Symmetrized second-rank tensor $\mathrm{T}_{[\mu\nu]} := \frac{1}{2}(\mathrm{T}_{\mu\nu}-\mathrm{T}_{\nu\mu})$: Antisymmetrized second-rank tensor
mielke-geometrodynamics_F01737	364	■ 0.591	`\mathrm{T}_{[\mu \nu]}:=\frac{1}{2}\left(\mathrm{~T}_{\mu \nu}-\mathrm{T}_{\nu \mu}\right)`	$\mathrm{T}_{[\mu \nu]}:=\frac{1}{2}\left(\mathrm{~T}_{\mu \nu}-\mathrm{T}_{\nu \mu}\right)$	$:=$: Equal by definition $\equiv$: Identically equal $\approx$: Isomorphic $\simeq$: Approximately equal $\sim$: Asymptotic expansion $\times$: Topological product $\otimes$: Direct product $\oplus$: Direct sum $\ltimes$: Semidirect product $\mathrm{G}_\circ$: Connected component of the group $\tilde{\mathrm{G}}$: Simply connected covering group $[\mathrm{A},\mathrm{B}] := \mathrm{AB}-\mathrm{BA}$: Commutator $\{\mathrm{A},\mathrm{B}\} := \mathrm{AB}+\mathrm{BA}$: Anticommutator $\mathrm{A}^{*}= \mathrm{C\,A}$: Complex-conjugate matrix $\mathrm{A}^{+} = \mathrm{A}^{*\mathrm{T}}$: Hermitian adjoint matrix $\overset{-1}{\pi}(\mathrm{N})$: Original domain of N with respect to the mapping $\pi : \mathrm{M} \to \mathrm{N}$ $\mathrm{T}_{(\mu\nu)} := \frac{1}{2}(\mathrm{T}_{\mu\nu}+\mathrm{T}_{\nu\mu})$: Symmetrized second-rank tensor $\mathrm{T}_{[\mu\nu]} := \frac{1}{2}(\mathrm{T}_{\mu\nu}-\mathrm{T}_{\nu\mu})$: Antisymmetrized second-rank tensor
mielke-geometrodynamics_F01738	365	■ 0.998	`2 \times 2`	2×2	2×2 Pauli matrices
mielke-geometrodynamics_F01739	365	■ 0.984	`\mathrm{C}(0,2)=\mathbb{H}`	$\mathrm{C}(0,2)=\mathbb{H}$	of the Clifford algebra $\mathrm{C}(0,2)=\mathbb{H}$ (field of quaternions).
mielke-geometrodynamics_F01740	365	■ 0.989	`4 \times 4`	4×4	4×4 Dirac matrices in the so-called Pauli realization:
mielke-geometrodynamics_F01741	366	■ 1.000	`4<2 \mathrm{p}+1<2 \mathrm{p}+2 \leq n=2 l+2`	$4<2 \mathrm{p}+1<2 \mathrm{p}+2 \leq n=2 l+2$	where $4 < 2\mathrm{p}+1 < 2\mathrm{p}+2 \leq n = 2l+2$; cf. KALINOWSKI (1984).

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mielke-geometrodynamics_F01742	366	0.997	`\emptyset`	$\emptyset$	(iii) Both X itself and the empty set $\emptyset$ are open.
mielke-geometrodynamics_F01743	366	1.000	`\mathrm{T}_{1}`	T_{1}	T_1: For each two distinct points $\mathrm{m}_1, \mathrm{m}_2 \in \mathrm{X}$, there exist neighborhoods $\mathrm{U}_\mathrm{i} \ni \mathrm{m}_\mathrm{i}$ of
mielke-geometrodynamics_F01744	366	1.000	`\mathrm{m}_{1}, \mathrm{~m}_{2} \in \mathrm{X}`	$\mathrm{m}_{1}, \mathrm{~m}_{2} \in \mathrm{X}$	T_1: For each two distinct points $\mathrm{m}_1, \mathrm{m}_2 \in \mathrm{X}$, there exist neighborhoods $\mathrm{U}_\mathrm{i} \ni \mathrm{m}_\mathrm{i}$ of
mielke-geometrodynamics_F01745	366	1.000	`\mathrm{U}_{\mathrm{i}} \ni \mathrm{m}_{\mathrm{i}}`	$\mathrm{U}_{\mathrm{i}} \ni \mathrm{m}_{\mathrm{i}}$	T_1: For each two distinct points $\mathrm{m}_1, \mathrm{m}_2 \in \mathrm{X}$, there exist neighborhoods $\mathrm{U}_\mathrm{i} \ni \mathrm{m}_\mathrm{i}$ of
mielke-geometrodynamics_F01746	366	1.000	`\mathrm{T}_{2}`	T_{2}	T_2: For any two distinct points $\mathrm{m}_1, \mathrm{m}_2 \in \mathrm{X}$ there exist neighborhoods $\mathrm{U}_\mathrm{i} \ni \mathrm{m}_\mathrm{i}$ that
mielke-geometrodynamics_F01747	366	0.566	`\mathrm{U}_{1} \bigcap \mathrm{U}_{2}=\varnothing`	$\mathrm{U}_{1} \bigcap \mathrm{U}_{2}=\varnothing$	are disjoint, namely $\mathrm{U}_1 \bigcap \mathrm{U}_2 = \emptyset$.
mielke-geometrodynamics_F01748	367	1.000	`\sigma: \mathrm{M} \rightarrow \mathrm{N}`	$\sigma: \mathrm{M} \rightarrow \mathrm{N}$	A diffeomorphism $\sigma : \mathrm{M} \rightarrow \mathrm{N}$ between manifolds of the same dimension (e.g., $\mathrm{M} \approx$
mielke-geometrodynamics_F01749	367	1.000	`\mathrm{M} \approx`	$\mathrm{M} \approx$	A diffeomorphism $\sigma : \mathrm{M} \rightarrow \mathrm{N}$ between manifolds of the same dimension (e.g., $\mathrm{M} \approx$
mielke-geometrodynamics_F01750	367	1.000	`\mathrm{N} \approx \mathbb{R}^{\mathrm{n}}`	$\mathrm{N} \approx \mathbb{R}^{\mathrm{n}}$	$\mathrm{N} \approx \mathbb{R}^\mathrm{n}$) is an injective mapping for which σ and the corresponding inverse mapping
mielke-geometrodynamics_F01751	367	0.980	`{ }^{-1}`	${ }^{-1}$	$\overset{-1}{\sigma}$ are C^∞ functions, i.e., every partial derivative of σ exists and is continuous in
mielke-geometrodynamics_F01752	367	0.990	`\sigma_{*}: \mathrm{T}_{\mathrm{m}}(\mathrm{M}) \rightarrow \mathrm{T}_{\sigma(\mathrm{m})}(\mathrm{N})`	$\sigma_{*}: \mathrm{T}_{\mathrm{m}}(\mathrm{M}) \rightarrow \mathrm{T}_{\sigma(\mathrm{m})}(\mathrm{N})$	m gives rise to a mapping $\sigma_* : \mathrm{T}_\mathrm{m}(\mathrm{M}) \rightarrow \mathrm{T}_{\sigma(\mathrm{m})}(\mathrm{N})$ within the space of tangent
mielke-geometrodynamics_F01753	367	0.999	`\sigma(\mathrm{s}(\mathrm{t}))`	$\sigma(\mathrm{s}(\mathrm{t}))$	vectors along the curve s(t) or $\sigma(\mathrm{s}(\mathrm{t}))$ if constructed according to the directional
mielke-geometrodynamics_F01754	367	1.000	`e^{f}`	e^{f}	derivative e^f. On the other hand, the transposed mapping of σ_* defines a linear
mielke-geometrodynamics_F01755	367	1.000	`\sigma_{*}`	σ_{*}	derivative e^f. On the other hand, the transposed mapping of σ_* defines a linear
mielke-geometrodynamics_F01756	367	1.000	`\sigma_{*}^{\mathrm{T}}: T_{\sigma(\mathrm{m})}^{*}( \mathrm{~N}) \rightarrow \mathrm{T}_{\mathrm{M}}^{*}( \mathrm{M})`	$\sigma_{*}^{\mathrm{T}}: T_{\sigma(\mathrm{m})}^{*}(\mathrm{~N}) \rightarrow \mathrm{T}_{\mathrm{M}}^{*}(\mathrm{M})$	mapping $\sigma_*^\mathrm{T} : T^*_{\sigma(\mathrm{m})}(\mathrm{N}) \rightarrow \mathrm{T}^*_\mathrm{M}(\mathrm{M})$ between the "dual" cotangent spaces. As such,
mielke-geometrodynamics_F01757	367	1.000	`{ }^{*} \alpha`	${ }^{*} \alpha$	Consequently, $\overset{*}{\sigma}\alpha$ denotes the inverse image of the differential form α. For cross
mielke-geometrodynamics_F01758	368	0.858	`(p, 0)`	$(p, 0)$	Completely *antisymmetric* covariant tensor fields of degree $(p, 0)$ are of prime impor-
mielke-geometrodynamics_F01759	369	0.982	`{ }^{1} \mathrm{df}=\vartheta^{\alpha} \partial_{\alpha} \mathrm{f}`	${ }^{1} \mathrm{df}=\vartheta^{\alpha} \partial_{\alpha} \mathrm{f}$	f, it is nothing but the total differential[1] $\mathrm{df} = \vartheta^\alpha \partial_\alpha \mathrm{f}$. This action is extended to
mielke-geometrodynamics_F01760	369	0.937	`\sigma_{*} \alpha^{(0)}:=\alpha^{(0)} \circ \sigma`	$\sigma_{*} \alpha^{(0)}:=\alpha^{(0)} \circ \sigma$	that for a 0-form is obtained by composition the $\sigma_*\alpha^{(0)} := \alpha^{(0)} \circ \sigma$.
mielke-geometrodynamics_F01761	369	1.000	`\varepsilon_{\alpha_{1} \ldots \alpha_{n}}`	$\varepsilon_{\alpha_{1} \ldots \alpha_{n}}$	Let $\varepsilon_{\alpha_1 \ldots \alpha_n}$ be the totally antisymmetric unit tensor (Levi-Civita tensor; see MTW,
mielke-geometrodynamics_F01762	369	0.999	`\varepsilon_{\hat{1} \ldots \tilde{n}}=+1`	$\varepsilon_{\hat{1} \ldots \tilde{n}}=+1$	p. 87) with the conventional choice of signs $\varepsilon_{\hat{1}\ldots\tilde{n}} = +1$. In a pseudo-Riemannian
mielke-geometrodynamics_F01763	369	1.000	`g_{\mathrm{ij}}`	g_{ij}	manifold endowed with the metric tensor g_ij, this unit tensor allows one to define the
mielke-geometrodynamics_F01764	369	0.997	`\mathrm{df}=\left(\partial \mathrm{f} / \partial x^{i} \right) d x^{i}`	$\mathrm{df}=\left(\partial \mathrm{f} / \partial x^{i} \right) d x^{i}$	to a coordinate basis, it would read $\mathrm{df} = (\partial \mathrm{f}/\partial x^i) dx^i$.
mielke-geometrodynamics_F01765	371	1.000	`\delta \alpha=0`	$\delta \alpha=0$	if and only if it is closed as well as *coclosed*, i.e., $\delta\alpha = 0$.
mielke-geometrodynamics_F01766	371	0.901	`\mathrm{B}(\mathrm{M}):=\mathrm{d} \mathrm{D}(\mathrm{M})`	$\mathrm{B}(\mathrm{M}):=\mathrm{d} \mathrm{D}(\mathrm{M})$	$\mathrm{B}(\mathrm{M}) := \mathrm{d}\,\mathrm{D}(\mathrm{M})$ of *exact* forms
mielke-geometrodynamics_F01767	371	1.000	`\mathrm{B}^{0}(\mathrm{M})=0`	$\mathrm{B}^{0}(\mathrm{M})=0$	On the other hand, we have $\mathrm{B}^0(\mathrm{M}) = 0$. Consequently,
mielke-geometrodynamics_F01768	372	0.995	`\mathrm{H}^{\mathrm{p}}(\mathrm{M})`	$\mathrm{H}^{\mathrm{p}}(\mathrm{M})$	manifolds), the dimensions of the spaces $\mathrm{H}^\mathrm{p}(\mathrm{M})$ are called the *Betti numbers*

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mielke-geometrodynamics_F01769	372	1.000	`\mathrm{t}=-1`	$\mathrm{t}=-1$	with the Betti numbers as coefficients, gives for $\mathrm{t}=-1$ the alternating sum
mielke-geometrodynamics_F01770	372	0.998	`\eta_{\mathrm{i}}`	η_{i}	with a countable basis η_{i}, a p-dimensional standard simplex is defined by the set
mielke-geometrodynamics_F01771	373	0.843	`\mathrm{c}: \Delta_{\mathrm{p}} \rightarrow \mathrm{M}`	$\mathrm{c}: \Delta_{\mathrm{p}} \rightarrow \mathrm{M}$	the C^{∞}-mapping $\mathrm{c}: \Delta_{\mathrm{p}} \rightarrow \mathrm{M}$. These simplices form a free $\mathbb{R}$-module $\mathrm{C}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$,
mielke-geometrodynamics_F01772	373	0.843	`\mathrm{C}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})`	$\mathrm{C}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$	the C^{∞}-mapping $\mathrm{c}: \Delta_{\mathrm{p}} \rightarrow \mathrm{M}$. These simplices form a free $\mathbb{R}$-module $\mathrm{C}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$,
mielke-geometrodynamics_F01773	373	1.000	`\partial \mathrm{c}`	$\partial \mathrm{c}$	whose elements are called p-chains. The boundary $\partial \mathrm{c}$ of a chain will be generated
mielke-geometrodynamics_F01774	373	0.997	`\mathrm{Z}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})=( \operatorname{Ker} \partial)_{\mathrm{p}}, \mathrm{B}_{ \mathrm{p}}(\mathrm{M}, \mathbb{R}):=(\operatorname{lm} \partial)_{\mathrm{p}}`	$\mathrm{Z}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})=(\operatorname{Ker} \partial)_{\mathrm{p}}, \mathrm{B}_{ \mathrm{p}}(\mathrm{M}, \mathbb{R}):=(\operatorname{lm} \partial)_{\mathrm{p}}$	tically." The spaces $\mathrm{Z}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})=(\operatorname{Ker}\partial)_{\mathrm{p}}, \mathrm{B}_{\mathrm{p}}(\mathrm{M}, \mathbb{R}):=(\operatorname{lm}\partial)_{\mathrm{p}}$ as well as
mielke-geometrodynamics_F01775	373	1.000	`\mathrm{N}_{(\mathrm{p})} \subset \mathrm{M}`	$\mathrm{N}_{(\mathrm{p})} \subset \mathrm{M}$	dimensional (compact) submanifold $\mathrm{N}_{(\mathrm{p})} \subset \mathrm{M}$ of M. Consequently, they can be
mielke-geometrodynamics_F01776	373	1.000	`\mathrm{N}_{(\mathrm{p})}`	$\mathrm{N}_{(\mathrm{p})}$	integrated, provided $\mathrm{N}_{(\mathrm{p})}$ admits a simplicial covering by the chain $\mathrm{c}_{\mathrm{p}} \in \mathrm{C}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$.
mielke-geometrodynamics_F01777	373	1.000	`\mathrm{c}_{\mathrm{p}} \in \mathrm{C}_{\mathrm{p}}( \mathrm{M}, \mathbb{R})`	$\mathrm{c}_{\mathrm{p}} \in \mathrm{C}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$	integrated, provided $\mathrm{N}_{(\mathrm{p})}$ admits a simplicial covering by the chain $\mathrm{c}_{\mathrm{p}} \in \mathrm{C}_{\mathrm{p}}(\mathrm{M}, \mathbb{R})$.
mielke-geometrodynamics_F01778	374	1.000	`\mathrm{H}^{\mathrm{P}}(\mathrm{M}) \times \mathrm{H}_{ \mathrm{p}}(\mathrm{M}, \mathbb{R}) \rightarrow \mathbb{R}`	$\mathrm{H}^{\mathrm{P}}(\mathrm{M}) \times \mathrm{H}_{\mathrm{p}}(\mathrm{M}, \mathbb{R}) \rightarrow \mathbb{R}$	ping $\mathrm{H}^{\mathrm{P}}(\mathrm{M}) \times \mathrm{H}_{\mathrm{p}}(\mathrm{M}, \mathbb{R}) \rightarrow \mathbb{R}$ from the Cartesian product of the cohomology and
mielke-geometrodynamics_F01779	375	1.000	`\mathrm{g} \in \mathrm{G}`	$\mathrm{g} \in \mathrm{G}$	(i) the elements $\mathrm{g} \in \mathrm{G}$ form an (analytic) manifold,
mielke-geometrodynamics_F01780	375	1.000	`\mathrm{G} \times \mathrm{G} \rightarrow \mathrm{G}`	$\mathrm{G} \times \mathrm{G} \rightarrow \mathrm{G}$	(ii) the mapping $\mathrm{G} \times \mathrm{G} \rightarrow \mathrm{G}$ originating from the pointwise multiplication
mielke-geometrodynamics_F01781	375	0.999	`\left(\mathrm{g}_{1}, \mathrm{~g}_{2}\right) \rightarrow \mathrm{g}_{1} \mathrm{~g}_{2}^{-1}`	$\left(\mathrm{g}_{1}, \mathrm{~g}_{2}\right) \rightarrow \mathrm{g}_{1} \mathrm{~g}_{2}^{-1}$	$(\mathrm{g}_1, \mathrm{g}_2) \rightarrow \mathrm{g}_1\mathrm{g}_2^{-1}$ of (group) elements is analytic as well.
mielke-geometrodynamics_F01782	375	1.000	`\mathrm{G}=\mathrm{GL}(\mathrm{N}, \mathbb{C})`	$\mathrm{G}=\mathrm{GL}(\mathrm{N}, \mathbb{C})$	The most general example is given by $\mathrm{G}=\mathrm{GL}(\mathrm{N}, \mathbb{C})$, i.e., by the general linear
mielke-geometrodynamics_F01783	375	0.999	`\mathrm{T}_{\mathrm{e}}(\mathrm{G})`	$\mathrm{T}_{\mathrm{e}}(\mathrm{G})$	The tangent space $\mathrm{T}_{\mathrm{e}}(\mathrm{G})$ attached to the unit element $\mathrm{e} \in \mathrm{G}$ of the group possesses
mielke-geometrodynamics_F01784	375	0.999	`\mathrm{e} \in \mathrm{G}`	$\mathrm{e} \in \mathrm{G}$	The tangent space $\mathrm{T}_{\mathrm{e}}(\mathrm{G})$ attached to the unit element $\mathrm{e} \in \mathrm{G}$ of the group possesses
mielke-geometrodynamics_F01785	375	0.952	`\xi=\left\{\xi^{\mathrm{j}}, \mathrm{j}=1, \ldots, \mathrm{~K}\right\}`	$\xi=\left\{\xi^{\mathrm{j}}, \mathrm{j}=1, \ldots, \mathrm{~K}\right\}$	the *Lie algebra* of $\mathfrak{g}$ of G (KN I., p. 38). Let $\xi=\{\xi^{\mathrm{j}}, \mathrm{j}=1, \ldots, \mathrm{K}\}$ be an appropriate
mielke-geometrodynamics_F01786	375	0.943	`\mathrm{c}_{\mathrm{ij}}^{\mathrm{k}}`	$\mathrm{c}_{\mathrm{ij}}^{\mathrm{k}}$	Here the real *structure constants* $\mathrm{c}_{\mathrm{ij}}^{\mathrm{k}}$ are characteristic of the group G. They are always
mielke-geometrodynamics_F01787	376	1.000	`\mathrm{A}=\mathrm{A}^{\mathrm{i}} \mathrm{I}_{ \mathrm{i}}`	$\mathrm{A}=\mathrm{A}^{\mathrm{i}} \mathrm{I}_{\mathrm{i}}$	defined for generic Lie-algebra-valued vectors $\mathrm{A}=\mathrm{A}^{\mathrm{i}}\mathrm{I}_{\mathrm{i}}$ and $\mathrm{B}=\mathrm{B}^{\mathrm{j}}\mathrm{I}_{\mathrm{j}}$ is linear and
mielke-geometrodynamics_F01788	376	1.000	`\mathrm{B}=\mathrm{B}^{\mathrm{j}} \mathrm{I}_{ \mathrm{j}}`	$\mathrm{B}=\mathrm{B}^{\mathrm{j}} \mathrm{I}_{\mathrm{j}}$	defined for generic Lie-algebra-valued vectors $\mathrm{A}=\mathrm{A}^{\mathrm{i}}\mathrm{I}_{\mathrm{i}}$ and $\mathrm{B}=\mathrm{B}^{\mathrm{j}}\mathrm{I}_{\mathrm{j}}$ is linear and
mielke-geometrodynamics_F01789	376	1.000	`\stackrel{\circ}{g}_{l s}`	$\stackrel{\circ}{g}_{l s}$	may be introduced that is determined by the *Cartan–Killing metric* $\stackrel{\circ}{g}_{ls}$ of the Lie
mielke-geometrodynamics_F01790	376	0.863	`\mathfrak{g l}(\mathrm{N}, \mathbb{C})`	$\mathfrak{g l}(\mathrm{N}, \mathbb{C})$	of $\mathfrak{g}$. In the case of the Lie algebra $\mathfrak{g}l(\mathrm{N}, \mathbb{C})$ of the general linear group, this product
mielke-geometrodynamics_F01791	376	0.737	`\mathrm{G}=\exp ^{\mathrm{Trg}}`	$\mathrm{G}=\exp ^{\mathrm{Trg}}$	det $\mathrm{G}=\exp^{\mathrm{Tr}\mathfrak{g}}$. The scalar product (C.12) is nondegenerate if and only if G is
mielke-geometrodynamics_F01792	377	0.995	`\xi^{\mathrm{j}}`	ξ^{j}	algebra. In terms of a real parametrization ξ^{j}, an arbitrary group element is generated
mielke-geometrodynamics_F01793	377	1.000	`\lambda_{\mathrm{j}}`	λ_{j}	Here the λ_{j} denote the generalized Gell-Mann matrices, which are Hermitian and